Combat Quickstart

A soldier which patrols an arena, spots you the moment it has a clear line of sight, chases you, shoots at you, runs to its camp when it is hurt, heals there and then walks back out to its patrol route. Every part of that loop is built from actions and considerations which ship with Wise Feline. There is **no custom AI code in it at all** and going through this page takes about ten minutes.

The scene ships with two of them, and the second one is the point of step 8: it is the same prefab and the same behavior asset, it needed no squad code, no coordination and no second behavior to maintain, and adding it took one drag and one reference. Everything from here to step 7 builds one soldier, because one soldier is where all of the ideas are.

The demo lives in `Assets/NoOpArmy.WiseFeline/Demos/CombatQuickstart` and its folder has a README with the same numbers you see here. If you would rather see it running before reading about it, open `CombatQuickstart.unity` and press **Play**.

> **NOTE**
>
> The chase action finds you through the AI Tags module, which is part of the ultimate edition. Everything else on this page works in every edition, and the chase can be rebuilt on the lite edition with `MoveToGameObjectAction` reading the `Player` key described below, at the cost of the line of sight check which needs a target.

What you will build

Five actions, in three action sets which are there only to keep the list tidy. Every action is interruptable and they all have the same priority, so the score is the only thing which decides what happens next.

Action	Built-in class	Wins when
Patrol	`MoveAndPatrol`	Nothing else scores above 0.25
Chase	`MoveToTargetWithTag`	A tagged player is within 15 m **and** in line of sight, and the soldier is healthy enough
Attack	`CreateGameObjectAction`	The player is inside about 3 m. Firing drains the soldier's own `Health` at 6 per second
Retreat	`MoveToGameObjectAction`	`Health` drops below roughly 35
Rest	`NoopAction`	It is standing at the camp and `Health` is not full yet. Heals at 10 per second

Nothing in the list above is a state or a branch of a tree. Nothing transitions to anything and nothing sits above anything else in an ordering. Each action asks a couple of questions about the world every 0.2 seconds, multiplies the answers into one number, and the brain runs the winner.

The loop is self sustaining without a player in it at all: attacking costs health, low health wins the retreat, resting at the camp refills it, and a full health bar collapses the rest score so patrolling wins again.

1. Open the demo and press Play

1. Open `Assets/NoOpArmy.WiseFeline/Demos/CombatQuickstart/CombatQuickstart.unity`.
2. Press **Play**. Both soldiers start walking their routes immediately, in opposite directions around the same rectangle.
3. Open **Window > NoOpArmy > Wise Feline** and select either `SoldierAlpha` or `SoldierBravo` in the hierarchy.

The Wise Feline window is now a live score board: five actions with the number each of them scored on the last think. Watching that list is the single most useful thing you can do while reading the rest of this page, because every claim below is visible in it as a number.

The rest of this page builds the same thing from an empty scene. If you only want to change the finished demo, skip to Make it yours.

> **TIP**
> If you want to experiment freely without losing the original, duplicate the `CombatQuickstart` folder before you start editing, or reimport the demo from the package.

2. The arena

A gray box is enough. The shapes have to read from above, nothing more.

- A **ground** plane, scaled to 40 by 40 meters.
- Four low **walls** forming a courtyard around the middle of the arena, with a gap at each corner, plus a couple of crates further out. They are 1.6 m tall and sunk 0.1 m into the ground, which matters because the sight ray below runs at foot height and would otherwise graze the bottom edge of a wall instead of hitting it.
- A **PatrolPoints** empty GameObject with four empty children at `(13, 0, 12)`, `(13, 0, -7)`, `(-5, 0, -7)` and `(-5, 0, 12)`. `MoveAndPatrol` walks the children of the object in order, so nothing else may be parented under it.
- A **PatrolPointsReversed** object holding those same four corners in the opposite order. It belongs to the second soldier in step 8 and nothing before that step reads it, so skip it for now if you are building along.

- A **Camp** empty GameObject in a corner at `(-15, 0, -15)`, well away from the route, with a flat colored patch under it so you can see where it is.
- A camera looking down at the arena from `(0, 32, -32)` rotated 42 degrees down.

The route is the whole staging trick of the demo: it runs *outside* the courtyard walls, so while the soldier patrols it usually cannot see the middle of the arena and only gets a sight line through the corner gaps. That is what makes line of sight, rather than distance, decide when the chase starts.

Then set up two user **layers** and bake the navigation mesh:

- Put the walls and the crates on a layer called `Obstacles` (in this project it lands on layer 12) and the player on a layer called `PlayerLayer` (layer 13). The names matter, the indices do not.
- Add a `NavMeshModifier` to every wall and crate with **Override Area** ticked and the area set to **Not Walkable**, otherwise the top face of a 1.6 m crate bakes as a little walkable island.
- Add a `NavMeshSurface` to the parent of the environment, set **Collect Objects** to **Children** and press **Bake**.

The agent itself is a capsule with a `NavMeshAgent` (speed 3.5, **stopping distance 2.5**) and a `NavMeshBasedAIMovement` component whose animator speed parameter is `Speed`. That stopping distance is the most behavior-defining number on the prefab: the chase action is finished when the agent is inside it, so 2.5 m is what "arrived at attack range" means.

All of the geometry sits under a child called `Visual`, and the `Animator` sits on that child too. Colliders, the NavMeshAgent, the blackboard and the brain stay on the root, so a real model can replace the placeholder capsule later by swapping one child. The animator controller declares the triggers the built-in actions send (`Walk`, `Run`, `Shoot`, `Rest`) and the float the movement component drives (`Speed`), so a rigged model with its own clips starts animating with no changes to the AI.

3. The blackboard and its starting values

Everything the AI knows about the world is four keys on one blackboard.

Create a blackboard definition with **Create > NoOpArmy > Wise Feline > BlackBoard Definition** and add:

Key	Type	Used by
`Health`	Float	The retreat and rest curves, the attack drain, the rest heal and the demo's damage script
`PatrolPoints`	GameObject	`MoveAndPatrol` reads the children of this object as its route

Key	Type	Used by
`Camp`	GameObject	`MoveToGameObjectAction` walks here, and the rest action measures the distance to it
`Player`	GameObject	The attack action's distance gate

Add a `BlackBoard` component to the soldier and assign the definition to it.

Keys start at `0` and `null`, so all four need starting values, and three of them are references to objects which only exist in the scene. An asset cannot reference a scene object, so the behavior asset's own changer cannot do this. Add the `BlackboardInitialValues` component to the soldier and fill in four entries:

Key	Type	Value
`Health`	Float	`100`
`PatrolPoints`	GameObject	drag the `PatrolPoints` object in
`Camp`	GameObject	drag the `Camp` object in
`Player`	GameObject	drag the player in

> **WARNING**
> Every one of those entries is a **set**, not an add. `Add Instead Of Set` and `Multiply by deltaTime for adds` are **on by default**, because the same entry type is also used for the per frame drain and heal further down this page. Leave them on and "start at 100 health" quietly becomes "add 100 times one frame's deltaTime to whatever is there already", which is a soldier that starts the game with about 1.7 health and immediately retreats.

4. Tags: how the soldier finds the player

The chase action does not have a reference to the player. It asks the tag system what is nearby.

- Add an `AITags` component to the player with the tag `Player`, and put the player's **collider on the same GameObject as that component**. `LineOfSightConsideration` compares the transform its ray hits against the transform of the target it was given, and those two only match when the collider and the tags are on one object.

- Add an `AITags` component to the soldier with the tag `Soldier`. Nothing in this demo searches for it, but tagging both sides is what lets you add a second agent later without touching anything else.
- Add an `AITagsManager` to the scene and set its world size to 100. It creates itself on demand, but a scene which uses tags should own one so its size is visible.

5. The behavior asset and the five actions

Create the behavior with **Create > NoOpArmy > Wise Feline > Agent Behavior**, open **Window > NoOpArmy > Wise Feline**, and add three action sets called `Patrol`, `Combat` and `Survival`. Then add the five actions with the values below. Leave **Is Interruptable** on and **Priority** at 0 everywhere: this behavior is decided entirely by scores.

Patrol (`MoveAndPatrol`, set *Patrol*)

Field	Value
Game Object Key Name	`PatrolPoints`
Finish After One Cycle	off
On Start Animation Trigger	`Walk`

Chase (`MoveToTargetWithTag`, set *Combat*)

Field	Value
Radius	`15`
Included Tags	`Player`
On Start Animation Trigger	`Run`

Attack (`CreateGameObjectAction`, set *Combat*)

Field	Value
Object To Create	the tracer prefab
Object Count To Create	`3`
Creation Delay	`0.4`
Create At The Same Time	off

Field	Value
Is Position Relative	on
Creation Position	`(0, 1.2, 0.6)` - the muzzle, in the agent's own space
Should Adapt Brain Rotation	on - the tracer inherits the agent's facing
On Start Animation Trigger	`Shoot`
Weight	`1.5`
Reset Changer On Start	on
Changer	every frame, `Health` **add** `-6`, multiply by deltaTime on, clamped to 0..100

The changer is the engine of the loop: shooting costs the soldier six health a second, so a firefight it never leaves will eventually push it into a retreat with no player involved at all. Think of it as a stand-in for taking incoming fire, and read Make it yours for how to replace it with real damage.

The weight of `1.5` is the one number here which is not a curve, and it is load bearing. An action's score is compensated for how many considerations it has by a factor of `1 - 1/considerationCount`, so Chase's three considerations get a bigger lift out of the same health value than Attack's two do. Without the weight, Chase overtakes Attack at about 64 health while the soldier is already standing at contact range, and because a chase which is already inside its stopping distance finishes instantly, the soldier would lock into a chase it has already completed and stop firing. The weight restores the intent - at contact range, shooting beats walking - without touching priorities or interruptability.

Retreat (`MoveToGameObjectAction`, set *Survival*)

Field	Value
Game Object Key Name	`Camp`
On Start Animation Trigger	`Run`

Rest (`NoopAction`, set *Survival*)

Field	Value
Stop Moving	on

Field	Value
On Start Animation Trigger	`Rest`
Changer	every frame, `Health` **add** `+10`, multiply by deltaTime on, clamped to 0..100

Finally add a `Brain` to the soldier, assign the behavior, and set **Think Duration** to `0.2` and **Update Targets Duration** to `0.5`. The brain re-scores five actions five times a second and refreshes the chase's target list twice a second, which is cheap and makes the reactions feel immediate.

6. The considerations and their curves

This is the part which actually decides the behavior. Nine considerations across five actions, each one a value read from the world, normalized between a min and a max range, and then passed through a curve.

Patrol - the baseline

Consideration	Need Target	Range	Curve
`ConstBoolConsideration`	off	-	`isTrue` on, `trueValue 0.25`

Why: patrol is the floor. It always scores exactly 0.25, so every other action on this page is really answering one question: "am I worth more than wandering around?"

Chase - the detection rule

Consideration	Need Target	Range	Curve
`TargetDistanceConsideration`	**on**	0..15, **inverse**, XZ plane	linear
`LineOfSightConsideration`	**on**	-	mask = `Obstacles` + `PlayerLayer`
`BlackboardFloatConsideration` (`Health`)	off	0..100	flat 0 up to 25, then rising to 1

Why the distance curve: inverted and capped at 15 meters, so a player at 15 m scores 0 and a player at 3 m scores 0.8. This consideration **is** the detection radius. There is no separate sensing system anywhere in this demo.

Why line of sight: it raycasts from the agent to the target and returns 1 or 0, so a player behind a wall multiplies the whole action to zero no matter how close they are. The mask contains the obstacles **and** the player's own layer, and that second part is not optional: the ray runs to infinity, so without the player's layer in the mask an obstacle standing *behind* the player would be the first thing hit and the soldier would report no sight line at a player in plain view. The ground stays out of the mask because the ray starts at the agent's pivot, which is at its feet. See LineOfSightConsideration for the full set of rules.

Why the health curve: a soldier at 20 health should not start a fight. The curve is flat 0 below 25 health, which multiplies the entire chase away.

Attack - only at contact

Consideration	Need Target	Range	Curve
`BlackboardGameObjectDistanceConsideration` (`Player`)	off	0..3.5, **inverse**, XZ plane	near 0 at 3.5 m, about 0.55 at 3 m, 0.93 at 2.5 m
`BlackboardFloatConsideration` (`Health`)	off	0..100	the same rising curve as the chase

Why: the curve is deliberately steep rather than a ramp. It is about 0.93 at the 2.5 m the chase's stopping distance delivers the soldier to and it falls off a cliff past 3 m, which is what makes "attack" mean adjacent rather than merely near. The distance is read from the `Player` blackboard key rather than from a target, because this action has no target list of its own. Aim comes from the chase having walked the agent into the player's face, plus **Should Adapt Brain Rotation** on the spawn.

Retreat - one curve which outbids everything

Consideration	Need Target	Range	Curve
`BlackboardFloatConsideration` (`Health`)	off	0..100, **inverse**	exactly 0 above 50 health, 0.92 at 25, 1 at 0

Why: one consideration means no compensation at all, so the curve value *is* the score. It stays at a hard zero while the soldier is healthy, passes the 0.25 patrol baseline at about 40 health, and beats a point blank firefight at about 35. This one curve is the entire "disengage when hurt" behavior, and it interrupts whatever else was running because every action here is interruptable.

Rest - closes the loop

Consideration	Need Target	Range	Curve
`BlackboardGameObjectDistanceConsideration` (`Camp`)	off	0..4, **inverse**, XZ plane	a plateau: about 0.9 from 3.4 m in, and 1 at the camp itself
`BlackboardFloatConsideration` (`Health`)	off	0..100, **inverse**	about 0.97 by 85 health, 0 at 100

Why the camp curve is a plateau and not a ramp: arriving anywhere inside the camp should count fully, otherwise the heal rate would depend on exactly where the NavMeshAgent happened to stop.

Why the health curve collapses at full health: that collapse is what ends the rest. Once `Health` is back at 100 the rest score drops under the 0.25 baseline, patrol wins again, and the soldier walks back out to its route. The loop closes itself.

7. Press Play and fight the soldier

Press **Play** with the Wise Feline window open and the soldier selected. You can watch the whole loop happen without touching anything: the soldier patrols at 0.25, the corner gap opens a sight line, chase climbs past the baseline, the soldier arrives and attack takes over, health drains at six a second, and somewhere below 35 the retreat curve spikes and takes the fight away from it.

But you can also just shoot it.

Control	What it does
W A S D or the **arrow keys**	Move at 6 m/s. Movement is relative to the world, because the camera is fixed
Mouse	Aim. The player turns towards wherever the cursor crosses the ground
Hold left mouse button	Fire a cyan tracer every 0.25 s, 8 damage each

The soldier fires amber tracers back at 10 damage a hit, and neither side can hit itself: a tracer passes straight through anything on its own team, including the shooter it spawned in front of. The player respawns at the starting spot when its health runs out, so the demo cannot end. The soldier has no death at all - it retreats and heals, which is the loop this demo is about.

Forcing the retreat. Nine hits, which is about two and a half seconds of holding the mouse button, puts the soldier under 35 health and the retreat curve takes over while you are still shooting at it. This is the part worth stopping on, because of what did *not* have to happen for it to work:

- No AI change of any kind. Not an action, not a consideration, not a curve, not a flag.
- No event, no message, no "on damaged" hook anywhere in the behavior.
- Nothing in the behavior asset knows that a player, a weapon or a projectile exists.

The demo's `QuickstartHealth` component subtracts the damage from the soldier's `Health` key and writes it straight back with `SetFloat`. That is the whole integration. The retreat and rest curves were already reading that number, so moving it is all it takes.

> **NOTE**
>
> The three scripts in the demo's `Scripts/` folder - the player controller, the tracer and the damage bridge - are **not AI code**. Delete all three and the behavior still runs exactly as it does now, with the soldier fighting nobody and the attack drain doing the job your bullets were doing.

8. The second soldier

Everything above is one agent. Here is what a second one costs.

1. Drag the soldier prefab into the scene again, at `(-5, 0, 14)` with a Y rotation of 180.
2. On its `BlackboardInitialValues`, point the `PatrolPoints` seed at `PatrolPointsReversed` instead of `PatrolPoints`.

That is the entire procedure. Same prefab, same behavior asset, same blackboard definition, same five actions and nine considerations. The one thing that differs between the two soldiers is which route object one seed points at.

Nothing had to be made safe for this. `Brain` clones the behavior asset in its own `Awake`, so each soldier evaluates its own copy of the five actions against its own blackboard: its own `Health`, its own target list, its own line of sight ray. Two agents sharing an asset never touch each other's state, and there is no step where you tell the framework how many agents there are going to be.

Press Play and watch a corner gap. The two walk the same rectangle in opposite directions, so they reach the gaps at different moments, and what comes out of that was not authored anywhere:

- One is mid firefight while the other is still patrolling twenty metres away, because a sight line opened for one of them and not the other.
- One breaks off to heal at 35 health and the other keeps shooting, because health is per agent and so is every score read off it.
- They cross twice a lap and ignore each other completely. Their tracers pass through each other too - the team filter on `QuickstartProjectile` is what makes the muzzle spawn safe, and it makes friendly fire impossible for free.

None of that is emergent in the hand-wavy sense. It is two agents reading the same world, at different positions, and reaching different conclusions about it - which is what a utility agent does whether there is one of them or fifty.

Where this stops. Two soldiers are two independent decisions, not a squad. Nothing here flanks, covers or calls for help, because utility AI arbitrates one agent's options and says nothing about coordination between agents. Shared intent needs shared data: a blackboard both of them read, or an influence map one of them writes into and the other scores against.

Adding more of them is the same two steps and scales as far as your NavMesh does - but note what they would all be doing. Every soldier here hunts the same player, so twenty of them is a crowd converging on you, not a battle. Two sides shooting at *each other* is a different demo: it needs an action which fires at the target the tag system chose, and the ones in the box today either move to a tag target (`MoveToTargetWithTag`) or spawn at the agent's own rotation (`CreateGameObjectAction`), with nothing joining the two.

Why utility AI made this easy

It is worth counting what the two usual alternatives would have cost here, a state machine and a behavior tree, because the difference is not a matter of taste.

Five behaviors as states is up to twenty directed transitions to author and maintain, and every one of them is a place to put a bug. Here there are no transitions at all: five actions, nine considerations and one comparison, highest score runs. A sixth state means edges from and to every existing one - the classic "why is this graph a hairball" moment. And "run away when you drop below 35 health" is a guard duplicated on every state which can lose health, or a global transition with its own priority rules; here it is one curve which outbids the others when the number is low, cutting short a patrol, a chase and a firefight identically because none of them are special.

A behavior tree is the closer comparison, and it is the genre default for a reason: this soldier fits one - a root selector with Retreat, then Combat, then Patrol, a condition decorator on each branch, and a sequence inside Combat which chases and then shoots. What it does not get for free is interruption: `MoveAndPatrol` with **Finish After One Cycle** off never finishes, so the patrol branch never returns and the root never looks at the world again on its own. Cutting one short needs a decorator which keeps watching its condition and aborts - observer aborts, reactive conditions and interrupt decorators are one idea under three names - plus a choice of how far the abort reaches: its own subtree, the lower priority branches beside it, or both.

That scope choice is the most notorious source of behavior tree bugs, and both mistakes are quiet: too narrow and the soldier finishes its three tracer burst before anything re-reads its health, too wide and a branch restarts from its first node and replays its entry effects. Here none of it was authored. Every action is interruptable and all five are re-scored every 0.2 seconds, so the retreat curve climbing past a point blank firefight *is* the interruption.

Priority in a tree is also positional and global, so a sixth behavior - the `Reload` action suggested in Make it yours - means picking its slot in that one left to right ordering and re-reading every abort scope it crosses. Adding one here is adding one action: its own considerations, the same priority 0, nothing else touched.

And a decorator is a gate: below 35 health, yes or no. Most of this behavior has no yes or no in it. A soldier at 40 health is *somewhat* willing to fight, and the same three meter gap is worth less to it at 40 health than at 90, because the attack score is distance multiplied by health rather than gated by it. The 35 itself is not written down anywhere; it is where the retreat curve crosses the attack score. Trees get there by bolting scoring onto themselves, which is why utility selectors exist in behavior tree land.

The player fighting back needed zero AI changes. Damage moves a float that the curves were already reading, so there was nothing to add: no new state, no new branch, no combat mode, no "under attack" flag, no event plumbing. `QuickstartHealth` is one `SetFloat` call and the behavior reacted to it correctly the first time it ran. That is the practical version of the utility AI pitch: the AI reads the world, so anything which changes the world is already an input.

Tuning is dragging curves. "Retreat earlier" is one keyframe dragged left in a curve editor, and the next Play session tells you in the score column whether you were right. In a state machine it is a constant in code, a recompile and a pass over every guard which compares against the same threshold, checking that none of them now contradict each other; in a tree it is a number in a decorator and a re-check of whether the branches it gates still abort each other the way you meant. The one thing here which was not a curve was the attack's weight of `1.5`, and even that is a slider: two actions shared a consideration and the compensation tilted the comparison, so the fix was a number in the inspector, not a special case in a transition table or a branch ordering.

The honest limit: utility AI arbitrates, it does not sequence. "Open the door, *then* walk through, *then* close it" is a sequence node and three children, and a column of scores says nothing of the sort. Staged plans whose steps have to run in a fixed order, and be read back in that order a year later, are what behavior trees are genuinely better at. The two mix happily: score the decision with Wise Feline, run the sequence inside the action. What you get here is everything above that: the choosing, which is the part that makes an agent look like it is thinking.

Make it yours

Three changes, in increasing order of how much they teach you.

Move a curve. Select the retreat action's health consideration and drag the keyframe where the curve leaves zero from `0.5` to `0.4` on the x axis. The consideration is inverted, so x is "how hurt am I" and that one drag means the soldier now starts thinking about leaving at 60 health instead of 50. Press Play: it breaks off after a couple of hits and spends most of its life walking to and from the camp. Push the same keyframe out to `0.8` instead and it fights down to almost nothing. Neither version involved code, and the score column in the Wise Feline window shows you exactly what the new curve did.

Swap the health drain for your damage system. The attack action's `-6` per second changer is a placeholder for taking incoming fire. Delete it and write your real damage into the same key instead:

```
// Anywhere in your damage code, on a GameObject which has a BlackBoard.
var board = GetComponent<BlackBoard>();
board.SetFloat("Health", Mathf.Clamp(board.GetFloat("Health") - amount, 0f, 100f));
```

That is the entire contract, and it is exactly what the demo's `QuickstartHealth` does. Every curve on this page keeps working, because none of them care where the number came from.

Swap proximity and line of sight for your own senses. The chase action's detection is two considerations: a distance and a raycast. If your game has a vision cone, a stealth value or a noise system, write one consideration that returns it, put it on the chase action next to the others, and delete the ones you replaced. Considerations multiply, so adding a sense is adding a factor and never a rewrite.

Other things worth trying: give a soldier a second patrol route and add a `BlackboardBoolConsideration` to pick between them, add a `Reload` action with a low weight and a `NoopAction` body, or duplicate `SoldierBravo` a few more times - step 8 is the whole procedure and it does not get longer.

Where to go next

- Considerations and Actions - how to write your own, and how scores are actually calculated.
- Built-in actions and considerations - the full list of what shipped in the box, including everything used on this page.
- Blackboards - the data layer, and starting values without code.
- The targeting system - what `Need Target` really does and how the best target is chosen.
- Influence maps - when "where should I go" needs a better answer than the nearest tagged object.
- The demos - the life simulation demos, which are the same ideas at ten times the scale.
- Overview - the utility theory behind all of it, now that you have seen it work.

Welcome to Wise Feline

Wise Feline is a utility AI system. Wise Feline allows you to make an AI which feels unscripted and immersive and allows you to do this in a fraction of the time it takes to do AI in a scripted and sequential manner. Wise Feline uses Utility theory and by extension the Utility AI approach championed by Dave Mark in the games industry which is used in games as big as The Sims and Guild Wars 2.

Philosophy

An AI system usually needs to perform two functions, listing the actions which can be taken and choosing an action from the list to execute. Utility AI does this by giving all possible actions a score between 0 and 1 and then chooses the action with the highest score.

Each action has a set of considerations which estimate the action's utility at the moment. Each consideration has a score between 0 and 1 based on the current parameters supplied to it. Each action's score is calculated by calculating the score of all of its considerations and then multiplying them by each other.

This is how humans take decisions. For example if you have actions below:

- Buying food
- Eating food
- Having a rest

Then the eating food action probably would have two considerations for calculating if the agent is hungry or not and also calculating if the agent has food or not. The consideration for being hungry is a continuous value between 0 and 1 but the having food available consideration is a binary which is either 0 or 1. The score of the action is calculated by multiplying these two so when the agent doesn't have food, the score for eating food will always be 0 since you don't have any food to eat.

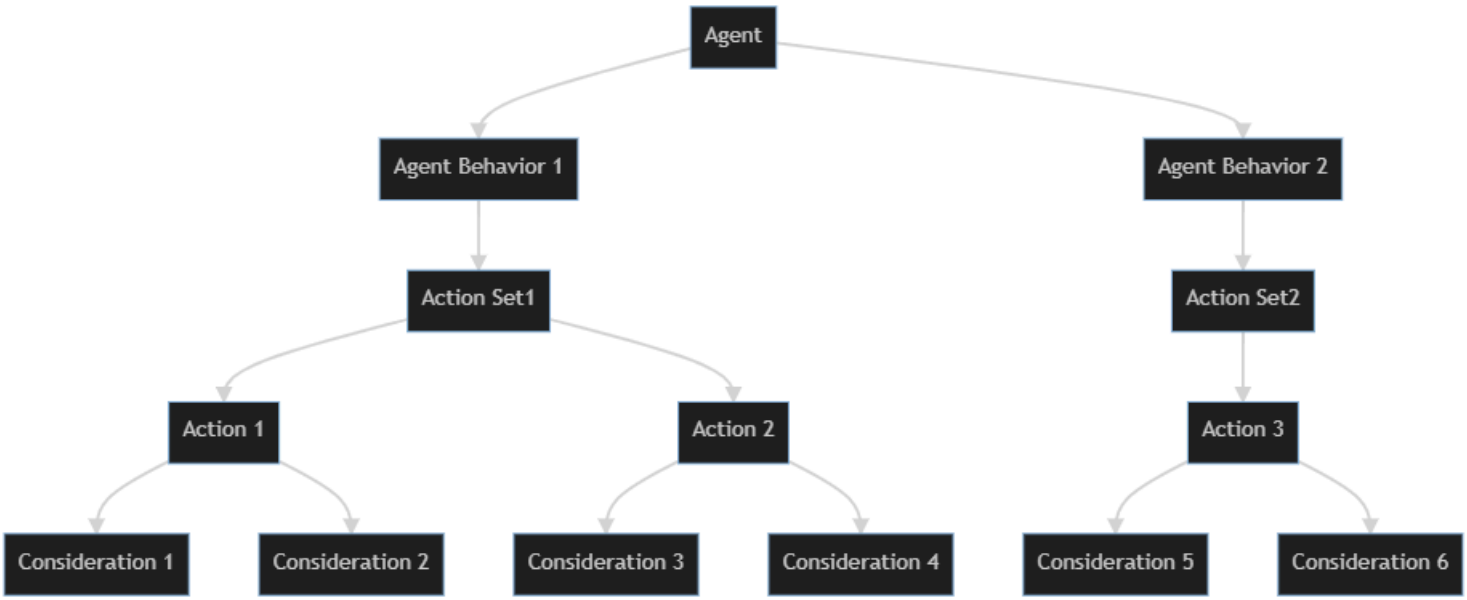
Buying food on the other hand has considerations for not being tired and not having food and being hungry and these should return their values in a way that if you are very hungry, the score becomes high even if you are tired. Resting action probably should only give a high score if you are tired but not that hungry.

Each consideration is a function of a curve whose x axis is the value that we read from the agent/environment and its y axis is the output score of the consideration. So for hunger we read the hunger value of the agent and return it and then its value is given to a curve as x axis and the resulting y axis is the actual score of the consideration. In this way we can make a consideration more sensitive to certain values so for example having a hunger of 0.5 can return 0.5 in the case of a linear curve and 0.9 or 0.1 in the case of custom curves.

These curves for considerations are the main way that designers make the agent react the way they want. For example if you want the agent to go find some food and eat no matter what if the hunger goes above 0.9, you add a consideration to buying food and having food which gives the score of 1 back if hunger is above 0.9 and you might even give a reverse consideration to actions which are not important when you are hungry so as hunger goes up it returns a lower score. The same hunger consideration but with different curves can be used for the purpose in different actions.

How the system works

This is the architecture diagram of the system



Any game object which wants to be controlled by utility AI should have a brain component attached. This component executes the main utility AI logic by executing all action considerations once in a while and choosing a high scoring action to do. Then it calls callbacks of the chosen action which can drive the agent in the world. The action code has access to components of the game object because the brain component of the game object is passed to it in its `OnInitialized()` callback.

Actions which an agent can do are organized inside action sets and have their own considerations listed under them with custom curves per action. A group of action sets themselves are stored in an asset

called Agent Behavior which is what you attach to a brain to use.

So the root of the tree is the brain component which has an Agent Behavior asset attached and then each agent behavior contains action sets as its children/branches and each action set contains a set of actions which contains their own considerations.

You define actions and considerations by deriving classes from ActionBase and ConsiderationBase.

To create each agent's AI you:

- Create an Agent Behavior by right clicking in the project view and choosing Create>NoOpArmy>Wise Feline>Agent Behavior.
- Name it whatever makes sense.
- Select the agent behavior asset you just created.
- Open the Wise Feline window by going to Window>NoOpArmy>Wise Feline window.
- Now add actions and considerations and action sets to the behavior.

Different pages of this manual describe each of these items in detail. Read them and come back to this for the bigger picture again to understand it well. There is also a demo which helps you by showing how to implement a semi-realistic sample.

Utility nodes in behavior trees

Behavior tree assets increasingly ship a utility selector or a utility evaluator node: a composite which asks its children for a score and runs the highest scoring one instead of taking them in order. If you already have one of those, the fair question before buying anything else is what a system where everything is scored adds on top of it.

This page answers exactly that question and nothing more. The theory is in the overview and the running proof is the Combat Quickstart.

What a utility node gives you

It is a real improvement over a fixed left to right ordering, and it is worth saying so plainly.

Inside that one node the children stop being a priority list. A score can be a continuous quantity instead of a yes or no, so "the more hurt I am the more I want cover" replaces "below 30 health, take cover", and adding a sixth child means writing another scoring function rather than finding the right slot for it in an ordering. Most of what is written about utility AI is true inside that node.

Two things bound it. The scoring is local: it decides between the children of one composite, at one point in one tree, and every other decision the agent makes is still the tree's structure. And the score usually comes from code, a method on each child returning a number, which is a fine place to start and the place tuning eventually has to leave.

What is different when the whole brain scores

Wise Feline has no tree for the scoring to sit inside. A Brain holds a list of actions, all of them are scored on every think, and the winner runs. Everything below follows from that one difference.

Curves are the authoring surface, not code. A score is not a number a method returns. Each consideration reads one value from the world, normalizes it into 0..1 against a min and max you set, and passes it through its own response curve; the action's score is those curves multiplied together. The curve is an editable graph in the Wise Feline window, so "retreat earlier" is a keyframe dragged left rather than a constant edited and recompiled. The same consideration class is also reused with a different curve on each action: one hunger value with one shape sends an agent looking for food, and with the inverse shape stops it from starting a fight.

The scores are visible while the game runs. Select an agent in Play mode with the window open and you get every action set, action and consideration with the number each of them scored on the last think, sortable by score. Debugging a decision is reading the column which produced it, rather than instrumenting a tree to find out why a branch was never taken.

Scoring picks the target too, not just the action. Any consideration can be marked `NeedTarget`. The action is then scored once per candidate in its target list and keeps the best one, so "which enemy do I shoot"

is the same multiplication as "do I shoot at all", and an action can lose to a different action because its best available target was poor. See the targeting system.

Interruption is not wired anywhere. Every action is re-scored on every think, so an action stops as soon as something else scores higher. There is no abort, observer or reactive decorator to configure and no scope to choose for it. The case which is most awkward in a tree is a low health rule which has to cut short a patrol, a chase and a firefight identically. Here that is one curve which outbids the others, because none of those three is special. Actions which must not be cut short opt out with IsInterruptable, which makes uninterruptable the exception rather than the wiring.

Anything which writes the world is already an input. Considerations can read blackboard keys, so a value written by your damage code, a pickup, a dialogue system or another agent becomes an AI input the moment it is written, with no event, hook or flag on the AI side. In the quickstart the player's bullets call a single `SetFloat` on a `Health` key, and the retreat is produced by curves which were already reading that number. Nothing in the behavior asset knows that a player, a weapon or a projectile exists.

The world model plugs into the same multiplication. The modules are not separate systems standing beside the AI. Influence maps provide a tactical value which a consideration scores and whose result it hands to the action through a blackboard key. Smart objects contribute the distance to a usable slot and whether it is free. AI tags supply the candidate lists that targeted scoring runs over. Memories and emotions can score how recently something happened. Every one of them ends up as one more consideration in the same product.

Grouping is a folder, not a position. Action sets organize actions in the editor and can be added to and removed from a Brain at runtime, so a jungle-specific or vehicle-specific set of choices enters and leaves the comparison as a unit. That changes which actions are scored and never which one is scored first, because nothing is first.

See it decide

The Combat Quickstart is the ten minute version of everything above. A soldier patrols, spots the player through a line of sight check, chases, shoots, retreats when hurt and heals at its camp: five actions, nine considerations, no custom AI code, and a live score column which explains every one of those changes as it happens. Then shoot the soldier yourself and watch it break off, with no AI change of any kind behind it.

Where a tree is still the better tool

Utility AI arbitrates; it does not sequence. "Open the door, then walk through, then close it" is a sequence node and three children, and a column of scores says nothing of the sort. Plans whose steps have to run in a fixed order, and be read back in that order a year later, are what behavior trees are genuinely good at.

Which is why this is not an argument for deleting your tree. The two mix happily: score the decision with Wise Feline and run whatever sequence you like inside the action it chose.

Considerations

Considerations video tutorial

Each action's score is calculated by calculating scores of all of its considerations and then multiplying them with each other. Considerations are defined in code by deriving from `ConsiderationBase` class. Like components each consideration should be in a file which shares the same name with the consideration class.

After defining a consideration in code, you can add it to actions of an Agent Behavior asset in the Wise Feline window.

To add considerations to actions

- Implement a consideration in a class. For example create a HungerConsideration.cs file and implement `HungerConsideration` inside it like this. The example code is posted below.
- Select the desired agent behavior asset and open the Wise Feline window at *Window>NoOpArmy>Wise Feline*.
- Select the action which you want to add the consideration to by clicking on it. You might need to select the appropriate action set first.
- Now in the considerations list for the action click Add Consideration. A searchable list of the consideration classes in your project opens, grouped by namespace. Type a part of the name to filter it and choose `HungerConsideration`. Abstract classes are not in the list because they cannot be instantiated.
- Select the consideration and set the curve and the parameters as you desire in the properties panel on the right.

The consideration can be defined like this:

```
public class HungerConsideration : ConsiderationBase
{
    private CatBehavior behavior;

    protected override void OnInitialized()
    {
        base.OnInitialized();
        behavior = Brain.GetComponent<CatBehavior>();
    }

    protected override float GetValue(Component target)
    {
        return behavior.Hunger;
```

```
    }
}
```

`OnInitialized()` is called when the consideration is initialized after the `Brain` component is created or the agent behavior asset is added to a brain at runtime.

`GetValue()` is called whenever the action wants to calculate its score.

What you return in GetValue should be a number between 0 and 1 (both inclusive) which will be fed to the curve of the consideration as the value for x axis and the y axis value at that point is the score of the consideration used in score calculations.

Managing considerations in the window

A consideration whose NeedTarget checkbox is checked shows a "T" tag on its row, between the name and the right edge. It is there while you are authoring the behavior and not only while the game runs, so the list tells you which considerations are scored once per target of the action and which ones are scored for the agent itself. Hovering the tag says the same in words.

A consideration you just added, duplicated or pasted becomes the selected row and is scrolled into view, so you can set its curve and its fields without clicking it first. Deleting a row selects its neighbour, the row under it when there is one and the row above it otherwise.

To remove a consideration, select it and press the Delete key, or right click it and choose Delete. To rename it, double click its row or press F2, type the name and press Enter; Escape cancels the rename. The Name field in the properties panel renames it as well, and the sub-asset under the behavior asset in the project view follows the name. Ctrl+D duplicates the selected consideration and the up and down arrow keys move the selection within the column.

Drag a consideration up or down to reorder it inside its action; a line shows where it will land. The right click menu has the same operations plus Move Up and Move Down, and Copy and Paste, which work between two behavior assets so you can reuse a consideration you already tuned. Every one of these edits is written to the asset right away and can be undone with Ctrl+Z.

For more information, see the API reference for the ConsiderationBase class.

Actions

Actions Video tutorial

Actions are the most important part of a utility AI system alongside considerations. An action is a concrete and most of the time atomic unit of work which an agent can do. Shoot, Find health pack, Find cover, Take cover in the found cover, eat food and Choose the best camera to shot the scene with are all examples of actions. However an action doesn't have to be low level and can be pretty high level, you can have a hierarchy of them in your game and for example have a set of actions for groups of NPCs. They can be things like: attack the target, spread members in the room, defend sensitive points, find weak spots to bomb every place and then some NPC actions which take command from the group so when the group chooses to attack the target, a member NPC might execute its shoot action or move toward target action or even cover the attackers action.

To create an action and add it to an action set in an agent behavior asset:

- Create a class derived from ActionBase.
- Name the action class and its file the same.
- Select an agent behavior asset and open the *Window>NoOpArmy>WiseFeline* window.
- Select an action set and then click Add Action. A searchable list of the action classes in your project opens, grouped by namespace. Type a part of the name to filter it and choose the action you created. Abstract classes are not in the list because they cannot be instantiated.

Here is an example action for going toward a food which should be saved at `GoTowardFood.cs`

```
public class GoTowardFood : ActionBase
{
    private CatBehavior behavior;

    protected override void OnInitialized()
    {
        base.OnInitialized();
        behavior = Brain.GetComponent<CatBehavior>();
    }

    protected override void OnStart()
    {
        base.OnStart();
        behavior.GetComponent<MeshRenderer>().material.color = Color.cyan;//for
visualization purposes only
    }

    protected override void UpdateTargets()
    {
```

```
        ClearTargets();
        AddTargets(FoodManager.instance.foods);//FoodManager has the list of all foods in
the world
    }

    protected override void OnUpdate()
    {
        base.OnUpdate();
        behavior.IncreaseHunger(0.01f * Time.deltaTime);
        behavior.IncreasePlayfulness(0.05f * Time.deltaTime);
        var foodPos = ChosenTarget.transform.position;//chosenTarget is the best food target
found by scoring
        behavior.MoveTowardPosition(foodPos);
    }
}
```

There are several things to note:

First there are a set of callbacks in an action which you can override. `OnInitialize()` is called as soon as the action is added to the brain or in the first frame if the behavior is in the brain when the game object is created. OnUpdate, OnLateUpdate, OnFixedUpdate, OnStart and .. are called in the same `MonoBehaviour` callbacks from the brain.

Secondly there are two types of actions, targeted and non-targeted. Targeted ones are those which work on a target like shoot and eat pack. This action is targeted. Non-targeted ones are those which don't have a target like idle or useHealthPackFromInventory. To make an action targeted, check the `NeedTarget` checkbox in at least one of its considerations. To learn more about this check the targetting system page.

To say more about targetting, If the action is targeted then the `chosenTarget` field has the best target in it which you should shoot at, eat or move toward or whatever which is right for your action. An action becomes a targeted one if at least one of its considerations are targeted and have the NeedTarget checkbox checked in the Wise Feline window. A targeted action means all considerations are calculated for all targets which you gave to the action in the `UpdateTargets()` callback and the one with the highest score is in the `chosenTarget` field. `UpdateTargets()` is called once per update target operation whose frequency is chosen in the `brain` component. The scores are calculated once per think operation whose frequency is again chosen in the brain component. Here you just define how the action will execute and that's it.

As you can see, to access the component attached to your game object, you can get them from the `brain` component of your game object which is passed to you in `OnInitialize()`. When filling the list of targets, you can fill it with whatever component type and you can find them using any method you want. Unity's own raycasting and overlap queries work fine for simpler games and samples, but for games with huge worlds and lots of agents, you should probably use custom data structures. We will probably

release helper modules for these later on. You don't have to clear and rebuild the list of targets every time either but in that case you should be careful to not add something multiple times and also to remove objects which are not good target candidates as well. Also if your action does not need any targets, you don't need to add any targets in `UpdateTargets()` but still have to implement an empty one.

Managing actions in the window

Clicking an action selects it, lists its considerations in the third column and shows the action in the properties panel on the right. An action you just added, duplicated or pasted becomes the selected row and is scrolled into view, so you can rename it or set its fields without clicking it first. Deleting a row selects its neighbour, the row under it when there is one and the row above it otherwise.

Between the name and the right edge of an action row there is a small tag which carries the priority and the weight of the action when they are not at their defaults, because neither of them is visible anywhere else in the three lists and both of them change which action gets to run. An action whose priority is not 0 shows "P" and the priority and an action whose weight is not 1 shows "W" and the weight, so an action with a priority of 2 and a weight of 0.5 reads "P2 W0.5". An action left at the defaults has no tag at all. The tag is there while you are authoring the behavior and not only while the game runs, and its tooltip spells the values out, in this case "priority 2, weight 0.5". In play mode the name of the target the action chose is added to the same tag, which the debugging page covers.

To remove an action, select it and press the Delete key, or right click it and choose Delete. To rename it, double click its row or press F2, type the name and press Enter; Escape cancels the rename. The Name field in the properties panel renames it as well, and the sub-asset under the behavior asset in the project view follows the name. Ctrl+D duplicates the selected action with its considerations, and the up and down arrow keys move the selection within the column.

Drag an action up or down to reorder it inside its action set; a line shows where it will land. The right click menu has the same operations plus Move Up and Move Down for when you would rather not drag, and Copy and Paste, which work between two behavior assets. Paste is only offered when what you copied fits the column you right clicked in, and right clicking the empty part of a column gives you the Add and Paste entries for that column. Every one of these edits is written to the asset right away and can be undone with Ctrl+Z.

IsInterruptable

Actions can set IsInterruptable in the behavior designer to false so other actions cannot interrupt them until they call finished. The only exception is an action with a higher priority which has ForceInterruptLowerPriorityActions set to true. In that case a higher priority action with ForceInterruptLowerPriorityActions set, can interrupt a lower priority action but not another action with equal or higher priority.

Priority can be used for grouping actions and if you are really adventurous it can be changed at runtime but you should be careful using it and observe the effects to see if it is what you want or not.

Score calculations

Let's call considerations with targets TCs and considerations without targets Cs. We have multiple of these per action. The score of an action without targeted considerations is calculated by multiplying the result of all Cs with each other. The action returns a score of 0 as soon as the first consideration returns a 0 score.

Actions with targeted considerations multiply all Cs but then multiply all TCs per target and arrive at a score per target. The target with the highest score is called T and T's TCs final score is multiplied with the Cs to arrive at the final score.

So if I have two Cs and multiplying them yields 0.8. Then I have 5 targets which the highest of them scored 0.6 after multiplying all of its TCs then 0.6 will be multiplied by 0.8 to get the final score for the best target for this action.

This score then will be compared with other actions to see if it wins the race to execute or not.

Additional multiplication factor

We use an additional multiplication factor based on the number of considerations when multiplying so you don't get punished for having many considerations.

Otherwise multiplying 1 by 0.9 returns 0.9 as the score but multiplying it by 11 0.9s will result in 0.313 which is not good.

Selection Mode

The selection mode of the brain component is very important on how the next action is selected so read that page to understand it well. Some selection modes care about priority of actions and some don't and you should know about them to be able to effectively use them.

For more information, see the API reference for the ActionBase class.

Targetting system

Actions can have targets but how does the targetting system work? If an action has one or more considerations whose `NeedTarget` field is set to true in the behavior designer UI then the action is an action with targets.

When the action gets calculated to have a score, each consideration which has its `NeedTarget` checked, gets calculated per target and the target will be sent to it. The rest of the considerations will be calculated only once for the action. Now the action has a list of targets and their scores and it will use the target with the highest score as its target and calculates its score by multiplying the score of that target with the score of considerations which did not have targets.

So if the Shoot action and GetHealthPack action both have 3 targets, Shoot action's final score will be like 0.96 and it has a target selected which is the target which caused the action to have the highest score and the GetHealthPack action will have say a score of 0.72 and it is the highest score it could get with the best target score. There might be a target for Shoot which it is far enough that the shoot action for that target would score 0.34 and lose to GetHealthPack.

How to choose which consideration should have targets?

Considerations which need to calculate something regarding the target you are considering should have NeedTarget checked in the designer UI. The distance to target consideration or what's the target's health consideration are two good examples. However the considerations which calculate something about yourself like how much ammo do I have don't need HasTarget set and will be calculated once for the agent itself.

How can I fill the list of targets in Update Targets()?

You can fill this in the action by using any mechanism which is good for your game. You can use raycasts, overlap functions, influence maps, AI Tag system or any custom system to find the targets. When filling this in, be careful to not give every possible object as a target and instead count the objects which can be potentially good targets in and keep the rest out. Calculating each targeted consideration per target takes time and you don't want to add objects which you already know will not be good targets. A melee action would not want anything far away to be added for example. If the action was rush toward enemy to do melee attack, you might want to include far away targets but if you want to just do the attack and you know it doesn't work more than 3 meters away, then probably doesn't make sense to include anything more than 3-4 meters away.

There are a few methods like `AddTarget()` `ClearTargets()` and `RemoveTarget()` which help you to fill the list of targets. The samples in the plugin have example actions which do this.

Built-in actions and considerations

Wise Feline contains a set of built-in actions and considerations which help you when prototyping things quickly or even building the real AI of your game. These can be found in the ActionAndConsiderationLibrary folder in the UtilityAI folder of the package.

Design philosophy

Built-in actions and considerations work on the assumption that you put all the data the AI needs in the blackboards of the GameObjects participating in the AI. The objects might be something in the environment which you can interact with or an agent or a player controlled object.

Sometimes the data might be stored as a set of tags if it is static info about the object or can benefit from spatial queries like give me the objects with this property in my 100 meters radius.

For these assumptions we have actions and considerations which can read/write blackboard data, tags and smart objects and we allow you to search influence maps without writing code. Also you can write into influence maps without writing code using the `InfluencerAgent`. You can create influence map views which are multiple maps combined by math operations without writing a single line of code.

We even have some animation integration which works with the built-in actions and considerations and answers some of your needs.

This all means that you can easily prototype and even complete your AI by writing a bit of code to integrate animations or doing very specific or performance sensitive behaviors. To prove the points above, we built the more complex life simulation demo with less than 10 custom actions and considerations combined and even those were not that much code and mostly added some visual effects/animations to the built-in ones.

Movement components

The movement actions need to move agents because without movement we could not make that many actions :) That however needs movement to be abstracted a bit so your agents don't have to follow our code too much. So we made multiple movement components which you can choose from.

The different ones implement an interface called `IAIMovement` so the actions don't differentiate between an agent which uses a `NavMeshAgent` component or a `CharacterController` or any other way to move.

The interface has two methods to move to a position and one for stopping to move. They are supposed to be called only once and then the movement component will move the agent there. The Methods aren't supposed to be called every frame. The movement methods do the same thing but the different overloads take the destination position by different ways. You can give a `GameObject` to this and ask to move to its position or give it a `Vector3` and ask to move to it. Feel free to use them in your own actions too.

These are the components so far.

CharacterControllerBasedAIMovement

You can attach this component to an agent to move it using built-in actions if the agent has a CharacterController component.

NavMeshBasedAIMovement

You can attach this component to an agent to move it using built-in actions if the agent has a NavMeshAgent component.

Animation integration

To integrate animations with your AI actions, your actions need to set animator's parameters so it knows what action is executing. We have a parameter in built-in actions which allow you to set a trigger when the action starts. We also allow you to set a velocity parameter in movement components to make them more useful in real-world scenarios and also to teach you how to potentially integrate more complex animation scenarios.

List of built-in Actions

We have created a group of actions which are general enough which can simulate many of the actions you need for your games. You will need to inherit from them to apply animations and other custom things but they are a good starting point to create your AI. We will make them more powerful as time goes.

Most of our actions inherit from `BlackboardActionBase` which allows you to modify the agent's blackboard over time.

BlackboardActionBase

> You will not use this action on its own most of the time but our other actions inherit from it.

This action has a `BlackboardChanger` which allows you to apply a change to blackboard values for the agent over time. The changer can apply a change once or multiple times with delay or every frame. It can even multiply the every frame changes of type float by delta time. The action has a boolean to reset the changer in `OnStart` which effectively is whenever the action is selected. You can also apply a delay at startup before starting the blackboard changes.

BlackboardActionWithSmartObjectTarget

Finds potential targets from smart objects and optionally claims and uses them. It also frees them in OnFinish if the claiming and using is switched on. Also the action will fail if it tries to claim/use the smart

object unsuccessfully.

This action doesn't execute any animations or movement and just tries to find targets using the smart object search parameters specified and then optionally claim and use the chosen target in `OnStart()` (when the action starts executing) and then free the chosen target in `OnFinish()`.

The search parameters are the smart object tags which the object should/should not have, the radius and the slot states of acceptable slots.

Read on smart objects for more information on how the search functionality works but basically you can say I want my object to have one of these tags and don't want it to have these other tags and you can also say if you want it to have any slot owned by you or any free slots or you don't care about the slots.

- You need at least one consideration with NeedTarget set to true to evaluate the targets of this and choose the best one.

BlackboardActionWithTagTarget

This action updates its targets based on AI tags system and executes blackboard changes which you choose. The action doesn't do anything with the targets but it is useful because it will not execute if one of the targets doesn't get a high enough score for this action to win in the brain as the selected action.

For example: Using this action and correct properties you can increase the warmth key of the blackboard if the character is in proximity of any object with the fire tag. You set the radius to say 5 and add fire to include tags and add a TargetDistance consideration which chooses the target if it is close enough and maybe add a BlackboardFloatConsideration to only choose it when you have low warmth. There are actions to move toward fire as well so don't worry about it.

- You need at least one consideration with NeedTarget set to true to evaluate the targets of this and choose the best one.

CreateGameObjectAction

This action creates gameobjects, can be used for shooting, providing loot and other similar things

It asks for the prefab of the object to create and the number of them needed and if it should create one after each delay or they all should be instantiated at once.

Also a checkbox `shouldAdaptBrainRotation` which if set then the object inherits the rotation of the agent creating it which is good for bullets and alike.

`creationPosition` is a world position by default, or an offset in the agent's own space when `isPositionRelative` is set. The relative mode plus `shouldAdaptBrainRotation` is how you get a muzzle: the objects appear in front of the agent and fly the way it is facing, wherever it happens to be standing.

The action succeeds once the last object of the batch has been created, and a new batch starts every time the action is selected again, so it can be used as a repeating attack rather than a one-off.

DestroySelfAction

Destroys the GameObject executing the action

This action takes a delay as a parameter and if the action is switched off before the delay finishes, the destruction will not happen.

DestroyTargetWithTagAction

Destroys the target object, can be used for killing, eating and similar actions.

The targets are found using the tag system.

The action takes the radius of the search and uses the agent's position as the center. The target should have at least one of the included tags and none of the excluded ones.

> You need at least one consideration with NeedTarget set so it can evaluate targets to see if they are a good fit to destroy or not

Movement actions

> These actions don't work if a movement component is not attached to the agent. See above for the list of them. The movement speed is whatever is set on the NavMeshAgent or any other movement component you attached to the agent.

MoveAndPatrol

Patrols the agent on the way points provided as children of a GameObject which is set on a blackboard key. Optionally can finish the action after one full cycle finished.

MoveToGameObjectAction

Moves the agent to the position of a GameObject set on a blackboard key. The action succeeds after it reaches the destination and fails if the movement method fails to move (returns false).

MoveToVector3Action

Moves the agent to the position of a Vector3 set on a blackboard key. The action succeeds after it reaches the destination and fails if the movement method fails to move (returns false).

MoveToTargetWithTag

Makes a list of potential targets using the AI Tag system and moves toward the chosen target if the action is selected to execute.

> Needs a consideration with NeedTarget set so targets can be evaluated and chosen.

MoveToSmartObject

Finds a set of smart objects as potential targets and if the action gets selected to execute then it moves toward the chosen target.

The search criteria can be specified for slots of the smart object to make sure we own a slot in the object/a free slot can be found. You can also choose none.

> Needs a consideration with NeedTarget set so targets can be evaluated and chosen.

NoopAction

This action is our favorite and doesn't do anything. It optionally can stop movement too.

It can be used in cases that you only desire the agent to stop moving and probably change some blackboard attributes. Let's say you want to cancel a movement action which is moving the agent somewhere and you don't want anything to happen after cancelation. This is the action for those times. This was the programming job of the life sim demo.

RotateTowardGameObjectAction

Rotates the agent toward the position of a GameObject which is set on a blackboard key.

The action takes the name of the GameObject key and calls `LookAt` on the agent's transform once, when the action starts. It does not need a movement component because it writes the rotation directly, and it does not need any targeted considerations because the object comes from the blackboard and not from a target list.

> There are two things to know about this action before you use it, both of which come from it being a very thin wrapper around `Transform.LookAt`.

It never finishes on its own. The action does not call `ActionSucceeded()` or `ActionFailed()`, so once it is selected it keeps being the current action until another action outscores it. Leave `IsInterruptable` on for it, and don't use it as a step which something else waits for. If you make it non-interruptable, the agent stays in it forever.

`LookAt` **rotates the whole body.** If the object on the key is higher or lower than the agent, the agent pitches to face it, which looks wrong on a character standing on the ground. Keep the object at the

agent's own height, or write your own action which only rotates around the Y axis, or simply let a movement action face the agent for you: agents which walk toward something end up looking at it anyway.

Also make sure the key holds an object before the action can be selected. A key which was never defined throws when it is read, and a key which is defined but still `null` does the same thing when the action asks for its transform. The `BlackboardInitialValues` component is the code-free way to fill it in.

List of built-in Considerations

Some considerations have specific requirements and some of them might only work with NeedTarget on or off. Also some considerations might return binary 0 or 1 values and some return a value between 0 and 1 and are continuous. Min and max range values matter for some of them as well.

These are all mentioned per consideration.

In a binary consideration, the values of your consideration curve between 0 and 1 don't matter because it always returns either 0 or 1.

And as said in the considerations page, if `NeedTarget` is false then the agent itself is sent to GetValue as target and if it is true, all targets of the action are sent to it one by one to get a score.

AITagConsideration

This consideration checks if the target or the executing agent has or doesn't have a specific tag. It takes a tag and also an operation enum which you can set to the desired tag and operation.

The ShouldHaveTag operation causes the consideration to return 1 if the object has the tag and 0 otherwise. The ShouldNotHaveTag operation returns 1 if the object does not have the tag and 0 otherwise.

- This consideration is binary.
- This consideration works both for targets and the executing agent.
- The target of the consideration should have the `AITags` component for this to work.

BlackboardBoolConsideration

Checks a boolean key in a blackboard. This returns 1 if the key exists and is true and if it doesn't exist or is false returns 0.

- This consideration is binary.
- This consideration works both for targets and the executing agent.
- The target of the consideration should have the `BlackBoard` component for this to work.

BlackboardFloatConsideration

Checks a float key in the blackboard and if it exists, returns its value, otherwise returns 0.

You can invert the value of this consideration and the value is always normalized between min range and max range.

- This consideration is continuous.
- This consideration works both for targets and the executing agent.
- The target of the consideration should have the `BlackBoard` component for this to work.

BlackboardGameObjectDistanceConsideration

Returns the distance between the executing agent and a GameObject which is specified in a key in the blackboard.

You can invert the value of this consideration and the value is always normalized between min range and max range.

For example, if you want distances less than 20 meters to be considered then you should set max range to 20. Any value larger than 20 will be normalized to 20 and returns 1 and 10 will return 0.5 and 0 will return 1. You can invert the results if you want lower distances to return values closer to 1 and higher ones move toward 0.

- This consideration is continuous.
- This consideration works both for targets and the executing agent.
- The target of the consideration should have the `BlackBoard` component for this to work.

BlackboardIntRangeConsideration

Checks an integer key in the blackboard and returns its value. If the key doesn't exist then it returns 0.

You should set the range to values which indicate the possible values of the int. if the int can be 1, 2 or 3 then the min range should be 1 and max range should be 3 and then 1 will return 0, 2 will return 0.5 and 3 will return 1. You can invert the result with the checkbox if needed as well.

- This consideration is continuous.
- This consideration works both for targets and the executing agent.
- The target of the consideration should have the `BlackBoard` component for this to work.

BlackboardIntValueConsideration

This consideration checks an int key in the blackboard and returns 1 if the value equals the expected value; otherwise, it returns 0.

- This consideration is binary.
- This consideration works both for targets and the executing agent.

- The target of the consideration should have the `BlackBoard` component for this to work.

BlackboardConsiderationKeyExists

This consideration returns 1 if the specified key name exists and otherwise returns 0

- This consideration is binary.
- This consideration works both for targets and the executing agent.
- The target of the consideration should have the `BlackBoard` component for this to work.

ConstBoolConsideration

Returns a constant value. This consideration is mostly used for testing.

If isTrue is set then the `trueValue` is returned, otherwise the `falseValue` is returned.

- This consideration is binary.
- This consideration doesn't care about NeedTarget and targets sent to it.

InfluenceMapConsideration

This consideration searches the influence map specified for a value and returns 1 if it finds something. Otherwise it returns 0.

Because the search is heavy we optionally store the result in a blackboard Vector3 key to be used by an action. If you leave the `queryResultKeyName` empty then we don't store the result anywhere.

The search uses the agent/target's position as the center and takes the radius in cell count as radius. The search condition and value has to be specified as well. For example if searchCondition is set to greater and searchValue to 0.3 then if a cell with values higher than 0.3 is found in the area noted by the radius and agent/target's position on the map, the consideration returns 1 and the map coordinates of the cell are stored in the blackboard key.

- This consideration is binary.
- This consideration works both with and without targets. The thing that targets affect is the center point of the search.
- The consideration only returns valid scores for SmartObjects if you set useNearestSlotPosition.

LineOfSightConsideration

Checks whether the agent can see the target or whether something is standing in the way.

It casts a ray from the position of the agent's `Brain` toward the position of the target, using the `layerMask` you set on it. If the ray hits nothing, or if the first thing it hits **is** the target, the consideration returns 1. If it hits anything else first, it returns 0 and the whole action is multiplied to zero.

This is what makes an enemy stop noticing a player who stepped behind a crate without you writing a sensing system: put it next to a `TargetDistanceConsideration` on a chase or an attack action and the distance becomes a detection radius which cover can block.

Setting the layer mask. This is the part which is worth reading twice, because a wrong mask fails quietly rather than loudly.

- Put the things which should block sight in the mask. Walls, crates, doors and closed shutters.
- Put **the target's own layer in the mask too**. The ray is cast to `Mathf.Infinity`, so if the target's layer is missing, an obstacle standing *behind* the target is the first thing the ray hits and the consideration reports "no line of sight" for a target which is standing in plain view.
- Leave the ground out of it. The ray starts at the `Brain`'s transform position, which for most characters is the pivot at their feet, so a mask which contains the floor can report a blocked view along almost any long sight line.
- Leave bullets, tracers and effects out of it. Anything on a mask layer which flies between the two agents blinks the sight line off for the frames it is in the way, which is exactly the moment you least want the AI to lose its target.
- A mask of `Nothing` hits nothing and therefore always returns 1. If the consideration seems to be doing nothing at all, check the mask first.

The target has to be the object the ray hits. The consideration compares the transform the ray hit with the transform of the target it was given, and those two are only the same object when the target's collider is on the same GameObject as the component which made it a target (its `AITags` component, for tag based actions). A collider on a child of the target, or on a model under it, does not match.

- This consideration is binary.
- This consideration is only meaningful with NeedTarget set to true. Without a target it is asked about the agent itself, which always returns 1.
- The target needs a collider, on the same GameObject, on a layer which is in the mask.

The Combat Quickstart uses this consideration as the whole of its "spotting" behavior and its scene shows the layer setup described above.

SmartObjectDistanceConsideration

Returns the distance to a target which is a smart object. You can choose to get the distance to a specific slot with a specific condition like the distance to the closest slot which is free.

If you check the box for useNearestSlotPosition then the consideration works only if the target is a smart object. The slotStatus allows you to ask for any slot's position or a slot which is free/owned by you.

For example if you choose FreeOrOwned then the distance to the closest slot which is either free or owned by the executing agent is returned.

As with other distance considerations, this can be calculated on the XZ plane and can be inverted. Also the distance score is affected by minRange and maxRange so if you want to give a score of 1 to distances higher than 50 meters then maxRange should be 50. In this case 60 meters will return 1, 25 meters will return 0.5 and 10 meters returns 0.2. You should check inverse if you want 10 meters to return 0.8 and 0 meters to return 1 and 50 meters and above to return 0.

- This consideration is continuous.
- This consideration is only meaningful with NeedTarget set to true and without targets always returns 0.
- The consideration only returns valid scores for SmartObjects if you set useNearestSlotPosition.

TargetDistanceConsideration

Returns the distance to a target. This normalizes to a value between 0 and 1 based on minRange and maxRange values so their value is extremely important for this to be meaningful.

If you want your score to check for distances less than 100 meters then you need to set maxValue to 100 so the consideration returns 1 for 100 meters away and greater. It will return 0.75 for 75 meters and so on. You can invert the consideration by checking inverse so it returns 1 for 0 meters and 0 for 100 meters and 0.25 for 75 meters.

- This consideration is continuous.
- This consideration is only meaningful with NeedTarget set to true and without targets always returns 0.

ValidGameObjectPathConsideration

This consideration returns 1 if a valid path from the agent's NavMeshAgent to the position of the GameObject stored in the key is present, otherwise returns 0. The NeedTarget causes the GameObject to be read from the target but always a path from the owning agent is calculated. So you can check if your agent has a valid path to a position which a target specifies but you cannot check if that target has a valid path to somewhere.

- This consideration is binary.
- This consideration works both for targets and the executing agent but only the reading of the vector3 happens on targets and the path is calculated for the agent itself anyways.
- The agent with the action owning this should have a NavMeshAgent component attached. The target of the consideration should have the `BlackBoard` component for this to work.

ValidPathConsideration

This consideration returns 1 if a valid path from the agent's NavMeshAgent to the position stored in the Vector3 key is present, otherwise returns 0. The NeedTarget causes the vector3 to be read from the target but always a path from the owning agent is calculated. So you can check if your agent has a valid

path to a position which a target specifies but you cannot check if that target has a valid path to somewhere.

- This consideration is binary.
- This consideration works both for targets and the executing agent but only the reading of the vector3 happens on targets and the path is calculated for the agent itself anyways.
- The agent with the action owning this should have a NavMeshAgent component attached. The target of the consideration should have the `BlackBoard` component for this to work.

Action Sets

An action set contains a set of actions and is mainly used like a directory for grouping similar actions together in the editor window and doesn't introduce any functional difference to your actions and their scoring. Other than visual organization uses, you can add an action set to a `Brain` component at runtime which is more specific than adding all action sets of an agent behavior asset.

To add an action set

- Select an agent behavior asset in the project view and open the *Window>NoOpArmy>Wise Feline* window or just double click on the behavior.
- Click Add Action Set button to add a new action set.

An action set can be deleted by selecting it and pressing the Delete key or by right clicking it and choosing Delete. It can be renamed by double clicking its row or by pressing F2 and typing the new name; Enter commits the name and Escape cancels. Changing the Name field in the properties panel on the right of the Wise Feline window renames it too, and the sub-asset shown under the behavior asset in the project view follows the name either way.

The right click menu of a row also has Duplicate, Copy, Paste, Move Up and Move Down. Ctrl+D duplicates the selected set and dragging a row up or down moves it, with a line showing where it will land. Copy and Paste work between two behavior assets, so you can open another behavior and paste an action set you copied from this one, with its actions and considerations.

Of course while selected you can add actions to it as well which is the main point of having action sets. Each agent behavior should have at least 1 action set.

Brain component

Brain is the main component in the Wise Feline system which allows you to direct a game object using your Agent Behavior assets. You should attach this component to any game object which you want to control using utility AI. The main parameter that the component takes is the Agent Behavior asset to use but there are additional parameters and methods which are discussed in the reference linked below.

Keep in mind that this component allows you to add and remove behaviors at runtime so you can add region specific behaviors or situation specific behaviors when an agent needs to consider them during gameplay. This makes your job much easier when designing action scores with considerations. If this feature did not exist you would have to have lots of flag considerations with 0 and 1 responses for the times that action was possible or not but thanks to this feature you can simply add jungle specific behaviors to agents only when they enter the jungle zone and remove them when they exit.

The component allows you to choose how often decision making (i.e. action scoring) runs and how often the agent updates its list of targets to consider. You see this by how often your actions and considerations are called to calculate their scores. However the `Update()`, `FixedUpdate()` and `LateUpdate()` callbacks of your current action with its current target (if applicable) will be called every `Update()`, `FixedUpdate()` and `LateUpdate()` no matter how low or high you set the frequency for thinking or updating targets.

For more information, see the API reference for the Brain class.

Agent Behavior asset

An Agent Behavior asset defines a list of actions which are scored and executed when the behavior is attached to a `Brain` component. To modify the agent behavior asset you use the Wise Feline window.

To create a new agent behavior:

- right click in the project view and go to the Create menu, then choose *NoOpArmy>Wise Feline>Agent Behavior*.
- Name the behavior whatever you want. Usually this is the name of the agent type or the name of a set of behaviors. For example this can be enemy behaviors, forest behaviors or more specifically enemy at forest behaviors.
- Then select the created asset and go to the *Window>NoOpArmy>Wise Feline* window.

Now here you can add action sets which behave as folders for organizing your actions. In each action set you can add actions and for each action you can define a set of considerations which affect how the action is scored at each think operation. Every change you make is written to the behavior asset as you make it, so you don't have to close the window to save your work, and every change can be undone with Ctrl+Z.

The Wise Feline window

Open the window at *Window>NoOpArmy>Wise Feline*, or double click an Agent Behavior asset in the project view. Its tab carries the Wise Feline icon, which is how you find it again in a layout with many tabs in it.

The window has four columns. The first three list the action sets of the behavior, the actions of the action set you selected and the considerations of the action you selected, and each of them has an Add button at the bottom: Add Action Set, Add Action and Add Consideration. Add Action needs an action set selected and Add Consideration needs an action selected, so those two are disabled until there is one. The fourth column is the Properties panel, which shows the item you clicked in any of the three lists.

The splitters between the columns can be dragged to give a column more room, and the sizes you set are remembered between sessions, so the window comes back the way you left it.

The toolbar shows the name of the behavior you are editing on the left. It reads "(read only)" after the name when the behavior cannot be written to, which is the case when your version control system keeps the asset locked and when the behavior only exists in memory. In play mode the title becomes something like "Debugging Soldier (Soldier Behavior)", naming the GameObject you selected and the behavior its `Brain` is running, and two toggles named Sort Actions and Freeze appear next to it. Those two toggles are only there while the game runs and they are described on the debugging page.

On the right of the toolbar there are three buttons. Documentation opens this manual in your browser, Discord opens the support server, and About opens a small window with the version of the package, the credits and a link to the product page.

When you have not selected anything the window can show, the columns are covered by a single line telling you what to select: an Agent Behavior asset or a GameObject with a `Brain` component in edit mode, and a GameObject with a `Brain` component in the scene in play mode. A column which has nothing in it shows a hint of its own instead, so an action set with no actions tells you to use Add Action rather than leaving you with an empty column.

Selecting a GameObject with a `Brain` component in edit mode shows the behavior asset assigned to that brain, and you can edit it there exactly as if you had selected the asset itself.

More details

You can attach agent behavior assets to `Brain` components at edit time or add/remove them dynamically to a `Brain` at runtime. Basically this is the data which is used at runtime to see what actions are available to an agent and how each action's score should be calculated.

The `Brain` component has properties about what agent behavior assets to use for the game object and how often it should execute the AI and the agent behavior asset contains what actions are available to execute and how their scores should be calculated.

Dynamic addition of behaviors at runtime

The brain's initialization completes in its `Start()` method. So if you want to add a behavior in a Start event, then you should do something like the code below or move the order of the brain's execution to be before your script.

```
//another component
void Start()
{
Brain brain = GetComponent<Brain>();
    if(brain.IsInitialized)
    {
        brain.AddBehavior(customDynamicBehavior);
    }
    else
    {
        //register a method to be called which adds the behavior
        brain.OnInitialized += OnAddCustomBehaviorAfterInitialize;
    }
}
```

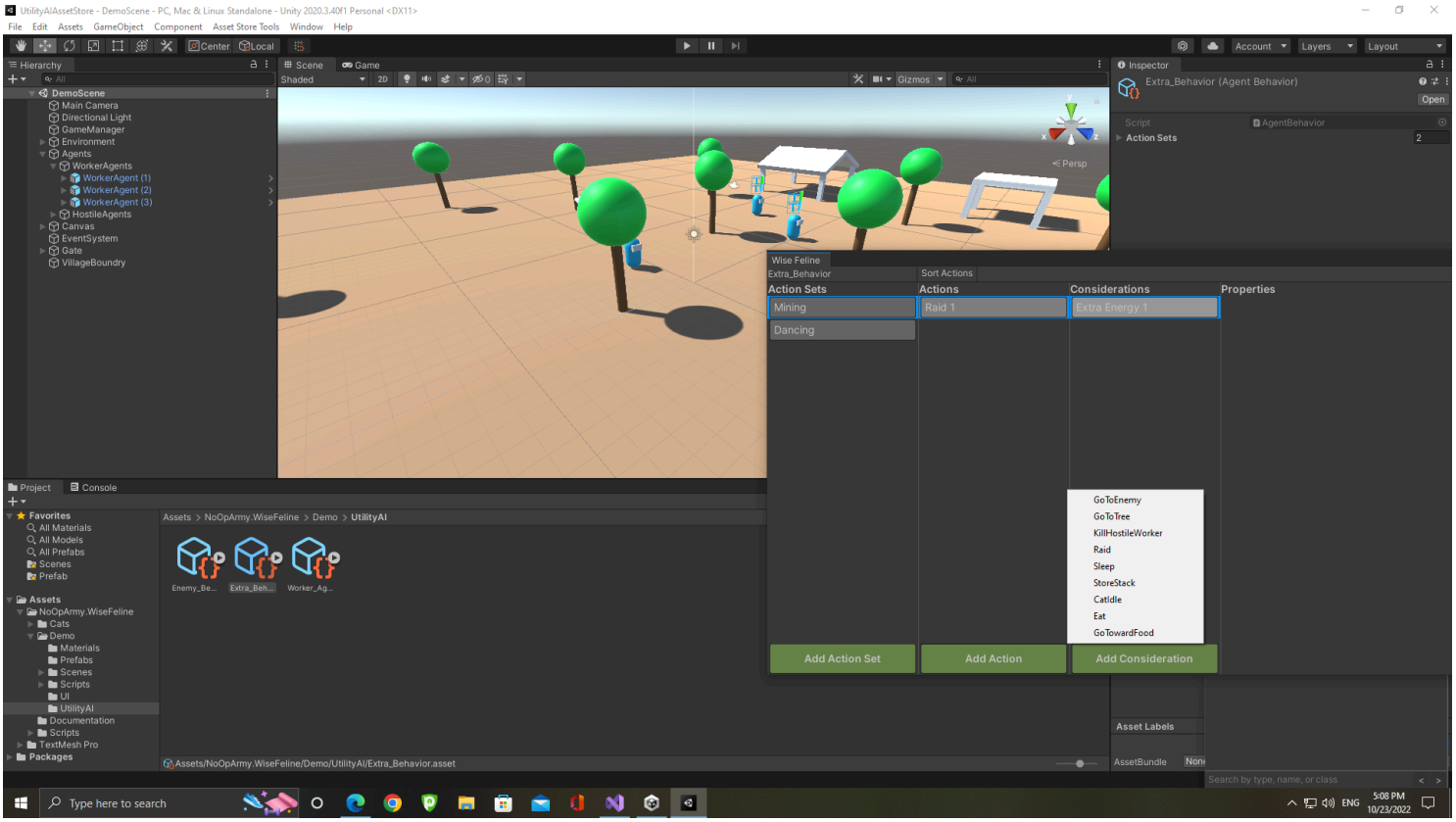
```
void OnAddCustomBehaviorAfterInitialize()
{
    GetComponent<Brain>().AddBehavior(customDynamicBehavior);
}
```

You don't need to be concerned with this if your code to add/remove behaviors happens after `Start()`.

For more information, see the API reference for the AgentBehavior class.

Wise Feline Editor Window

The editor window can be opened at Window>NoOpArmy>Wise Feline



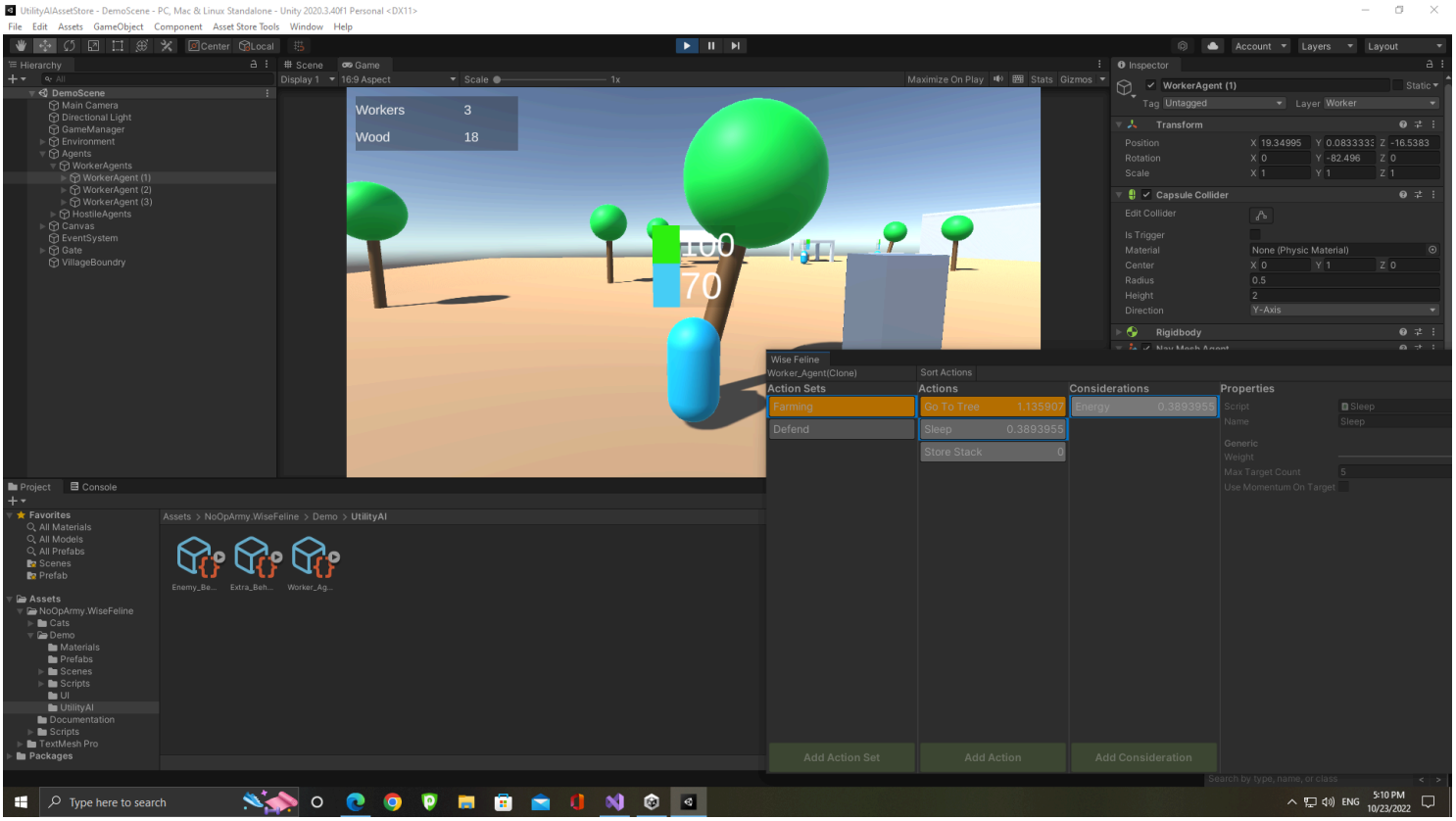
When the window is open you can select an Agent Behavior asset and then edit the actions, considerations and action sets available in the asset.

At runtime the window can be used to debug the scores of actions to see if curves and waits are set correctly to get the results you want. You just need to select game objects which have the brain attached in the hierarchy/scene and then open this window to see how its scores are being calculated.

How to Debug

You can debug your AI scoring by selecting a game object which has a `Brain` component attached at runtime while the Wise Feline window is open. The window then shows the list of all action sets and actions available to the agent and shows you what score each of the actions got. The title of the window tells you which game object you are looking at and which behavior it is running. Action sets and behaviors which were added at runtime with `AddBehavior` or `AddActionSet` are shown too, so the window lists what the brain really has and not only what the asset had at edit time.

The window looks like this



As you can see the left most column is the action sets and then you have actions and then considerations. Clicking an action set lists its actions and clicking an action lists its considerations. The window updates whenever the selected brain thinks, which is the think duration you set on the `Brain` component, not once per frame.

What the rows tell you

Every row shows the score it got in the last think with three decimals, and a bar behind the label whose width follows that score, so you can see the ranking without reading the numbers. An action's bar is measured against the highest scoring action of the set you are looking at, so the winner always has a full bar and the others are drawn relative to it. A consideration's bar is the score itself, which is always between 0 and 1. Each action set row shows the highest score among its actions, and its bar is measured against the highest of those set scores, so the first column tells you which set is winning.

The action which is running and the action set it belongs to are highlighted, and the highlight stays on for as long as the action runs, including uninterruptable actions which the brain is not allowed to switch away from. The window takes the running action from the brain itself, so an action the brain decided to keep rather than to start again stays highlighted too.

An action with targets shows the name of the target it chose in the small tag on its row, in parentheses. The same tag carries the priority of the action when it is not 0 and its weight when it is not 1, written as "P" and "W" followed by the value. Both of those change which action gets to run: the weight is already multiplied into the score you are reading on the row, and the priority is what the selection mode of the brain and the interruption rules work with. So an action with a priority of 2 which is shooting at a soldier reads "P2 (Soldier)". Considerations which score targets, the ones with NeedTarget checked, are marked with a "T" tag. None of these tags are specific to play mode; they are drawn while you are authoring the behavior as well.

Sort Actions and Freeze

Two toggles appear in the toolbar while you are in play mode.

Sort Actions reorders the actions of the selected action set by score after every think, so the action which is winning is always at the top of the column. It only changes the order of the rows on screen and never the order stored in the asset, and turning it off puts the rows back in the order the action set has them in. Freeze stops the window from applying new scores. Scores change every think and the interesting one is often gone before you have read it, so freeze the window and take your time. Turning Freeze off applies the scores of the last think right away, without waiting for the next one.

Both toggles take effect the moment you click them, and both of them go away when you leave play mode.

The properties panel in play mode

The properties panel on the right is read only while the game runs. It shows the settings of the item you selected but you cannot change them from there.

When you select a consideration, a section under its properties shows the response curve of that consideration with a marker at the point it last read. The caption under the curve prints that point as the normalized input and the score which came out of it, for example "Input 0.250, output 0.062". The line under the caption prints the value the consideration actually returned and the range it is normalized against, for example "Raw value 25 in the range 0 to 100". Those two lines together are the quickest way to tell a curve with the wrong shape from an input which is not the number you expected: if the raw value is not what you thought the agent had, the curve is not the problem.

Under that there is a Score history graph, which keeps one sample per think for the last 64 thinks of the consideration's score. Selecting an action shows the same graph for the score of the action. An action's

score is the product of its considerations times its weight and is not bounded by 1, so that graph scales itself to the largest value it has seen instead of to a fixed top.

When the window has nothing to show

In play mode the window replaces the columns with one line saying why there is nothing to look at.

Selecting a prefab in the project view instead of the agent in the scene gives you "Soldier is a prefab asset. Select the agent in the scene to debug it.", with the name of what you selected at the front of it. A prefab has no scores; only an instance which is running does. Selecting nothing, or a GameObject without a `Brain` component, gives you "Select a GameObject with a Brain component in the scene to debug it.".

A brain initializes in its `Start()`, so an agent you just spawned, or one whose GameObject is inactive, has nothing to show yet. That case gives you "The Brain on Soldier is not initialized yet. A Brain initializes in Start, so an inactive GameObject never gets there.". The window waits for that brain, so as soon as it initializes the columns fill in by themselves and you do not have to select the agent again. A brain which is initialized but has no behavior at all gives you "The Brain on Soldier has no behavior to show.".

If the agent is destroyed while you are watching it, the window notices within half a second and says "The agent was destroyed. Select another GameObject with a Brain component to keep debugging." instead of leaving the last scores on the screen.

Profiling and optimizations

Profiling

Profiling the AI might be a bit tricky because both thinking and looking for targets happen once in a while but if they are heavy, they can create spikes. We have markers called `WiseFelineThink` and `WiseFelineGetTargets` which you can look for in the CPU profiler to see how much time they take but it is more likely that your actions are more important to optimize than these unless you have really expensive considerations and in an order which causes them to always execute.

Optimization

To optimize your AI, please do the least amount of both Think and Get Targets as you can and set the order of your considerations in a way that the most expensive considerations which contain raycasts and path findings get executed last. In this way if the other considerations cause the score to be 0, the expensive ones will be skipped.

Also keep in mind that sometimes you can create game specific world databases using octrees and similar structures which allows you to do your queries faster. We are going to eventually help in this as well after the main AI features are done.

Design Resources

There is a good amount of resources regarding utility AI from Dave Mark's book to his videos and other things which all can be found in this wiki page of a utility AI sandbox project here.

We are not associated with that project or Mr. Mark but his book and these videos were very useful to us so we thought it might help you guys too and hopefully we give back a bit to him by sending a few potential book purchases over.

The demos

This page covers the two life simulation demos. If you are new to the package, start with the Combat Quickstart instead: it is one behavior asset shared by two agents and it has its own walkthrough.

The description for both life sim demos should be read in order. Even the light demo's description is not repeated in the complete one's so don't skip a section unless you are very familiar with everything and can easily jump around the tutorials.

Light Demo

This video describes what this sample is and how it works.

There is a Demos folder in the asset which contains either 1 or 2 demos depending on the version of the package you have. One of them is the light demo which is described here and the other one is the complete demo described in the next section.

How to run the demo

Simply running the scene in the demos/LifeSimLight/Scenes folder runs the demo. You can use the fly cam to observe the agents. Just use WASD to move and hold right click and move the mouse to look around.

Important folders

- All actions and considerations use the built-in actions and considerations which rely on blackboards and our movement components to work. Read the related pages linked here to learn more about them.
- Scripts contains all the custom code for the sample which is a simple script just setting some blackboard keys and the fly cam.
- Prefabs Contains all prefabs, behavior definitions and blackboard definitions.
- The Art folder outside this demo and in the Demos folder contains all the artwork

The AI design

The agents use the same AI and it is pretty simple.

In the Prefabs/Agent/AI demo you can find the blackboard definition and the agent behavior files related to the agent in the demo.

The blackboard definition contains 4 floats:

- Hunger which shows how hungry the agent is.
- Energy which shows how tired an agent is.
- Bladder which shows how much the agent needs to go to the WC.

- Entertainment which shows how bored the agent is.

It also contains 4 GameObject keys which tell the agent which WC, TV, Refrigerator and Bed belong to this agent. `LightDemoAgent.cs` sets the object keys based on what the designer dragged to its slots in the inspector.

```
myBlackBoard.SetGameObject("Bed", bedGameObject);
myBlackBoard.SetGameObject("WC", wcGameObject);
myBlackBoard.SetGameObject("Refrigerator", refrigeratorGameObject);
myBlackBoard.SetGameObject("TV", tvGameObject);
```

Before doing this it also needs to get the blackboard component and set the values of the float attributes to something random so all different agents don't start doing the same thing like this

```
myBlackBoard = GetComponent<BlackBoard>();
myBlackBoard.SetFloat("Hunger", Random.Range(0f, 1f));
myBlackBoard.SetFloat("Energy", Random.Range(0f, 1f));
myBlackBoard.SetFloat("Bladder", Random.Range(0f, 1f));
myBlackBoard.SetFloat("Entertainment", Random.Range(0f, 1f));
```

This randomization needs to be done in a smarter way but this is simple enough and good enough for the sample. You probably want to set these based on some parameters like starting hour of the simulation or a scenario/quest/any other thing.

The same `LightDemoAgent.cs` calculates some personalized values per agent to change the 4 float attributes using them over time so an agent gets hungry faster and another needs more sleep.

```
void Start()
{
    personalizedHunger = Random.Range(0.01f, 0.02f);
    personalizedEnergy = Random.Range(0.01f, 0.02f);
    personalizedBladder = Random.Range(0.01f, 0.02f);
    personalizedEntertainment = Random.Range(0.01f, 0.02f);
}
```

Then every frame it does something similar to this for all attributes

```
float hunger = myBlackBoard.GetFloat("Hunger");
if (hunger > 0)
        myBlackBoard.SetFloat("Hunger", hunger - personalizedHunger * Time.deltaTime);
```

As you can see it reads the current hunger from the blackboard and then decreases it by the personalized amount and sets it on the blackboard again. For all values 1 means they are full and there is no need to do an action to increase them and 0 means they are not full. You might have designed it differently so a full bladder means bad and an empty one is good but we wanted them to be uniform so a full bladder = 1 and means you don't need to go to the bathroom.

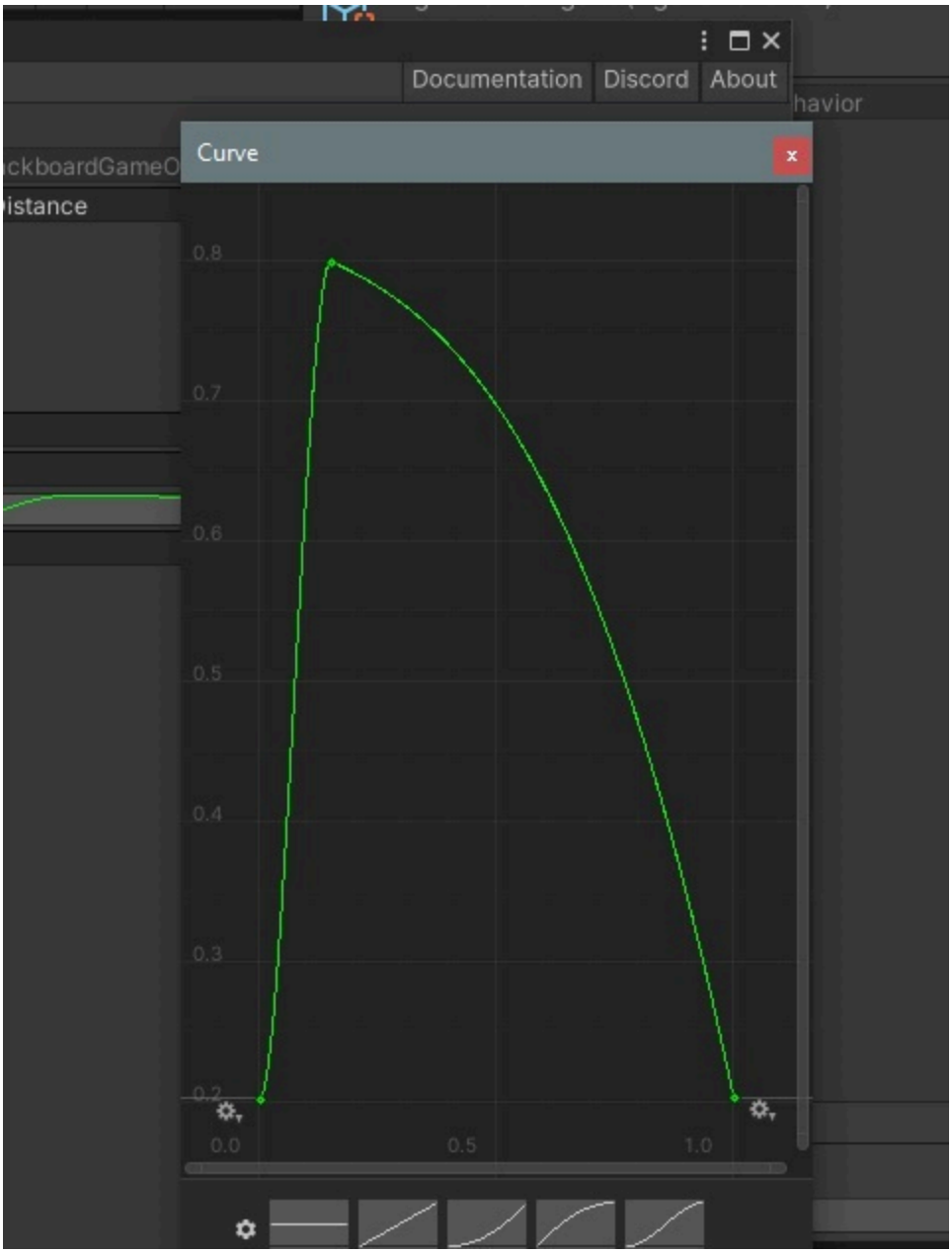
Now let's look at the AI behavior made for the utility AI of this agent in Prefabs/Agent/AI/LightDemoAgent.asset.

It contains 8 actions, four of which are for moving toward a place like the TV for entertainment or the bed for sleeping which are named MoveToX and 4 others which actually do the thing like watching TV or sleeping.

All movement actions use `MoveToGameObjectAction` which moves to a gameobject specified by a Gameobject key on the blackboard. All other actions use `BlackboardActionBase` which allows them to change blackboard keys. This is the action that almost all other built-in actions inherit from.

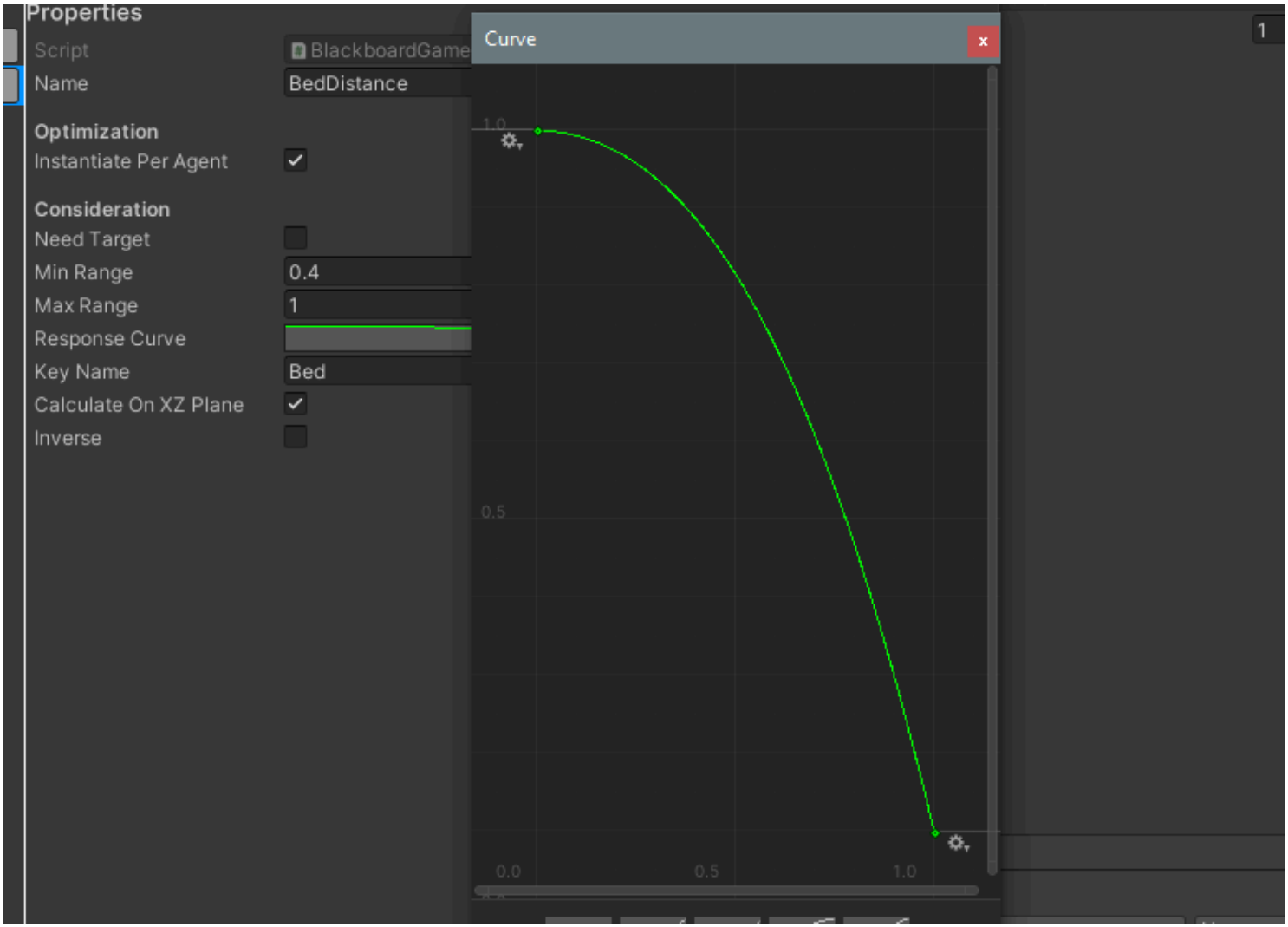
All actions use two considerations, one is for checking the float for their attribute on the blackboard which is the same on both and the other is a distance check consideration which has different curves so when the agent is far from the object, the MoveTo scores higher and when it gets close, the actual action scores higher.

For example MoveToBed uses a BlackboardFloatConsideration which it named EnergyFloat and a BlackboardGameObjectDistance Consideration which it uses to check the distance with the bed. The curve is like this



This curve means for values close to the max range which is set to 10 it will return a low score and for values close to 0 it returns a very low score as well but for stuff in the middle it returns high values.

The SleepAction shares the EnergyFloat consideration which checks the float attribute in the blackboard but its BedDistance has a very different curve.



The range is different and is between 0.4 and 1 meters and the closer to 0.4 it is, it returns a higher value so as soon as we are close to the 0.4 then this scores very high so this action is chosen instead of MoveToBed.

All other actions and their move ones are designed similar to sleep and move to bed and you can take a look at them yourself.

Complete life simulation / Immersive world demo

This is a Video Tutorial of a good part of the complete sample but you still need to read this for the description of every action and consideration.

This sample uses influence maps, smart objects, utility AI, AI tags, blackboards and all other features to create an immersive world.

How to run the demo

Simply running the scene in the demos/LifeSimDemo/Scenes folder runs the demo.

- You can use the fly cam to observe the agents. Just use WASD to move and hold right click and move the mouse to look around.
- You can also focus on characters by clicking on them. There is a camera button to follow that object in the UI and also the UI shows attributes of the agent and its current action.
- Use the Tab key to move between agents and focus on them one by one. To focus on the next type of agent like go from humans to cats or from cats to mice, use Shift+Tab.

Important folders

- Materials contains all the materials assigned to the different characters and objects.
- UI contains the atlas for the UI icons you see on the screen.
- SmartObjects contains all smart object definitions which are used in the project. These define how many slots an object has and where the slot positions are and things like that.
- Prefabs contains all prefabs used in the game and their ai definitions like agent behaviors, blackboard definitions, influence maps templates and ... Smart object definitions are not defined here and are in their own folder. For example Prefabs/Cat contains 4 prefabs for the base cat and the 3 different colors of cat and an AI folder containing the cat's influence map template, blackboard definition for all cat attributes and the agent behavior file for the Utility AI's Brain component.
- Scripts contains a few general scripts like the follow cam and the main game/simulation manager and folders containing other scripts.
- Scripts/UI contains the code for custom UI controls and the main UI Manager which updates the UI.
- Scripts/SmartObjectBehaviors contains the custom SmartObjectBehavior classes we implemented for the car and mouse hole smart objects.
- Scripts/Agents contains the main agent code for all agent types outside their AI code which changes their blackboard attributes over time. It also contains the code required for cat companionship behaviors like cats greeting each other or humans petting cats.
- Scripts/Actions contains all of our custom actions for the demo.
- Scripts/Considerations contains all considerations for the demo.
- The Art folder outside this demo and in the Demos folder contains all the artwork

The AI design

The AI of this is much more complex because many more things are going on so we describe it in parts.

Trees

We want to spawn oaks for humans to gather and mice to eat. Trees and oaks are the things you need to look into.

- `OakTree.cs` is attached to trees and spawns oaks from the trees. The script finds where to spawn the next oak by looking at oaks influence map and then if it found a free place, it spawns an oak.The

check is like this `if (oakMap.SearchForValueWithRandomStartingPoint(0.5f, SearchCondition.Less, oakMap.WorldToMapPosition(transform.position), 3, out resultPosition))`
- The Oak prefab has an `InfluencerAgent` attached which writes the oak's template on the oaks map only once. They also have an AITags component so it has a tag which allows others to query it.

These oaks are eaten by mice and humans and are gathered by humans as their job so new ones are needed so the tree generates new ones every few seconds which you can check in `OakTree.cs`

Mice

Mice have their houses/mouse holes and live in there, come out and eat oaks and will try to run away if cats try to hunt them. Since cats eat mice, our mice are the only agents which breed more of themselves. Actually the holes generate new mice when needed.

We are going to look at the hole and the mice behavior now. The Prefabs/Mouse folder contains both the MouseAgent and the MiceHole prefabs. It also contains the AI folder which contains the influence map template, blackboard definition and agent behavior assets for the mouse and the smart object definition for the hole.

The hole is a smart object which has 3 slots for 3 mice and it has a behavior attached called MouseBreedingBehavior. The SmartObject component makes this prefab a smart object and it needs a smart object definition to know how many slots it has and it also needs to add the behaviors to the behaviors list of the SmartObject component. The breeding behavior is pretty simple so let's take a quick look at it.

```
private void Start()
{
    StartCoroutine(CheckForSpawningFirstMouse());
}

public IEnumerator CheckForSpawningFirstMouse()
{
    while (true)
    {
        yield return new WaitForSeconds(Random.Range(0, 15));

        bool hasAtLeastOneMouse = false;
        foreach (var item in smartObject.GetSlots())
        {
            if (item.owner != null)
            {
                hasAtLeastOneMouse = true;
                break;
            }
```

```
         }

        if (!hasAtLeastOneMouse)
        {
            InstantiateNewMouse();
        }
    }
}
```

This code simply breeds new mice whenever the house is empty but we also have code in the behavior so whenever the mouse uses the house, if it is not hungry, it breeds a new one and becomes fully hungry.

```
public override void OnUsedByAgent(GameObject agent, SmartObject smartObject, int slot)
{
    base.OnUsedByAgent(agent, smartObject, slot);
    if (agent.GetComponent<BlackBoard>().GetFloat("Hunger") > 0.4f)//The mouse can only
breed if it is not hungry :)
        StartCoroutine(BreedNewMouse(agent));

}

IEnumerator BreedNewMouse(GameObject agent)
{
    yield return new WaitForSeconds(3);

    if (smartObject.GetFreeWithoutOwnerSlotIndex() != -1)//if there is a free slot
    {
        if (agent != null) // May be killed by cats or humans
            agent.GetComponent<BlackBoard>().SetFloat("Hunger", 0);

        InstantiateNewMouse();
     }
}


private void InstantiateNewMouse()
{
    var smartObject = GetComponent<SmartObject>();
    var slotIndex = smartObject.GetFreeWithoutOwnerSlotIndex();
    var slotPosition = smartObject.GetSlotPosition(slotIndex);
    var newMouse = Instantiate(mousePrefab, slotPosition, Quaternion.identity);

    smartObject.SetOwner(newMouse, slotIndex);
```

```
        newMouse.GetComponent<MouseAgent>().SetHome(smartObject);
    }
```

The `OnUsedByAgent()` method is called by the smart object on its behaviors when one of its slots are being used by an agent. Smart objects can be claimed/reserved and then used and slots can have owners. These houses can have 3 slots for 3 mice to live in them and when those die, the slots become free. Here when a mouse starts using a slot it means she came back home to sleep. We check if the mouse's hunger level is not that bad then it can breed a new mouse. Of course we check to see if a free slot exists or not. If no free slots exist `smartObject.GetFreeWithoutOwnerSlotIndex()` returns -1 so we make sure that is not the case. To learn more about smart objects you can read their docs here.

The mouse itself has only a few attributes in its blackboard definition.

- Attacked is a bool which is set to true when a cat attacks the mouse so it freezes in its place.
- Hunger is a float which shows the mouse's hunger level. Breeding behavior of the hole sets this to 0 when the mouse is breeding and eating oak increases it.
- OakLocation is a Vector3 set by the influence map system described below. This is a location which has a higher density of oaks for eating.
- LowDangerPosition is another Vector3 set by another influence map described below. This is a place with less cats and humans which we can move to.

The MouseAgent prefab uses a BlackBoard component with this definition to read/write these values. It also has an `InfluencerAgent` which puts its influence on the Mouse map when it moves on. This map is used by humans and cats and other systems to know where the mice are. It also uses an `AITags` to assign itself a tag so others can query the AI Tags system for the Mouse tag and find mouse agents. It also has a Brain component which executes the utility AI logic for the mouse and we describe below. The custom script `MouseAgent` is attached to the prefab which does the following activities.

- It tells the main manager script when it is created and when it is destroyed so it can be added/removed in the list of trackable objects for the camera.
- Has methods to get/set its house which is used by the breeding behavior above and other code.
- Set its hunger to 0.1 at start. Mice don't get hungry over time in this script.
- Handles the selection logic for the UI which is described later on and is the same for all agent types.

The brain component references an agent behavior asset with a couple of actions and their considerations. The file is Prefabs/Mouse/AI/MouseAgentBehavior.asset. This behavior contains the following actions and their considerations.

Sleep

This action is for the mouse sleeping and uses NoopAction which does nothing so the mouse just stays where it is. The code for the action truly does nothing other than an optional stopping of the movement

if you set its flag to true. This is the `OnStart()` which gets executed whenever the action is selected to execute.

```
protected override void OnStart()
{
    base.OnStart();
    if(StopMoving)
    {
        var movement = Brain.GetComponent<IAIMovement>();
        if (movement != null)
            movement.StopMoving();
    }
}
```

If you want to know more about the movement components and how they move the agents, you can read more at the built-in actions and considerations page.

The only consideration used for this action is the TimeOfDayConsideration which is set to return 1 between 3AM and 9AM. This is one of the 3 custom considerations in the simulation project which does not use a built-in consideration. The code of the `GetValue()` method is like this

```
protected override float GetValue(Component target)
{
    int currentHour = DayAndNightCycle.Instance.GetHour();
    if (currentHour >= MinAcceptableHourRange && currentHour < MaxAcceptableHourRange)
    {
        return 1;
    }
    return 0;
}
```

As you can see it simply reads the current hour from the DayAndNightCycle script and if it is between the specified hours, it returns 1 and otherwise it returns 0.

Search For Oak

This action searches for a high density of oak in an area to move to it. The action itself is actually just a `MoveToVector3Action` which takes a blackboard key to move to. However the key is shared between the action and its `InfluenceMapConsideration` which searches the oaks map around the mouse position and puts the resulting high density position in the same key. As a result of this we only do the map search once and if the search finds a good enough point and the mouse is hungry enough, this action scores high and the movement toward the point happens.

The action itself can be read about in the Built-in page so we will not repeat how it moves to the vector3 specified but the considerations are worth taking a look at.

The HungerConsideration is a `BlackboardFloatConsideration` which checks a float value in the blackboard to find its score. It uses the hunger value and the closer to 1 it is, it returns a higher value but the inverse checkbox is checked so actually it returns higher scores when the score is close to 0. This is a built-in consideration as well. This consideration returns 0.7 at max so the action can never score higher than 0.7. We discuss the reason when we look at the other actions but basically there are other actions which are more priority than this like eating an oak if it is close enough which should execute if their score is high so this search action should never score higher than them.

The other consideration is of the type `InfluenceMapConsideration` and searches the Oak map in 20 meters radius for values greater than 0.8 so if we have a position close to an oak in our 20 meters, this will find it and puts it in the blackboard key specified which is set to the same OakLocation key that the action uses for movement. This consideration only returns 1 if the influence map search is successful and otherwise returns 0 so the action will not execute if we cannot find a point with a value higher than 0.8 in our 20 meter radius.

> The order of considerations matters since if the first consideration returns 0 in non-editor builds of the project, the rest of the considerations are not executed so you should put more expensive considerations like influence map searches after the simpler ones like blackboard checks. Expensive things are the things which take memory/CPU like path finding, influence map searches, ray casts and collision checks and other spatial queries.

Move To Target Oak

This action moves to a specific oak we targeted and uses the `MoveToTargetWithTag` which is a built-in action. The Search For Oak action searches the area for a high density of oak and moves toward it but this one chooses a specific oak to move to and eat. `MoveToTargetWithTag` asks for a tag to find potential targets and a radius to look for them and uses the `AITags` system to do the search. Here we look in our 8 meters radius for objects with the Oak tag. Nothing reserves an oak once a mouse picks it, so two mice can head for the same one and the loser retargets. The Move To Oak For Gathering section under the human agent explains why the demo leaves this in and how to prevent it.

If we wanted to we could exclude objects which have another tag like oaks which are dirty or ruined or ... or we could look for a more general tag like mouse food and set the tag both on oaks and walnuts and maybe other things which mice eat which i don't like to mention here.

The considerations of this action are different from the search one so when in the correct situations, this scores higher than the search. The radius is smaller here so between 20 to 8 meters, the search action will obviously win if it can find something but if we have something in our 8 meters then this action will have targets so its considerations are executed to find a good target. The HungerConsideration is similar

to the search for oak action's hunger consideration and only checks the blackboard to see how hungry the mouse is. However its max value is 0.9 unlike the one for the search action which at most returns 0.7. The curve shapes are almost the same but the max point chosen for them is different in the curve so the curve for this one goes to a higher point. The other consideration which has NeedTarget set to true is the TargetDistance consideration and for targets closer than 4 meters, it returns a value closer to 1, the closer it is to you so a target with 2 meters of distance will score higher than one in 3 meters.

Actions with Considerations with NeedTarget set to true execute their considerations with NeedTarget set once per target and then choose the target with the highest score. This target is put in `chosenTarget` field of the `ActionBase` class which is accessible to all actions which you implement. then if the action with its best target wins the competition with other actions and gets selected, you know which target is your best one and you should use. The considerations without NeedTarget set execute only once for the action and its brain and are not supposed to look at targets to return their score. You can learn more about this at the targetting page.

The influence map consideration for the search action and the target distance consideration for this action can return scores up to 1 but since the hunger consideration for this action can return higher values than the one in the search action, this action wins if we have a target close by. This is very important because when we have a target in close proximity, the influence map search can find a good position too. The next action has a similar setup to these two as well but the hunger for that returns values up to 1.

Eat Oak

EatOak action uses `DestroyTargetWithTag` action which is a built-in one and destroys its chosen target found by its tag. The built-in actions have a changer argument which changes the blackboard for the GameObject which the brain is attached to. We use the changer here to add a value of 0.2 once to the Hunger key of the blackboard so as a result this eating action increases the Hunger attribute by 0.2 (1 means full and no hunger actually) and also destroys the oak so it essentially does anything needed for eating. If you wanted to show a VFX, you could put it in the `OnDestroy()` method of a script attached to the oak or could inherit from this action and override `OnFinish()` to show the effect.

The considerations of this action are set in a way that when we are close enough to an oak, it scores higher than Move To Oak and Search For Oak. There are 2 considerations. The TargetDistance one looks for targets which are at most 1 meter away and the closer they are, it returns a higher score. And the BlackboardFloatConsideration looks at the Hunger key but for most values it returns one unless we are really close to 1 and don't need to it. This means that if we are close to an oak by accident and we are half full, we still will eat it since we can have more food and it is close so why not. This is probably in line with what a real mouse would do as well. The fact that both of the considerations for this action return values up to 1 means that when we are close to an oak, Move To Oak scores 0.9 and Search For Oak scores 0.7 but Eat Oak scores 1.

Move To Home

The mouse moves to its home which is set at birth time. This is a custom action which we take a look at its code.

It first caches its movement component which is a built in component. This one uses NavmeshBasedMovement component since it uses a NavMeshAgent to move around and not a CharacterController.

```
protected override void OnInitialized()
{
    agentMovement = Brain.GetComponent<IAIMovement>();
}
```

This is good practice to cache components which you want to use a lot to save yourself calling GetComponent every time this wants to execute.

```
protected override void UpdateTargets()
{
}

protected override void OnStart()
{
    base.OnStart();
    var home = Brain.GetComponent<MouseAgent>().GetHome();

    if (!agentMovement.MoveToPosition(home.transform.position, ReachToDistanceSuccessfully))
        ActionFailed();
}
```

We don't have targets in this action so `UpdateTargets()` is empty and actually is not called either. It would have been called if at least one of the considerations had `NeedTarget` set to true.

In `OnStart()` which is called each time the action is selected to execute once, we simply get our home from the `MouseAgent` component and ask the movement component to move to it. If it returns false it means we failed so we fail the action. In this simulation this will almost never happen since the path finding to our home should fail for this to not succeed. The movement function asks for a callback to call when it succeeds and the callback contains the following code.

```
void ReachToDistanceSuccessfully()
{
    var home = Brain.GetComponent<MouseAgent>().GetHome();
    var mySlotIndex = home.GetFirstOwnedSlotIndex(Brain.gameObject);
    // This should not happen unless we have force removals/takeovers which we don't yet.
    // However the mouse commits suicide if this happens which is extreme but whatever,
```

```
unrealistic things which don't happen in the simulation don't need messing up with.
    if (mySlotIndex == -1)
    {
        Destroy(Brain.gameObject);
        ActionFailed();
        return;
    }

    var slotPos = home.GetSlotPosition(mySlotIndex);
    if (home.Claim(mySlotIndex, Brain.gameObject))
    {
        Brain.transform.position = slotPos;
        home.Use(mySlotIndex, Brain.gameObject);
    }


    ActionSucceeded();
}
```

As you can see this method simply checks if our slot is still ours and if yes, first tries to claim the slot of the smart object which should succeed and then sets our position to the exact position of the slot since the NavMeshAgent moves very close to it but not exactly on it. Then it calls the `Used()` method of the smart object which calls `OnUsed` in all behaviors and causes the breeding behavior described above to make a new mouse if this mouse's hunger value is high enough and free slots exist in this house.

As you can see in the joke comment in the code, if we add something to the simulation so the mice can forcefully take each other's houses then our mouse commits suicide and destroys itself since it has no home to return to but this doesn't happen in our simulation since ownerships are respected here and there is some good law and order coded in the sim :)

By the way whenever you call succeeded or failed on an action, it no longer executes and the brain starts thinking again to evaluate all actions and their considerations to see which one it should select as the next best action to execute.

Now let's see when the mouse wants to go home. There are only 2 considerations. There is a hunger consideration which is a `BlackboardFloatConsideration` and checks hunger and its curve returns values close to 1 if we are near full so a hungry mouse never goes home. The other consideration is a custom one which checks if the home has free slots to go back to or not. The `GetValue()` for the consideration looks like this

```
protected override float GetValue(Component target)
{
    MouseAgent mouse = target.GetComponent<MouseAgent>();
    if (mouse != null && mouse.HomeHasFreeSlot())
```

```
            return 1;
        else
            return 0;
    }
```

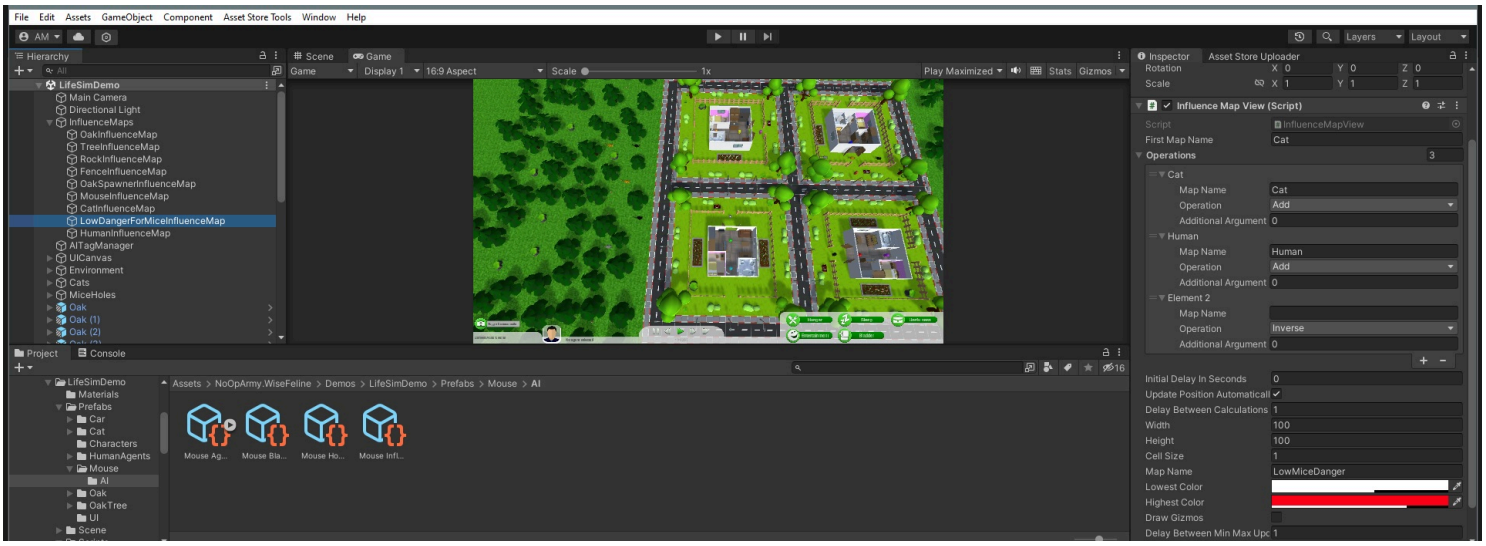
As you can see it is a very simple code which simply checks if the mouse has a home and if the home has empty slots to make more children or not. If there are no free slots then the mouse cannot make more mice so does not have the motivation to go back. It does not sound good but keeps the simulation interesting by keeping the mice outside when there are too many of them so cats can eat them and then they can make more of themselves. We are not trying to code any moral or social or ... messages into this. We just tried to make it look interesting so keep your interpretations to yourselves.

As a result of these considerations which both can return up to 1, if a mouse's home has free slots and it is mostly full, it will go home to make more mice. Maybe we should have called it an incubator or something else. They sleep anywhere and eat anywhere and only make babies at home? Whatever.

Before making this action happen only when we need to make new mice, we did not have the move to low danger locations action and mice would go home a lot. As a result cats searched for high density areas for mice and would roam around mice holes and ate them as soon as they came out. It was not that interesting so we made this action more rare and added more intelligence to mice.

Move To Low Danger Location

This action uses a built in `MoveToVector3Action` and shares its blackboard key with its influence map consideration. The interesting thing is that the influence map that the search consideration looks at is the map of humans and cats combined and a low danger position is defined as a position which the least amount of cats and humans are around it. This action has two considerations which one of them is the same `BlackboardFloatConsideration` which looks at the hunger value and after your hunger is higher than half or so its value goes up fast and for 0.8 or so and higher returns 0.8 which is its highest value. The other consideration is an `InfluenceMapConsideration` which checks the LowMiceDanger map in its 40 meter radius for values higher than 0.6. If you look at the object for the map



It is an [influence map view] which combines other maps. It starts with the cats map and adds cats and humans to it and then inverses it so higher values are for the places which are actually empty of cats and humans. It uses cats twice because they are more dangerous for mice.

As a result this consideration returns 1 when it can find somewhere which is relatively empty of humans and cats. The found position is put in the blackboard under the same LowDangerPosition key which the action uses to move to its position.

Under Attack

This action is for mice to freeze when they are under attack from cats so the hunt is more beautiful. The action has only one consideration which checks the bool key called Attacks in the blackboard and if true returns 1. As a result when under attack this usually gets the highest score since it does not care about anything else including hunger which most other mice actions check.

The action is a NoOp like the one described above and causes the mouse to freeze and get eaten by the cat.

Cats

Cats are maybe the most complex agents or at least are on par with human agents so you are in for a treat. They seek each other to greet and rub their heads. They hunt mice to eat. They sleep and get petted by humans and ... They have less actions than humans but more custom code in their actions. All this said everything is relatively simple and much simpler than other AI algorithms so don't worry and read on.

The cats have 4 prefabs. 1 for the base cat prefab and 3 prefabs for gray, black and orange cats. The differences are very minimal and other than the meshes, they just have their colors as AI Tags so they can greet their friendly factions and not other factions. They don't fight so don't worry about that.

The cat prefabs have an `AITags` component which assigns them a Cat tag and a tag based on their color which is Gray, Orange or Black. They have an `InfluencerAgent` which adds their influence using their

template to the Cat influence map. They have a Brain which executes the utility AI and also the `NavMeshBasedAIMovement` and `NavMeshAgent` components for movement. There is a `Blackboard` component which has a definition defining all blackboard keys for the cat. The keys are

- Hunger which shows how hungry the cat is. The closer to 0 it is, the cat is more hungry.
- Emotion which shows how much entertainment and love from others the cat needs. The closer to 0 it is, the need is higher for greetings and petting behaviors.
- MouseLocation is a vector3 used by the search action and its influence map consideration to find the locations which have a high density of mouse in them.
- TargetCat, Waiting and Companionship are used in the companionship system used later on which is the system governing how cats greet each other and how humans pet them.

Let's look at all the actions and their considerations one by one.

Morning Sleep

This uses the `NoopAction` and just stops the movement of the cat which uses the same `NavMeshAgent` and `NavmeshBasedMovement` that the mice use too. The only consideration is the `TimeOfDay` consideration which for this one returns 1 when the time range is between 6AM and 10AM.

> This manual is created to be read in order so read the mouse one first to know more about the actions and considerations used.

Night Sleep

The only difference between this and the morning sleep is the time range that the cat goes to sleep. The time range for the `TimeOfDay` consideration is between 4PM and 8PM.

Search For Mice

This action uses `MoveToVector3Action` to move to a vector3 key on the blackboard attached to the cat but the key is shared with one of its considerations which searches the influence map for mice to find a place which has a lot of mice in it. The considerations are

- HungerConsideration which is a `BlackboardFloatConsideration` and checks the Hunger key in the cat's blackboard and for the values closer to 0 returns a higher value but its max value is about 0.7 so considerations for actually eating a mouse get higher score than this when the mouse is close enough.
- The other one is an `InfluenceMapConsideration` which looks in the mouse map in the 30 meter radius of the cat to find values greater than 0.8 which means there are a lot of mice in that area. If successful then it puts the position in the MouseLocation key which the action uses to move to. This is similar to the search for oak action in the mouse prefab above which has more description.

All mice have an `InfluencerAgent` component and put their template on the map where they are so the map allows us to find them pretty easily. If we wanted to we could try to find mice which are alone or places which mice exist but no cats are around but no need to make it more complex for the sample really.

Hunt Mouse

This action has custom code and is defined in `HuntingMouseAction.cs`. This said the custom action inherits from `MoveToTargetWithTag` and essentially tries to move to objects tagged with Mouse but does some additional things on top.

The important parts of the code are

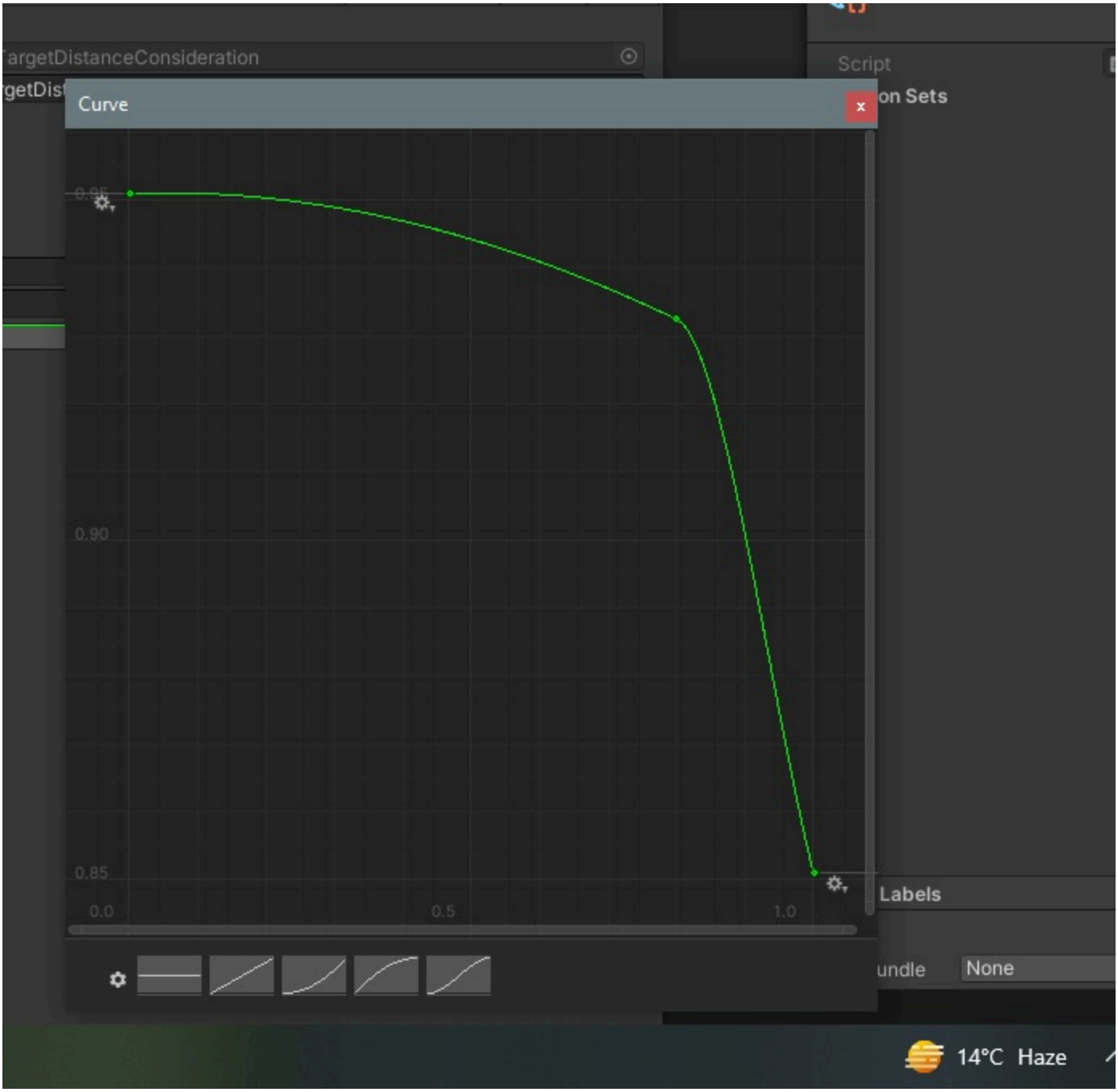
```
protected override void OnStart()
{
    base.OnStart();
    trailRenderer.startColor = Color.red;
    trailRenderer.endColor = Color.red;
}

protected override void OnFinish()
{
    base.OnFinish();

    trailRenderer.startColor = Color.white;
    trailRenderer.endColor = Color.white;
}
```

As you can see the only modifications the action does are making the trail renderer of cats red when they start hunting and turning it back to white when the action finishes. In this way you can easily add your code on top of the built-in actions or your other custom made ones.

The action is set to look for objects tagged Mouse in its 12 meter radius and will move toward the best of them if chosen. When arrived it calls `ActionSucceeded()` which causes `OnFinish()` to get called. `OnFinish()` is called in the case of failures or even another action getting a higher score and forcing this to change as well so there is no situation that the action changes and `OnFinish()` not getting called.

There are two considerations. HungerConsideration which looks at the Hunger float on the blackboard and returns values up to about 0.9. The curve returns 1 for higher values which means less hungry but the inverse is checked so actually the closer to 0 you are, this returns a higher value. This does not return 1 so the actual mouse eating behavior has a chance of scoring higher. The other consideration is a TargetDistance which considers targets up to 12 meters and the closer a target is, it returns a higher value and actually returns something close to 0.95 for a big range of values because it makes sense for it

and it wants to move toward the target so 2 meters and 5 are not that different for this action. The curve looks like this:



We don't show all curves here but I encourage you to examine them and see what shapes we used for each action. Dave Mark's book and talks in the resources section talk about these a lot as well.

Eat Mouse

This action has some custom code and inherits from `DestroyTargetWithTag` which destroys its target and finds targets using their tags. The custom code on top of that other than changing the trail renderer's color checks the mouse to see if it is already being attacked by another cat or not. If not it sets the Attacked boolean to true in the mouse's blackboard and then destroys it. Here it is the code of the action

```
protected override void OnStart()
{
    base.OnStart();
    trailRenderer.startColor = Color.red;
    trailRenderer.endColor = Color.red;
    MouseAgent mouseAgent = ChosenTarget?.GetComponent<MouseAgent>();
    if (!mouseAgent.WaitToBeEaten())
        ActionFailed();
}


protected override void OnFinish()
{
    base.OnFinish();

    trailRenderer.startColor = Color.white;
    trailRenderer.endColor = Color.white;
}
```

As you can see the important part of the code is what `WaitToBeEaten()` returns and its code looks like this:

```
public bool WaitToBeEaten()
{
    if (myBlackboard.GetBool("Attacked"))
        return false;

    myBlackboard.SetBool("Attacked", true);
    GetComponent<Brain>().Think();

    return true;
}
```

This code checks to see if the mouse is already being attacked, then it returns false and we will not try to take food from another cat. Otherwise we mark it as attacked and nobody else will try to take it from us.

The main work happens in the parent class which is `DestroyTargetWithTagAction` and its code looks like this:

```
protected override void UpdateTargets()
{
    ClearTargets();
    AddTargets(AITagsManager.Instance.GetTagsAroundPoint(Brain.transform.position, radius,
```

```
includedTags, excludedTags));
}

protected override void OnStart()
{
    base.OnStart();
    startTime = Time.time;
}

protected override void OnUpdate()
{
    base.OnUpdate();
    if (Time.time - startTime >= delayForDestruction)
    {
        if (ChosenTarget != null)
        {
            GameObject.Destroy(ChosenTarget.gameObject);
            ActionSucceeded();
        }
        else //Something else already destroyed that object
        {
            ActionFailed();
        }
    }
}
```

The code is pretty simple. In `UpdateTargets()` we clear all targets and add the new ones using the AI Tags system. Then after a target and the action are chosen, in `OnStart()` we just set start time and based on the delay, when the time to kill finally arrives, we destroy the target if it still exists and call it a success and otherwise fail the action.

In the example of `EatMouseAction` in the child class's `OnStart()` we might fail the action if the mouse is being attacked by another cat and the `OnUpdate()` will never get called and the other cat will destroy the mouse.

Also the action's changer is set to increase the Hunger by 0.5 (fill its bar by half) which is the result of eating a mouse.

Seek Cats

There is a system for cats to greet each other and for humans to pet them and the system requires work from both parties involved so the code is a bit complex. The system is used in this action and the rest of the cat actions and the petting action in humans. So we describe it here.

There is an interface which both humans and cats implement and it starts a companionship, ends it, waits for one to start and checks if a cat/human is busy in another companionship act. It does not differ between a human petting a cat or two cats rubbing their heads with each other.

```
/// <summary>
/// agents like other cats and humans which want to do friendly actions with cats need to implement this interface
/// The reason is that the petting behavior or cat greetings can only happen in sequence and a cat can be in one of these at a time
/// </summary>
public interface ICatCompanionship
{
    /// <summary>
    /// Tells a cat that this cat is being petted by a human or being greeted by a cat and should wait for this action to finish
    /// </summary>
    /// <param name="otherCat"></param>
    public abstract void WaitForOtherCompanionship(GameObject otherCat);

    /// <summary>
    /// Is the agent busy in another Companionship
    /// </summary>
    /// <returns></returns>
    public abstract bool IsBusy();

    /// <summary>
    /// A Companionship gets started involving this agent
    /// </summary>
    public abstract void StartCompanionship();

    /// <summary>
    /// Ends a Companionship action and this agent is free to do another now
    /// </summary>
    public abstract void EndCompanionship();
}
```

The Start and End methods are called by the cat/human which initiates the companionship on the other cat and `IsBusy()` is naturally called to check if the other party is busy with another cat/human or not. `WaitForOtherCompanionship()` is called by the other cat/human to tell the cat to stop moving and wait for the companionship to happen.

The seek action itself has custom code but it inherits from MoveToTargetWithTag. Its code looks like this:

```
protected override void OnInitialized()
{
    base.OnInitialized();
    CatAgent catAgent = Brain.GetComponent<CatAgent>();
    if (catAgent != null)
        includedTags = includedTags.Concat(catAgent.SeekingCatTag).ToArray();
}

        protected override void OnStart()
        {
            base.OnStart();

            var blackboard = Brain.GetComponent<BlackBoard>();

            targetCat = ChosenTarget.GetComponent<CatAgent>();

            if (targetCat.IsBusy())
                ActionFailed();


            targetCat.WaitForOtherCompanionship(Brain.gameObject);
            blackboard.SetGameObject("TargetCat", ChosenTarget.gameObject);
            blackboard.SetBool("Companionship", true);
        }

        protected override void OnUpdate()
        {
            base.OnUpdate();

            if ((targetCat.transform.position - Brain.transform.position).magnitude < 1f)
                ActionSucceeded();
        }
```

First I want to show you how simply we implemented factions. Each `CatAgent` component has a `SeekingCatTag` array which allows it to say cats of which category it seeks. Gray cats only seek the Gray tag and the orange and black ones seek both Orange and Black in their cat prefabs so here we add the tags set there to the list of tags which this action will try to find and move to in `OnInitialized()`.

Then in `OnStart()` we check if the other cat is busy `if (targetCat.IsBusy())` and if yes we fail the action. Otherwise we tell the target to wait for us using the interface methods mentioned above and set the other cat as our TargetCat in our blackboard and set the bool which causes the companionship action to score high. Companionship is the action used in cats for greeting and in humans to pet a cat. The last thing we do is that in our update, whenever we are close to the other tag, finish the action by declaring success.

The `CatAgent` component's waiting function and `IsBusy()` are defined like this

```
public void WaitForOtherCompanionship(GameObject otherCat)
{
    myBlackboard.SetBool("Waiting", true);
    myBlackboard.SetGameObject("TargetCat", otherCat);
    GetComponent<IAIMovement>().StopMoving();
}

public bool IsBusy()
{
    return myBlackboard.GetBool("Waiting") || myBlackboard.GetBool("Companionship");
}
```

We return that we are busy if we started a companionship in our seek action or if another cat called us to wait and our Waiting is set. `WaitForOtherCompanionship()` simply stops the cat's movement component and sets Waiting and TargetCat.

The agent has other important functions which are described in the next action.

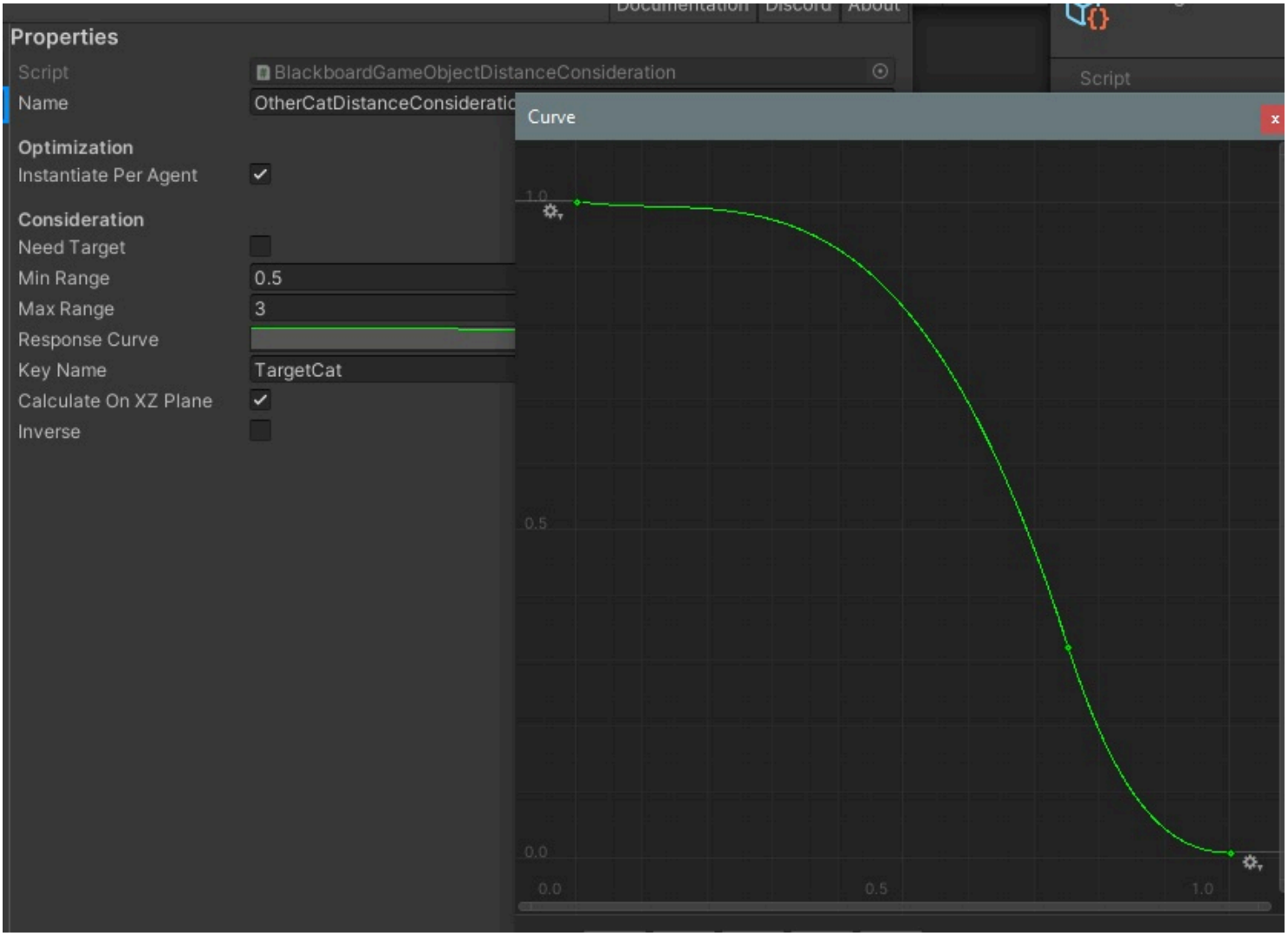
The considerations for the seek action are many but they are very simple. There is a TargetDistance which considers targets up to 50 meters and at max returns 0.8 for targets which are at its 0.1 mark which is 5 meters when you multiply it by the 50 meter max distance. Then it has a BlackboardBoolConsideration which checks the Waiting key and only returns 1 when it is false. The EmotionConsideration is a BlackboardFloatConsideration which checks the Emotion key and returns high values up to 1 for emotions more than 0.1 and has inverse checked so essentially for anything lower than 0.9 it returns a high value. Cats are very much into playing so this makes sense. The IsCatFreeConsideration has NeedTarget set to true so works on the targets of the seek action and its code looks like this

```
protected override float GetValue(Component target)
{
    CatAgent cat = target.GetComponent<CatAgent>();
    if (cat != null && cat.IsBusy() || target.gameObject == Brain.gameObject ||
target.GetComponent<BlackBoard>().GetFloat("Emotion") > 0.6f)
        return 0;
     else
         return 1;
}
```

If we are busy or the target is ourselves or the target doesn't have a low enough emotion, then this fails. We don't want to greet targets which have an already high emotion. Since the consideration is binary only the 0 and 1 positions of the curve matter which return 0 for 0 and 1 for 1.

Companionship

This action is the actual greeting action of the cats and actually is used in humans when they pet cats too. It happens when a cat's Companionship bool key in the blackboard is set to true and its distance to another cat is low enough. The considerations are just a bool for the Companionship key which returns 1 if the bool is set. And a BlackboardGameObjectConsideration which checks the distance to a GameObject set in the TargetCat key and for distances between 0.5 and 3 meters it returns a value between 1 and 0. The curve returns a high value up to half of the max distance and looks like this



Keep in mind that this action is executed on both greeting cats and the human and the cat he/she is petting.

Let's look at the action code itself.

```
protected override void OnInitialized()
{
    base.OnInitialized();
    myCompanionship = Brain.GetComponent<ICatCompanionship>();
    myBlackboard = Brain.GetComponent<BlackBoard>();
```

```
}

protected override void OnStart()
{
    base.OnStart();
    targetCompanionship =
myBlackboard.GetGameObject("TargetCat").GetComponent<ICatCompanionship>();
    targetCompanionship.StartCompanionship();

}

protected override void OnUpdate()
{
    base.OnUpdate();

    if (myCompanionship is CatAgent)
    {
        if (myBlackboard.GetFloat("Emotion") >= 0.9f)
            ActionSucceeded();
    }
    else if (myCompanionship is HumanAgent)
    {
        if (myBlackboard.GetFloat("Entertainment") >= 0.9f)
            ActionSucceeded();
    }
}

protected override void OnFinish()
{
    base.OnFinish();

    targetCompanionship.EndCompanionship();
}
```

In `OnInitialized()` we get our `ICompanionship` component and it might be a CatAgent or HumanAgent. As said before this action is used on both for greeting/petting a cat. In `OnStart()` we get our target's `Icompanionship` based on what is set in the blackboard in the seek action. Then we call its `StartCompanionship()` and in `OnFinish()` we call its `EndCompanionship()`.

`OnUpdate()` is a bit more interesting since it checks the Emotion or Entertainment key of the blackboard depending on the fact that we are actually attached to a human or a cat and if it is greater than 0.9, the action is succeeded. The `is` keyword in C# can check object types and since we know only these two classes implemented the interface we got using `GetComponent<T>()`, we can know if we are a human or a cat by checking this. We could have a boolean for the designer to set or check the tags of the object but this is also a way to check this.

The changer of the action is set to repeat an action with delay and to add 0.1 once per second to the Emotion of the cat executing this action so eventually 0.9 is reached. Obviously the one attached to a human will increase its own Entertainment and not Emotion.

Now let's look at `StartCompanionship()` and `EndCompanionship()` in `CatAgent`.

```
public void StartCompanionship()
{
    if (myBlackboard.GetBool("Companionship") == false)
    {
        myBlackboard.SetBool("Companionship", true);
        myBlackboard.SetBool("Waiting", false);
        GetComponent<Brain>().Think();
     }
}

public void EndCompanionship()
{
    myBlackboard.SetBool("Companionship", false);
}
```

The Waiting key should be true when we are waiting for a companionship to start but when it gets started, it should become false and Companionship key should be set to true. When the companionship finishes then the Companionship key should be set to false which is what these two methods do. These key changes cause the other cat to go from waiting action to Companionship to increase its Emotion.

Wait For Companionship Cat

This action happens when its single consideration which checks the Waiting key in the blackboard returns 1 and does nothing. It uses `BlackboardBoolConsideration` with a curve which returns 1 when the Waiting boolean key is set to true. And the `NoopAction.cs` does nothing and it even does not stop movement since the previous action which was seeking another cat or the `WaitForCompanionship()` method called by the other human/cat already stopped its movement. The Emotion of this cat doesn't change in waiting but after this the cat will go to Companionship which will increase its Emotion.

Other methods of CatAgent component

```
private void Awake()
{
    indicator = GetComponent<SelectedIndicator>();
    myBlackboard = GetComponent<BlackBoard>();

    personalizedHunger = Random.Range(minPersonalizedValue, maxPersonalizedValue);
    personalizedEmotion = Random.Range(minPersonalizedValue, maxPersonalizedValue);
```

```
}

void Start()
{
    myBlackboard.SetFloat("Hunger", Random.Range(0, 0.7f));
    myBlackboard.SetFloat("Emotion", Random.Range(0, 0.7f));
    myBlackboard.SetBool("Companionship", false);

    LifeSimDemoManager.Instance.AddFollowableAgent(indicator, AgentType.Cat);
}

private void OnDestroy()
{
    if (LifeSimDemoManager.Instance != null)
        LifeSimDemoManager.Instance.RemoveFollowableAgent(indicator, AgentType.Cat);
}

void Update()
{
    float hunger = myBlackboard.GetFloat("Hunger");
    if (hunger > 0)
        myBlackboard.SetFloat("Hunger", hunger - personalizedHunger * Time.deltaTime);

    float emotion = myBlackboard.GetFloat("Emotion");
    if (emotion > 0)
        myBlackboard.SetFloat("Emotion", emotion - personalizedEmotion * Time.deltaTime);

    // Check for mouse click or touch input
    if (Input.GetMouseButtonDown(0))
    {
        // Cast a ray from the camera to the mouse position
        Ray ray = Camera.main.ScreenPointToRay(Input.mousePosition);
        RaycastHit hit;

        // Check if the ray hits the object with a collider
        if (Physics.Raycast(ray, out hit))
        {
            // Check if the hit object is the one you want
            if (hit.collider.gameObject == gameObject)
            {
                LifeSimDemoManager.Instance.SetAsSelectedAgent(indicator, AgentType.Cat);
            }
        }
    }
}
```

The selection and indicator logic is explained later on in the general simulation code section but the rest of the code is pretty easy and similar to the light simulation's human code. We set initial values for Emotion, Hunger and other blackboard keys and then find some random personalized values for the cat. In Update we reduce Hunger and Emotion by these random and personalized values so all cats don't tend to do the same thing all the time. There are other ways to achieve this like more random selection behaviors in the brain but we chose to use highest score for ease of understanding in the demos.

Human agents

The human agent's behavior has similarities to the light simulation's human agents above so read up on that one first. The HumanAgent prefab has its own general code in `HumanAgent.cs`. It has the `Brain` for Utility AI, a `Blackboard` component, a `NavMeshAgent` and a `NavMeshBasedAIMovement`, and of course an `InfluencerAgent` to add its influence to the Human map.

The blackboard for humans is in the AI folder in Prefabs/HumanAgent/AI alongside the agent behavior and the blackboard definition.

- Hunger is the attribute which fills when the agent eats food and depletes over time.
- Energy is the attribute which fills when the agent sleeps and depletes over time.
- Bladder is the attribute which fills when the agent goes to the WC and depletes over time.
- Entertainment is the attribute which fills when the agent watches TV or pets cats and depletes over time.
- Usefulness is the attribute which fills when the agent works and depletes over time.
- Job is an int key which shows what the job of the agent is.
- OakLocation is a Vector3 which is set by an influence map consideration and used by the actions that the human uses to gather or eat oaks.
- TargetCat is a GameObject key which is used by the system which pets cats to know what cat to pet.
- Companionship is a bool key which causes the human agent to give a high score to the petting action.

The influence map template is nothing special to look at so let's go through the AI behaviors.

> Many of the actions set an animation parameter name for the On Start Animation Trigger and we actually have an animator with those parameters but Unity did not allow us to submit the plugin with Mixamo animations so we have capsules for the characters in the demo. You can use this feature of the built-in actions to drive your animations.

Move To WC

Most of the move actions use smart objects but the WC one does not move to a smart object because we wanted it to be different. Smart Objects have ownerships so agents will not use each other's beds but they'll use each other's WCs thanks to this difference.

This action uses the built-in `MoveToTargetWithTag` action and tries to move to an object in 30 meters with the tag WC. As said above it also sets a trigger in the animator with the name Walking in its `OnStart()`. Describing how animations work is beyond the scope of this document and you can read Unity's documentation on that. This might not be enough for your animation needs. Even the fact that the NavMeshBasedAIMovement allows you to set the velocity of the movement in an animation parameter might not be enough either but in these cases you can inherit from the action and do whatever you need there and call the `base` methods to do the movement.

The considerations of the action are a TargetDistance consideration which considers up to 100 meters and gets executed per target that the tagging system finds and also a consideration which checks the Bladder float key on the blackboard. The blackboard one returns up to 0.7 and the target distance one returns up to about 0.85 so the actual Use WC can score higher than this.

Use WC

This action uses the `BlackboardActionWithTagTarget` class which finds all objects with the tag WC in 2 meters radius and if it is chosen it simply increases the Bladder float key on the blackboard by 0.5 every second.

It has two considerations

- A BlackboardFloatConsideration which checks the Bladder and it has inverse checked so actually instead of returning 1 for values closer to 1, it will inverse the curve and returns values close to 1 for the times that Bladder is closer to 0.
- The other one is a TargetDistance consideration which for distances between 0.5 and 1.5 meters returns a score which is close to one for the lower half of the distance and naturally anything less than 0.5 will have the 0.5 meters score which is 1.

This action is chosen over the Move To WC one because its curve will return scores up to 1 and this will score higher when close to a WC. The Move To WC action cannot return a value much higher than 0.7 for score because one of its considerations returns 0.7 as its max value.

Move To Bed

This action uses `MoveToSmartObject` and it checks 100 meters and the object should have a Bed tag and of course be Owned by the agent. The considerations are naturally a check for the energy of the agent and another for the target distance check. The distance check uses `SmartObjectDistanceConsideration` and sets the required slot status as Owned.

When the slot status contains owned then the smart object system checks the actual ownership of the agent toward that smart object.

Sleep

Each agent has his/her own place to sleep. Either a bed or furniture is his/her sleeping place. The action uses `BlackboardActionWithSmartObjectTarget` class and the slot status is set to Owned. The radius is 1 meters and the tag is Bed for the search for the target in the smart object system. The action increases by 0.05 every second. The considerations are SmartObjectDistanceConsideration and the usual BlackboardFloatConsideration. The float checks the Energy key on the blackboard and has inverse checked. The distance checker has the slot status is set to Owned so the distance check cares if the object is owned by the agent or not. The action itself had this too so each bed will only score high and only gets included in the list of targets for their owners. And I'm being very very loose when I call all of them beds. Some agents sleep on the couch!

Move To Entertainment Unit

This action finds smart objects with a specific tag which in this case is Entertainment and then moves to the chosen target. The class that it uses is `MoveToSmartObject`. This action will fail if no free slots can be found on the smart object and also will fail if there is no path/possibility of movement to the object by the agent's movement component.

The action has two considerations very similar to the ones the Move To WC above had but the ownership requirements of the smart object would mean that no two agents will use the same Entertainment slot on an object. The considerations are

- A EntertainmentFloat consideration which is a `BlackboardFloatConsideration` checking the Entertainment float key on the blackboard and returning up to 0.5. It has inverse checked so an empty Entertainment will return 0.5.
- And a TargetDistanceConsideration which for up to 100 meters returns a value between 0 and 0.8 and naturally for closer distances will return a higher value so the closest target is chosen.

The reason that the non-targeted Entertainment float check returns 0.5 at maximum is that going to an entertainment object is one of the least important things when other options are going to WC, eating, sleeping and ...

Entertainment

Similar to Use WC, simply increases the Entertainment attribute of the agent. `BlackboardActionWithSmartObjectTarget` is the action class used and it finds smart object targets and the changer of the action increases Entertainment by 0.1 per second. The target search happens with the smart object system in 3 meters for objects which have the Entertainment tag. This action does not choose a target which does not have a free slot just like the Move To Entertainment so a TV used by somebody else will not be used by this agent.

There are two considerations which one of them is a targeted consideration checking target distance called SmartObjectDistanceConsideration. This consideration can optionally check the distance with the closest slot of a specific status instead of the position of the GameObject. We use the slot position here because the slot is far away from the GameObject transform. This one doesn't use ownership but beds

and furniture which agents sleep on use ownership. The other one is a BlackboardFloatConsideration which checks the Entertainment key of the blackboard. The curve returns up to 1 and has the inverse checked. This means this action will be chosen over the movement one when it scores high since that one will never score 1.

Go To Refrigerator

This action is the same as Move To WC and moves to a target which has the Food tag. The differences are the tags and the blackboard key checked and used which is Hunger and the curve shapes.

Eat

This is very similar to Use WC as well and just increases Hunger attribute. The considerations are for checking Hunger level and target distance as well. Check Use WC for more details.

Drive

This is the first job action that we are encountering. All humans have jobs either as oak gatherers, programmers or drivers. Each job requires at least two actions like Go To Car/Go to laptop and then drive/programming.

The Job key in the blackboard tells what job the agent has and changing it would actually change the job of the agent. 0 is programming, 1 is driving and 2 is oak gathering.

This action is the driving part after the human went to his/her vehicle. The class used for the action is `DriveAction` which is a custom action which inherits from `BlackboardActionWithSmartObjectTarget`. The action that this inherits from finds targets by using the smart object system and does nothing else other than being able to change the blackboard. The custom action just claims and uses the smart object and waits for it to finish driving and the going home time arrives and then it finishes the action successfully. The `CarBehavior` which is described below is a smart object behavior with a coroutine which simply moves the car around.

```
protected override void OnStart()
{
    base.OnStart();
    carSmartObject = ChosenTarget.GetComponent<SmartObject>();
    carBehavior = ChosenTarget.GetComponent<CarBehavior>();

    if (carSmartObject.Claim(0, Brain.gameObject))
    {
        carSmartObject.Use(0, Brain.gameObject);
    }
}

protected override void OnUpdate()
{
```

```
        base.OnUpdate();
        //waits for the driving to finish and final hour of the job arrives and then finishes
    the action successfully
        while (carBehavior.IsDriving)
            return;

        while (DayAndNightCycle.Instance.GetHour() < endDrivingTime)
            return;

        carSmartObject.FreeAllSlots();

        ActionSucceeded();

    }
```

As you can see the `OnUpdate()` which is the important part simply waits for the driving and time to finish and then frees the object and declares success.

The considerations are:

- BlackboardIntValueConsideration which checks a blackboard key to be a specific int value. It is a binary consideration which returns 1 only when the int key is equal to ExpectedValue. Here Job is checked to be 1 (means driver)
- The SmartObjectDistanceConsideration checks up to 3 meters and it returns 1 for almost all of them. Keep in mind that this will only consider targets added by the action so a far away car will not be considered at all. The search radius is 3 meters in the action.
- There is a Time Of Day consideration which only returns 1 between 10:00-18:00.

Car behavior

The car is a smart object which is used by human agents which have the driver job. The CarSmartObject prefab uses the NavMeshAgent component for movement. It has a SmartObjectSlotOwnerSetter component which sets the owners of slots based on what you put in its SlotOwners array. It has a CarBehavior smart object behavior script and it also has a SmartObject component which references a smart object definition with one slot and a reference to the driving behavior (described below) as its behavior.

`CarBehavior.cs`'s main methods which are overridden for the behavior are like this

```
    public override void OnClaimedByAgent(GameObject agent, SmartObject smartObject, int slot)
    {
        base.OnClaimedByAgent(agent, smartObject, slot);
        agent.GetComponent<HumanAgent>().StartDrivingState(smartObject);
    }
```

```
public override void OnUsedByAgent(GameObject agent, SmartObject smartObject, int slot)
{
    base.OnUsedByAgent(agent, smartObject, slot);
    StartCoroutine(Driving());
}

public override void OnFreedByAgent(GameObject agent, SmartObject smartObject, int slot)
{
    base.OnFreedByAgent(agent, smartObject, slot);
    smartObject.FreeAllSlots();

    agent.GetComponent<HumanAgent>().EndDrivingState(smartObject);
}
```

As you can see it just tells the human agent to go to driving state when it is claimed and when freed tells it to go back to normal state. The complex driving logic is in the `Driving()` coroutine called in the `OnUsedByAgent()`

```
private IEnumerator Driving()
{
    isDriving = true;
    navAgent.SetDestination(firstPosition.transform.position);

    while (!IsReached())
    {
        yield return new WaitForSeconds(1);
    }

    navAgent.SetDestination(secondPosition.transform.position);

    while (!IsReached())
    {
        yield return new WaitForSeconds(1);
    }

    navAgent.SetDestination(parkPosition.transform.position);

    while (!IsReached())
    {
        yield return new WaitForSeconds(1);
    }

    isDriving = false;
}
```

```
private bool IsReached()
{
    if (!navAgent.pathPending)
    {
        if (navAgent.remainingDistance <= navAgent.stoppingDistance)
        {
            if (!navAgent.hasPath || navAgent.velocity.sqrMagnitude == 0f)
            {
                return true;
            }
        }
    }
    return false;
}
```

As you can see it is not in fact complex. It starts moving toward a first position and then when reached moves to the second and then to the parking position and then stops. `IsReached()` simply checks to see if it arrived at the destination or not.

The reason that it drives like a drunk driver who never was a good driver to begin with, is that NavMeshAgent's movement logic is not made for driving and similar behavior and is more aligned with how a creature would move. Also we did not set street paths so it behaves like this and it is an insult to drunk drivers actually!

Anyways this is how this code drives the car when the agent is acting as driver.

Go To Car

This action uses `MoveToSmartObject` class to move to the car for driving it. Each car is owned by an agent and the ownership is set by the `SmartObjectSlotOwnerSetter` component on the cars. The action searches for smart objects with the tag Car in 100 meters.

The considerations are:

- DriveJobValue which uses `BlackboardIntValueConsideration` which checks to see if an int key is the expected value or not and is a binary consideration. This one returns 1 for Job being equal to 1. The curve returns something a bit higher than 0.7 so the actual driving action above has a bigger chance when the agent is close to the car.
- There is a TimeOfDay consideration which returns 1 for the time between 10AM to 6PM.
- And of course a distance consideration for targets which always returns 1 because we don't need to differentiate between the targets other than checking ownership which the action does when adding targets. We have to have a consideration which scores all targets anyways so we have to include something. Theoretically a distance one can become useful later on in the demo so we

added this. Otherwise a constant 1 would be enough. We know each agent only has one car and we have no buying/selling.

Eat Oak

Some of the humans go out to do their oak gathering job and while close to an oak and hungry, they can eat one. It will look intelligent and logical to do so. The action class used is `DestroyTargetWithTag` because the action needs to find oak targets and if one of them is selected, it destroys the target and its changer increases Hunger by 0.15. The action finds targets by looking for objects with the Oak tag in its 3 meters radius using the tag system

The considerations are a float which checks Hunger and if it is relatively low, it returns values up to 1. And one which checks the target's distance which checks up to 3 meters and its curve returns something close to 1 for almost all values. This means that if hungry and close to an oak, the agent will eat it no matter what. We could check if there are any mice around or not and other things like only do this if very hungry or only if the time to go home is very far away from now but as is, our humans will eat oaks until they are more than half full.

Go To Oak Hunt

This action moves to a location which the influence map says there are lots of oaks in. The considerations naturally are a check for the int value of the job which returns 0.5 if the Job key is 2. There is an `InfluenceMapConsideration` which searches the Oak map for values greater than 0.7 in its 100 meters radius. This will put the result in the OakLocation key to move to. There is also a time of day consideration which is only returning 1 in the time range of 6AM to 4PM.

This action returns at most 0.5 so the more important action of actually gathering or eating oak can score higher in close ranges.

Move To Oak For Gathering

This uses `MoveToTargetWithTag` and tries to find objects tagged Oak in the 10 meter radius and move to the selected one if the action is selected. This action happens after the previous action brings the agent close enough to oaks. The considerations are designed in a way so if oaks are close by then this will score higher. The radius for the previous action was 100 meters.

The considerations are a time of day consideration so this happens only from 6:00 to 16:00 and a check to see if Job key is equal to 2. And an important one which checks target distances. It checks the target up to 10 meters and for close ones other than very close ones returns up to 0.9. The job key one returns 0.7 so it has less score than the eating/gathering ones.

Oaks are plain tagged objects, so nothing reserves an oak once an agent picks it. Two gatherers near the same cluster can both walk to the closest oak; whoever arrives first gathers it, and the other agent's Gather Oak action fails and it retargets the next one. The demo leaves this in because the oak actions exist to show the tag system and influence maps, and a failed gather that retargets is harmless here. In a

real game you would prevent it in one of two ways. Make the oak a SmartObject with a single slot and use `BlackboardActionWithSmartObjectTarget` as the programming and sleep actions do, so the slot claim keeps a second agent from choosing it (see Smart Objects). Or keep tags and have the move action add a Claimed tag to its chosen target in `OnStart`, remove it in `OnFinish`, and list Claimed in `excludedTags` on both oak actions. Mice have the same race when eating oaks, and the same fix applies.

Gather Oak

This is a simple `DestroyTargetWithTag` which has a small radius and looks for objects tagged Oak in 2 meters. It has a delay for destroying the oak, which looks good and differentiates it from eating as well. The changer on the action increases Usefulness key of the blackboard by 0.1. The considerations naturally are a Job key check to be 2 and a target distance and these return values up to 0.8 combined so this is chosen over the above actions other than eating oak if the agent is hungry.

Programming

This action uses `BlackboardActionWithSmartObjectTarget` to find its laptop and use it and the action's changer increases usefulness by 0.03 per second. The considerations are a check for the Job key to be 0, a target distance check and a time of day check like the other job considerations. The distance one checks ownership of the target too. It checks between 0.5 to 2 meters and returns up to 1. Since programmers usually start late, their time of day for work is between 12AM to 12PM. The Job key checker checks to see whether the Job value is 0 (the programmer's number) or not, and if it is 0 then it returns 0.5 on the curve.

Go To Laptop

This action is for moving toward the laptop for programming. This action uses MoveToSmartObject class and searches for targets in 100 meters and searches for objects which have the Laptop tag. The considerations are the same as the programming job with different curves and values.

The Job key checker returns 0.5 when the value is correct. The distance one doesn't matter so the distance returns 1 for all of them. The reason that it doesn't matter is that there is only one laptop owned by us but there is only one.

Seek Cat

The custom action is `SeakingCatAction` and it inherits from `MoveToTargetWithTag`. This simply finds the best cat in the area and then sets Companionship and TargetCat keys in its blackboard so it can pet the cat. Read more in the description of this action in the cats section. Keep in mind that humans pet all cats and don't add their preferences for specific cats. It is amazing that with only a few lines of code we have this relatively complex faction system and the cats and humans can easily change factions and do everything else like changing the faction of others. We can even have events like holidays well even gray cats greet others and then go back to their usual after the event.

The considerations are a check for the human's own Entertainment key which returns up to 1 and has inverse checked, a target distance check which returns at least 0.5 and an `IsCatFreeConsideration` which checks if the cat is interested in doing this with us and is free or not. This consideration is described in detail in the cat section but the main code for it is this.

```
CatAgent cat = target.GetComponent<CatAgent>();
if (cat != null && cat.IsBusy() || target.gameObject == Brain.gameObject ||
target.GetComponent<BlackBoard>().GetFloat("Emotion") > 0.6f)
    return 0;
else
    return 1;
```

Pet Cat

This action is the same `CatCompanionshipAction` which cats use and only gets a high score when a target cat exists and is set in the blackboard. The only difference is that the changer increases the Entertainment value for the human and not Emotion which is something cats have and apparently humans don't :)

The considerations are only two which check the two relevant blackboard keys and both of them if they have the correct value, will return 1.

The interface is implemented in the HumanAgent like this

```
public void StartCompanionship()
{
    if (myBlackboard.GetBool("Companionship") == false)
    {
        myBlackboard.SetBool("Companionship", true);
        GetComponent<Brain>().Think();
    }
}

public void EndCompanionship()
{
    myBlackboard.SetBool("Companionship", false);
}

public void WaitForOtherCompanionship(GameObject otherCat)
{
}

public bool IsBusy()
{
```

```
        return true;
    }
```

As you can see since the human will not be at the receiving end of this, `IsBusy()` and `WaitForOtherCompanionship()`'s implementation don't matter.

Kill Mouse

Kill Mouse is another action like Eat Oak to make humans smarter. It uses `DestroyTargetWithTagAction` and if the human is close to a mouse, then the human destroys the mouse. The action checks for objects with the tag Mouse in 1.5 meters radius. The only consideration is a distance check always returning one for all targets so if mice are super close to a human, they get killed.

The HumanAgent code

The code of the `HumanAgent` class is actually first initializing blackboard values.

```
    private void Awake()
    {
        indicator = GetComponent<SelectedIndicator>();
        myBlackboard = GetComponent<BlackBoard>();

        myBlackboard.SetFloat("Hunger", Random.Range(0.3f, 0.7f));
        myBlackboard.SetFloat("Energy", Random.Range(0.3f, 0.7f));
        myBlackboard.SetFloat("Bladder", Random.Range(0.3f, 0.7f));
        myBlackboard.SetFloat("Entertainment", Random.Range(0.3f, 0.7f));
        myBlackboard.SetFloat("Usefulness", Random.Range(0.3f, 0.7f));
        myBlackboard.SetInt("Job", job);
        myBlackboard.SetBool("Companionship", false);

    }
```

The job value is read from the job exposed to the inspector so you can change the jobs of the agents in the inspector.

```
    void Start()
    {
        personalizedHunger = Random.Range(minPersonalizedValue, maxPersonalizedValue);
        personalizedEnergy = Random.Range(minPersonalizedValue / 2f, maxPersonalizedValue / 2f);
        personalizedBladder = Random.Range(minPersonalizedValue, maxPersonalizedValue);
        personalizedEntertainment = Random.Range(minPersonalizedValue, maxPersonalizedValue);
        personalizedUsefulness = Random.Range(minPersonalizedValue / 2, maxPersonalizedValue
    / 2);
```

```
        LifeSimDemoManager.Instance.AddFollowableAgent(indicator, AgentType.Human);
    }
```

We set the personalized random values for the changes to main attributes so the agents act differently from each other and different simulation runs are different. This is what we do with all other agents including the light simulation as well.

The update function simply decreases all attributes based on the chosen personalized values.

```
    void Update()
    {
        float hunger = myBlackboard.GetFloat("Hunger");
        if (hunger > 0)
            myBlackboard.SetFloat("Hunger", hunger - personalizedHunger * Time.deltaTime);

        float energy = myBlackboard.GetFloat("Energy");
        if (energy > 0)
            myBlackboard.SetFloat("Energy", energy - personalizedEnergy * Time.deltaTime);

        float bladder = myBlackboard.GetFloat("Bladder");
        if (bladder > 0)
            myBlackboard.SetFloat("Bladder", bladder - personalizedBladder * Time.deltaTime);

        float entertainment = myBlackboard.GetFloat("Entertainment");
        if (entertainment > 0)
            myBlackboard.SetFloat("Entertainment", entertainment - personalizedEntertainment
    * Time.deltaTime);

        float usefulness = myBlackboard.GetFloat("Usefulness");
        if (usefulness > 0)
            myBlackboard.SetFloat("Usefulness", usefulness - personalizedUsefulness
    * Time.deltaTime);


        // Check for mouse click or touch input
        if (Input.GetMouseButtonDown(0))
        {
            // Cast a ray from the camera to the mouse position
            Ray ray = Camera.main.ScreenPointToRay(Input.mousePosition);
            RaycastHit hit;

            // Check if the ray hits the object with a collider
            if (Physics.Raycast(ray, out hit))
            {
                // Check if the hit object is the one you want
```

```
            if (hit.collider.gameObject == gameObject)
            {
                LifeSimDemoManager.Instance.SetAsSelectedAgent(indicator, AgentType.Human);
            }
        }
    }
}
```

It has some code regarding the selection logic like other agents which is discussed below as well.

The car driving behavior calls these two methods.

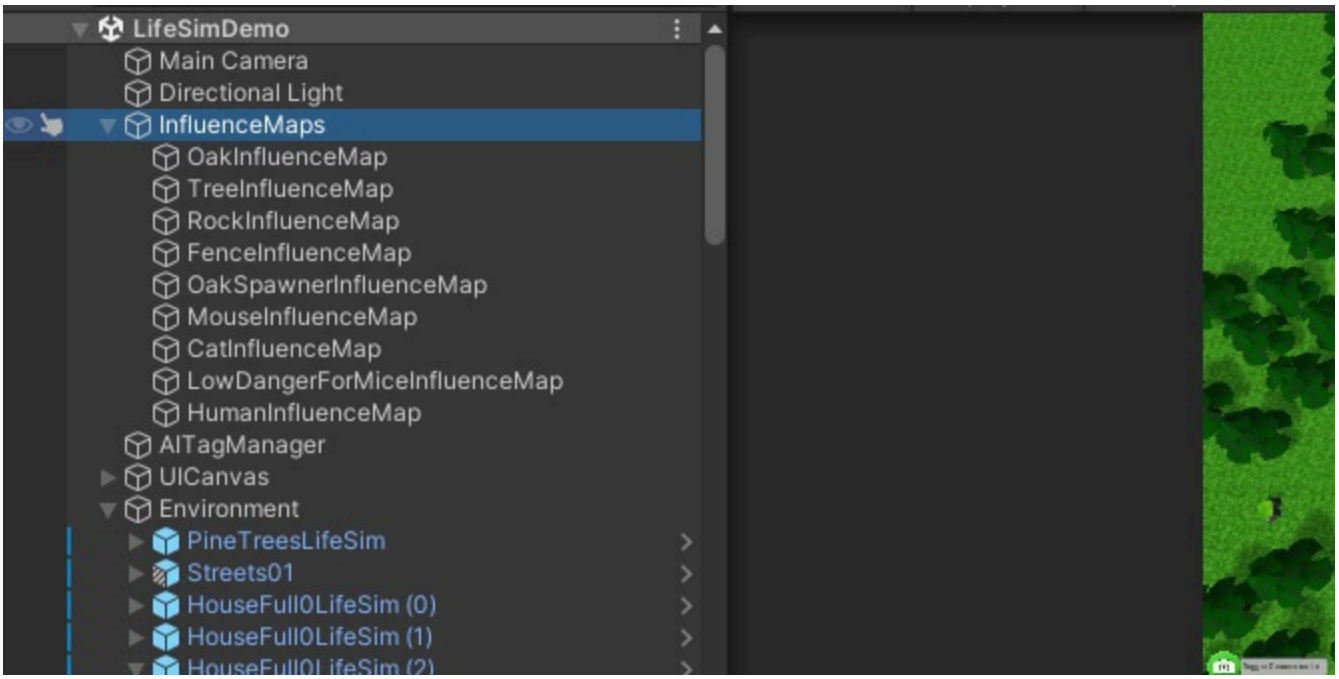
```
internal void StartDrivingState(SmartObject smartObject)
{
    humanMesh.SetActive(false);
    GetComponent<NavMeshAgent>().enabled = false;
    transform.SetParent(smartObject.transform);
    transform.position = smartObject.transform.position;
    transform.LookAt(smartObject.transform.forward);
}

internal void EndDrivingState(SmartObject smartObject)
{
    humanMesh.SetActive(true);
    GetComponent<NavMeshAgent>().enabled = true;
    transform.SetParent(null);
    transform.position = smartObject.GetSlotPosition(0);
}
```

The methods simply hide the agent when it starts driving and makes it a child of the car and does the reverse when driving finishes.

The rest is the `ICatCompanionship` interface which is described in the actions. And this concludes our tour of the AI in the codebase.

Maps

We use several influence maps and influence map views as it got mentioned in the review above. All of the maps and views are children of the InfluenceMaps object.



General simulation code

The simulation needs some additional code for the follow camera and updating the UI and advancing time of day. Here we are going to take a very brief look at those scripts to make your job a bit easier.

- `CameraFollow.cs` is a very simple camera follow code which tolerates having a null target as well.
- `DayAndNightCycle.cs` moves time forward in a 24 hour day and changes its rotation too so you can attach it to your directional light which changes the sky color as well.
- `LifeSimDemoManager.cs` This class manages the change of simulation speed, has the code for changing the agent camera follows using Tab and Shift+Tab and other general things. There is no AI code in this class.
- `SelectedIndicator.cs` Is the code which selects the agent you click on and turns on the indicator on top of the agent's head. The agents themselves check this too and call methods in the `LifeSimDemoManager` to change the UI and do some other things. Needless to say the way that we code this can be improved a lot but this has nothing to do with the AI.
- `UIMenu.cs` Manages most UI code to show the correct icon and update the attribute bars and ... The event managers are attached in the editor and not in code.
- `SliderController.cs` Is the custom bar for the attributes of agents.
- `GameSpeedButton.cs` This handles the looks of the button which shows what speed the game is running on.

And after these many pages this is finally over. We are going to improve this tutorial and improve it and make more interesting stuff for you. Thanks for reading and please rate and review the assets if you liked them. Also send us your feedback on discord or using email.

Blackboards

Blackboards video tutorial

A Blackboard is a data structure which allows you to share data between different components of your AI without having direct dependencies between them. A blackboard is a `Dictionary` like data structure which allows you to add data by defining it as a set of key value pairs.

Let's say you want to know what is the safest location to go to, you can define a safe location key in the blackboard and store the `Vector3` of the location in it. Our implementation of blackboards is more strongly typed than a `Dictionary<string,object>` and you should define all keys with the data type of their values before being able to set/get them.

Getting Started

To define a blackboard you first need to create a blackboard definition. To do that right click in your project and click on Create>NoOpArmy>Wise Feline>BlackBoard Definition. Then you need to assign that definition to a `BlackBoard` component. Then you can set or get the value of the keys you defined in the definition.

You have multiple methods in the `BlackBoard` component like `GetFloat(key)`, `SetFloat()`, `GetVector3(key)` and `SetVector3(key,value)`. You have these pairs for all data types and you can set and get the values in your code as you need.

Starting values without code

Keys start empty. A float starts at 0, a bool at false, a string empty and a GameObject at `null`, and an agent whose health starts at 0 or whose patrol route key is `null` does not behave the way you designed it.

The `BlackboardInitialValues` component solves this without any code. Add it to the same GameObject as the `BlackBoard` component and fill in a list of values in the inspector, and they are written into the blackboard when the scene starts. Each entry is a key type, a key name and the value to write, which is the same little editor you already know from the blackboard changer of the built-in actions.

This is also the only code-free way to put a **reference to a scene object** into a blackboard. Behavior assets and their changers live in the project, and an asset cannot reference something which only exists in a scene, but a component in the scene can. So the object holding your patrol points, the camp an agent retreats to, the player, a spawn point or a door can all simply be dragged into this list.

> **WARNING**
>
> Entries here are almost always **sets**, but `Add Instead Of Set` and `Multiply by deltaTime for adds` are **on by default**, because the same entry type is also used by actions which change a value a little bit every frame. Turn both of them off for starting values. If you leave them on, "start at 100 health" becomes "add 100 multiplied by one frame's deltaTime to whatever is there already", and the agent starts the game with about 1.7 health instead.

Two more things worth knowing:

- The key has to already be defined, with the same type, in the blackboard definition which the `BlackBoard` component uses. Writing an undefined key throws.
- The values are applied in `Start()`, because the `BlackBoard` component defines its keys in `Awake()`. The component also carries a negative execution order so that its `Start()` runs before the `Start()` of normally ordered components. That matters more than it looks: actions such as `MoveAndPatrol` read their GameObject key once, when the `Brain` initializes them in its own `Start()`, and without the ordering the route would sometimes come up empty depending on which `Start()` happened to run first.

The Combat Quickstart uses this component for all four of its keys and is a complete worked example.

How can I use this?

Your AI can use this in considerations and actions and other parts of your game. Let's say you have an action like GoToLocation, you can take a blackboard key which you know contains a `Vector3`.

A small tutorial

- Create a definition and call it b.
- Add a float key called speed.
- Create a new script called `TestBlackBoard` and attach it with a `BlackBoard` component to a GameObject.
- Set the definition in the blackboard to the one you made above.
- Open the `TestBlackBoard` script you made and add some code like this.

```
public class TestBlackBoard : MonoBehaviour
{
        private BlackBoard board;

        void Start()
        {
```

```
                board = GetComponent<BlackBoard>();
                board.SetFloat("speed", 3);
                var speed = board.GetFloat("speed");
                print(speed);
        }
}
```

The code should print 3.

> This module is only available in the ultimate edition of the Wise Feline package

AI Tags module

AI tags is a module which allows you to attach tags to objects at edit time and also at runtime. It also allows you to query all objects with tags in a sphere to decide based on the result or do anything with them. You can use this sub-system with Utility AI and it has a sample consideration or as a fully separate module outside of your utility AI.

How to use AI Tags

Add the `AITags` component to GameObjects which should have tags. You can add tags in the inspector or at runtime using the available methods.

Then you can use methods of the `AITagsManager` singleton to get objects around a point and check their tags. Also if you know an object has the `AITags` component, you can check its tags in code by getting the component directly.

`AiTagsManager` singleton is created automatically if you don't add it to a component in your scene. It has methods for getting objects around a point which include and/or exclude some tags. At its most complex form you can give it a list of tags which at least one of them should exist in an object to qualify and also another array for tags which if any of them on the object then the object disqualifies from being returned.

Use-Cases

A few small examples might help. Imagine you want to implement generic behaviors like I want my agent to eat all plants and avoid all animals. You can avoid hard coding this by tagging all plants with both plant and their specific plant name as tags and also tag all animals with animal and their specific animal name and then use the search functionality of the AI Tags system to find all plants or all animals.

Also imagine the game has a dragon which can set things on fire and you don't want to specifically code all components for this, you can simply add firable to things which can be set on fire, even better you can add the tag at runtime to something after some time, let's say there is an ice monster which can not be set on fire unless there are no minions around. You can add the firable tag to it at runtime after all of its minions died.

> This module is only available in the ultimate edition of the Wise Feline package or the separate Influence Maps package

Influence maps module

Great video about influence maps by Dave Mark

Agents need to know what is going on around them in the world to be able to make good decisions. Having a toolbox which allows you to know spatial information about the world like:

- Where are the main paths which enemies use?
- Where is the best position to throw my grenade?
- Where is the busiest area of the map around here?
- Where can I find some friendly forces to join?
- Where are the most trees located so I can spawn some rabbits?

is very useful.

Influence maps allow you to survey the environment of your game and find spatial info in a way which raycasts and octrees don't allow. An Influence map gives you info on what parts of the map something exists too much and on what part it doesn't exist. It can give you max and min points and busy and sparse parts. It can give you info not only on things which are present in the map but also about the events which happened before or will happen later.

The way influence maps work is that the map is a 2d array which agents propagate their influence on. For example you can have a map which all enemies propagate their position and attack range on and another where trees propagate their existence on.

Now agents can find out where they should flee to using the map points with smaller values in the enemies map and they also can find where they should fire at, by finding points with higher values.

You can search for higher and lower values in a circle in a specific part of the map or in the whole map. You can also combine multiple maps and then search on the resulting map so for example you can find a point where there are both lots of trees and enemies to set the trees there on fire to burn your enemies alive. Combining this system with AI tags, Smart Objects, The octree and Utility AI allows you to do very smart and immersive things in your games which most game designers only dream of.

Components and programming

In general this system can be used both as a set of components and also as a set of pure .NET classes without any components. The `InfluencerAgent` component allows you to propagate yourself on a map and the InfluenceMapComponent component allows you to create a map by just attaching it to a

GameObject. The `InfluenceMapView` component allows you to create a map by combining other maps and the `InfluenceMapCollection` component is a singleton which allows you to find maps by their name.

Each agent can propagate its value to the map but the value should be a circle with gradient values most of the time. It doesn't make sense to calculate all values every time. For this reason, you create a set of influence map templates which agents use as their stamp to propagate their value. In this way an enemy can propagate 1 in its position and reduce its influence value up to some distance like 15 meters to 0. The fall off of the value is chosen by the curve that you define in the template. The template values are calculated once and are just copied to the positions which need them afterwards.

To create your first influence map

- Right click in the project view and go to Create>NoOpArmy>WiseFeline>Influence Map Template and name your template linear. Then choose a linear curve and a value of 50 for the radius.
- Add two gameObjects to the scene called agent and map.
- Add an `InfluenceMapComponent` component to the map and an `InfluencerAgent` to the agent. Name them map enemies (this is the name that you can use to get the map from InfluenceMapCollection).
- Select the agent GameObject and then select the map and the linear template for the agent. You can choose a map by dragging it or entering its name.
- Select the map GameObject and then set the size and resolution of the map to some values like 1000 for width and height and 1 for cell size.

Now if you press play and move the agent around, you can see the propagation of the agent on the map in scene view gizmos. If the gizmos are not shown, check the box to draw them in the `InfluenceMapComponent` component.

InfluenceMapComponent

This component allows you to create an influence map with the size and resolution specified and can draw the values of the map as gizmos in the scene view for you. It also has methods to search the map for values and also convert positions from map coordinates to world coordinates and vice versa.

The `InfluenceMapComponent` component has all methods you need in most cases but also gives you access to the map in the `Map` property and the map has all of the search functionality you need.

> Influence maps are 2D arrays and they have a different resolution from your world so you need to convert positions of your GameObjects from world coordinates to map coordinates to use them with influence maps. Also when you get a coordinate from an influence map search or some other operation, you need to convert it from map coordinates to world coordinates so it can be used by your GameObjects.

InfluenceMapCollection

The `InfluenceMapCollection` singleton allows you to get maps by their name and maps created using the `InfluenceMapComponent` and `InfluenceMapView` components automatically register themselves to it. This component is created automatically if you don't have one in the scene.

How the search works

The search functionality takes a value and an enum which is a comparison operator. For example if you give it 0.6 and greater, then it will try to find a value bigger than 0.6 in the radius and stops as soon as it finds one. It starts searching from a random position in the specified search area so different agents don't end up choosing the same location.

Videos

These are the imap videos

https://www.youtube.com/watch?v=hnnQ7eEkeCQ https://www.youtube.com/watch?v=9Lk9ibddw60&t=118s https://www.youtube.com/watch?v=O2bDqOW1i5c https://www.youtube.com/watch?v=kuAwzUzzzIg&t=19s

> This module is only available in the ultimate edition of the Wise Feline package

Smart Objects module

Our video tutorial for this

Agents need to know what objects are available in the environment and what they can do with them. We usually hard-code these facts in the agent code and look for specific objects and the agent code itself executes a behavior/interaction with the object.

Smart objects allow you to look for an object which can do a specific task/behavior or fulfil a specific need. Instead of the agent executing a behavior on the object, the object executes a behavior on the agent and dictates what it can do and how it is done.

For example you can look for things which can heal you or give you immunity and then both a healthpack and an armor can advertise themselves to you as objects which can get the job done. Then your AI can choose which one to use based on different parameters like how good they are or how dangerous it is to move to them.

Instead of the agent executing an animation to equip the armor and then change its own properties, the armor has a behavior which takes the agent as a parameter and gets executed when the agent calls `Use()` on the armor object. The method can then play an equipping animation on the agent and change its components to achieve the desired effect.

In this way Smart objects allow you a degree of flexibility which can be very useful for games with dynamic environments. For example if a character needs warmth during winter and at the moment, clothing, fire and vehicles provide it with different qualities, your designers can easily add a magic hat to the game which advertises itself for giving the best type of warmth and it can have a behavior which attaches itself to the agent and executes an animation on the agent which puts the hat on the head of the agent. Then it plays the happiness sound effect of the agent. This all can happen without agent code being changed so it is very useful for mods, updates and in general very dynamic logic and is heavily used in games like The Sims.

Smart objects can also have multiple slots which can be used by multiple agents at the same time. An obvious example is a car which can be used by multiple agents but also a bed or a house can be used like that.

Technical details

- A GameObject which is declared as smart object has a SmartObject component attached.
- A SmartObject component has a smart object definition assigned.

- A smart object definition asset is created from the right click menu of the project view and defines all the behaviors a smart object supports and all of its slots and their properties.
- The `SmartObjectManager` singleton manages all smart objects and allows you to query for smart objects which have specific behaviors and/or tags.

As said before a smart object is made of a set of behaviors and a set of slots which can be used by agents. Each smart object slot can be reserved and then used by an agent and when it is used, behaviors of the smart object get executed for that agent. Both the smart object and the agent can free a slot.

Smart Object slots

A slot is a reservable and usable part of a smart object. A freezer usually has 1 slot because it can only be used by one agent at a time while a car has probably 4.

Smart Object definitions

A smart object is defined by a `SmartObjectDefinition` scriptable object asset and is attached to a GameObject using a `SmartObject` component. The definition asset defines how many slots the object has and their properties and the component defines the behaviors which the smart object can do. A smart object should have at least 1 slot. Optionally each slot can have additional custom behaviors or only support the behaviors the main object supports. Also each slot has a position offset from the object it is defined for and the agent's ideal position from the smart object position can be set as well. You can even choose the agent's ideal rotation toward the slot.

Smart Object Behaviors

A smart object behavior can be defined by inheriting from the `SmartObjectBehaviorBase` class. You can override virtual calls for the time that an agent successfully reserves a slot of the object and when the agent uses a slot or frees it successfully. The agent's GameObject with the `SmartObject` component and the slot index are passed to the call and the behavior can execute whatever functionality it wants on the agent or the smart object itself.

Behaviors are attached to smart object's GameObject as a component and assigned in the SmartObject component.

The agent has no requirements and any GameObject can reserve and use smart objects. If an agent is destroyed which reserved or was using a slot of a smart object, the slot is automatically considered free.

This is the behavior base class's code.

```
    /// <summary>
    /// Behaviors of smart objects inherit from this class.
    /// A smart object behavior operates on the agent which reserves, uses or frees a
slot in a smart objects. These events are defined in this base class as virtual methods
```

```
to override.
    /// </summary>
    public class SmartObjectBehaviourBase : MonoBehaviour
    {
        /// <summary>
        /// Called when a slot of the object is successfully used by an agent
        /// </summary>
        /// <param name="agent"></param>
        /// <param name="smartObject"></param>
        /// <param name="slot"></param>
        public virtual void OnUsedByAgent(GameObject agent, SmartObject smartObject,
int slot)
        {

        }

        /// <summary>
        /// Called when a slot of the smart object is successfully claimed by an object
        /// </summary>
        /// <param name="agent"></param>
        /// <param name="smartObject"></param>
        /// <param name="slot"></param>
        public virtual void OnClaimedByAgent(GameObject agent, SmartObject smartObject,
int slot)
        {

        }

        /// <summary>
        /// Called when an agent frees the smart object
        /// </summary>
        /// <param name="agent"></param>
        /// <param name="smartObject"></param>
        /// <param name="slot"></param>
        public virtual void OnFreedByAgent(GameObject agent, SmartObject smartObject,
int slot)
        {

        }
    }
```

As you can see there are three methods to override which give you all the information you need to execute your logic and since this is a component, you can get any other info you need from the inspector or other objects.

SmartObject component

You attach this component to each GameObject which you want to declare as a smart object. The definition is assigned to this component and the behaviors are assigned here as well. Behaviors can be both slot specific and general. The general ones work on all slots.

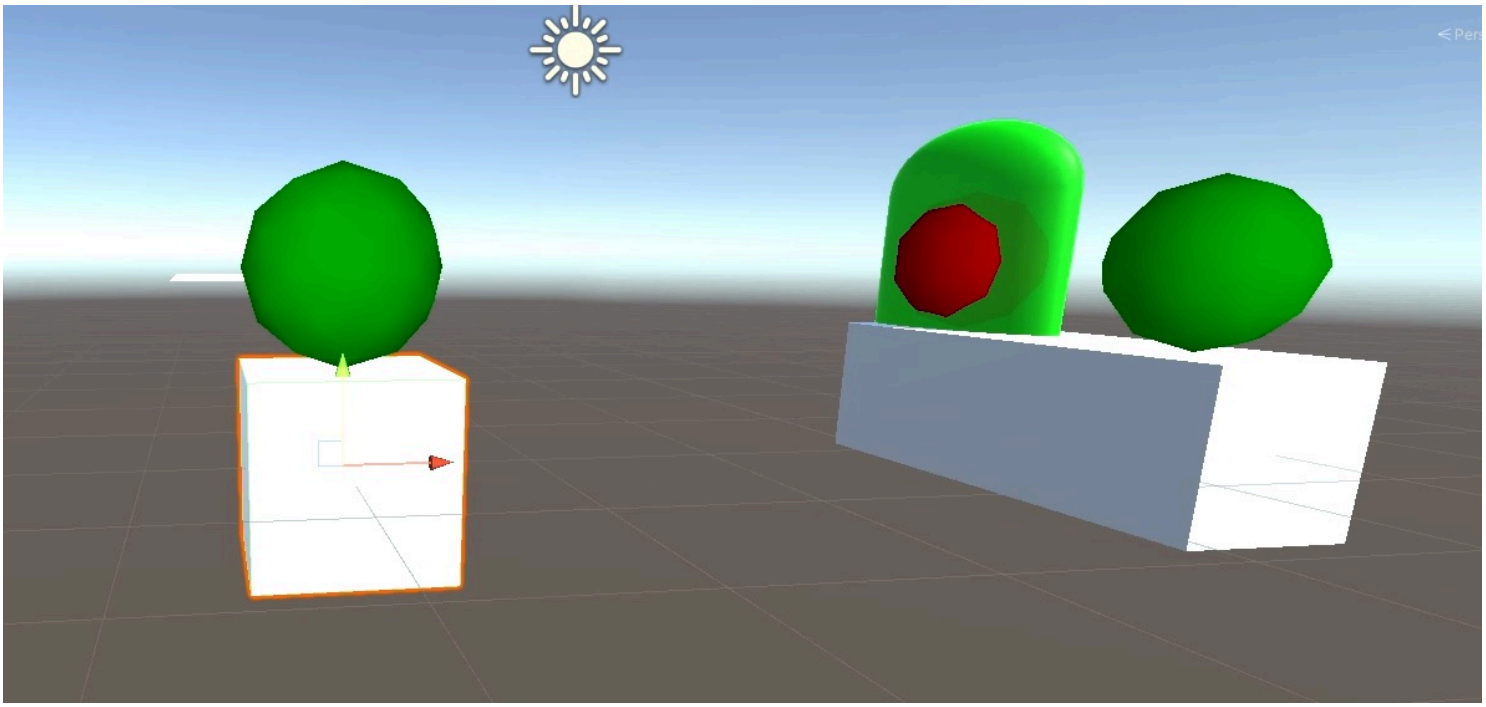
There are some other settings regarding tags which the smart object has, whether the object should update its position in the octree for queries periodically or not and ...

SmartObjectsManager

This singleton is created automatically if it is not attached to an object in the scene and allows you to find smart objects around a point and with a radius and filter the acceptable smart objects by the number of free slots they have and the behaviors they support. You can also filter them by including and excluding smart objects with specific tags. More details on how to query for smart objects are provided below.

Gizmos

Smart Objects have gizmos which show you where each slot is and if it is free (green), claimed (blue) or in use (red).



The object on the right has two slots which one of them is in-use by the capsule. The other slot and the slot of the other object are free so they are green. They are drawn on their location based on their offset too.

A simple demo

The demo folder of the smart objects folder contains a simple demo which has two agents, a car and a refrigerator. One of the agents looks for something to eat first and grabs the fridge and the fridge's eat behavior causes it to turn green and then after that finishes it searches for something drivable and goes to the car. The other agent at the start searches for something drivable and goes to the car. The car waits for both agents to sit and then moves to a destination.

Let's go through the different parts together.

Smart Object behaviors

We have only two behaviors defined for this demo. Eating and driving. Here is the eating behavior

```
    /// <summary>
    /// This behavior give the agent some food to eat
    /// This is a simple demo so we just change the color of the agent to green
    /// </summary>
    public class EatFoodBehaviour :
NoOpArmy.WiseFeline.SmartObjects.SmartObjectBehaviourBase
    {

        /// <summary>
        /// Called when the agent reserves this so nobody else touches it
        /// </summary>
        /// <param name="agent"></param>
        /// <param name="smartObject"></param>
        /// <param name="slot"></param>
        public override void OnClaimedByAgent(GameObject agent, SmartObject smartObject,
int slot)
        {
            base.OnClaimedByAgent(agent, smartObject, slot);
            //Just change our own color to indicate reserving happened
            GetComponent<Renderer>().material.color = Color.blue;
            agent.transform.LookAt(transform.position);
        }

        /// <summary>
        /// Called when the agent using us frees us
        /// </summary>
        /// <param name="agent"></param>
        /// <param name="smartObject"></param>
        /// <param name="slot"></param>
        public override void OnFreedByAgent(GameObject agent, SmartObject smartObject,
int slot)
```

```
        {
            base.OnFreedByAgent(agent, smartObject, slot);
            //Just change color and size to defaults
            GetComponent<Renderer>().material.color = Color.white;
            transform.localScale = Vector3.one;
        }

        /// <summary>
        /// Called when the agent uses us
        /// </summary>
        /// <param name="agent"></param>
        /// <param name="smartObject"></param>
        /// <param name="slot"></param>
        public override void OnUsedByAgent(GameObject agent, SmartObject smartObject,
int slot)
        {
            base.OnUsedByAgent(agent, smartObject, slot);
            //Just do change some colors on ourself and the agent so we can see the object
is used and the agents which ate before driving are observable
            agent.GetComponent<Renderer>().material.color = Color.green;
            agent.transform.Translate(0, 0, 1);
            transform.localScale = Vector3.one * 0.5f;
            //free the object
            smartObject.Free(slot, agent);
        }
    }
```

As you can see this behavior just changes colors of itself and the agent to help you observe it is doing something. When claimed, it turns blue and when freed it turns white. Also when used it turns the agent to green to show it is not hungry anymore.

The driving behavior is more involved but essentially is no different.

```
    public class DrivingBehaviour : SmartObjectBehaviourBase
    {
        /// <summary>
        /// The direction this will drive toward
        /// </summary>
        [Tooltip("The direction this will drive toward")]
        public Vector3 movementDirection;

        /// <summary>
        /// Driving speed
        /// </summary>
        public float speed = 5;
```

```
        /// <summary>
        /// The destination on the XZ plane
        /// </summary>
        public Vector3 destinationXZ;

        /// <summary>
        /// The slots which are claimed by a user. We use this to check if all slots are
filled or not.
        /// We could use a counter or get all slots and see if all are claimed or not but
this can be useful later on
        /// </summary>
        private List<int> claimedSlots = new List<int>();

        public override void OnClaimedByAgent(GameObject agent, SmartObject smartObject,
int slot)
        {
            base.OnClaimedByAgent(agent, smartObject, slot);
            claimedSlots.Add(slot);
        }

        public override void OnFreedByAgent(GameObject agent, SmartObject smartObject,
int slot)
        {
            base.OnFreedByAgent(agent, smartObject, slot);
            agent.transform.SetParent(null);
            claimedSlots.Remove(slot);
        }

        public override void OnUsedByAgent(GameObject agent, SmartObject smartObject,
int slot)
        {
            agent.transform.SetParent(smartObject.transform);
            base.OnUsedByAgent(agent, smartObject, slot);
            //if all slots are filled, drive into the night
            if (AreAllslotsFilled(smartObject))
            {
                StartCoroutine(Drive(smartObject));
            }
        }

        IEnumerator Drive(SmartObject obj)
        {
            //super simple driving logic
            var targetY = Quaternion.LookRotation(movementDirection).eulerAngles.y;
            while (Mathf.Abs(obj.transform.eulerAngles.y - targetY) > 0.05f)
```

```
            {
                obj.transform.Rotate(0, 90 * Time.deltaTime, 0);
                obj.transform.position += movementDirection * speed * Time.deltaTime;
                yield return null;
            }

            //Take control of the camera
            Camera.main.transform.SetParent(obj.transform);

            //Move toward a destination
            while (Vector3.Distance(new Vector3(obj.transform.position.x, destinationXZ.y,
obj.transform.position.z), destinationXZ) > 1)
            {
                obj.transform.position += movementDirection * speed * Time.deltaTime;
                yield return null;
            }

            //Arrived at destination. Free the camera and all agents
            Camera.main.transform.SetParent(null);
            obj.FreeAllSlots();
        }

        public bool AreAllslotsFilled(SmartObject smartObject)
        {
            return smartObject.GetSlotCount() == claimedSlots.Count;
        }
    }
```

This behavior waits for all of its slots to get occupied and then starts moving at the direction specified in the inspector toward a destination and when arrived at the destination, frees all of its slots. One interesting thing it does is that it even takes control of the camera when it starts moving and this shows you the flexibility of this system where a behavior can do almost anything. In the video tutorial linked above, we quickly made a space ship using this behavior.

Agent code

Now let's take a look at the agent code.

```
    /// <summary>
    /// This script allows your demo agent to eat food and then drive somewhere.
    /// you can choose for each agent to do either both or one of them or do neither which
is not useful.
    /// </summary>
    public class SmartObjectDemoPlayer : MonoBehaviour
    {
```

```
        /// <summary>
        /// Should the agent go eat something before driving somewhere
        /// </summary>
        [Tooltip("Should the agent go eat something before driving somewhere")]
        public bool shouldEat;

        /// <summary>
        /// Should the agent find something drivable and drive somewhere
        /// </summary>
        [Tooltip("Should the agent find something drivable and drive somewhere")]
        public bool shouldDrive = true;

        IEnumerator Start()
        {
            if (shouldEat)
                yield return StartCoroutine(Eat());
            if (shouldDrive)
                yield return StartCoroutine(Drive());
        }

        IEnumerator Eat()
        {
            List<(SmartObject smartObject, SmartObject.FilterResult data)> objects =
SmartObjectsManager.Instance.GetSmartObjectsAroundPoint<EatFoodBehaviour>
(transform.position, 30, true);
            if (objects.Count > 0)
            {
                int slot = objects[0].smartObject.GetFreeSlotIndex();
                if (slot != -1)
                {
                    float w = 0;
                    var initialPosition = transform.position;
                    while (w < 1)
                    {
                        transform.position = Vector3.Lerp(initialPosition,
objects[0].smartObject.transform.position, w);
                        w += Time.deltaTime / 2;
                        yield return null;
                    }

                    var b1 = objects[0].smartObject.Claim(slot, this.gameObject);
                    if (b1)
                    {
                        yield return new WaitForSeconds(1);
                        var b2 = objects[0].smartObject.Use(slot, this.gameObject);
                    }
```

```
                }
            }
        }

        IEnumerator Drive()
        {
            List<(SmartObject smartObject, SmartObject.FilterResult data)> objects =
SmartObjectsManager.Instance.GetSmartObjectsAroundPoint<DrivingBehaviour>
(transform.position, 30, true);
            if (objects.Count > 0)
            {
                int slot = objects[0].smartObject.GetFreeSlotIndex();
                if (slot != -1)
                {
                    float w = 0;
                    var initialPosition = transform.position;

                    while (w < 1)
                    {
                        transform.position = Vector3.Lerp(initialPosition,
objects[0].smartObject.GetAgentSlotPosition(slot).position, w);
                        w += Time.deltaTime / 2;
                        yield return null;
                    }

                    var b1 = objects[0].smartObject.Claim(slot, this.gameObject);
                    if (b1)
                    {
                        yield return new WaitForSeconds(1);
                        var b2 = objects[0].smartObject.Use(slot, this.gameObject);
                    }
                }
            }
        }
    }
```

The main code which you need to pay attention to are the two coroutines for eating and driving. They both use the `SmartObjectsManager.Instance.GetSmartObjectsAroundPoint<T>()` method to find smart objects which support the behavior they want and then find the free slots of the smart object found by calling `GetFreeSlotIndex()` method of the `SmartObject` component. Then they try to claim and use the object with some movement code added on top.

The rest of the code in this is ordinary Unity code and a quite simple one for that matter.

How these connect?

- You create smart object definitions by right clicking in the project window and then going to *Create>NoOpArmy>Wise Feline>SmartObject Definition*.
- You fill in the number of slots and their offsets in the definition.
- Then attach a `SmartObject` component to the GameObject you want to make a smart object and assign this definition to it.
- For adding behaviors you first write them and then attach them as components to the same GameObject.
- At last you assign the behaviors in the SmartObject component as general or slot specific behaviors.

The agent/player doesn't need any code other than its own behavior attached to it.

Remembrance

Remembrance is a memory and emotions module for your AI agents so they can remember what happened in the world and react to it. Your NPCs can hold grudges against players and other NPCs, return a favor or even try to seduce you or bribe you. They can remember what happened by whom to whom and where and when. You can also attach any custom data to your memories.

How it works

NPCs or any other objects in your game record their memories in a container and then the container can be searched for values with different criteria. They can also set float emotion values per object or in general.

Memory

There are two components that you can use or you can use the underlying data containers. Each of the listed components has an `AddMemory()` method which records a new memory to be remembered. They all also have multiple overloads of a querying method which gives you the closest memory to a specific time with additional arguments which limit the acceptable memories. For example you can return only memories with a specific target or of a specific type or both. There are methods caching the last recorded memory with a specific action id or target for specific common use-cases as well. They usually are of the form `GetLastXXX()`

Memory component

This is the component you want most of the times.

This component is more advanced and can remove memories from the middle of the list and uses a `List<T>` as its underlying data container. When the maximum capacity is reached, the first item in the list is removed after each addition. This component has the forgetting functionality where you can specify in each memory, after how many seconds it should be forgotten and the component forgets that memory after the specified seconds.

SimpleMemory component

This component is simpler and uses a `Queue<T>` as its underlying data container and when reaching capacity, dequeues the first item in the queue. This component cannot forget memories and cannot remove any memory other than the first one in the queue but if it is enough for your use-cases then it has better performance in scenarios where item removal from the list becomes a bottleneck in your game.

Since queues don't have indexing, all searches use the `foreach` construct in C# and are slower than for loops. A foreach structure calls `MoveNext()` and `Current()` methods of the queue to get to the next object

and then read it.

MemoryData

The `MemoryData` class is the type that the methods in the component accept and return.

Properties of a memory

A memory is stored using the `MemoryData` class and it contains a set of properties which can be used for querying

- **MemoryType** This is an integer value which can be used to find out what type the memory is based on your custom types in your game. For example if your `additionalData` can be of multiple types, this can be used to see what type it should be casted to.
- **StartTime** The time that this memory started. The constructor sets this automatically.
- **EndTime** The time that this memory ended. Calling `End(float time)` on the `MemoryData` object sets this automatically.
- **ForgetAfterSeconds** If greater than 0 then the memory will be forgotten after this many seconds.
- **Position** The position that this memory happened in. If you need additional data like room name/level name/bounds, you can put them in additional data or other optional fields.
- **ActionType** This field is of type `System.Type` in .NET and allows you to set the type of the class which caused this memory as an action so you know for example if the shooting was using the `Spell` class or the `FireBall` class. This could be derived from other fields in theory but having it is handy especially in the Utility AI integration.
- **Initiator** The object which initiated the memory. You can put any Unity object including GameObjects and components in this as a reference. It can also be null.
- **Target** The target of this memory, for example the killed object in a murder memory. This can be null if the memory has no targets.
- **ActionId** This can be used for a unique id of an action instance. For example if you want to store exactly which magic spell of an NPC hit you, you can store its id here and put the NPC itself in the initiator field and the spell type in the ActionType field.
- **AdditionalData** is any custom data that you want to attach to this memory. Usually you should find out what this can be casted to based on `ActionType`, `actionId` or `memoryType`

Emotions

There is a component called `EmotionComponent` and an Emotion Collection asset which define what emotions exist and allow you to set and get their values. Emotions can be general or per object. For example your NPC might be generally sad but happy toward a tree they really like or they can have an emotion only for objects, like affection but it might not be meaningful in your game mechanics to have a general affection emotion for the agents.

All emotions are floats between 0 and 1 and define how they are combined with a similar emotion. For example if two objects share an emotion like hate toward each other, then you can ask them to align their hates so if I hate my friend by 0.9 and he hates me by 0.6, first of all we are probably not friends and secondly, there are methods to let our hates affect each other so both are set at the higher value or are averaged based on the settings in the emotions collection.

Debuggers

Both modules have their own specific debugger windows to help you see what values exist in an agent's memory or what emotions it holds.

Wise Feline Den

The Wise Feline Den free package has a set of features helping you make better game AI and other systems faster. These are heavily used in our other products like Utility AI and Influence maps.

We have

- Combat Scripts
- Movement Scripts
- Blackboards
- Data Recorder
- and more

All of these are available in the menu at the left and can be learned and used to your advantage. Check them to see what great stuff we are giving away for free. Level up your game with these awesome tools.

Cookbooks

The cookbooks will be showing you how to implement a specific mechanic, an algorithm or a game genre in Utility AI and other parts of our tool set in both Unity and Unreal.

Some features might be available in one of the engines or in one version of our plugin but not the other but even in those cases we describe potential alternatives or how you can resolve the problem. Don't get too fixated on having a specific feature in a specific way and instead focus on the behavior you want to achieve. This said we strive for feature parity and we almost always have it to a very large extent but the engines themselves are very different with different sets of features and philosophies. To see the rest of the series click here.

Wise Feline Utility AI for different game genres

Different algorithms are more suitable in different places and here we want to take an overview of usefulness of utility AI for different game mechanics and genres. Of course all types of algorithms and toolsets can be used in many different situations but it is important to use what is more useful, more efficient or faster/easier to use for the job. We made this library and use it for our internal projects and are selling it too so you know our biases too and everybody else has theirs too but we try to be clear and objective as much as we can.

Features which Utility AI can add to almost all types of games

Live worlds

Utility AI allows you to make worlds whose NPCs live for themselves and have their own desires, goals and act in their own and their societies' interest.

Deep Worlds

Having complex AI and live worlds would mean that the worlds become very deep and when you play in them, you feel immersed in a universe or situation or point in time unlike anything else you've ever seen.

Story telling systems

The best stories have conflict, events and situations. The best tool to create these situations is utility AI because it is not rigid and it can react to what happened in the environment. Especially if you combine it with tags, smart objects and influence maps then it can use the environment effectively and know a great deal about the world which helps it make decisions which seem real and in context.

Personality Changes

A character's personality usually changes in real life and in stories. We love to see these and these changes surprise us. They bring us pain and pleasure. We also get excited about them. In general when something unexpected or unknown happens, feelings of hope and fear increase.

Utility AI can easily make this possible without you trying too much to offer specific changes in the AI design. If your AI has personality attributes and the considerations check those attributes to decide what to do, then you can alter the behavior by changing the attributes.

Let's say you decide if a healer NPC goes near a danger zone to help a wounded soldier or not by checking its health and braveness and if he has a health pack or not. With Utility AI you can easily change a healer's braveness the more it saves people or the more he doesn't get killed or even better when somebody important to the NPC dies or a dramatic event happens. Imagine a dragon attacking a city and you want it to cause all NPCs to get frightened, you can reduce the braveness of characters by

half and the very weak souls can remain in a shelter enough to die of starvation. It is very hard to code this without utility AI.

Player generated behaviors even for NPCs

Since utility AI by nature is very parameterized and all actions are considered all the time, if you allow players to define events, items or NPCs behavior by modifying their attributes then they can change behaviors enough that a new sort of user generated content is made.

Consider the healer example in the personality changes part, if a player can craft their own items and the item can increase people's braveness then you can have a pill which people take to get brave and this can be worth a lot. You can buy 100 of them for your soldiers. That is if you think it is a good strategy of course.

To keep the game balanced you can increase the cost of larger changes which the items affect or require lowering of character XP for the crafter or ask them to sacrifice something big/do a hard quest for the ability to make powerful items. Let's say there is a crafting ability and the first increase in braveness requires one reduction in crafting but the second increase requires 2 and then third one requires 4 and it goes up exponentially. In this way you can enable players to make content for the game in a way that is very meaningful and valuable but without utility AI designing and using these sorts of things is very hard.

Easy behavior modification

Adding new behaviors or modifying the current ones is much easier with Utility AI since you can add the action to the list of actions and order doesn't matter. Other than order, the fact that you add it to a single place and you cannot forget a transition makes it easier too.

The AI still needs to have correct considerations and curves for a correct score but this exists for every action no matter when it is added and also allows you to compartmentalize the score calculation per action which helps. Every consideration is explicit so if the new action should not happen when the character has a gun, no order or lack of existence of a transition indicates that implicitly but the action itself has a consideration for that. Every programmer or designer with experience knows explicit things are easier to know about and change since they are in front of you and you cannot miss them.

Natural behavior

Because Utility AI considers all actions all the time, the behavior tends to become more natural and considering all the context which is possible but harder to get using other ways of making AI.

Immersive Worlds

When characters of a world act as if their own life matters and the events in the world really change things around, you feel that things are real and get immersed in it. Specially when events have lasting

impact, you take them more seriously. Utility AI helps you to incorporate these things in a much easier way than AIs with rigid structures.

Usually events either change attitudes and personal characteristics of people or change their environment and utility AI can easily deal with both thanks to curve shape changes, reading of attributes from variables dynamically and also things like tags and influence maps which help it to make decisions based on the environment. If you want to cause players to be afraid of a place which is attacked on a regular basis, it is enough to put this data somewhere in a knowledge base and then add actions which do something in relation to these sorts of data and modify other actions based on them.

For example, bring water home action can avoid those areas and guards can guard toward that direction more heavily but also running away action can be added for the time people are in the dangerous areas at night. Doing this in a scripted AI system with rigid structures like behavior trees or state machines is really hard and tuning something like a goal oriented system for correct behavior is next to impossible. This becomes much harder if you want these to be dynamic and not predefined.

Utility AI helps you to compartmentalize each action's considerations and scoring mechanism and easily make them affected by these events and other changes. It is still work to do but much less than the other options.

Examples in Different game genres

Third person shooters and FPS games

Shooters did not need very complex AIs back then where the actions were limited and the cover points were pretty well-known and static but these days that the maps are dynamic and the number of weapons is huge. It would be very helpful if your AI could decide its action based on some heuristics to see what action is the best in this moment in time.

Utility AI allows you to do that so your NPCs can play and decide like a human does and appear very smart and interesting to play against, by your actual players. The NPC can surprise the players and force them to apply new tactics just like when they play against real humans.

Adventure games

The main point of most adventure games is to make you feel you are going through a journey and the more natural and immersive it becomes, it's better. Utility AI allows you to make NPCs which are not very simple and stupid and have their own desires, have personalities and modify their behavior based on what you will do and what happened.

RPG games

Role playing games intend to simulate a situation which you can immerse yourself in. Their ultimate goal is to help you feel that you are in an environment or situation. However the AI for NPCs falls short of this

by having stupid dialog or a very strict and limited behavior. Utility AI is the cure for this to give the NPCs their own desires and lives.

Utility AI allows you to give your NPCs lots of behaviors and desires so they act in a believable manner. You can give them considerations which check what other NPCs and players are doing and the personality of the NPC itself when it wants to choose what to do.

In RPGs events and quests matter a lot. Utility AI can help your NPCs a lot to react based on what events and quests have happened or are going on, and act based on those. Another good thing is that if a designer adds a new quest which causes say a rain of fire, it can easily add an action to the list of actions for NPCs to go home or to a shelter when this happens without getting worried that if it is executing in the correct place in a behavior tree, or if all needed transitions are added in the FSM of the characters.

Card and board games

This genre also benefits a lot from being able to code the AI in the way that a human thinks. Imagine that you are making a poker or blackjack. How the AI should decide what card to play next or what to do in general? Should it draw another card or not? Utility AI allows you to encode how a player plays into your AI.

You can code considerations which highlight what situations a card is a good one to play. Then you allow them all to compete with each other and the best action for the situation is chosen. You don't have to exactly say when this action should happen which is very hard when you don't know what the opponent will exactly do. In a simple platformer you have a set of limited actions which you will work against but in a card game or a board game either you have to have a huge amount of memory and analyze the complete space of the game which works with games with less randomness like chess or go but not as well for games with randomness in their rules but a very good AI which is fun to play with can be made using actions and considerations relatively easily if you research how human players think when choosing their next move and how they assess usefulness of each action.

Simulation games

This is the genre that had one of its most famous games made using Utility AI. The Sims uses Utility AI and you can find lots of materials about it online from GDC talks to university papers. Utility AI is literally simulating how people make decisions so it is best suited to simulate how intelligent entities behave in a world. All other algorithms should try to mimic how the brain works but utility AI is simulating it directly. It is possible to mimic the behavior with other algorithms partially but utility AI is the best way to do this.

MMOs and MMORPGS

MMORPGs are essentially role playing games so whatever applies to RPGs applies to these as well. Only worlds are deeper and expectations are higher because NPC behaviors are compared to human behaviors and they usually are immersion breakers and not immersion makers. Utility AI can help you break this and make NPCs which are deep, have many actions and act according to their interests and

what happened in the world and not based on a simple and small tree. Also you can share the behavior between NPCs with minor personality changes since the behaviors are not rigid anyways and just different NPCs decide a bit more differently compared to others.

A brave soldier and his coward comrade can use the same behavior but the brave one has a higher accuracy of attacking and tends to choose more risky behaviors.

Sandbox games

Sandbox games are known for lots of choices and possibilities. A good AI for a sandbox game should be able to deal with unexpected situations and act smartly or interestingly in them. The best AI to react to dynamic unknown situations is Utility AI.

Survival games

Again a survival game is a specific sort of sandbox game which somehow is an RPG too which is about surviving so whatever applies to them, will apply to these as well.

Metaverses or metaverse games

The idea of the metaverse is that different experiences made by different players can be mixed up and interact with each other. The only AI algorithm which can work in an unexpected environment given that the way it calculates considerations and executes actions is compatible with the environment is utility AI. The rules are not hard coded and are softly specified so you check if any enemy object is around and shoot at them and having low health or lots of enemies and no cover gives the shooting action a high score. The only key is to identify enemies in a way that all experiences can specify it. Using tags is a good way of doing this. If that is the case that all enemies of your NPC/agent have an enemy tag attached to them then your AI can react to them even if it doesn't know how exactly they would behave. Faction names or even a memory of what actions an agent took can be used for this as well. This will not be perfect and not as good as when you exactly know what will be the behavior of those enemies but still Utility AI is the only system which allows you to do this.

In theory any AI which checks conditions which are calculated based on dynamic parameters will be more adaptable to experiences which not all of them are authored by you. But in action a state machine or behavior tree or even a planner will only work if the conditions are known to produce the best result for that situation. Their transitions have order and are more explicit. This can be helpful in a very scripted situation but not when the situation is unknown or has too many parameters. They don't work based on heuristics and that is the major difference between them and utility AI.

If you have a state machine which transitions from a state based on heuristics and not only when it is sure that this specific transition to another state is the best one then it is effectively limited utility AI which each action can only be switched to a selected number of others and not all of them. Also if a behavior tree's nodes are selected in a similar manner then it is the same thing and the only difference is that the actions and their score comparisons have implicit orderings and each action will only activate

after a threshold or if you basically have a root with all actions as its children evaluated in parallel and then one selected based on their score, then it is utility AI.

In the same spirit, Utility AI can be turned into a behavior tree or state machine or can become a bit more rigid in its structure if you use appropriate considerations. Imagine a consideration which gives a score higher than 0 for an action only if the currently executing action is shooting, that can be used to have some actions effectively only available in the graph of your decisions as children of the shooting action. Usually such a graph doesn't exist in utility AI or at least it is not easy to draw but if you want to turn your utility AI into that, you can.

But that is not the point. The point is not to show each AI algorithm can more or less implement the others if you stretch it with some degree of success. It is not a useful practice and it is better to use the tool which is suitable for that specific sort of way that you want to offer your AI behaviors. And in this sense and compared to other known algorithms, utility AI works best in unknown environments where the decisions don't have definitive answers. Wait the real world is like that! :) And this is another difference of utility AI compared to the others. It is derived from a theory which tries to decode human behavior in the real world which is what we want to simulate and not an abstraction created to be able to model a decision tree with accuracy.

> This is one of the cookbooks showing you how to implement a specific mechanic, an algorithm or a game genre in Utility AI and other parts of our tool set

Cool downs

Once in a while you need to implement a cooldown for an action so it doesn't execute for x number of seconds after it executes once. To implement such a mechanic you need to remember when an action got executed and how much you should wait before you can execute it again. To do this the action should have a consideration which returns 0 before the cooldown time finishes and returns 1 after that so the rest of the score is not affected by it.

It can be easily hardcoded if you want to do it for a specific action. Let's call the action Attack1 and let's say we want to execute it at most once per 5 seconds so with 5 seconds of cooldown. You can store a last executed time in the action itself and set it to current time when action gets activated and then have a consideration which subtracts the current time from the stored time in the action and if less than 5 returns 0 and otherwise returns 1. The curve for the consideration should return 0 for 0 and 1 for one as well so a linear curve would be perfect.

The method to override in Unity is `OnStart()` and in Unreal it is `OnActivated()`. These get called when an action is selected to execute once.

This said you can implement it more flexibly by using a blackboard and in this way you can have an item which if the character eats or an action which he does and reduces the cooldown. For example he can eat a potion to reduce his tiredness and Attack1 requires tiredness of less than 10. Let's say tiredness is between 0 and 100 and it decreases by 15 per second when not attacking. Attack1 increases tiredness by 90 as well. There is also a pray action which has a 50% chance of reducing tiredness to 0 or increasing it by 10.

There is a consideration for the Attack1 which checks the blackboard float key of tiredness and if less than 10 returns 1 and otherwise returns 0. A code in a component of the agent decreases tiredness by 15 per second if the attack actions are not being executed. Eat fruit action decreases tiredness by say 50 and its consideration gives it a high score when you are tired and a fruit is close by. The pray action gets a random number and if less than 0.5 then it increases tiredness by 5 and otherwise sets it to 0. Its consideration will see if there are no fruits close by and we are tired and the situation requires a not tired character.

Now to make it more interesting, some characters which have a more religious/spiritual attribute can use this more effectively and we can have a cool down for praying itself so an agent can use it once per few battles to not make the battle scenes look like a church in chaos. Also if this was based on player choices then it could improve based on how many temples a player built in a city building game or how much the player sent his army to wreak havoc on cities in a strategic game. To be clear, sending armies to ruin

cities should decrease the effectiveness of prayers and other spiritual spells and actions in most games unless you are making a game about an evil God.

Wise Feline Den

The Wise Feline Den free package has a set of features helping you make better game AI and other systems faster. These are heavily used in our other products like Utility AI and Influence maps.

We have

- Combat Scripts
- Movement Scripts
- Blackboards
- Data Recorder
- and more

All of these are available in the menu at the left and can be learned and used to your advantage. Check them to see what great stuff we are giving away for free. Level up your game with these awesome tools.

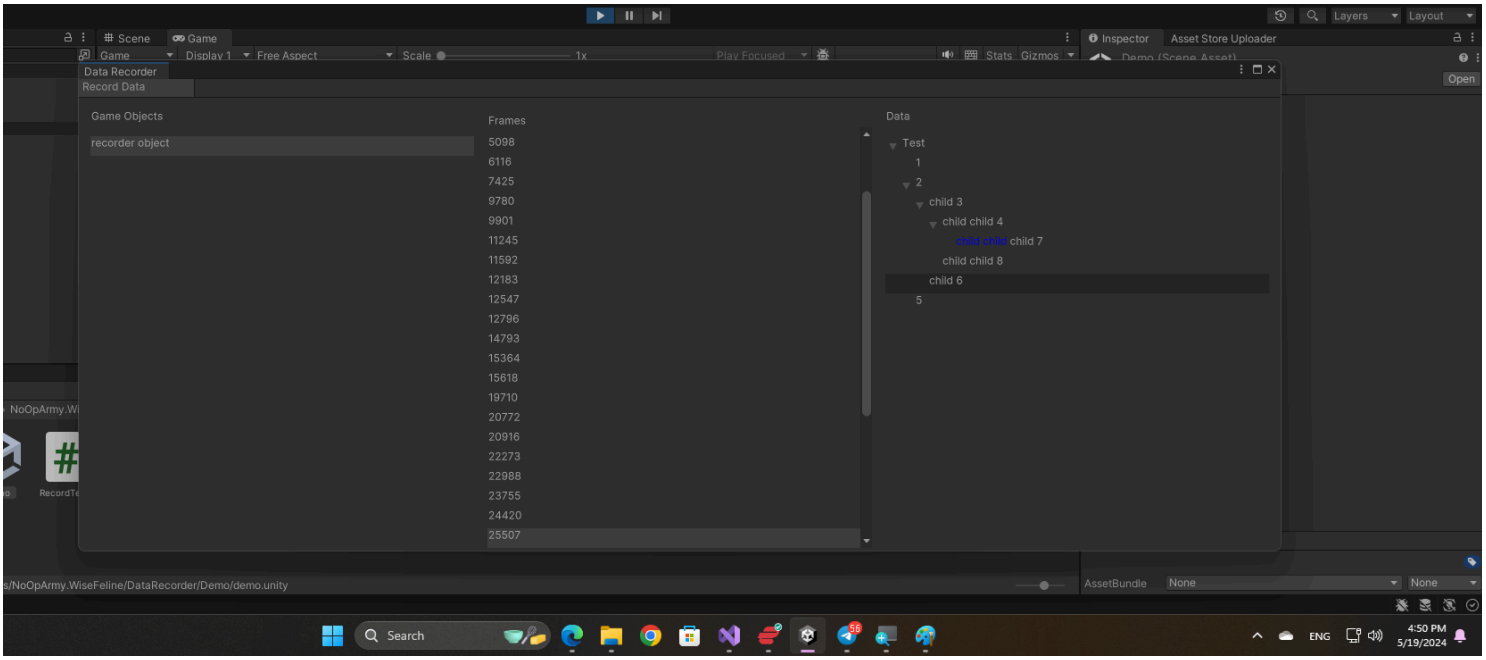
Data Recorder

The Data Recorder allows you to record data from GameObjects over time and then look at them during or after the play session.

The window can be opened by going to the *Window>NoOpArmy>Data Recorder* menu. You have to attach `DataRecorder` component to GameObjects you want to record and your other scripts can use it to record data every frame.

The data format

The data is a set of text records which can use rich text and can have a level. They are shown in a tree like structure.



As you can see here, the left most panel is the list of objects with data recorders on them, the middle panel is the frames and the right side one shows the data. The data is a tree of sub-records created based on levels you give to them.

How to use

After attaching a `DataRecorder` component, you can get it and call its methods to add records and their sub-records.

You can use `GetCurrentFrameRecord()` method of the DataRecorder to get/create the frame data for the current frame. Any frame you call this, you'll have an entry in the data recorder. The index this function returns can be used to add sub records which are shown in the data section.

The `RecordTester` script in the demo folder of the `DataRecorder` looks like this.

```
private DataRecorder recorder;

        void Start()
        {
            recorder = GetComponent<DataRecorder>();
        }

        private void Update()
        {
            if (UnityEngine.Random.value < 0.999f)
                return;
            int index = recorder.GetCurrentFrameRecord();

            var sub0 = new SubRecord();
            sub0.text = $"Test";
            sub0.indentationLevel = 0;
            recorder.AddSubRecord(sub0, index);

            var sub = new SubRecord();
            sub.text = $"1";
            sub.indentationLevel = 1;
            recorder.AddSubRecord(sub, index);
            var sub2 = new SubRecord();
            sub2.text = $"2";
            sub2.indentationLevel = 1;
            recorder.AddSubRecord(sub2, index);

            var sub3 = new SubRecord();
            sub3.text = $"child 3";
            sub3.indentationLevel = 2;
            recorder.AddSubRecord(sub3, index);

            var sub4 = new SubRecord();
            sub4.text = $"child child 4";
            sub4.indentationLevel = 3;
            recorder.AddSubRecord(sub4, index);

            var sub7 = new SubRecord();
            sub7.text = $"<color=blue>child child</color> child 7";
            sub7.indentationLevel = 4;
            recorder.AddSubRecord(sub7, index);

            var sub8 = new SubRecord();
            sub8.text = $"child child 8";
            sub8.indentationLevel = 3;
            recorder.AddSubRecord(sub8, index);
```

```
        var sub6 = new SubRecord();
        sub6.text = $"child 6";
        sub6.indentationLevel = 2;
        recorder.AddSubRecord(sub6, index);

        var sub5 = new SubRecord();
        sub5.text = $"5";
        sub5.indentationLevel = 1;
        recorder.AddSubRecord(sub5, index);
    }
```

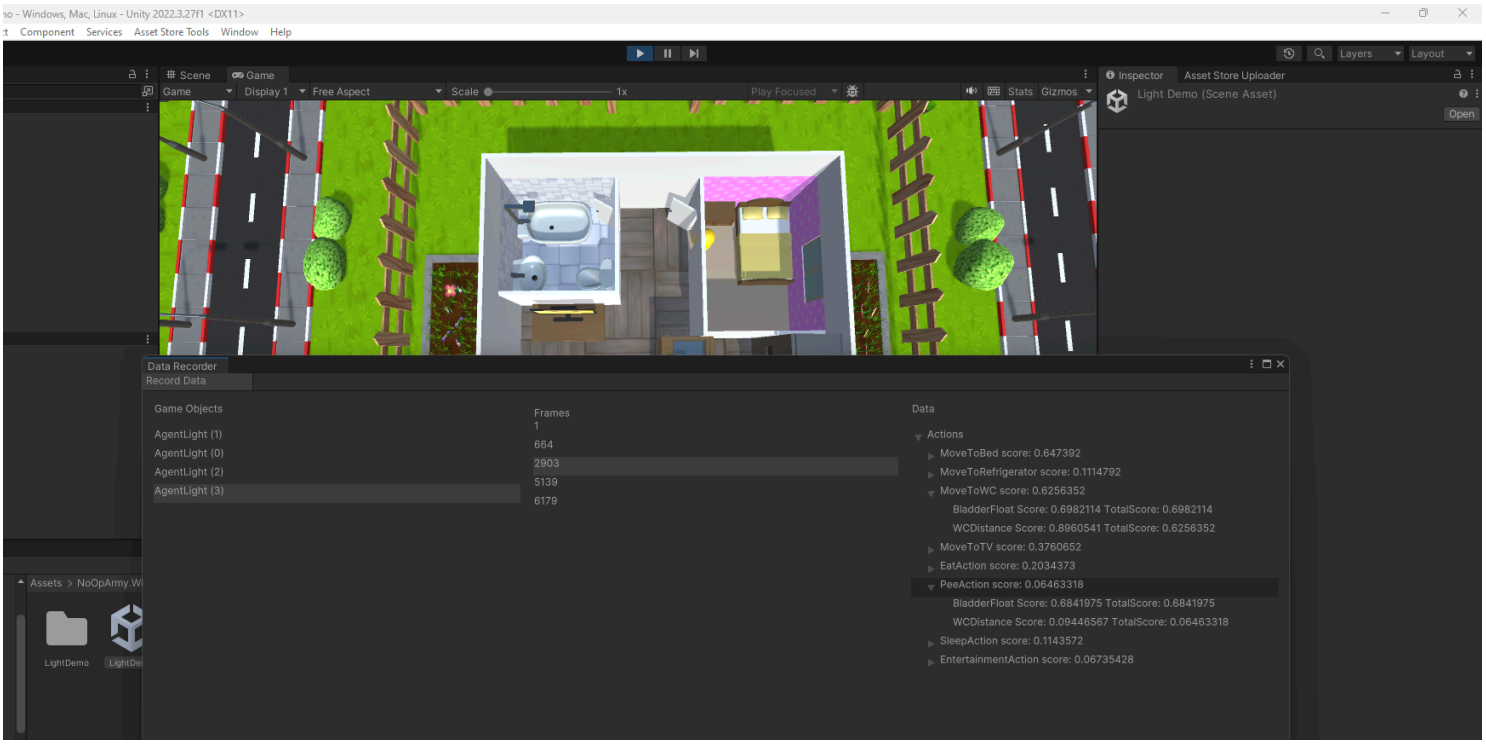
As you can see `AddSubRecord()` is used to add sub-records per frame and each sub-record has a level which tells which level of the tree it should be in. If you log an attack and then 3 related info about it, then their level should be attack's level + 1. Think of it as indentation level.

Also as you can see, you can use Unity's `<color=blue>text</color>` and other tags to make your text color something special or make it bold/italic/ any other thing which Unity supports.

You can integrate this into any system of your game which can benefit from recording hierarchical data over time and being able to look at it afterwards.

Utility AI integration

The `Brain` component integrates with Data Recorder so if you attach a `DataRecorder` to a GameObject with a `Brain` then the actions and considerations and their scores will be recorded whenever the brain thinks.



If you don't have our other packages, we encourage you to take a look at them to add very immersive AI to your game.

Blackboards

Blackboards video tutorial

A Blackboard is a data structure which allows you to share data between different components of your AI without having direct dependencies between them. A blackboard is a `Dictionary` like data structure which allows you to add data by defining it as a set of key value pairs.

Let's say you want to know what is the safest location to go to, you can define a safe location key in the blackboard and store the `Vector3` of the location in it. Our implementation of blackboards is more strongly typed than a `Dictionary<string,object>` and you should define all keys with the data type of their values before being able to set/get them.

Getting Started

To define a blackboard you first need to create a blackboard definition. To do that right click in your project and click on Create>NoOpArmy>Wise Feline>BlackBoard Definition. Then you need to assign that definition to a `BlackBoard` component. Then you can set or get the value of the keys you defined in the definition.

You have multiple methods in the `BlackBoard` component like `GetFloat(key)`, `SetFloat()`, `GetVector3(key)` and `SetVector3(key,value)`. You have these pairs for all data types and you can set and get the values in your code as you need.

Starting values without code

Keys start empty. A float starts at 0, a bool at false, a string empty and a GameObject at `null`, and an agent whose health starts at 0 or whose patrol route key is `null` does not behave the way you designed it.

The `BlackboardInitialValues` component solves this without any code. Add it to the same GameObject as the `BlackBoard` component and fill in a list of values in the inspector, and they are written into the blackboard when the scene starts. Each entry is a key type, a key name and the value to write, which is the same little editor you already know from the blackboard changers.

This is also the only code-free way to put a **reference to a scene object** into a blackboard. Assets live in the project and an asset cannot reference something which only exists in a scene, but a component in the scene can. So the object holding your patrol points, a camp an agent retreats to, the player, a spawn point or a door can all simply be dragged into this list.

> **WARNING**
>
> Entries here are almost always **sets**, but `Add Instead Of Set` and `Multiply by deltaTime for adds` are **on by default**, because the same entry type is also used for changes which are applied a little bit every frame. Turn both of them off for starting values. If you leave them on, "start at 100 health" becomes "add 100 multiplied by one frame's deltaTime to whatever is there already", and the agent starts the game with about 1.7 health instead.

Two more things worth knowing:

- The key has to already be defined, with the same type, in the blackboard definition which the `BlackBoard` component uses. Writing an undefined key throws.
- The values are applied in `Start()`, because the `BlackBoard` component defines its keys in `Awake()`. The component also carries a negative execution order so that its `Start()` runs before the `Start()` of normally ordered components, which is what makes it safe for anything that reads a key during its own `Start()`.

How can I use this?

Your AI can use this in considerations and actions and other parts of your game. Let's say you have an action like GoToLocation, you can take a blackboard key which you know contains a `Vector3`.

A small tutorial

- Create a definition and call it b.
- Add a float key called speed.
- Create a new script called `TestBlackBoard` and attach it with a `BlackBoard` component to a GameObject.
- Set the definition in the blackboard to the one you made above.
- Open the `TestBlackBoard` script you made and add some code like this.

```
public class TestBlackBoard : MonoBehaviour
{
        private BlackBoard board;

        void Start()
        {
                board = GetComponent<BlackBoard>();
                board.SetFloat("speed", 3);
                var speed = board.GetFloat("speed");
                print(speed);
```

```
            }
    }
```

The code should print 3.

Movement scripts

You should be able to move your AI characters easily and we implemented an interface with two implementations for it. There is

- NavMesh based one.
- CharacterController based one.

The interface is `IAIMovement`

```
/// <summary>
/// This interface is implemented by all AI movement types which allow an agent to choose a target and move toward it.
/// </summary>
public interface IAIMovement
{

    /// <summary>
    /// Should contain the movement destination
    /// </summary>
    Vector3 Destination { get; }

    /// <summary>
    /// Moves the character to a position specified by a Vector3 and calls a callback when arrived
    /// </summary>
    /// <param name="position"></param>
    /// <param name="callback"></param>
    /// <returns>True if the movement can happen, false otherwise. Path Finding failures and other issues can prevent the movement to happen.</returns>
    bool MoveToPosition(Vector3 position, Action callback = null);

    /// <summary>
    /// Moves the character to the position of a GameObject and calls a callback when arrived
    /// </summary>
    /// <param name="position"></param>
    /// <param name="callback"></param>
    /// <returns>True if the movement can happen, false otherwise. Path Finding failures and other issues can prevent the movement to happen.</returns>
    bool MoveToPosition(GameObject position, Action callback = null);

    /// <summary>
    /// Stops the movement no matter if we arrived at the set destination or not
    /// </summary>
```

```
    void StopMoving();
}
```

CharacterControllerBasedAIMovement

This one should be used if your NPC has a `CharacterController` attached.

NavMeshBasedAIMovement

You should use this component to move your character if it has a `NavMeshAgent` component attached.

Combat Scripts

We've implemented a few scripts so you can implement range and melee attacks with different types of health and damage structures with armor and other stuff. The scripts are pretty simple and easy to use and customize. We are going to list them below.

> All of these scripts are in DenUtilityComponents folder in the Den folder

Attack

This component has methods for doing range and melee attacks and it can either apply the damage to all hit objects or just the first one. It can optionally change the material color of the GameObject which it is attached to when the attack starts and can delay the actual attack so the player has a chance to respond to the telegraphed attack.

This is a sample signature of one of the methods `Coroutine MeleeAttackWithDelay(Vector3 offset, float radius, float damageAmount, float castDelay, Action Done = null, bool applyToAllHits = false)`. The offset is offset from the object's own position. There are range attacks too.

The way checks work

The checks use no layers and use `CheckSphere()` like this `int count = Physics.OverlapSphereNonAlloc(position, radius, results);` so it does not allocate.

In the future we can support layers or other shapes if needed.

IDamagable

Any script which wants to take damage from this system should implement this interface. It only has two methods for increasing health/healing and applying damage.

```
/// <summary>
    /// Any component which wants to receive damage can implement this interface
    /// </summary>
    public interface IDamagable
    {
        /// <summary>
        /// Applies damage to the GameObject
        /// </summary>
        /// <param name="amount"></param>
        /// <param name="damager"></param>
        void ApplyDamage(float amount, GameObject damager);

        /// <summary>
```

```
        /// Heals the GameObject
        /// </summary>
        /// <param name="amount"></param>
        /// <param name="healer"></param>
        void IncreaseHealth(float amount, GameObject healer);
    }
```

The `Health` component described below does implement this interface.

Health

This component implements a health system for your game objects which need it. For effects it only can change material colors for debugging purposes but all of its methods are overridable and whenever the health changes it fires its `OnHealthChanged` events.

It can automatically destroy the GameObject when the health reaches 0 and it can add a delay before doing this.

Other Scripts

These are a set of useful scripts which are not in a specific category

Faction scripts

The `Faction` component allows you to assign a faction to an agent. The faction is an int id but there is also a scriptable object to define each id's faction.

You create the scriptable object using NoOpArmy>Wise Feline>Faction Collection menu and each faction has an id, a name and a description.

The Id in the faction collection should be the same as the id in the `Faction` component. Later on we will add more to this system by adding a faction manager system and ...

NPCZoneTrigger

This script can be used by NPCs to know when a player or another NPC enters their zone. You attach NPCZoneTrigger to a trigger and set the NPC which it should notify and then if any GameObject other than that NPC enters/leaves the collider, it notifies the NPC.

The NPC should implement an interface called `INPCZoneMessageReceiver` for this to work. This interface has two methods for the time that an object enters and leaves the zone.

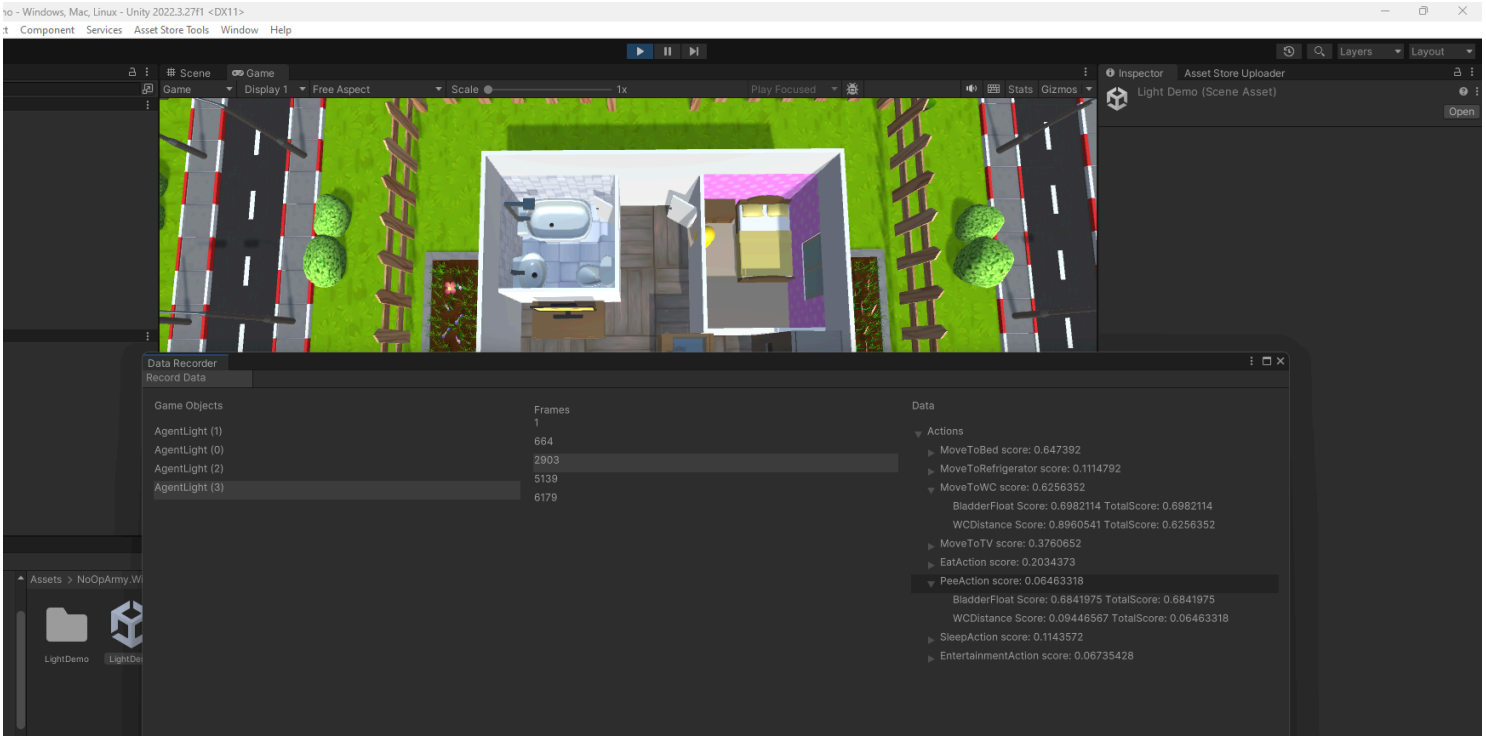
This can be used to create dialog for NPCs or set their blackboard values to know somebody else is in the area/left the area.

DestroyObjectWithDelay

Destroys the object with some delay. Mostly used for magic spells or projectiles to destroy them if a collision or something similar hasn't destroyed them.

Utility AI integration

The `Brain` component integrates with Data Recorder so if you attach a `DataRecorder` to a GameObject with a `Brain` then the actions and considerations and their scores will be recorded whenever the brain thinks.



If you don't have our other packages, we encourage you to take a look at them to add very immersive AI to your game.

> This module is only available in the ultimate edition of the Wise Feline package or the separate Influence Maps package

Influence maps module

Great video about influence maps by Dave Mark

Agents need to know what is going on around them in the world to be able to make good decisions. Having a toolbox which allows you to know spatial information about the world like:

- Where are the main paths which enemies use?
- Where is the best position to throw my grenade?
- Where is the busiest area of the map around here?
- Where can I find some friendly forces to join?
- Where are the most trees located so I can spawn some rabbits?

is very useful.

Influence maps allow you to survey the environment of your game and find spatial info in a way which raycasts and octrees don't allow. An Influence map gives you info on what parts of the map something exists too much and on what part it doesn't exist. It can give you max and min points and busy and sparse parts. It can give you info not only on things which are present in the map but also about the events which happened before or will happen later.

The way influence maps work is that the map is a 2d array which agents propagate their influence on. For example you can have a map which all enemies propagate their position and attack range on and another where trees propagate their existence on.

Now agents can find out where they should flee to using the map points with smaller values in the enemies map and they also can find where they should fire at, by finding points with higher values.

You can search for higher and lower values in a circle in a specific part of the map or in the whole map. You can also combine multiple maps and then search on the resulting map so for example you can find a point where there are both lots of trees and enemies to set the trees there on fire to burn your enemies alive. Combining this system with AI tags, Smart Objects, The octree and Utility AI allows you to do very smart and immersive things in your games which most game designers only dream of.

Components and programming

In general this system can be used both as a set of components and also as a set of pure .NET classes without any components. The `InfluencerAgent` component allows you to propagate yourself on a map and the InfluenceMapComponent component allows you to create a map by just attaching it to a

GameObject. The `InfluenceMapView` component allows you to create a map by combining other maps and the `InfluenceMapCollection` component is a singleton which allows you to find maps by their name.

Each agent can propagate its value to the map but the value should be a circle with gradient values most of the time. It doesn't make sense to calculate all values every time. For this reason, you create a set of influence map templates which agents use as their stamp to propagate their value. In this way an enemy can propagate 1 in its position and reduce its influence value up to some distance like 15 meters to 0. The fall off of the value is chosen by the curve that you define in the template. The template values are calculated once and are just copied to the positions which need them afterwards.

To create your first influence map

- Right click in the project view and go to Create>NoOpArmy>WiseFeline>Influence Map Template and name your template linear. Then choose a linear curve and a value of 50 for the radius.
- Add two gameObjects to the scene called agent and map.
- Add an `InfluenceMapComponent` component to the map and an `InfluencerAgent` to the agent. Name them map enemies (this is the name that you can use to get the map from InfluenceMapCollection).
- Select the agent GameObject and then select the map and the linear template for the agent. You can choose a map by dragging it or entering its name.
- Select the map GameObject and then set the size and resolution of the map to some values like 1000 for width and height and 1 for cell size.

Now if you press play and move the agent around, you can see the propagation of the agent on the map in scene view gizmos. If the gizmos are not shown, check the box to draw them in the `InfluenceMapComponent` component.

InfluenceMapComponent

This component allows you to create an influence map with the size and resolution specified and can draw the values of the map as gizmos in the scene view for you. It also has methods to search the map for values and also convert positions from map coordinates to world coordinates and vice versa.

The `InfluenceMapComponent` component has all methods you need in most cases but also gives you access to the map in the `Map` property and the map has all of the search functionality you need.

> Influence maps are 2D arrays and they have a different resolution from your world so you need to convert positions of your GameObjects from world coordinates to map coordinates to use them with influence maps. Also when you get a coordinate from an influence map search or some other operation, you need to convert it from map coordinates to world coordinates so it can be used by your GameObjects.

InfluenceMapCollection

The `InfluenceMapCollection` singleton allows you to get maps by their name and maps created using the `InfluenceMapComponent` and `InfluenceMapView` components automatically register themselves to it. This component is created automatically if you don't have one in the scene.

How the search works

The search functionality takes a value and an enum which is a comparison operator. For example if you give it 0.6 and greater, then it will try to find a value bigger than 0.6 in the radius and stops as soon as it finds one. It starts searching from a random position in the specified search area so different agents don't end up choosing the same location.

Videos

These are the imap videos

https://www.youtube.com/watch?v=hnnQ7eEkeCQ https://www.youtube.com/watch?v=9Lk9ibddw60&t=118s https://www.youtube.com/watch?v=O2bDqOW1i5c https://www.youtube.com/watch?v=kuAwzUzzzIg&t=19s

InfluenceMapView

An InfluenceMapView component allows you to create a map by combining other maps. For example you can make a threat view by adding this component to a GameObject and then give it the enemies map and bases map and tell it to add them. The InfluenceMapView can have a name and be found using the `InfluenceMapCollection` class as well. It also asks you how often it should be recalculated from the maps it has and there is no limit in number of maps and operations, however the more maps you add to it, the heavier it will be to calculate.

Adding, subtracting and other operations on maps

You can add two maps to each other to search on both of them combined. You also can subtract them from each other and do other operations as well.

These methods are available

- *Add* which adds another map to the map calling the add and either returns a new map or fills the data in the map you supplied.
- *Subtract* which subtracts another map from the map calling the subtract and either returns a new map or fills the data in the map you supplied.
- *Multiply* which multiplies a number with the map calling the multiply and either returns a new map or fills the data in the map you supplied.
- *Invert* which inverts the map calling the invert and either returns a new map or fills the data in the map you supplied.
- *Normalize* which normalizes the values of the map between 0 and 1 based on its minimum and maximum.
- *GetCellsArray* Returns the raw array that the map uses. Be careful to not modify this and just use it to calculate other operations which we did not write.
- *SetCellsArray* sets the array of cells of a map.

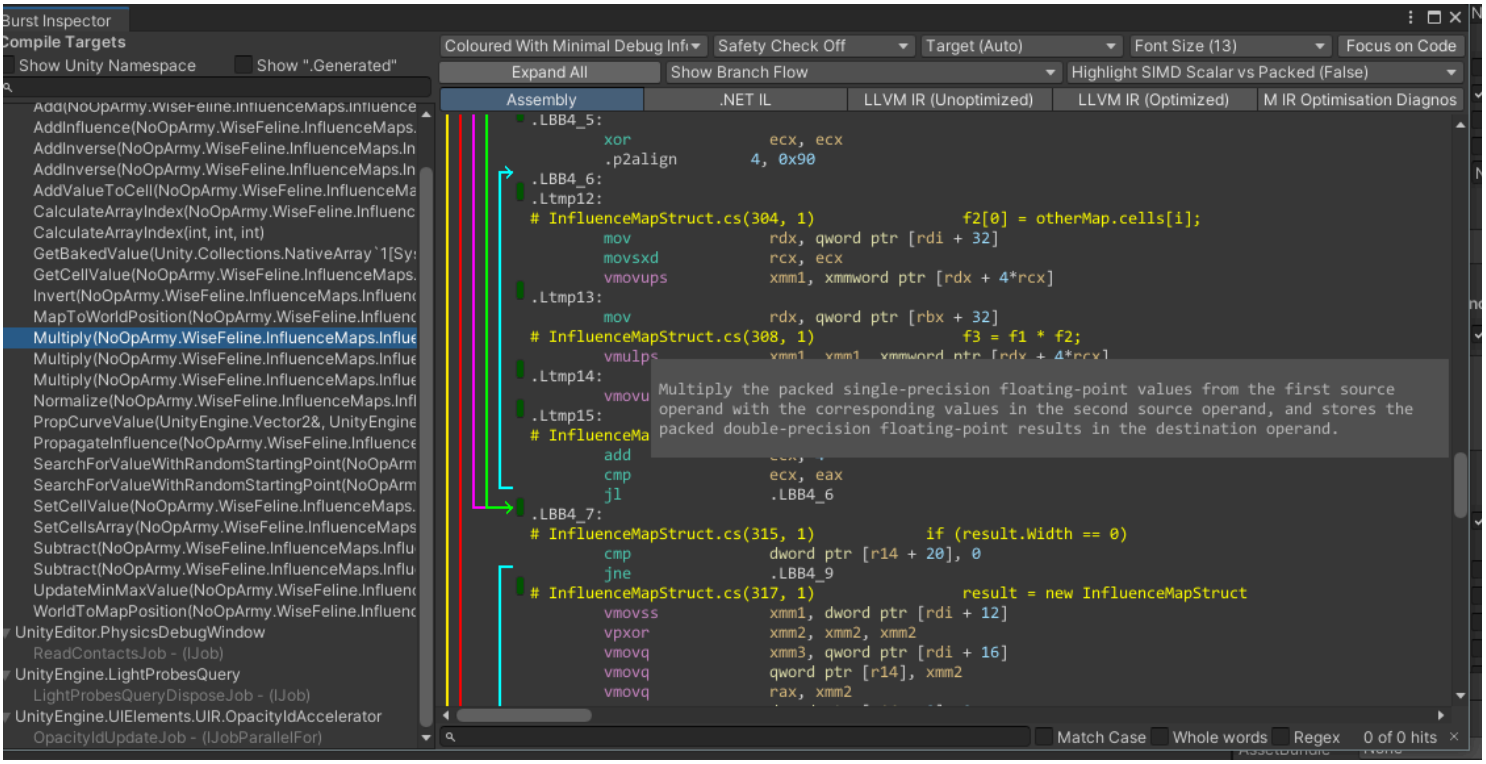
Burst support

Influence maps support burst and the operations become a lot faster. To enable burst support you need to be on Unity 2022 or later and do the following:

- Install burst in your project from the package manager.
- Define WF_BURST in the other settings part of the player settings of your project.

All of the components will work as before but instead of the `InfluenceMap` class they'll use `InfluenceMapStruct` struct under the hood. The struct has a reference to a native array and lots of burst compilable methods. You can technically use this in jobs and ECS/entities but we don't directly support any of them.

All of the Add/Multiply/... map operations use float4 and generate SIMD packed instructions which means doing multiple operations using a single instruction.



Sample utility AI integration

There is a consideration in the actions and considerations library folder which searches an influence map and writes the result in a blackboard and returns a score based on the fact that the search was successful or not. In this way the action doesn't have to recalculate the search and can use the value in the blackboard to do its operation.

Let's say you want to see if there is a good place to throw a grenade at. If you add the InfluenceMapConsideration to it and ask it to search the enemies map for a value higher than 0.8 and the action scores high enough, you can read the position this consideration found from the blackboard key you entered instead of re-executing the search function.

Demos

There are a few small demos located at the demos folder of the influence maps folder. Take a look at them and also the immersive demo for ultimate which uses influence maps

Remembrance

Remembrance is a memory and emotions module for your AI agents so they can remember what happened in the world and react to it. Your NPCs can hold grudges against players and other NPCs, return a favor or even try to seduce you or bribe you. They can remember what happened by whom to whom and where and when. You can also attach any custom data to your memories.

How it works

NPCs or any other objects in your game record their memories in a container and then the container can be searched for values with different criteria. They can also set float emotion values per object or in general.

Memory

There are two components that you can use or you can use the underlying data containers. Each of the listed components has an `AddMemory()` method which records a new memory to be remembered. They all also have multiple overloads of a querying method which gives you the closest memory to a specific time with additional arguments which limit the acceptable memories. For example you can return only memories with a specific target or of a specific type or both. There are methods caching the last recorded memory with a specific action id or target for specific common use-cases as well. They usually are of the form `GetLastXXX()`

Memory component

This is the component you want most of the times.

This component is more advanced and can remove memories from the middle of the list and uses a `List<T>` as its underlying data container. When the maximum capacity is reached, the first item in the list is removed after each addition. This component has the forgetting functionality where you can specify in each memory, after how many seconds it should be forgotten and the component forgets that memory after the specified seconds.

SimpleMemory component

This component is simpler and uses a `Queue<T>` as its underlying data container and when reaching capacity, dequeues the first item in the queue. This component cannot forget memories and cannot remove any memory other than the first one in the queue but if it is enough for your use-cases then it has better performance in scenarios where item removal from the list becomes a bottleneck in your game.

Since queues don't have indexing, all searches use the `foreach` construct in C# and are slower than for loops. A foreach structure calls `MoveNext()` and `Current()` methods of the queue to get to the next object

and then read it.

MemoryData

The `MemoryData` class is the type that the methods in the component accept and return.

Properties of a memory

A memory is stored using the `MemoryData` class and it contains a set of properties which can be used for querying

- **MemoryType** This is an integer value which can be used to find out what type the memory is based on your custom types in your game. For example if your `additionalData` can be of multiple types, this can be used to see what type it should be casted to.
- **StartTime** The time that this memory started. The constructor sets this automatically.
- **EndTime** The time that this memory ended. Calling `End(float time)` on the `MemoryData` object sets this automatically.
- **ForgetAfterSeconds** If greater than 0 then the memory will be forgotten after this many seconds.
- **Position** The position that this memory happened in. If you need additional data like room name/level name/bounds, you can put them in additional data or other optional fields.
- **ActionType** This field is of type `System.Type` in .NET and allows you to set the type of the class which caused this memory as an action so you know for example if the shooting was using the `Spell` class or the `FireBall` class. This could be derived from other fields in theory but having it is handy especially in the Utility AI integration.
- **Initiator** The object which initiated the memory. You can put any Unity object including GameObjects and components in this as a reference. It can also be null.
- **Target** The target of this memory, for example the killed object in a murder memory. This can be null if the memory has no targets.
- **ActionId** This can be used for a unique id of an action instance. For example if you want to store exactly which magic spell of an NPC hit you, you can store its id here and put the NPC itself in the initiator field and the spell type in the ActionType field.
- **AdditionalData** is any custom data that you want to attach to this memory. Usually you should find out what this can be casted to based on `ActionType`, `actionId` or `memoryType`

Emotions

There is a component called `EmotionComponent` and an Emotion Collection asset which define what emotions exist and allow you to set and get their values. Emotions can be general or per object. For example your NPC might be generally sad but happy toward a tree they really like or they can have an emotion only for objects, like affection but it might not be meaningful in your game mechanics to have a general affection emotion for the agents.

All emotions are floats between 0 and 1 and define how they are combined with a similar emotion. For example if two objects share an emotion like hate toward each other, then you can ask them to align their hates so if I hate my friend by 0.9 and he hates me by 0.6, first of all we are probably not friends and secondly, there are methods to let our hates affect each other so both are set at the higher value or are averaged based on the settings in the emotions collection.

Debuggers

Both modules have their own specific debugger windows to help you see what values exist in an agent's memory or what emotions it holds.

Queries

Both of the components and all the containers have multiple overloads of the `GetClosestMemoryToTime()` method. This method in its simplest form only takes a time and gives you the closest memory to that time.

The order of all the methods here is O(N) because to find the closest time in a list which might not be sorted by time, you have to go through all items. Sorted versions will be added later on which will have a O(N/2) order but because the number of memories is not high in each container, this is not a huge concern.

Other overloads of the method limit the acceptable memories so for example the overload which takes an `ActionType` and an initiator and a target, only searches in the memories which have the specified actionType, initiator and target and returns the one which is closest to the specified time from that list.

The time is based on whatever unit you have your game for time and is provided by you yourself when creating `MemoryData` objects or calling `End()` on them. If you send 0, then the earliest memory with the specified conditions will be returned. If you specify now's time like `Time.time` then the closest memory to the current moment will be returned. However you can use any other time like between 11:00 and 13:00 in the school driver which wants to know which children did not come home with him and he did not see them.

Complex unanticipated queries

If none of the overloads provided satisfy your condition, there is an overload which takes a predicate as a lambda and can check anything you want. A predicate is a function taking an argument of type MemoryData and returning a bool which indicates if this object/memory is acceptable or not.

For example let's say you want to know the number of enemies killed below water to go down there and find them to loot them as a fish robot which observes the battlefield. You would write a query like

```
var underWaterKills = GetComponent<Memory>().GetClosestMemoryToTime(item => item.MemoryType
== 2 && item.Position.y < seaLevel),Time.time);
```

A `MemoryType` of 2 means killing actions in our custom game code. And the code above gives you the code to get all kill actions under water, recorded in your head.

Emotions

The emotions module allows your agents to have multiple emotions. Emotions can be general or toward a specific object which can be anything from a chair to a zone or another agent.

Each emotion is a floating point value between 0 and 1 and cannot be outside this range. All emotions are named and can be combined with each other. So if you feel love toward an object by a value of 0.8 and they feel love toward you by 0.5, these can be combined by making both emotions to change to a third value. The third value can be the smaller of the two or the larger one or the average. Using more complex formulas makes it hard to understand and usually it is not needed and if you need those, nothing forces you to use the combining feature of this module. We will implement more complex combinations if they are needed in your games.

EmotionComponent

This component should be attached to any agent which wants to have emotions.

It takes two emotion collections for general emotions and per object emotions. Collections are just the name of emotions and how they are combined with a similar emotion in another agent.

This component has methods to increase/decrease emotions or set them.

You can use it like this.

```
GetComponent<EmotionComponent>().AddEmotion(angerIndex, -0.1f);
```

As you can see, we are using something called `angerIndex` to decrease anger by -0.1. The emotion index is so you don't supply string names of emotions for every get and set. This is similar to how animation parameters allow you to use hashes instead of parameter names. To get the index of each emotion you should use `GetGeneralEmotionIndex()` or `GetPerObjectEmotionIndex()`. This is because the collection of per object emotions is different from general emotions. You can assign the same collection to both though.

This is how the sample takes the index of each emotion

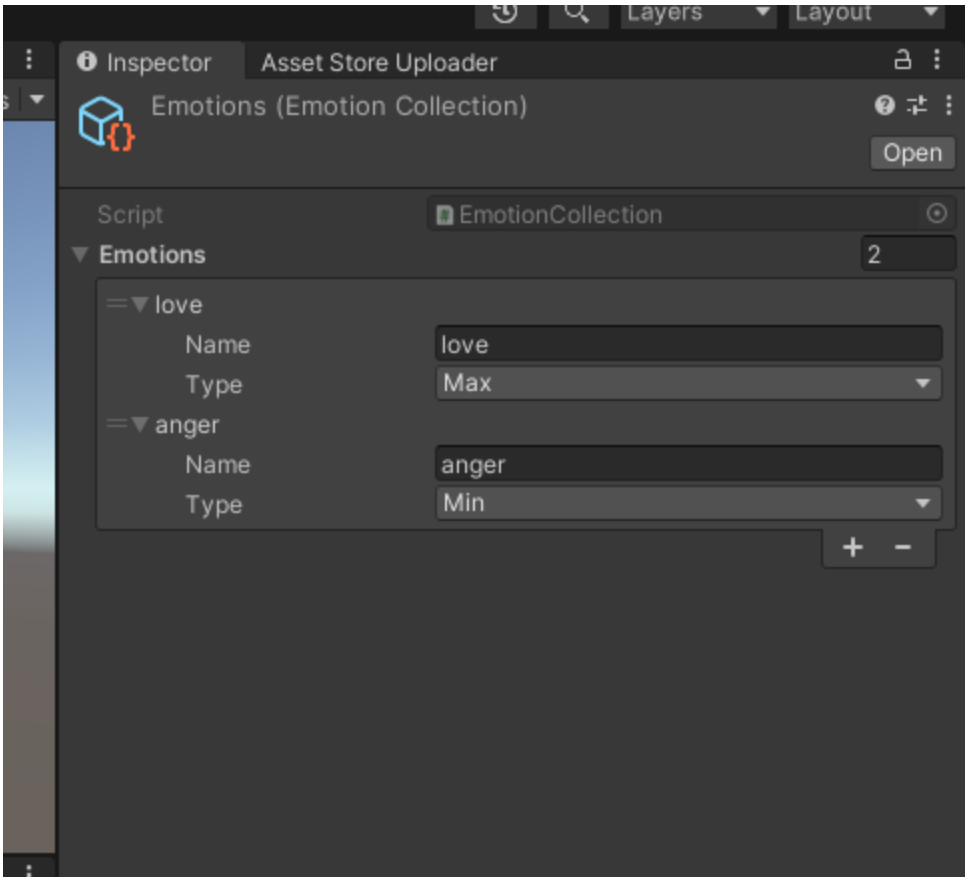
```
emotion = GetComponent<EmotionComponent>();
loveIndex = emotion.GetGeneralEmotionIndex("love");
angerIndex = emotion.GetGeneralEmotionIndex("anger");
emotion.SetEmotion(loveIndex, 1);
emotion.SetEmotion(angerIndex, 0.4f);
```

As you can see it stores the index in a variable and then uses it to set the emotion to a value. You should store these and then re-use them when adding/setting emotions. They are O(N) and the lists are small but still, the whole point is to re-use them to avoid the search.

Emotion Collections

You create these scriptable objects using right click in the project window. You should go to Create>Wise Feline>Emotion collection.

Each collection lists all relevant emotions and how they would be combined with another collection when two objects try to combine their emotions so their feelings toward another affect each other.



Emotion combination

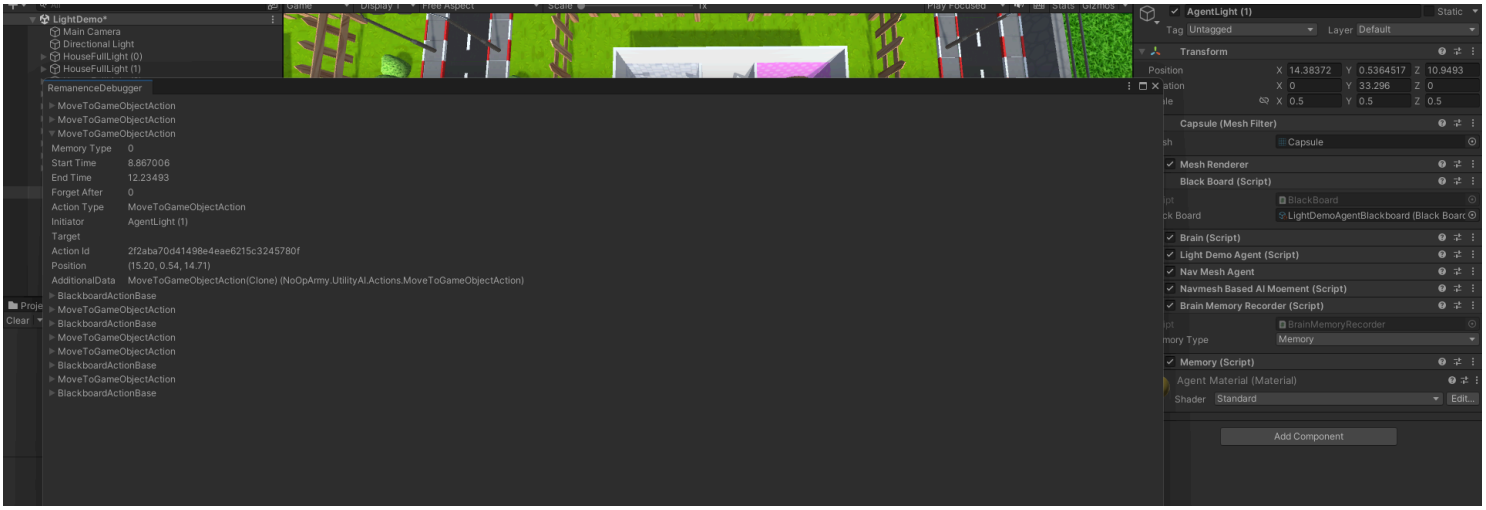
The combination is not automatic and you need to call the `Effect(GameObject other)` method of the `EmotionComponent` for it to calculate. The calculation is bidirectional and affects both objects' feeling toward each other.

Remembrance Debuggers

There are two debugger windows helping you figure out what exists in the memory of an agent or what feelings it has.

Remembrance memory debugger

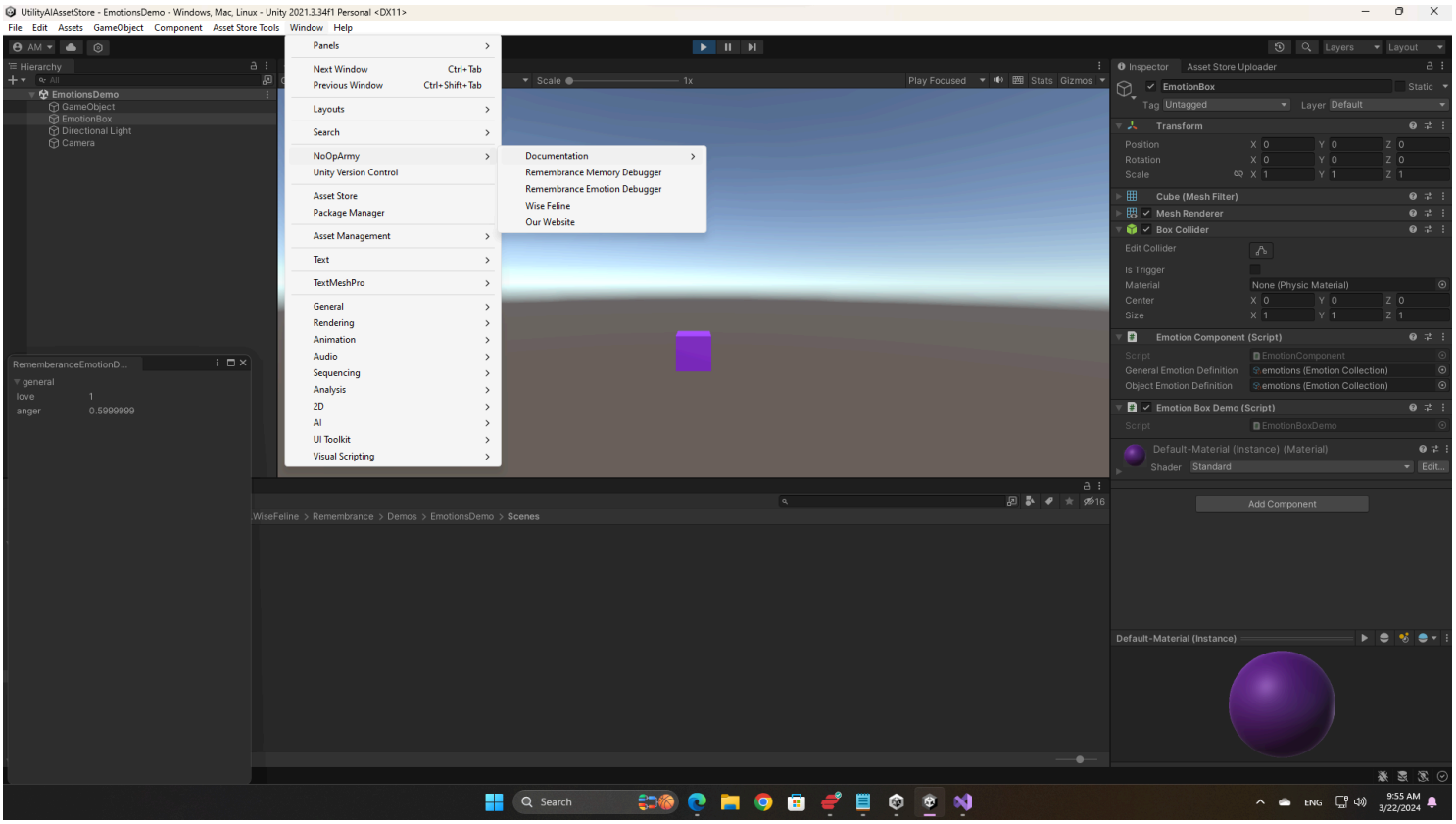
You can open this window from the Window>NoOpArmy>Remembrance Memory Debugger menu.



When a GameObject with a `Memory` or `SimpleMemory` is selected, this window shows all of its memories at runtime. You can inspect what the agent remembers, what will be forgotten and when and what data is stored in each memory.

Remembrance Emotion Debugger

There is a window at Window>NoOpArmy>Remembrance Emotion Debugger menu.



This window shows all general emotions and per object emotions of an agent.

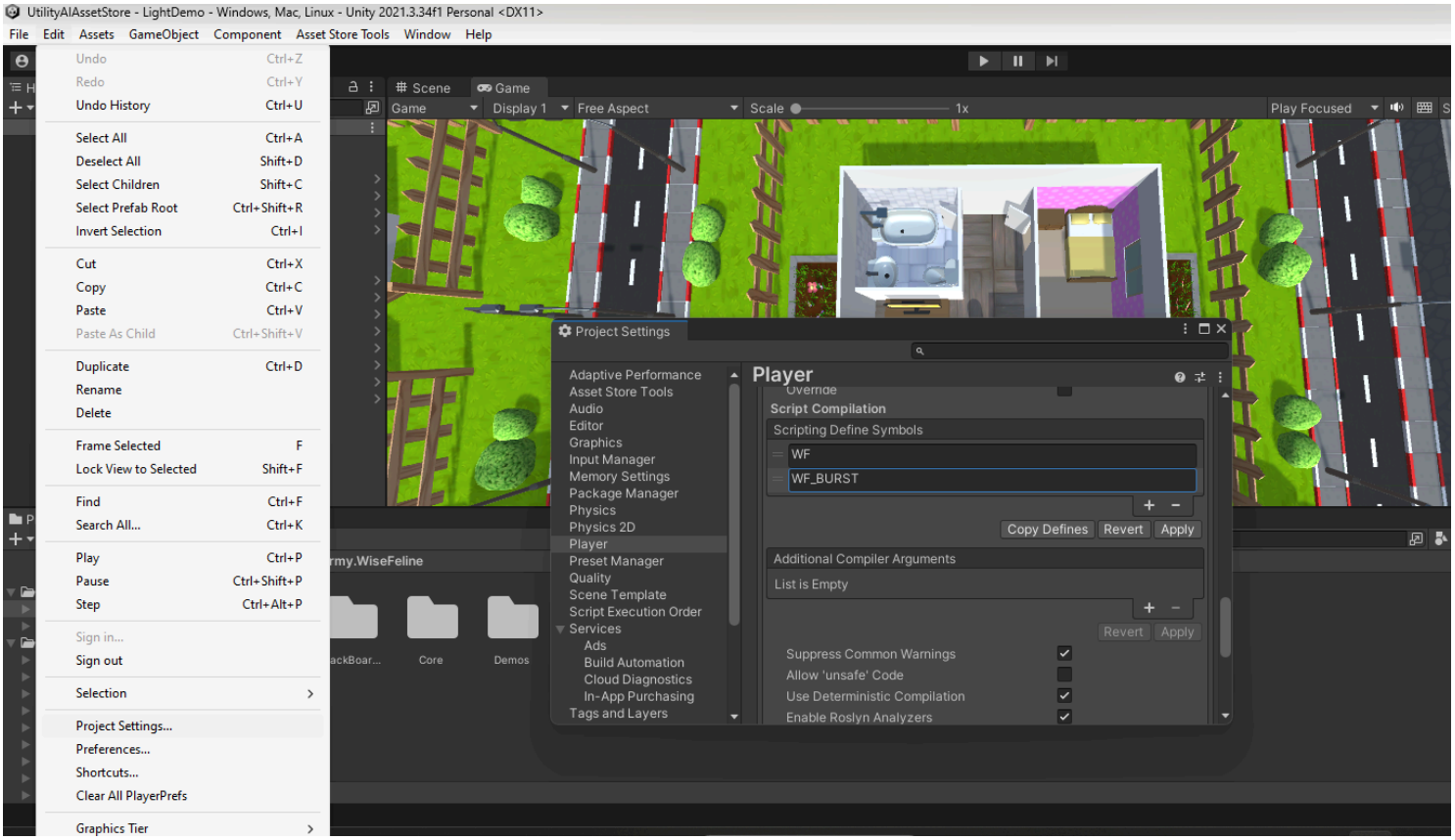
Low level containers

The components above use two container classes to do all of their insertion, removal and search operations and these can be used instead of the components directly by your classes. You might desire this if you want to use a different `MemoryData` class with different fields but you still want to use this library for searching and other functionalities.

The `Memory` component uses `MemoryContainer<T>` which uses lists and the `SimpleMemory` component uses `CircularMemoryContainer<T>` which uses queues. The components are mostly wrappers around these with a few values to be set in the inspector. They call the methods of the containers and return the result back. The `Memory` component also calls a method in its `Update()` for the forgetting feature.

Utility AI integration

For this to become available, you should add the WF define to the list of defines in your player settings. Go to the edit>Project Settings menu and then choose the player tab, in other settings you can add define symbols in the Scripting Define Symbols section.



If you attach the `BrainMemoryRecorder` component to an agent with a Utility AI brain (both Lite and Ultimate), then it records all action changes as memories. It records the current action as a new memory and ends the previous action. The type of the action is recorded in the `type` field of the memory and the target and initiator are set to the agent and its target respectively.

There is also a sample consideration available which gives a score back based on if an action happened in the last x seconds or not.

Namespace NoOpArmy

Classes

Singleton<T>

Base class for the manager components that Wise Feline looks up by type (tags, smart objects, influence maps, recording). Put one in the scene to configure it, or let the first Instance access create one on demand.

Class Singleton<T>

Namespace: NoOpArmy

Assembly: APIRef.dll

Base class for the manager components that Wise Feline looks up by type (tags, smart objects, influence maps, recording). Put one in the scene to configure it, or let the first Instance access create one on demand.

```
public class Singleton<T> : MonoBehaviour where T : MonoBehaviour
```

Type Parameters

T

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← Singleton<T>

Derived

AITagsManager, RecordManager, InfluenceMapCollection, SmartObjectsManager

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) ,

Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() ,
Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) ,
Component.GetComponentsInParent<T>(bool) ,
Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() ,
Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) ,
Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() ,
Component.GetComponentIndex() , Component.CompareTag(string) ,
Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,

Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) , Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) , Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) , Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Properties

Instance

The single instance of this manager, or null while the application is quitting. Resolves in order: the cached instance, an instance already present in the loaded scenes, and finally a new `DontDestroyOnLoad` GameObject created for it.

```
public static T Instance { get; }
```

Property Value

T

Methods

Awake()

```
protected void Awake()
```

OnDestroy()

```
protected void OnDestroy()
```

Namespace NoOpArmy.SmartObjects.Sample

Classes

BlackboardSmartObjectArea

BlackboardSmartObjectBehavior

BlackboardSmartObjectCharacter

DrivingBehaviour

EatFoodBehaviour

This behavior give the agent some food to eat This is a simple demo so we just change the color of the agent to green

SmartObjectDemoPlayer

This script allows your demo agent to eat food and then drive somewhere. you can choose for each agent to do either both or one of them or do neither which is not useful.

Class BlackboardSmartObjectArea

Namespace: NoOpArmy.SmartObjects.Sample

Assembly: APIRef.dll

```
public class BlackboardSmartObjectArea : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← BlackboardSmartObjectArea

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) , Component.SendMessageUpwards(string, object, SendMessageOptions) , Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,

Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,

Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Class BlackboardSmartObjectBehavior

Namespace: NoOpArmy.SmartObjects.Sample

Assembly: APIRef.dll

```
public class BlackboardSmartObjectBehavior : SmartObjectBehaviourBase
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← SmartObjectBehaviourBase ← BlackboardSmartObjectBehavior

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) , Component.SendMessageUpwards(string, object, SendMessageOptions) ,

Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,

Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

changerOnClaim

Blackboard changes to apply at claim to the agent.

```
[Tooltip("Blackboard changes to apply at claim to the agent.")]
public BlackBoardChangeer changerOnClaim
```

Field Value

BlackBoardChangeer

changerOnFree

Blackboard changes to apply at free to the agent.

```
[Tooltip("Blackboard changes to apply at free to the agent.")]
public BlackBoardChangeer changerOnFree
```

Field Value

BlackBoardChangeer

changerOnUse

Blackboard changes to apply at use to the agent.

```
[Tooltip("Blackboard changes to apply at use to the agent.")]
public BlackBoardChangeer changerOnUse
```

Field Value

BlackBoardChangeer

Methods

OnClaimedByAgent(GameObject, SmartObject, int)

Called when a slot of the smart object is successfully claimed by an object

```
public override void OnClaimedByAgent(GameObject agent, SmartObject smartObject, int slot)
```

Parameters

`agent` GameObject

`smartObject` SmartObject

`slot` int

OnFreedByAgent(GameObject, SmartObject, int)

Called when an agent frees the smart object

```
public override void OnFreedByAgent(GameObject agent, SmartObject smartObject, int slot)
```

Parameters

`agent` GameObject

`smartObject` SmartObject

`slot` int

OnUsedByAgent(GameObject, SmartObject, int)

Called when a slot of the object is successfully used by an agent

```
public override void OnUsedByAgent(GameObject agent, SmartObject smartObject, int slot)
```

Parameters

`agent` GameObject

`smartObject` SmartObject

`slot` int

Class BlackboardSmartObjectCharacter

Namespace: NoOpArmy.SmartObjects.Sample

Assembly: APIRef.dll

```
public class BlackboardSmartObjectCharacter : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← BlackboardSmartObjectCharacter

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) , Component.SendMessageUpwards(string, object, SendMessageOptions) , Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,

Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,

Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Class DrivingBehaviour

Namespace: NoOpArmy.SmartObjects.Sample

Assembly: APIRef.dll

```
public class DrivingBehaviour : SmartObjectBehaviourBase
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← SmartObjectBehaviourBase ← DrivingBehaviour

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) , Component.SendMessageUpwards(string, object, SendMessageOptions) ,

Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,

Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

destinationXZ

The destination on the XZ plane

```
public Vector3 destinationXZ
```

Field Value

Vector3

movementDirection

The direction this will drive toward

```
[Tooltip("The direction this will drive toward")]
public Vector3 movementDirection
```

Field Value

Vector3

speed

Driving speed

```
public float speed
```

Field Value

float

Methods

AreAllslotsFilled(SmartObject)

```
public bool AreAllslotsFilled(SmartObject smartObject)
```

Parameters

`smartObject` SmartObject

Returns

bool

OnClaimedByAgent(GameObject, SmartObject, int)

Called when a slot of the smart object is successfully claimed by an object

```
public override void OnClaimedByAgent(GameObject agent, SmartObject smartObject, int slot)
```

Parameters

`agent` GameObject

`smartObject` SmartObject

`slot` int

OnFreedByAgent(GameObject, SmartObject, int)

Called when an agent frees the smart object

```
public override void OnFreedByAgent(GameObject agent, SmartObject smartObject, int slot)
```

Parameters

`agent` GameObject

`smartObject` SmartObject

`slot` int

OnUsedByAgent(GameObject, SmartObject, int)

Called when a slot of the object is successfully used by an agent

```
public override void OnUsedByAgent(GameObject agent, SmartObject smartObject, int slot)
```

Parameters

`agent` GameObject

`smartObject` SmartObject

`slot` int

Class EatFoodBehaviour

Namespace: NoOpArmy.SmartObjects.Sample

Assembly: APIRef.dll

This behavior give the agent some food to eat This is a simple demo so we just change the color of the agent to green

```
public class EatFoodBehaviour : SmartObjectBehaviourBase
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← SmartObjectBehaviourBase ← EatFoodBehaviour

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() ,

Component.GetComponentIndex() , Component.CompareTag(string) ,
Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,

Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Methods

OnClaimedByAgent(GameObject, SmartObject, int)

Called when the agent reserves this so nobody else touches it

```
public override void OnClaimedByAgent(GameObject agent, SmartObject smartObject, int slot)
```

Parameters

`agent` GameObject

`smartObject` SmartObject

`slot` int

OnFreedByAgent(GameObject, SmartObject, int)

Called when the agent using us, frees us

```
public override void OnFreedByAgent(GameObject agent, SmartObject smartObject, int slot)
```

Parameters

`agent` GameObject

`smartObject` SmartObject

`slot` int

OnUsedByAgent(GameObject, SmartObject, int)

Called when the agent uses us

```
public override void OnUsedByAgent(GameObject agent, SmartObject smartObject, int slot)
```

Parameters

`agent` GameObject

`smartObject` SmartObject

`slot` int

Class SmartObjectDemoPlayer

Namespace: NoOpArmy.SmartObjects.Sample

Assembly: APIRef.dll

This script allows your demo agent to eat food and then drive somewhere. you can choose for each agent to do either both or one of them or do neither which is not useful.

```
public class SmartObjectDemoPlayer : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← SmartObjectDemoPlayer

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) ,

Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,

Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

shouldDrive

Should the agent find something drivable and drive somewhere

```
[Tooltip("Should the agent find something drivable and drive somewhere")]
public bool shouldDrive
```

Field Value

bool

shouldEat

Should the agent go eat something before driving somewhere

```
[Tooltip("Should the agent go eat something before driving somewhere")]
public bool shouldEat
```

Field Value

bool

Namespace NoOpArmy.UtilityAI.Actions

Classes

BlackboardActionBase

Actions which want to do some blackboard changes along side their custom code can derive from this

BlackboardActionWithSmartObjectTarget

Finds potential targets from smart objects and optionally claims and uses them. It frees them inOnFinish if the claiming and using is on. Also the action will fail if it tries to claim/use the smart object unsuccessfully.

BlackboardActionWithTagTarget

This action updates its targets based on AI tags system and executes blackboard changes which you choose. The action doesn't do anything with the targets but will not execute if one of the targets don't get a high score. You need at least one consideration with NeedTarget set to true for this to evaluate the targets.

CreateGameObjectAction

This action creates gameobjects, can be used for shooting, providing loot and other similar things

DestroySelfAction

Destroys the GameObject executing the action

DestroyTargetWithTagAction

Destroys the target object, can be used for killing, eating and similar actions It finds the target to destroy using the tags manager and the tags specified in the action

MoveAndPatrol

Patrols the agent on the way points provided as children of a GameObject which is set on a blackboard key

MoveToGameObjectAction

Moves the agent to the position of a GameObject set on a blackboard key.

MoveToSmartObject

Finds a set of smart objects as potential trgets and if the action gets selected to execute then it moves toward the chosen target.

MoveToTargetWithTag

Makes a list of potential targets using the AI Tag system and moves toward the chosen target if the action is selected to execute. Needs a consideration with NeedTarget set so targets can be evaluated and chosen.

MoveToVector3Action

Moves the agent to the position of a Vector3 set on a blackboard key.

NoopAction

This action is our favorite and doesn't do anything. It optionally can stop movement too.

RotateTowardGameObjectAction

Rotates the agent toward the position of a GameObject set on a blackboard key.

Class BlackboardActionBase

Namespace: NoOpArmy.UtilityAI.Actions

Assembly: APIRef.dll

Actions which want to do some blackboard changes along side their custom code can derive from this

```
public class BlackboardActionBase : ActionBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ActionBase ← BlackboardActionBase

Derived

BlackboardActionWithSmartObjectTarget, BlackboardActionWithTagTarget, CreateGameObjectAction, DestroySelfAction, DestroyTargetWithTagAction, MoveAndPatrol, MoveToGameObjectAction, MoveToSmartObject, MoveToTargetWithTag, MoveToVector3Action, NoopAction, RotateTowardGameObjectAction

Inherited Members

ActionBase._priority , ActionBase._isInterruptable , ActionBase.forceInterruptLowerPriorityActions , ActionBase._maxTargetCount , ActionBase._useMomentumOnTarget , ActionBase.Considerations , ActionBase.Brain , ActionBase.IsInitialized , ActionBase.Score , ActionBase.Weight , ActionBase.ChosenTarget , ActionBase.AddTarget(Component) , ActionBase.AddTargets<T>(T[]) , ActionBase.AddTargets<T>(List<T>) , ActionBase.ClearTargets() , ActionBase.RemoveTarget(Component) , ActionBase.Clone(ActionSet) , ActionBase.OnLateUpdate() , ActionBase.OnFixedUpdate() , ActionBase.ActionSucceed() , ActionBase.ActionSucceeded() , ActionBase.ActionFailed() , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,

Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

OnStartAnimationTrigger

```
[Tooltip("During initialization, we attempt to find the animation controller in the game
object hierarchy and set this trigger at the start of the action.")]
public string OnStartAnimationTrigger
```

Field Value

string

ResetChangerOnStart

Resets the changer whenever this action is selected. Mostly useful for changers which execute a specific number of times

```
[Tooltip("Resets the changer whenever this action is selected. Mostly useful for changers
which execute a specific number of times")]
public bool ResetChangerOnStart
```

Field Value

bool

blackBoard

The blackboard to use for changes

```
protected BlackBoard blackBoard
```

Field Value

BlackBoard

changer

The change you want to happen to the blackboard during this action

```
[Tooltip("The change you want to happen to the blackboard during this action")]
public BlackBoardChangeer changer
```

Field Value

BlackBoardChangeer

changerStartingDelayInSeconds

The start delay before changes start taking effect.

```
[Tooltip("The start delay before changes in seconds")]
public float changerStartingDelayInSeconds
```

Field Value

float

forceUpdateTargetsOnFinish

If true, upon completion of this action, it will prompt the brain to immediately update the target for all actions.

```
[Tooltip("If true, upon completion of this action, it will prompt the brain to immediately
update the target for all actions.")]
public bool forceUpdateTargetsOnFinish
```

Field Value

bool

Methods

OnFinish()

Called when the action is finished, either by failing or succeeding in achieving a desired behaviour

```
protected override void OnFinish()
```

OnInitialized()

Called when the action is initialized

```
protected override void OnInitialized()
```

OnStart()

Should be used like MonoBehaviour's Start for the action

```
protected override void OnStart()
```

OnUpdate()

Should be used like MonoBehaviour's Update for the action

```
protected override void OnUpdate()
```

UpdateTargets()

Fires when you need to update the list of targets

```
protected override void UpdateTargets()
```

Class BlackboardActionWithSmartObjectTarget

Namespace: NoOpArmy.UtilityAI.Actions

Assembly: APIRef.dll

Finds potential targets from smart objects and optionally claims and uses them. It frees them inOnFinish if the claiming and using is on. Also the action will fail if it tries to claim/use the smart object unsuccessfully.

```
public class BlackboardActionWithSmartObjectTarget : BlackboardActionBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ActionBase ← BlackboardActionBase ← BlackboardActionWithSmartObjectTarget

Inherited Members

BlackboardActionBase.changerStartingDelayInSeconds , BlackboardActionBase.forceUpdateTargetsOnFinish , BlackboardActionBase.ResetChangerOnStart , BlackboardActionBase.changer , BlackboardActionBase.OnStartAnimationTrigger , BlackboardActionBase.blackBoard , BlackboardActionBase.OnInitialized() , BlackboardActionBase.OnUpdate() , ActionBase._priority , ActionBase._isInterruptable , ActionBase.forceInterruptLowerPriorityActions , ActionBase._maxTargetCount , ActionBase._useMomentumOnTarget , ActionBase.Considerations , ActionBase.Brain , ActionBase.IsInitialized , ActionBase.Score , ActionBase.Weight , ActionBase.ChosenTarget , ActionBase.AddTarget(Component) , ActionBase.AddTargets<T>(T[]) , ActionBase.AddTargets<T>(List<T>) , ActionBase.ClearTargets() , ActionBase.RemoveTarget(Component) , ActionBase.Clone(ActionSet) , ActionBase.OnLateUpdate() , ActionBase.OnFixedUpdate() , ActionBase.ActionSucceed() , ActionBase.ActionSucceeded() , ActionBase.ActionFailed() , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,

Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

ClaimAndUseSmartObject

Should the action claim and use the smart object in OnStart and then Free the object in OnFinish?

```
[Tooltip("Should the action claim and use the smart object in OnStart and then Free the
object in OnFinish?")]
public bool ClaimAndUseSmartObject
```

Field Value

bool

SlotStatus

Finds smart objects with the chosen slot status.

```
[Tooltip("Finds smart objects with the chosen slot status.")]
public SmartObjectsManager.SearchQueryOptions SlotStatus
```

Field Value

SmartObjectsManager.SearchQueryOptions

excludedTags

The tags which if one of them is on the smart object, it is excluded from the search

```
[Tooltip("The tags which if one of them is on the smart object, it is excluded from
the search")]
public string[] excludedTags
```

Field Value

string[]

includedTags

The tags which at least one of them should be on the smart object so it is included in the search results

```
[Tooltip("The tags which at least one of them should be on the smart object so it is
included in the search results")]
public string[] includedTags
```

Field Value

string[]

searchRadius

Radius of the search for smart objects

```
public float searchRadius
```

Field Value

float

Methods

OnFinish()

Called when the action is finished, either by failing or succeeding in achieving a desired behaviour

```
protected override void OnFinish()
```

OnStart()

Should be used like MonoBehaviour's Start for the action

```
protected override void OnStart()
```

UpdateTargets()

Fires when you need to update the list of targets

```
protected override void UpdateTargets()
```

Class BlackboardActionWithTagTarget

Namespace: NoOpArmy.UtilityAI.Actions

Assembly: APIRef.dll

This action updates its targets based on AI tags system and executes blackboard changes which you choose. The action doesn't do anything with the targets but will not execute if one of the targets don't get a high score. You need at least one consideration with NeedTarget set to true for this to evaluate the targets.

```
public class BlackboardActionWithTagTarget : BlackboardActionBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ActionBase ← BlackboardActionBase ← BlackboardActionWithTagTarget

Inherited Members

BlackboardActionBase.changerStartingDelayInSeconds , BlackboardActionBase.forceUpdateTargetsOnFinish , BlackboardActionBase.ResetChangerOnStart , BlackboardActionBase.changer , BlackboardActionBase.OnStartAnimationTrigger , BlackboardActionBase.blackBoard , BlackboardActionBase.OnInitialized() , BlackboardActionBase.OnStart() , BlackboardActionBase.OnUpdate() , BlackboardActionBase.OnFinish() , ActionBase._priority , ActionBase._isInterruptable , ActionBase.forceInterruptLowerPriorityActions , ActionBase._maxTargetCount , ActionBase._useMomentumOnTarget , ActionBase.Considerations , ActionBase.Brain , ActionBase.IsInitialized , ActionBase.Score , ActionBase.Weight , ActionBase.ChosenTarget , ActionBase.AddTarget(Component) , ActionBase.AddTargets<T>(T[]) , ActionBase.AddTargets<T>(List<T>) , ActionBase.ClearTargets() , ActionBase.RemoveTarget(Component) , ActionBase.Clone(ActionSet) , ActionBase.OnLateUpdate() , ActionBase.OnFixedUpdate() , ActionBase.ActionSucceed() , ActionBase.ActionSucceeded() , ActionBase.ActionFailed() , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,

Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

excludedTags

Tags which if any of them are on the AITags component of the object, the object cannot be considered a potential target

```
[Tooltip("Tags which if any of them are on the AITags component of the object, the object
cannot be considered a potential target")]
public string[] excludedTags
```

Field Value

string[]

includedTags

Tags which at least one of them should be in the AITags component of the object to be considered a valid target.

```
[Tooltip("Tags which at least one of them should be in the AITags component of the object to
be considered a valid target.")]
public string[] includedTags
```

Field Value

string[]

radius

The radius of the search for tags

```
[Header("Search properties")]
[Tooltip("The radius of the search for tags")]
public float radius
```

Field Value

float

Methods

UpdateTargets()

Fires when you need to update the list of targets

```
protected override void UpdateTargets()
```

Class CreateGameObjectAction

Namespace: NoOpArmy.UtilityAI.Actions

Assembly: APIRef.dll

This action creates gameobjects, can be used for shooting, providing loot and other similar things

```
public class CreateGameObjectAction : BlackboardActionBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ActionBase ← BlackboardActionBase ← CreateGameObjectAction

Inherited Members

BlackboardActionBase.changerStartingDelayInSeconds , BlackboardActionBase.forceUpdateTargetsOnFinish , BlackboardActionBase.ResetChangerOnStart , BlackboardActionBase.changer , BlackboardActionBase.OnStartAnimationTrigger , BlackboardActionBase.blackBoard , BlackboardActionBase.OnFinish() , ActionBase._priority , ActionBase._isInterruptable , ActionBase.forceInterruptLowerPriorityActions , ActionBase._maxTargetCount , ActionBase._useMomentumOnTarget , ActionBase.Considerations , ActionBase.Brain , ActionBase.IsInitialized , ActionBase.Score , ActionBase.Weight , ActionBase.ChosenTarget , ActionBase.AddTarget(Component) , ActionBase.AddTargets<T>(T[]) , ActionBase.AddTargets<T>(List<T>) , ActionBase.ClearTargets() , ActionBase.RemoveTarget(Component) , ActionBase.Clone(ActionSet) , ActionBase.OnLateUpdate() , ActionBase.OnFixedUpdate() , ActionBase.ActionSucceed() , ActionBase.ActionSucceeded() , ActionBase.ActionFailed() , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) ,

Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

createAtTheSameTime

Should we create all objects at the same time or once after each delay

```
[Tooltip("Should we create all objects at the same time or once after each delay")]
public bool createAtTheSameTime
```

Field Value

bool

creationDelay

Delay to create an object

```
[Tooltip("Delay to create an object")]
public float creationDelay
```

Field Value

float

creationPosition

The position offset to create the object

```
[Tooltip("The position offset to create the object")]
public Vector3 creationPosition
```

Field Value

Vector3

isPositionRelative

Is the position specified from origin or the position of the executing game object.

```
[Tooltip("Is the position specified from origin or the position of the executing
game object.")]
public bool isPositionRelative
```

Field Value

bool

objectCountToCreate

How many objects to create

```
[Tooltip("How many objects to create")]
public int objectCountToCreate
```

Field Value

int

objectToCreate

The GameObject to create

```
[Tooltip("The GameObject to create")]
public GameObject objectToCreate
```

Field Value

GameObject

shouldAdaptBrainRotation

Should the object adapt the brain's object rotation

```
[Tooltip("Should the object adapt the brain's object rotation")]
public bool shouldAdaptBrainRotation
```

Field Value

bool

Methods

OnInitialized()

Called when the action is initialized

```
protected override void OnInitialized()
```

OnStart()

Should be used like MonoBehaviour's Start for the action

```
protected override void OnStart()
```

OnUpdate()

Should be used like MonoBehaviour's Update for the action

```
protected override void OnUpdate()
```

UpdateTargets()

Fires when you need to update the list of targets

```
protected override void UpdateTargets()
```

Class DestroySelfAction

Namespace: NoOpArmy.UtilityAI.Actions

Assembly: APIRef.dll

Destroys the GameObject executing the action

```
public class DestroySelfAction : BlackboardActionBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ActionBase ← BlackboardActionBase ← DestroySelfAction

Inherited Members

BlackboardActionBase.changerStartingDelayInSeconds , BlackboardActionBase.forceUpdateTargetsOnFinish , BlackboardActionBase.ResetChangerOnStart , BlackboardActionBase.changer , BlackboardActionBase.OnStartAnimationTrigger , BlackboardActionBase.blackBoard , BlackboardActionBase.OnInitialized() , BlackboardActionBase.OnFinish() , ActionBase._priority , ActionBase._isInterruptable , ActionBase.forceInterruptLowerPriorityActions , ActionBase._maxTargetCount , ActionBase._useMomentumOnTarget , ActionBase.Considerations , ActionBase.Brain , ActionBase.IsInitialized , ActionBase.Score , ActionBase.Weight , ActionBase.ChosenTarget , ActionBase.AddTarget(Component) , ActionBase.AddTargets<T>(T[]) , ActionBase.AddTargets<T>(List<T>) , ActionBase.ClearTargets() , ActionBase.RemoveTarget(Component) , ActionBase.Clone(ActionSet) , ActionBase.OnLateUpdate() , ActionBase.OnFixedUpdate() , ActionBase.ActionSucceed() , ActionBase.ActionSucceeded() , ActionBase.ActionFailed() , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,

Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

delayForDestruction

How much time should happen before destruction happens

```
[Tooltip("How much time should happen before destruction happens")]
public float delayForDestruction
```

Field Value

float

Methods

OnStart()

Should be used like MonoBehaviour's Start for the action

```
protected override void OnStart()
```

OnUpdate()

Should be used like MonoBehaviour's Update for the action

```
protected override void OnUpdate()
```

UpdateTargets()

Fires when you need to update the list of targets

```
protected override void UpdateTargets()
```

Class DestroyTargetWithTagAction

Namespace: NoOpArmy.UtilityAI.Actions

Assembly: APIRef.dll

Destroys the target object, can be used for killing, eating and similar actions It finds the target to destroy using the tags manager and the tags specified in the action

```
public class DestroyTargetWithTagAction : BlackboardActionBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ActionBase ← BlackboardActionBase ← DestroyTargetWithTagAction

Inherited Members

BlackboardActionBase.changerStartingDelayInSeconds , BlackboardActionBase.forceUpdateTargetsOnFinish , BlackboardActionBase.ResetChangerOnStart , BlackboardActionBase.changer , BlackboardActionBase.OnStartAnimationTrigger , BlackboardActionBase.blackBoard , BlackboardActionBase.OnInitialized() , BlackboardActionBase.OnFinish() , ActionBase._priority , ActionBase._isInterruptable , ActionBase.forceInterruptLowerPriorityActions , ActionBase._maxTargetCount , ActionBase._useMomentumOnTarget , ActionBase.Considerations , ActionBase.Brain , ActionBase.IsInitialized , ActionBase.Score , ActionBase.Weight , ActionBase.ChosenTarget , ActionBase.AddTarget(Component) , ActionBase.AddTargets<T>(T[]) , ActionBase.AddTargets<T>(List<T>) , ActionBase.ClearTargets() , ActionBase.RemoveTarget(Component) , ActionBase.Clone(ActionSet) , ActionBase.OnLateUpdate() , ActionBase.OnFixedUpdate() , ActionBase.ActionSucceed() , ActionBase.ActionSucceeded() , ActionBase.ActionFailed() , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,

Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

delayForDestruction

How much time should pass before destruction happens

```
[Tooltip("How much time should pass before destruction happens")]
public float delayForDestruction
```

Field Value

float

excludedTags

The tags which existence of any of them on the target object means it is an invalid target

```
[Tooltip("The tags which existence of any of them on the target object means it is an
invalid target")]
public string[] excludedTags
```

Field Value

string[]

includedTags

The tags which existence of any of them on the target object means it is a valid target

```
[Tooltip("The tags which existence of any of them on the target object means it is a
valid target")]
public string[] includedTags
```

Field Value

string[]

radius

```
public float radius
```

Field Value

float

Methods

OnStart()

Should be used like MonoBehaviour's Start for the action

```
protected override void OnStart()
```

OnUpdate()

Should be used like MonoBehaviour's Update for the action

```
protected override void OnUpdate()
```

UpdateTargets()

Fires when you need to update the list of targets

```
protected override void UpdateTargets()
```

Class MoveAndPatrol

Namespace: NoOpArmy.UtilityAI.Actions

Assembly: APIRef.dll

Patrols the agent on the way points provided as children of a GameObject which is set on a blackboard key

```
public class MoveAndPatrol : BlackboardActionBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ActionBase ← BlackboardActionBase ← MoveAndPatrol

Inherited Members

BlackboardActionBase.changerStartingDelayInSeconds , BlackboardActionBase.forceUpdateTargetsOnFinish , BlackboardActionBase.ResetChangerOnStart , BlackboardActionBase.changer , BlackboardActionBase.OnStartAnimationTrigger , BlackboardActionBase.blackBoard , BlackboardActionBase.OnUpdate() , BlackboardActionBase.OnFinish() , ActionBase._priority , ActionBase._isInterruptable , ActionBase.forceInterruptLowerPriorityActions , ActionBase._maxTargetCount , ActionBase._useMomentumOnTarget , ActionBase.Considerations , ActionBase.Brain , ActionBase.IsInitialized , ActionBase.Score , ActionBase.Weight , ActionBase.ChosenTarget , ActionBase.AddTarget(Component) , ActionBase.AddTargets<T>(T[]) , ActionBase.AddTargets<T>(List<T>) , ActionBase.ClearTargets() , ActionBase.RemoveTarget(Component) , ActionBase.Clone(ActionSet) , ActionBase.OnLateUpdate() , ActionBase.OnFixedUpdate() , ActionBase.ActionSucceed() , ActionBase.ActionSucceeded() , ActionBase.ActionFailed() , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,

Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

FinishAfterOneCycle

Should the action finishes after going through the points once?

```
[Tooltip("Should the action finishes after going through the points once")]
public bool FinishAfterOneCycle
```

Field Value

bool

gameObjectKeyName

Name of the key in the blackboard of the GameObject which contains all patrol points as children

```
[Tooltip("Name of the key in the blackboard of the GameObject which contains all patrol
points as children")]
public string gameObjectKeyName
```

Field Value

string

Methods

OnInitialized()

Called when the action is initialized

```
protected override void OnInitialized()
```

OnStart()

Should be used like MonoBehaviour's Start for the action

```
protected override void OnStart()
```

UpdateTargets()

Fires when you need to update the list of targets

```
protected override void UpdateTargets()
```

Class MoveToGameObjectAction

Namespace: NoOpArmy.UtilityAI.Actions

Assembly: APIRef.dll

Moves the agent to the position of a GameObject set on a blackboard key.

```
public class MoveToGameObjectAction : BlackboardActionBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ActionBase ← BlackboardActionBase ← MoveToGameObjectAction

Inherited Members

BlackboardActionBase.changerStartingDelayInSeconds , BlackboardActionBase.forceUpdateTargetsOnFinish , BlackboardActionBase.ResetChangerOnStart , BlackboardActionBase.changer , BlackboardActionBase.OnStartAnimationTrigger , BlackboardActionBase.blackBoard , BlackboardActionBase.OnUpdate() , BlackboardActionBase.OnFinish() , ActionBase._priority , ActionBase._isInterruptable , ActionBase.forceInterruptLowerPriorityActions , ActionBase._maxTargetCount , ActionBase._useMomentumOnTarget , ActionBase.Considerations , ActionBase.Brain , ActionBase.IsInitialized , ActionBase.Score , ActionBase.Weight , ActionBase.ChosenTarget , ActionBase.AddTarget(Component) , ActionBase.AddTargets<T>(T[]) , ActionBase.AddTargets<T>(List<T>) , ActionBase.ClearTargets() , ActionBase.RemoveTarget(Component) , ActionBase.Clone(ActionSet) , ActionBase.OnLateUpdate() , ActionBase.OnFixedUpdate() , ActionBase.ActionSucceed() , ActionBase.ActionSucceeded() , ActionBase.ActionFailed() , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,

Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

gameObjectKeyName

Name of the GameObject key in the blackboard

```
[Tooltip("Name of the GameObject key in the blackboard")]
public string gameObjectKeyName
```

Field Value

string

Methods

OnInitialized()

Called when the action is initialized

```
protected override void OnInitialized()
```

OnStart()

Should be used like MonoBehaviour's Start for the action

```
protected override void OnStart()
```

UpdateTargets()

Fires when you need to update the list of targets

```
protected override void UpdateTargets()
```

Class MoveToSmartObject

Namespace: NoOpArmy.UtilityAI.Actions

Assembly: APIRef.dll

Finds a set of smart objects as potential trgets and if the action gets selected to execute then it moves toward the chosen target.

```
public class MoveToSmartObject : BlackboardActionBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ActionBase ← BlackboardActionBase ← MoveToSmartObject

Inherited Members

BlackboardActionBase.changerStartingDelayInSeconds , BlackboardActionBase.forceUpdateTargetsOnFinish , BlackboardActionBase.ResetChangerOnStart , BlackboardActionBase.changer , BlackboardActionBase.OnStartAnimationTrigger , BlackboardActionBase.blackBoard , BlackboardActionBase.OnUpdate() , ActionBase._priority , ActionBase._isInterruptable , ActionBase.forceInterruptLowerPriorityActions , ActionBase._maxTargetCount , ActionBase._useMomentumOnTarget , ActionBase.Considerations , ActionBase.Brain , ActionBase.IsInitialized , ActionBase.Score , ActionBase.Weight , ActionBase.ChosenTarget , ActionBase.AddTarget(Component) , ActionBase.AddTargets<T>(T[]) , ActionBase.AddTargets<T>(List<T>) , ActionBase.ClearTargets() , ActionBase.RemoveTarget(Component) , ActionBase.Clone(ActionSet) , ActionBase.OnLateUpdate() , ActionBase.OnFixedUpdate() , ActionBase.ActionSucceed() , ActionBase.ActionSucceeded() , ActionBase.ActionFailed() , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,

Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

SlotStatus

Criteria that the slots should meet to be considered a potential target

```
public SmartObjectsManager.SearchQueryOptions SlotStatus
```

Field Value

SmartObjectsManager.SearchQueryOptions

excludedTags

The tag which none of them should be on the smart object so the object is considered a target.

```
public string[] excludedTags
```

Field Value

string[]

includedTags

The tags which at least one of them should be on the smart object to be valid

```
public string[] includedTags
```

Field Value

string[]

searchRadius

Radius of the search for SmartObjects

```
[Tooltip("Radius of the search for SmartObjects")]
public float searchRadius
```

Field Value

float

Methods

OnFinish()

Called when the action is finished, either by failing or succeeding in achieving a desired behaviour

```
protected override void OnFinish()
```

OnInitialized()

Called when the action is initialized

```
protected override void OnInitialized()
```

OnStart()

Should be used like MonoBehaviour's Start for the action

```
protected override void OnStart()
```

UpdateTargets()

Fires when you need to update the list of targets

```
protected override void UpdateTargets()
```

Class MoveToTargetWithTag

Namespace: NoOpArmy.UtilityAI.Actions

Assembly: APIRef.dll

Makes a list of potential targets using the AI Tag system and moves toward the chosen target if the action is selected to execute. Needs a consideration with NeedTarget set so targets can be evaluated and chosen.

```
public class MoveToTargetWithTag : BlackboardActionBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ActionBase ← BlackboardActionBase ← MoveToTargetWithTag

Inherited Members

BlackboardActionBase.changerStartingDelayInSeconds , BlackboardActionBase.forceUpdateTargetsOnFinish , BlackboardActionBase.ResetChangerOnStart , BlackboardActionBase.changer , BlackboardActionBase.OnStartAnimationTrigger , BlackboardActionBase.blackBoard , BlackboardActionBase.OnUpdate() , BlackboardActionBase.OnFinish() , ActionBase._priority , ActionBase._isInterruptable , ActionBase.forceInterruptLowerPriorityActions , ActionBase._maxTargetCount , ActionBase._useMomentumOnTarget , ActionBase.Considerations , ActionBase.Brain , ActionBase.IsInitialized , ActionBase.Score , ActionBase.Weight , ActionBase.ChosenTarget , ActionBase.AddTarget(Component) , ActionBase.AddTargets<T>(T[]) , ActionBase.AddTargets<T>(List<T>) , ActionBase.ClearTargets() , ActionBase.RemoveTarget(Component) , ActionBase.Clone(ActionSet) , ActionBase.OnLateUpdate() , ActionBase.OnFixedUpdate() , ActionBase.ActionSucceed() , ActionBase.ActionSucceeded() , ActionBase.ActionFailed() , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,

Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

excludedTags

Tags which none of them should be present on the AITags component of the object to be a potential target.

```
public string[] excludedTags
```

Field Value

string[]

includedTags

Tags which at least one of them should be present on the AITags component of the object to be a potential target.

```
public string[] includedTags
```

Field Value

string[]

radius

The radius of the search

```
[Tooltip("The radius of the search")]
public float radius
```

Field Value

float

Methods

OnInitialized()

Called when the action is initialized

```
protected override void OnInitialized()
```

OnStart()

Should be used like MonoBehaviour's Start for the action

```
protected override void OnStart()
```

UpdateTargets()

Fires when you need to update the list of targets

```
protected override void UpdateTargets()
```

Class MoveToVector3Action

Namespace: NoOpArmy.UtilityAI.Actions

Assembly: APIRef.dll

Moves the agent to the position of a Vector3 set on a blackboard key.

```
public class MoveToVector3Action : BlackboardActionBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ActionBase ← BlackboardActionBase ← MoveToVector3Action

Inherited Members

BlackboardActionBase.changerStartingDelayInSeconds , BlackboardActionBase.forceUpdateTargetsOnFinish , BlackboardActionBase.ResetChangerOnStart , BlackboardActionBase.changer , BlackboardActionBase.OnStartAnimationTrigger , BlackboardActionBase.blackBoard , BlackboardActionBase.OnUpdate() , BlackboardActionBase.OnFinish() , ActionBase._priority , ActionBase._isInterruptable , ActionBase.forceInterruptLowerPriorityActions , ActionBase._maxTargetCount , ActionBase._useMomentumOnTarget , ActionBase.Considerations , ActionBase.Brain , ActionBase.IsInitialized , ActionBase.Score , ActionBase.Weight , ActionBase.ChosenTarget , ActionBase.AddTarget(Component) , ActionBase.AddTargets<T>(T[]) , ActionBase.AddTargets<T>(List<T>) , ActionBase.ClearTargets() , ActionBase.RemoveTarget(Component) , ActionBase.Clone(ActionSet) , ActionBase.OnLateUpdate() , ActionBase.OnFixedUpdate() , ActionBase.ActionSucceed() , ActionBase.ActionSucceeded() , ActionBase.ActionFailed() , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,

Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

positionKeyName

Name of the Vector3 key in the blackboard

```
[Tooltip("Name of the Vector3 key in the blackboard")]
public string positionKeyName
```

Field Value

string

Methods

OnInitialized()

Called when the action is initialized

```
protected override void OnInitialized()
```

OnStart()

Should be used like MonoBehaviour's Start for the action

```
protected override void OnStart()
```

UpdateTargets()

Fires when you need to update the list of targets

```
protected override void UpdateTargets()
```

Class NoopAction

Namespace: NoOpArmy.UtilityAI.Actions

Assembly: APIRef.dll

This action is our favorite and doesn't do anything. It optionally can stop movement too.

```
public class NoopAction : BlackboardActionBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ActionBase ← BlackboardActionBase ← NoopAction

Inherited Members

BlackboardActionBase.changerStartingDelayInSeconds , BlackboardActionBase.forceUpdateTargetsOnFinish , BlackboardActionBase.ResetChangerOnStart , BlackboardActionBase.changer , BlackboardActionBase.OnStartAnimationTrigger , BlackboardActionBase.blackBoard , BlackboardActionBase.OnInitialized() , BlackboardActionBase.OnUpdate() , BlackboardActionBase.OnFinish() , ActionBase._priority , ActionBase._isInterruptable , ActionBase.forceInterruptLowerPriorityActions , ActionBase._maxTargetCount , ActionBase._useMomentumOnTarget , ActionBase.Considerations , ActionBase.Brain , ActionBase.IsInitialized , ActionBase.Score , ActionBase.Weight , ActionBase.ChosenTarget , ActionBase.AddTarget(Component) , ActionBase.AddTargets<T>(T[]) , ActionBase.AddTargets<T>(List<T>) , ActionBase.ClearTargets() , ActionBase.RemoveTarget(Component) , ActionBase.Clone(ActionSet) , ActionBase.OnLateUpdate() , ActionBase.OnFixedUpdate() , ActionBase.ActionSucceed() , ActionBase.ActionSucceeded() , ActionBase.ActionFailed() , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) ,

Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Methods

OnStart()

Should be used like MonoBehaviour's Start for the action

```
protected override void OnStart()
```

UpdateTargets()

Fires when you need to update the list of targets

```
protected override void UpdateTargets()
```

Class RotateTowardGameObjectAction

Namespace: NoOpArmy.UtilityAI.Actions

Assembly: APIRef.dll

Rotates the agent toward the position of a GameObject set on a blackboard key.

```
public class RotateTowardGameObjectAction : BlackboardActionBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ActionBase ← BlackboardActionBase ← RotateTowardGameObjectAction

Inherited Members

BlackboardActionBase.changerStartingDelayInSeconds , BlackboardActionBase.forceUpdateTargetsOnFinish , BlackboardActionBase.ResetChangerOnStart , BlackboardActionBase.changer , BlackboardActionBase.OnStartAnimationTrigger , BlackboardActionBase.blackBoard , BlackboardActionBase.OnUpdate() , BlackboardActionBase.OnFinish() , ActionBase._priority , ActionBase._isInterruptable , ActionBase.forceInterruptLowerPriorityActions , ActionBase._maxTargetCount , ActionBase._useMomentumOnTarget , ActionBase.Considerations , ActionBase.Brain , ActionBase.IsInitialized , ActionBase.Score , ActionBase.Weight , ActionBase.ChosenTarget , ActionBase.AddTarget(Component) , ActionBase.AddTargets<T>(T[]) , ActionBase.AddTargets<T>(List<T>) , ActionBase.ClearTargets() , ActionBase.RemoveTarget(Component) , ActionBase.Clone(ActionSet) , ActionBase.OnLateUpdate() , ActionBase.OnFixedUpdate() , ActionBase.ActionSucceed() , ActionBase.ActionSucceeded() , ActionBase.ActionFailed() , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,

Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

gameObjectKeyName

Name of the GameObject key in the blackboard

```
[Tooltip("Name of the GameObject key in the blackboard")]
public string gameObjectKeyName
```

Field Value

string

Methods

OnInitialized()

Called when the action is initialized

```
protected override void OnInitialized()
```

OnStart()

Should be used like MonoBehaviour's Start for the action

```
protected override void OnStart()
```

UpdateTargets()

Fires when you need to update the list of targets

```
protected override void UpdateTargets()
```

Namespace NoOpArmy.UtilityAI.Considerations

Classes

BlackboardBoolConsideration

Checks a boolean key in a blackboard. This reutnrs 1 if the key exists and is true and if it doesn't exist or is false returns 0.

BlackboardConsiderationKeyExists

This consideration returns 1 if the specified key name exists and otherwise returns 0

BlackboardFloatConsideration

Checks afloat key in the blackboard and if it exists, returns its value, otherwise returns 0

BlackboardGameObjectDistanceConsideration

Returns the distance between the executing agent and a GameObject which is specified in a key in the blackboard.

BlackboardIntRangeConsideration

Checks an integer key in the blackboard and returns its value. If the key doesn't exist then it returns 0

BlackboardIntValueConsideration

This consideration checks an int key in the blackboard and returns 1 if the value equals the expected value; otherwise, it returns 0.

ConstBoolConsideration

Returns a constant value. This consideration is mostly used for testing

InfluenceMapConsideration

This consideration searches the influence map specified for a value and returns 1 if it finds something. Otherwise it returns 0

LineOfSightConsideration

Checks if we have a line of sight to our target or something blocks the view

SmartObjectDistanceConsideration

Returns the distance to a target which is a smart object. You can choose to get the distance to a specific slot with a specific condition like the distance to the closes slot which is free.

TargetDistanceConsideration

Returns the distance to a target. This normalizes to a value between 0 and 1 based on minRange and maxRange values

ValidGameObjectPathConsideration

This consideration returns 1 if a valid path from the agent's NavmeshAgent to the position of the GameObject stored in the key is present, otherwise returns 0 The NeedTarget causes the GameObject to be read from the target but always a path from the owning agent is calculated

ValidPathConsideration

This consideration returns 1 if a valid path from the agent's NavmeshAgent to the position stored in the Vector3 key is present, otherwise returns 0 The NeedTarget causes the Vector3 to be read from the target but always a path from the owning agent is calculated

Enums

BlackboardConsiderationKeyExists.KeyType

The blackboard key type

Class BlackboardBoolConsideration

Namespace: NoOpArmy.UtilityAI.Considerations

Assembly: APIRef.dll

Checks a boolean key in a blackboard. This reutnrs 1 if the key exists and is true and if it doesn't exist or is false returns 0.

```
public class BlackboardBoolConsideration : ConsiderationBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ConsiderationBase ← BlackboardBoolConsideration

Inherited Members

ConsiderationBase.instantiatePerAgent , ConsiderationBase.minRange , ConsiderationBase.maxRange , ConsiderationBase.Score , ConsiderationBase.Brain , ConsiderationBase.Clone() , ConsiderationBase.OnInitialized() , ConsiderationBase.UpdateRange(float, float) , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,

Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) , Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) , Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) , Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) , Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) , Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) , Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

keyName

Name of the boolean key to check

```
[Tooltip("Name of the boolean key to check")]
public string keyName
```

Field Value

string

Methods

GetValue(Component)

This method should return the current value of the consideration which is used for multiplication with other considerations to calculate the score of the action

```
protected override float GetValue(Component target)
```

Parameters

`target` Component

If the consideration has a target then this value is the target component which the consideration should be calculated for. The actual type of this depends on the type of the component that the action stores in its target list in the UpdateTargets call. If the consideration doesn't have a target, then the Brain component of the executing agent itself is passed to the method.

Returns

float

A value between minRange and maxRange which indicates the current value/score of the consideration

Class BlackboardConsiderationKeyExists

Namespace: NoOpArmy.UtilityAI.Considerations

Assembly: APIRef.dll

This consideration returns 1 if the specified key name exists and otherwise returns 0

```
public class BlackboardConsiderationKeyExists : ConsiderationBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ConsiderationBase ← BlackboardConsiderationKeyExists

Inherited Members

ConsiderationBase.instantiatePerAgent , ConsiderationBase.minRange , ConsiderationBase.maxRange , ConsiderationBase.Score , ConsiderationBase.Brain , ConsiderationBase.Clone() , ConsiderationBase.OnInitialized() , ConsiderationBase.UpdateRange(float, float) , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,

Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) , Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) , Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) , Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) , Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) , Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) , Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

keyName

Name of the key in the blackboard

```
[Tooltip("Name of the key in the blackboard")]
public string keyName
```

Field Value

string

keyType

Type of the key to check for existence

```
[Tooltip("Type of the key to check for existence")]
public BlackboardConsiderationKeyExists.KeyType keyType
```

Field Value

BlackboardConsiderationKeyExists.KeyType

Methods

GetValue(Component)

This method should return the current value of the consideration which is used for multiplication with other considerations to calculate the score of the action

```
protected override float GetValue(Component target)
```

Parameters

`target` Component

If the consideration has a target then this value is the target component which the consideration should be calculated for. The actual type of this depends on the type of the component that the action stores in its target list in the UpdateTargets call. If the consideration doesn't have a target, then the Brain component of the executing agent itself is passed to the method.

Returns

float

A value between minRange and maxRange which indicates the current value/score of the consideration

Enum BlackboardConsiderationKeyExists.KeyType

Namespace: NoOpArmy.UtilityAI.Considerations

Assembly: APIRef.dll

The blackboard key type

```
public enum BlackboardConsiderationKeyExists.KeyType
```

Fields

Bool = 2

Float = 0

GameObject = 3

Int = 1

Object = 5

UnityObject = 4

Vector3 = 6

Class BlackboardFloatConsideration

Namespace: NoOpArmy.UtilityAI.Considerations

Assembly: APIRef.dll

Checks afloat key in the blackboard and if it exists, returns its value, otherwise returns 0

```
public class BlackboardFloatConsideration : ConsiderationBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ConsiderationBase ← BlackboardFloatConsideration

Inherited Members

ConsiderationBase.instantiatePerAgent , ConsiderationBase.minRange , ConsiderationBase.maxRange , ConsiderationBase.Score , ConsiderationBase.Brain , ConsiderationBase.Clone() , ConsiderationBase.OnInitialized() , ConsiderationBase.UpdateRange(float, float) , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) , Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,

Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

keyName

Name of the key in the blackboard

```
[Tooltip("Name of the key in the blackboard")]
public string keyName
```

Field Value

string

Methods

GetValue(Component)

This method should return the current value of the consideration which is used for multiplication with other considerations to calculate the score of the action

```
protected override float GetValue(Component target)
```

Parameters

`target` Component

If the consideration has a target then this value is the target component which the consideration should be calculated for. The actual type of this depends on the type of the component that the action stores in its target list in the UpdateTargets call. If the consideration doesn't have a target, then the Brain component of the executing agent itself is passed to the method.

Returns

float

A value between minRange and maxRange which indicates the current value/score of the consideration

Class BlackboardGameObjectDistanceConsideration

Namespace: NoOpArmy.UtilityAI.Considerations

Assembly: APIRef.dll

Returns the distance between the executing agent and a GameObject which is specified in a key in the blackboard.

```
public class BlackboardGameObjectDistanceConsideration : ConsiderationBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ConsiderationBase ← BlackboardGameObjectDistanceConsideration

Inherited Members

ConsiderationBase.instantiatePerAgent , ConsiderationBase.minRange , ConsiderationBase.maxRange , ConsiderationBase.Score , ConsiderationBase.Brain , ConsiderationBase.Clone() , ConsiderationBase.OnInitialized() , ConsiderationBase.UpdateRange(float, float) , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,

Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

keyName

Name of the key in the blackboard

```
[Tooltip("Name of the key in the blackboard")]
public string keyName
```

Field Value

string

Methods

GetValue(Component)

This method should return the current value of the consideration which is used for multiplication with other considerations to calculate the score of the action

```
protected override float GetValue(Component target)
```

Parameters

`target` Component

If the consideration has a target then this value is the target component which the consideration should be calculated for. The actual type of this depends on the type of the component that the action stores in its target list in the UpdateTargets call. If the consideration doesn't have a target, then the Brain component of the executing agent itself is passed to the method.

Returns

float

A value between minRange and maxRange which indicates the current value/score of the consideration

Class BlackboardIntRangeConsideration

Namespace: NoOpArmy.UtilityAI.Considerations

Assembly: APIRef.dll

Checks an integer key in the blackboard and returns its value. If the key doesn't exist then it returns 0

```
public class BlackboardIntRangeConsideration : ConsiderationBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ConsiderationBase ← BlackboardIntRangeConsideration

Inherited Members

ConsiderationBase.instantiatePerAgent , ConsiderationBase.minRange , ConsiderationBase.maxRange , ConsiderationBase.Score , ConsiderationBase.Brain , ConsiderationBase.Clone() , ConsiderationBase.OnInitialized() , ConsiderationBase.UpdateRange(float, float) , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,

Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) , Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) , Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) , Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) , Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) , Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) , Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

keyName

Name of the key in the blackboard

```
[Tooltip("Name of the key in the blackboard")]
public string keyName
```

Field Value

string

Methods

GetValue(Component)

This method should return the current value of the consideration which is used for multiplication with other considerations to calculate the score of the action

```
protected override float GetValue(Component target)
```

Parameters

`target` Component

If the consideration has a target then this value is the target component which the consideration should be calculated for. The actual type of this depends on the type of the component that the action stores in its target list in the UpdateTargets call. If the consideration doesn't have a target, then the Brain component of the executing agent itself is passed to the method.

Returns

float

A value between minRange and maxRange which indicates the current value/score of the consideration

Class BlackboardIntValueConsideration

Namespace: NoOpArmy.UtilityAI.Considerations

Assembly: APIRef.dll

This consideration checks an int key in the blackboard and returns 1 if the value equals the expected value; otherwise, it returns 0.

```
public class BlackboardIntValueConsideration : ConsiderationBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ConsiderationBase ← BlackboardIntValueConsideration

Inherited Members

ConsiderationBase.instantiatePerAgent , ConsiderationBase.minRange , ConsiderationBase.maxRange , ConsiderationBase.Score , ConsiderationBase.Brain , ConsiderationBase.Clone() , ConsiderationBase.OnInitialized() , ConsiderationBase.UpdateRange(float, float) , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) ,

Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) , Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) , Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) , Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) , Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) , Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) , Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) , Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

expectedValue

The consideration returns 1 if the value equals the expected value; otherwise, it returns 0.

```
[Tooltip("The consideration returns 1 if the value equals the expected value; otherwise, it returns 0.")]
public int expectedValue
```

Field Value

int

keyName

Name of the key in the blackboard

```
[Tooltip("Name of the key in the blackboard")]
public string keyName
```

Field Value

string

Methods

GetValue(Component)

This method should return the current value of the consideration which is used for multiplication with other considerations to calculate the score of the action

```
protected override float GetValue(Component target)
```

Parameters

`target` Component

If the consideration has a target then this value is the target component which the consideration should be calculated for. The actual type of this depends on the type of the component that the action stores in its target list in the UpdateTargets call. If the consideration doesn't have a target, then the Brain component of the executing agent itself is passed to the method.

Returns

float

A value between minRange and maxRange which indicates the current value/score of the consideration

Class ConstBoolConsideration

Namespace: NoOpArmy.UtilityAI.Considerations

Assembly: APIRef.dll

Returns a constant value. This consideration is mostly used for testing

```
public class ConstBoolConsideration : ConsiderationBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ConsiderationBase ← ConstBoolConsideration

Inherited Members

ConsiderationBase.instantiatePerAgent , ConsiderationBase.minRange , ConsiderationBase.maxRange , ConsiderationBase.Score , ConsiderationBase.Brain , ConsiderationBase.Clone() , ConsiderationBase.OnInitialized() , ConsiderationBase.UpdateRange(float, float) , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) , Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,

Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

falseValue

The value to return if isTrue is not set.

```
public float falseValue
```

Field Value

float

isTrue

Is this consideration in the true condition or not

```
public bool isTrue
```

Field Value

bool

trueValue

The value to return if isTrue is set.

```
public float trueValue
```

Field Value

float

Methods

GetValue(Component)

This method should return the current value of the consideration which is used for multiplication with other considerations to calculate the score of the action

```
protected override float GetValue(Component target)
```

Parameters

`target` Component

If the consideration has a target then this value is the target component which the consideration should be calculated for. The actual type of this depends on the type of the component that the action stores in its target list in the UpdateTargets call. If the consideration doesn't have a target, then the Brain component of the executing agent itself is passed to the method.

Returns

float

A value between minRange and maxRange which indicates the current value/score of the consideration

Class InfluenceMapConsideration

Namespace: NoOpArmy.UtilityAI.Considerations

Assembly: APIRef.dll

This consideration searches the influence map specified for a value and returns 1 if it finds something. Otherwise it returns 0

```
public class InfluenceMapConsideration : ConsiderationBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ConsiderationBase ← InfluenceMapConsideration

Inherited Members

ConsiderationBase.instantiatePerAgent , ConsiderationBase.minRange , ConsiderationBase.maxRange , ConsiderationBase.Score , ConsiderationBase.Brain , ConsiderationBase.Clone() , ConsiderationBase.UpdateRange(float, float) , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) , Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,

Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

mapName

The name of the map to consider.

```
[Tooltip("The name of the map to consider.")]
public string mapName
```

Field Value

string

queryResultKeyName

Stores the result of the influence map query to be used by the actions in a Vector3 blackboard key. The results will be in x and z components in influence map cells coordinates. Leave this empty if you don't need it.

```
[Tooltip("Stores the result of the influence map query to be used by the actions in a Vector3 blackboard key. The results will be in x and z components in influence map cells coordinates. Leave this empty if you don't need it.")]
public string queryResultKeyName
```

Field Value

string

radius

The radius to search in map cells

```
[Tooltip("The radius to search in map cells")]
public int radius
```

Field Value

int

searchCondition

The search condition

```
public SearchCondition searchCondition
```

Field Value

SearchCondition

searchValue

The value to search for.For example if this is 0.6 and condition is greater, then the search functionality will look for a value bigger than 0.6 in the radius around the target

```
public float searchValue
```

Field Value

float

Methods

GetValue(Component)

This method should return the current value of the consideration which is used for multiplication with other considerations to calculate the score of the action

```
protected override float GetValue(Component target)
```

Parameters

`target` Component

If the consideration has a target then this value is the target component which the consideration should be calculated for. The actual type of this depends on the type of the component that the action stores in its target list in the UpdateTargets call. If the consideration doesn't have a target, then the Brain component of the executing agent itself is passed to the method.

Returns

float

A value between minRange and maxRange which indicates the current value/score of the consideration

OnInitialized()

Fired when the consideration is initialized

```
protected override void OnInitialized()
```

Class LineOfSightConsideration

Namespace: NoOpArmy.UtilityAI.Considerations

Assembly: APIRef.dll

Checks if we have a line of sight to our target or something blocks the view

```
public class LineOfSightConsideration : ConsiderationBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ConsiderationBase ← LineOfSightConsideration

Inherited Members

ConsiderationBase.instantiatePerAgent , ConsiderationBase.minRange , ConsiderationBase.maxRange , ConsiderationBase.Score , ConsiderationBase.Brain , ConsiderationBase.Clone() , ConsiderationBase.OnInitialized() , ConsiderationBase.UpdateRange(float, float) , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) , Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,

Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) , Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) , Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) , Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) , Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) , Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

layerMask

The layer mask for the raycast

```
[Tooltip("The layer mask for the raycast")]
public LayerMask layerMask
```

Field Value

LayerMask

Methods

GetValue(Component)

This method should return the current value of the consideration which is used for multiplication with other considerations to calculate the score of the action

```
protected override float GetValue(Component target)
```

Parameters

`target` Component

If the consideration has a target then this value is the target component which the consideration should be calculated for. The actual type of this depends on the type of the component that the action stores in its target list in the UpdateTargets call. If the consideration doesn't have a target, then the Brain component of the executing agent itself is passed to the method.

Returns

float

A value between minRange and maxRange which indicates the current value/score of the consideration

Class SmartObjectDistanceConsideration

Namespace: NoOpArmy.UtilityAI.Considerations

Assembly: APIRef.dll

Returns the distance to a target which is a smart object. You can choose to get the distance to a specific slot with a specific condition like the distance to the closes slot which is free.

```
public class SmartObjectDistanceConsideration : ConsiderationBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ConsiderationBase ← SmartObjectDistanceConsideration

Inherited Members

ConsiderationBase.instantiatePerAgent , ConsiderationBase.minRange , ConsiderationBase.maxRange , ConsiderationBase.Score , ConsiderationBase.Brain , ConsiderationBase.Clone() , ConsiderationBase.OnInitialized() , ConsiderationBase.UpdateRange(float, float) , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) ,

Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) , Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) , Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) , Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) , Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) , Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) , Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) , Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

SlotStatus

The search query to use for finding the smart object if useNearestSlotPosition is true

```
[Tooltip("The search query to use for finding the smart object if useNearestSlotPosition
is true")]
public SmartObjectsManager.SearchQueryOptions SlotStatus
```

Field Value

SmartObjectsManager.SearchQueryOptions

inverse

Invert the final score (0.8 --> 0.2

```
[Tooltip("Invert the final score ( 0.8 --> 0.2")]
public bool inverse
```

Field Value

bool

useNearestSlotPosition

If true, score is computed using the best slot position, not the smart object position.

```
[Tooltip("If true, score is computed using the best slot position, not the smart
object position.")]
public bool useNearestSlotPosition
```

Field Value

bool

Methods

GetValue(Component)

This method should return the current value of the consideration which is used for multiplication with other considerations to calculate the score of the action

```
protected override float GetValue(Component target)
```

Parameters

`target` Component

If the consideration has a target then this value is the target component which the consideration should be calculated for. The actual type of this depends on the type of the component that the action stores in its target list in the UpdateTargets call. If the consideration doesn't have a target, then the Brain component of the executing agent itself is passed to the method.

Returns

float

A value between minRange and maxRange which indicates the current value/score of the consideration

Class TargetDistanceConsideration

Namespace: NoOpArmy.UtilityAI.Considerations

Assembly: APIRef.dll

Returns the distance to a target. This normalizes to a value between 0 and 1 based on minRange and maxRange values

```
public class TargetDistanceConsideration : ConsiderationBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ConsiderationBase ← TargetDistanceConsideration

Inherited Members

ConsiderationBase.instantiatePerAgent , ConsiderationBase.minRange , ConsiderationBase.maxRange , ConsiderationBase.Score , ConsiderationBase.Brain , ConsiderationBase.Clone() , ConsiderationBase.OnInitialized() , ConsiderationBase.UpdateRange(float, float) , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,

Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) , Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) , Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) , Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) , Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) , Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) , Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

inverse

Invert the final score (0.8 --> 0.2

```
[Tooltip("Invert the final score ( 0.8 --> 0.2")]
public bool inverse
```

Field Value

bool

Methods

GetValue(Component)

This method should return the current value of the consideration which is used for multiplication with other considerations to calculate the score of the action

```
protected override float GetValue(Component target)
```

Parameters

`target` Component

If the consideration has a target then this value is the target component which the consideration should be calculated for. The actual type of this depends on the type of the component that the action stores in its target list in the UpdateTargets call. If the consideration doesn't have a target, then the Brain component of the executing agent itself is passed to the method.

Returns

float

A value between minRange and maxRange which indicates the current value/score of the consideration

Class ValidGameObjectPathConsideration

Namespace: NoOpArmy.UtilityAI.Considerations

Assembly: APIRef.dll

This consideration returns 1 if a valid path from the agent's NavmeshAgent to the position of the GameObject stored in the key is present, otherwise returns 0 The NeedTarget causes the GameObject to be read from the target but always a path from the owning agent is calculated

```
public class ValidGameObjectPathConsideration : ConsiderationBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ConsiderationBase ← ValidGameObjectPathConsideration

Inherited Members

ConsiderationBase.instantiatePerAgent , ConsiderationBase.minRange , ConsiderationBase.maxRange , ConsiderationBase.Score , ConsiderationBase.Brain , ConsiderationBase.Clone() , ConsiderationBase.UpdateRange(float, float) , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) ,

Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) , Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) , Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) , Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) , Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) , Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) , Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) , Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

navMeshPath

```
public NavMeshPath navMeshPath
```

Field Value

NavMeshPath

vector3KeyName

The GameObject key to read

```
[Tooltip("The GameObject key to read")]
public string vector3KeyName
```

Field Value

string

Methods

GetValue(Component)

This method should return the current value of the consideration which is used for multiplication with other considerations to calculate the score of the action

```
protected override float GetValue(Component target)
```

Parameters

`target` Component

If the consideration has a target then this value is the target component which the consideration should be calculated for. The actual type of this depends on the type of the component that the action stores in its target list in the UpdateTargets call. If the consideration doesn't have a target, then the Brain component of the executing agent itself is passed to the method.

Returns

float

A value between minRange and maxRange which indicates the current value/score of the consideration

OnInitialized()

Fired when the consideration is initialized

```
protected override void OnInitialized()
```

Class ValidPathConsideration

Namespace: NoOpArmy.UtilityAI.Considerations

Assembly: APIRef.dll

This consideration returns 1 if a valid path from the agent's NavmeshAgent to the position stored in the Vector3 key is present, otherwise returns 0 The NeedTarget causes the Vector3 to be read from the target but always a path from the owning agent is calculated

```
public class ValidPathConsideration : ConsiderationBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ConsiderationBase ← ValidPathConsideration

Inherited Members

ConsiderationBase.instantiatePerAgent , ConsiderationBase.minRange , ConsiderationBase.maxRange , ConsiderationBase.Score , ConsiderationBase.Brain , ConsiderationBase.Clone() , ConsiderationBase.UpdateRange(float, float) , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,

Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) , Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) , Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) , Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) , Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) , Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) , Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

navMeshPath

```
public NavMeshPath navMeshPath
```

Field Value

NavMeshPath

vector3KeyName

```
public string vector3KeyName
```

Field Value

string

Methods

GetValue(Component)

This method should return the current value of the consideration which is used for multiplication with other considerations to calculate the score of the action

```
protected override float GetValue(Component target)
```

Parameters

`target` Component

If the consideration has a target then this value is the target component which the consideration should be calculated for. The actual type of this depends on the type of the component that the action stores in its target list in the UpdateTargets call. If the consideration doesn't have a target, then the Brain component of the executing agent itself is passed to the method.

Returns

float

A value between minRange and maxRange which indicates the current value/score of the consideration

OnInitialized()

Fired when the consideration is initialized

```
protected override void OnInitialized()
```

Namespace NoOpArmy.UtilityAI.Sample

Classes

PlayWithBlackboard

Class PlayWithBlackboard

Namespace: NoOpArmy.UtilityAI.Sample

Assembly: APIRef.dll

```
public class PlayWithBlackboard : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← PlayWithBlackboard

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) , Component.SendMessageUpwards(string, object, SendMessageOptions) , Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,

Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,

Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

blackboard

The blackboard to read/write to

```
public BlackBoard blackboard
```

Field Value

BlackBoard

changer

```
public BlackBoardChangeer changer
```

Field Value

BlackBoardChangeer

colorChange

```
public BlackBoardChange colorChange
```

Field Value

BlackBoardChange

Namespace NoOpArmy.WiseFeline

Classes

AIObject

This is the base class of Wise Feline AI assets. You never need to use this directly.

ActionBase

AI actions should derive from this class to define a custom action which can be added to the set of actions for an agent

ActionSet

Action sets are like directories which group a set of actions together but you don't need to deal with them in code unless you are making debugging/editor tools or highly advanced systems An Agent Behavior asset is made of a set of ActionSet objects which each of them contain a set of actions.

AgentBehavior

This class is used to serialize a group of action sets as a scriptable object. You usually create one of these by going to the Assets>Create>NoOpArmy>WiseFeline>AgentBehavior menu and then select it and open the Window>NoOpArmy>WiseFeline to add action sets, actions and considerations to it and modify their parameters.

Attack

Attach thiscomponent to any GameObject which needs to attack others

BlackboardInitialValues

Writes a list of values which you author in the inspector into the BlackBoard component which is on the same GameObject. Blackboard keys start at their default values which are 0, false, an empty string or null, so any key which needs a starting value has to be written before the agent starts thinking and this component does that without any code. It is also the only code-free way to put a reference to an object which lives in a scene into a blackboard, because assets like behaviors and their changers cannot reference scene objects while a component which lives in the scene can. So patrol point holders, camps, the player and other scene objects can simply be dragged into the list below. The values are applied in Start because the BlackBoard component defines its keys in Awake. The negative execution order matters: actions like MoveAndPatrol read their GameObject key once, when the Brain initializes them in its own Start, so the seeding has to be finished before any Start of a default-ordered component runs. Without it the order of two Starts on the same GameObject is unspecified and a patrol route would silently come up empty.

Brain

You should attach this component to any GameObject which wishes to be an AI agent controlled by a set of actions

CharacterControllerBasedAIMovement

This component allows you to move the character to a destination. This component implements the IAIMovement interface so generic actions can use it. The component uses the CharacterController component to move and does not use path finding and a NavmeshAgent

ConsiderationBase

Considerations should inherit from this class. Each action has a list of these and multiplies the value of calling their GetValue methods to calculate its score. Each consideration is either a consideration for the target of the action or the action itself. The target based ones will be executed once per target when the action score is being calculated for that target.

DestroyObjectWithDelay

Faction

This component specifies the faction of the GameObject used by Attack, Health and other components

FactionData

FactionsCollection

Health

Use this component to add health to a GameObject

NPCZoneTrigger

NavMeshBasedAIMovement

Movement component allows you to move an agent on the NavMesh. This component implements the IAIMovement interface so generic actions can use it.

ReflectionUtilities

Contains utility methods for working with data types defined as actions and considerations and ... You should not need to use this unless you are making debug/editor tools for Wise Feline

Structs

ActionData

This struct represents the runtime info for a an action and its final score

ActionScoringData

Interfaces

IDamagable

Any component which wants to receive damage can implement this interface

INPCZoneMessageReceiver

Enums

Brain.ActionSelectionAlgorithm

Different action selection modes

Class AIObject

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

This is the base class of Wise Feline AI assets. You never need to use this directly.

```
public class AIObject : ScriptableObject
```

Inheritance

object ← Object ← ScriptableObject ← AIObject

Derived

ActionBase, ActionSet, ConsiderationBase

Inherited Members

ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) , Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) , Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,

Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

Name

The name of the asset

```
public string Name
```

Field Value

string

guid

The unique id of the asset.

```
[HideInInspector]
public string guid
```

Field Value

string

Methods

SetGuid(string, bool)

Sets the Guid of the object. We do not ensure the IDs are unique and assume the application developer generates IDs with the guarantees he/she needs.

```
public void SetGuid(string newId, bool canReplace = false)
```

Parameters

`newId` string

The ID to assign

`canReplace` bool

Can the ID be replaced if it is alredy set, defaults to false

Remarks

This method should onlybe called once per object

Class ActionBase

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

AI actions should derive from this class to define a custom action which can be added to the set of actions for an agent

```
public abstract class ActionBase : AIObject
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ActionBase

Derived

BlackboardActionBase

Inherited Members

AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,

Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) , Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) , Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) , Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) , Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) , Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) , Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

ChosenTarget

The target with the best score. The type of this component depends on the type that you yourself store in the list in the UpdateTargets callback. If the action doesn't have any targets and target based considerations then this field doesn't matter and it value should be considered undefined.

```
public Component ChosenTarget
```

Field Value

Component

_isInterruptable

Is this action interruptable mid-execution by another action which has a higher score

```
public bool _isInterruptable
```

Field Value

bool

_maxTargetCount

Maximum number of targets which this action should consider

```
[SerializeField]
protected int _maxTargetCount
```

Field Value

int

_priority

This is the priority of the action. Actions will be chosen for execution if their score is non-zero even if actions with lower priorities have higher scores

```
public int _priority
```

Field Value

int

_useMomentumOnTarget

Should the action add a 25% score bonus to the current target to not change the target multiple times too quickly when scores are too close.

```
[SerializeField]
protected bool _useMomentumOnTarget
```

Field Value

bool

forceInterruptLowerPriorityActions

Can this action forcefully interrupt, non-interruptable actions which are lower priority.

```
public bool forceInterruptLowerPriorityActions
```

Field Value

bool

Properties

Brain

The brain component which the action is executing for.

```
protected Brain Brain { get; }
```

Property Value

Brain

Considerations

List of the considerations for the action. The score of the action is the result of multiplication of the scores of all of these considerations

```
public List<ConsiderationBase> Considerations { get; }
```

Property Value

List <ConsiderationBase>

IsInitialized

Is the action initialized?

```
public bool IsInitialized { get; protected set; }
```

Property Value

bool

Score

The last calculated score of the action in the last think operation.

```
public float Score { get; }
```

Property Value

float

Weight

The wait of the action which is multiplied by the score to allow you to prioritize some actions

```
public float Weight { get; }
```

Property Value

float

Methods

ActionFailed()

```
protected void ActionFailed()
```

ActionSucceed()

```
[Obsolete("Use ActionSucceeded")]
protected void ActionSucceed()
```

ActionSucceeded()

```
protected void ActionSucceeded()
```

AddTarget(Component)

Adds a component to the targets list of the action

```
protected void AddTarget(Component t)
```

Parameters

`t` Component

AddTargets<T>(List<T>)

Adds an array of targets to the list of targets for the action

```
protected void AddTargets<T>(List<T> targets) where T : Component
```

Parameters

`targets` List<T>

Type Parameters

`T`

AddTargets<T>(T[])

Adds an array of targets to the list of targets for the action

```
protected void AddTargets<T>(T[] targets) where T : Component
```

Parameters

`targets` T[]

Type Parameters

`T`

ClearTargets()

Clears the list of targets for the action

```
protected void ClearTargets()
```

Clone(ActionSet)

Clones the action for execution at runtime.

```
public ActionBase Clone(ActionSet set)
```

Parameters

`set` ActionSet

Returns

ActionBase

OnFinish()

Called when the action is finished, either by failing or succeeding in achieving a desired behaviour

```
protected virtual void OnFinish()
```

OnFixedUpdate()

Should be used like MonoBehaviour's FixedUpdate for the action

```
protected virtual void OnFixedUpdate()
```

OnInitialized()

Called when the action is initialized

```
protected virtual void OnInitialized()
```

OnLateUpdate()

Should be used like MonoBehaviour's LateUpdate for the action

```
protected virtual void OnLateUpdate()
```

OnStart()

Should be used like MonoBehaviour's Start for the action

```
protected virtual void OnStart()
```

OnUpdate()

Should be used like MonoBehaviour's Update for the action

```
protected virtual void OnUpdate()
```

RemoveTarget(Component)

Removes a component from the list of targets for the action

```
protected void RemoveTarget(Component t)
```

Parameters

`t` Component

UpdateTargets()

Fires when you need to update the list of targets

```
protected abstract void UpdateTargets()
```

Struct ActionData

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

This struct represents the runtime info for a an action and its final score

```
public struct ActionData
```

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

ActionData(ActionSet, ActionBase, float, int, int)

```
public ActionData(ActionSet set, ActionBase action, float score, int priority, int index)
```

Parameters

`set` ActionSet

`action` ActionBase

`score` float

`priority` int

`index` int

Fields

Action

Which action this instance refers to.

```
public ActionBase Action
```

Field Value

ActionBase

ActionSet

The action set this instance refers to.

```
public ActionSet ActionSet
```

Field Value

ActionSet

Index

index of the action in the main list

```
public int Index
```

Field Value

int

Priority

Priority of the action

```
public int Priority
```

Field Value

int

Score

The score of the action in the last think operation.

```
public float Score
```

Field Value

float

Struct ActionScoringData

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

```
public struct ActionScoringData
```

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Constructors

ActionScoringData(float, int, int)

```
public ActionScoringData(float score, int priority, int index)
```

Parameters

`score` float

`priority` int

`index` int

Fields

Index

index of the action in the main list

```
public int Index
```

Field Value

int

Priority

Priority of the action

```
public int Priority
```

Field Value

int

Score

The score of the action in the last think operation.

```
public float Score
```

Field Value

float

Class ActionSet

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

Action sets are like directories which group a set of actions together but you don't need to deal with them in code unless you are making debugging/editor tools or highly advanced systems An Agent Behavior asset is made of a set of ActionSet objects which each of them contain a set of actions.

```
public class ActionSet : AIObject
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ActionSet

Inherited Members

AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) , Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) , Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,

Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

Behavior

The agent behavior that this instance belongs too.

```
[HideInInspector]
public AgentBehavior Behavior
```

Field Value

AgentBehavior

Properties

Actions

The list of actions which are contained in this action set.

```
public List<ActionBase> Actions { get; }
```

Property Value

List <ActionBase>

Methods

Clone(AgentBehavior)

Clones the action set and all of its actions to be used at runtime for an agent.

```
public ActionSet Clone(AgentBehavior behavior)
```

Parameters

`behavior` AgentBehavior

Returns

ActionSet

Class AgentBehavior

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

This class is used to serialize a group of action sets as a scriptable object. You usually create one of these by going to the Assets>Create>NoOpArmy>WiseFeline>AgentBehavior menu and then select it and open the Window>NoOpArmy>WiseFeline to add action sets, actions and considerations to it and modify their parameters.

```
[CreateAssetMenu(menuName = "NoOpArmy/Wise Feline/AgentBehavior")]
public class AgentBehavior : ScriptableObject
```

Inheritance

object ← Object ← ScriptableObject ← AgentBehavior

Inherited Members

ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) , Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,

Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

OnThinkDone

Fires when an action is done

```
public Action<ActionData> OnThinkDone
```

Field Value

Action <ActionData>

SelectedAction

The action currently selected

```
[HideInInspector]
public ActionBase SelectedAction
```

Field Value

ActionBase

SelectedActionSet

The action set selected at the moment in the UI

```
[HideInInspector]
public ActionSet SelectedActionSet
```

Field Value

ActionSet

Properties

ActionSets

The list of action sets in the asset

```
public List<ActionSet> ActionSets { get; }
```

Property Value

List <ActionSet>

Methods

Clone()

Clones the behavior for runtime execution by a Brain component.

```
public AgentBehavior Clone()
```

Returns

AgentBehavior

GetEmpty()

Creates an in-memory behavior with no action sets, used as the placeholder for a brain that was not given a behavior asset. Every call returns a fresh instance; there is no shared one, so nothing survives into the next Play Mode session when Domain Reload is disabled.

```
public static AgentBehavior GetEmpty()
```

Returns

AgentBehavior

Class Attack

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

Attach thiscomponent to any GameObject which needs to attack others

```
public class Attack : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← Attack

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) ,

Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,

Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

drawDebugLines

Draws lines for spehre checks

```
[Tooltip("Draws lines for spehre checks")]
public bool drawDebugLines
```

Field Value

bool

maxAttackCountInSingleCast

We allocate an array of this size and no colliders after this count will be allocated but this is allocated only once

```
[Tooltip("We allocate an array of this size and no colliders after this count will be
allocated but this is allocated only once")]
public int maxAttackCountInSingleCast
```

Field Value

int

Methods

CanAttack()

```
public bool CanAttack()
```

Returns

bool

MeleeAttack(Vector3, float, float, bool)

Executes a melee attack

```
public void MeleeAttack(Vector3 offset, float radius, float damageAmount, bool
applyToAllHits = false)
```

Parameters

`offset` Vector3

`radius` float

`damageAmount` float

`applyToAllHits` bool

MeleeAttackWithDelay(Vector3, float, float, float, Action, bool)

Executes a melee attack with delay

```
public Coroutine MeleeAttackWithDelay(Vector3 offset, float radius, float damageAmount,
float castDelay, Action Done = null, bool applyToAllHits = false)
```

Parameters

`offset` Vector3

`radius` float

`damageAmount` float

`castDelay` float

`Done` Action

`applyToAllHits` bool

Returns

Coroutine

RangeAttack(Vector3, float, float, bool)

Executes a range attack

```
public void RangeAttack(Vector3 center, float radius, float damageAmount, bool
applyToAllHits = false)
```

Parameters

`center` Vector3

`radius` float

`damageAmount` float

`applyToAllHits` bool

RangeAttackWithDelay(Vector3, float, float, float, Action, bool)

executes a range attack with delay

```
public Coroutine RangeAttackWithDelay(Vector3 center, float radius, float damageAmount,
float castDelay, Action Done = null, bool applyToAllHits = false)
```

Parameters

`center` Vector3

`radius` float

`damageAmount` float

`castDelay` float

`Done` Action

`applyToAllHits` bool

Returns

Coroutine

SetAttackDelayColor(bool, Color)

Sets the fact that we should change color on attack delay and the color used

```
public void SetAttackDelayColor(bool shouldChangeColor, Color color)
```

Parameters

`shouldChangeColor` bool

`color` Color

ShootProjectile(GameObject, Vector3)

Shoots a projectile

```
public void ShootProjectile(GameObject projectile, Vector3 offset)
```

Parameters

`projectile` GameObject

`offset` Vector3

ShootProjectile(GameObject, Vector3, Action, float)

Shoots a projecitle

```
public IEnumerator ShootProjectile(GameObject projectile, Vector3 offset, Action Done,
float delay)
```

Parameters

`projectile` GameObject

`offset` Vector3

`Done` Action

`delay` float

Returns

IEnumerator

StopAttack()

```
public void StopAttack()
```

Class BlackboardInitialValues

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

Writes a list of values which you author in the inspector into the BlackBoard component which is on the same GameObject. Blackboard keys start at their default values which are 0, false, an empty string or null, so any key which needs a starting value has to be written before the agent starts thinking and this component does that without any code. It is also the only code-free way to put a reference to an object which lives in a scene into a blackboard, because assets like behaviors and their changers cannot reference scene objects while a component which lives in the scene can. So patrol point holders, camps, the player and other scene objects can simply be dragged into the list below. The values are applied in Start because the BlackBoard component defines its keys in Awake. The negative execution order matters: actions like MoveAndPatrol read their GameObject key once, when the Brain initializes them in its own Start, so the seeding has to be finished before any Start of a default-ordered component runs. Without it the order of two Starts on the same GameObject is unspecified and a patrol route would silently come up empty.

```
[DefaultExecutionOrder(-100)]
[RequireComponent(typeof(BlackBoard))]
public class BlackboardInitialValues : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← BlackboardInitialValues

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) ,

Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) ,
Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) ,
Component.GetComponentsInChildren<T>(bool, List<T>) ,
Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) ,
Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) ,
Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() ,
Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) ,
Component.GetComponentsInParent<T>(bool) ,
Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() ,
Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) ,
Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() ,
Component.GetComponentIndex() , Component.CompareTag(string) ,
Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,

Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

values

The values which are written to the blackboard of this GameObject when the scene starts. Every entry has a key type, a key name and the value to write and the key should already be defined with the same type in the blackboard definition which the BlackBoard component uses, otherwise applying the value throws. Usually you want to turn "Add Instead Of Set" off for these entries because they are starting values and not changes, specially since additive changes are multiplied by deltaTime when "Multiply by deltaTime for adds" is on.

```
[Tooltip("The values which are written to the blackboard of this GameObject when the scene
starts. Every key should already be defined with the same type in the blackboard definition.
Usually you want to turn \"Add Instead Of Set\" off for these entries because they are
starting values and not changes.")]
public List<BlackBoardChange> values
```

Field Value

List <BlackBoardChange>

Class Brain

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

You should attach this component to any GameObject which wishes to be an AI agent controlled by a set of actions

```
public class Brain : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← Brain

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) ,

Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,

Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

_thinkDuration

The wait period between calculations of action scores

```
public float _thinkDuration
```

Field Value

float

_updateTargetsDuration

The wait period between calls to update target lists. Since updating this list is usually heavy, you should do this less often than score calculations and deal with null targets in your actions.

```
public float _updateTargetsDuration
```

Field Value

float

actionSelectionAlgorithm

The algorithm the brain uses for selecting the best action

```
[Tooltip("The algorithm the brain uses for selecting the best action")]
public Brain.ActionSelectionAlgorithm actionSelectionAlgorithm
```

Field Value

Brain.ActionSelectionAlgorithm

context

Assign any component which your AI always needs to this to avoid calling GetComponent in all actions/considerations

```
public Component context
```

Field Value

Component

itemCountForRandomActionSelection

The top N items are considered for random action selection and this value is what N is.

```
[Tooltip("The top N items are considered for random action selection and this value is what
N is.")]
public int itemCountForRandomActionSelection
```

Field Value

int

scoreDeltaForMidExecutionActionChange

If the score difference between the chosen action and the currently executing one is higher than this, then the actions changes to the chosen one

```
[Tooltip("If the score difference between the chosen action and the currently executing one
is higher than this, then the actions changes to the chosen one")]
public float scoreDeltaForMidExecutionActionChange
```

Field Value

float

Properties

Behavior

```
public AgentBehavior Behavior { get; }
```

Property Value

AgentBehavior

IsInitialized

Is the brain initialized yet

```
public bool IsInitialized { get; }
```

Property Value

bool

currentAction

Gets the currently executing action class

```
public ActionBase currentAction { get; }
```

Property Value

ActionBase

dataRecorderRecordIndex

```
public int dataRecorderRecordIndex { get; }
```

Property Value

int

Methods

AddActionSet(ActionSet)

Adds an action set to the set of behaviors this brain has available to score and execute

```
public void AddActionSet(ActionSet set)
```

Parameters

`set` ActionSet

The action set to add

Remarks

Duplicat actions which already exist in the brain will be added again too so you should design your action sets acordingly. You usually do not need multiple instances of an action in the same brain

AddBehavior(AgentBehavior)

Adds actions inside a behavior agent asset to the actions of this brain

```
public void AddBehavior(AgentBehavior agentBehavior)
```

Parameters

`agentBehavior` AgentBehavior

The agent behavior asset to add

Remarks

Duplicate actions which already exist in the brain will be added again too. You should design your behaviors so they don't have duplicate actions You usually do not need multiple instances of an action in the same brain

PauseExecutingActions(bool)

Pauses/Resumes executing the current action.

```
public void PauseExecutingActions(bool pause)
```

Parameters

`pause` bool

pauses ecution if true and resumes it if false

PauseThinking(bool)

Puases/Resumes thinking which changes action scores

```
public void PauseThinking(bool pause)
```

Parameters

`pause` bool

Pauses the thinking if true and resumes it if false.

RemoveActionSet(ActionSet)

Removes a specific action set from the list of behaviors of this brain

```
public void RemoveActionSet(ActionSet set)
```

Parameters

`set` ActionSet

RemoveBehavior(AgentBehavior)

Removes actions of a behavior from this brain

```
public void RemoveBehavior(AgentBehavior agentBehavior)
```

Parameters

`agentBehavior` AgentBehavior

Think()

Calculates all action scores and selects the next action to execute. This might or might not change the currently executing action

```
public void Think()
```

UpdateActionsTargets()

Updates the target list of all actions so the updated targets can be used while thinking

```
public void UpdateActionsTargets()
```

Events

OnActionFailed

This event gets fired when an action fails

```
public event Action<ActionBase> OnActionFailed
```

Event Type

Action<ActionBase>

OnActionSucceeded

This event gets fired when an action succeeds

```
public event Action<ActionBase> OnActionSucceeded
```

Event Type

Action<ActionBase>

OnBehaviorListModified

Fires when the list of behaviors available to this brain is modified by calling AddBehavior() or RemoveBehavior()

```
public event Action OnBehaviorListModified
```

Event Type

Action

OnCurrentActionChanged

Fires when the executing action changes. The first parameter is the previous action and the second one is the action we are changing too If the first parameter is null then either the previous action is fully finished successfully or with a failure or this is the first action to execute

```
public event Action<ActionBase, ActionBase> OnCurrentActionChanged
```

Event Type

Action <ActionBase, ActionBase>

OnInitialized

This event is called when the brain gets fully initialized for you to be able to add your behaviors.

```
public event Action OnInitialized
```

Event Type

Action

Enum Brain.ActionSelectionAlgorithm

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

Different action selection modes

```
public enum Brain.ActionSelectionAlgorithm : byte
```

Fields

`HighestScore = 0`

Choose the action with the highest score

`RandomConsideringPriority = 1`

Randomly choose from the top N without considering priority

`RandomWithoutConsideringPriority = 2`

Randomly choose from the top N without considering priority so only actions with priorities as high as the top one will be considered

`WeightedRandomConsideringPriority = 3`

Weighted Randomly choise from the top N without considering priority

`WeightedRandomWithoutConsideringPriority = 4`

Weighted Randomly choose from the top N without considering priority so only actions with priorities as high as the top one will be considered

Class CharacterControllerBasedAIMovement

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

This component allows you to move the character to a destination. This component implements the IAIMovement interface so generic actions can use it. The component uses the CharacterController component to move and does not use path finding and a NavmeshAgent

```
[RequireComponent(typeof(CharacterController))]
public class CharacterControllerBasedAIMovement : MonoBehaviour, IAIMovement
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← CharacterControllerBasedAIMovement

Implements

IAIMovement

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) ,

Component.GetComponentsInParent<T>(bool) ,
Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() ,
Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) ,
Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() ,
Component.GetComponentIndex() , Component.CompareTag(string) ,
Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,

Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) , Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

CalculateOnXZPlane

```
public bool CalculateOnXZPlane
```

Field Value

bool

animatorSpeedParameter

```
public string animatorSpeedParameter
```

Field Value

string

arrivalDistance

The distance to destination which means we arrived

```
[Tooltip("The distance to destination which means we arrived")]
public float arrivalDistance
```

Field Value

float

speed

Movement speed of the character used by the CharacterController.SimpleMove() function

```
[Tooltip("Movement speed of the character used by the CharacterController.SimpleMove()
function")]
public float speed
```

Field Value

float

Properties

Destination

Should contain the movement destination

```
public Vector3 Destination { get; }
```

Property Value

Vector3

Methods

MoveToPosition(GameObject, Action)

```
public bool MoveToPosition(GameObject target, Action callback)
```

Parameters

`target` GameObject

`callback` Action

Returns

bool

MoveToPosition(Vector3, Action)

Moves the character to a position specified by a Vector3 and calls a callback when arrived

```
public bool MoveToPosition(Vector3 position, Action callback)
```

Parameters

`position` Vector3

`callback` Action

Returns

bool

> Ture if the movement can happen, false otherwise. Path Finding failures and other issues can prevent the movement to happen.

StopMoving()

Stops the movement no matter if we arrived at the set destination or not

```
public void StopMoving()
```

Class ConsiderationBase

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

Considerations should inherit from this class. Each action has a list of these and multiplies the value of calling their GetValue methods to calculate its score. Each consideration is either a consideration for the target of the action or the action itself. The target based ones will be executed once per target when the action score is being calculated for that target.

```
public abstract class ConsiderationBase : AIObject
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ConsiderationBase

Derived

BlackboardBoolConsideration, BlackboardConsiderationKeyExists, BlackboardFloatConsideration, BlackboardGameObjectDistanceConsideration, BlackboardIntRangeConsideration, BlackboardIntValueConsideration, ConstBoolConsideration, InfluenceMapConsideration, LineOfSightConsideration, SmartObjectDistanceConsideration, TargetDistanceConsideration, ValidGameObjectPathConsideration, ValidPathConsideration, AITagConsideration

Inherited Members

AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,

Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

instantiatePerAgent

If true, a new instance of the consideration will be created per action per instance of the brain component which needs this. Otherwise a single instance will be used for all of them. If you don't use fields in the class and the GetValue method is stateless, then turn this off to allocate less memory.

```
[Header("Optimization")]
[SerializeField]
[Tooltip("If you don't need a new instance of this class per brain turn this off, this will
share a unique class with all agents.")]
protected bool instantiatePerAgent
```

Field Value

bool

maxRange

Maximum value that the consideration can return from GetValue

```
[SerializeField]
protected float maxRange
```

Field Value

float

minRange

Minimum value that the consideration can return from GetValue

```
[SerializeField]
protected float minRange
```

Field Value

float

Properties

Brain

The Brain component that this consideration is being attached to one of its actions

```
protected Brain Brain { get; }
```

Property Value

Brain

Score

The score of this consideration returned from the last GetValue call.

```
public float Score { get; }
```

Property Value

float

Methods

Clone()

Returns a cloned copy of the consideration or the main instance if instancing is turned off.

```
public ConsiderationBase Clone()
```

Returns

ConsiderationBase

GetValue(Component)

This method should return the current value of the consideration which is used for multiplication with other considerations to calculate the score of the action

```
protected abstract float GetValue(Component target)
```

Parameters

`target` Component

If the consideration has a target then this value is the target component which the consideration should be calculated for. The actual type of this depends on the type of the component that the action stores in its target list in the UpdateTargets call. If the consideration doesn't have a target, then the Brain component of the executing agent itself is passed to the method.

Returns

float

A value between minRange and maxRange which indicates the current value/score of the consideration

OnInitialized()

Fired when the consideration is initialized

```
protected virtual void OnInitialized()
```

UpdateRange(float, float)

Updates the range of acceptable values from GetValue

```
public void UpdateRange(float min, float max)
```

Parameters

`min` float

`max` float

Class DestroyObjectWithDelay

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

```
public class DestroyObjectWithDelay : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← DestroyObjectWithDelay

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) , Component.SendMessageUpwards(string, object, SendMessageOptions) , Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,

Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,

Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

delayInSeconds

```
public float delayInSeconds
```

Field Value

float

Class Faction

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

This component specifies the faction of the GameObject used by Attack, Health and other components

```
public class Faction : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← Faction

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) ,

Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,

Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

faction

```
public int faction
```

Field Value

int

Class FactionData

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

```
[Serializable]
public class FactionData
```

Inheritance

object ← FactionData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

description

```
public string description
```

Field Value

string

factionName

```
public string factionName
```

Field Value

string

id

```
public int id
```

Field Value

int

Class FactionsCollection

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

```
[CreateAssetMenu(menuName = "NoOpArmy/Wise Feline/Factions Collection")]
public class FactionsCollection : ScriptableObject
```

Inheritance

object ← Object ← ScriptableObject ← FactionsCollection

Inherited Members

ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) , Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) , Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) , Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) , Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) , Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) , Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,

Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

factions

```
public List<FactionData> factions
```

Field Value

List<FactionData>

Class Health

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

Use this component to add health to a GameObject

```
public class Health : MonoBehaviour, IDamagable
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← Health

Implements

IDamagable

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) ,

Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() ,
Component.GetComponentIndex() , Component.CompareTag(string) ,
Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,

Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

HealthPoints

The health points of the character

```
[Tooltip("The health points of the character")]
public float HealthPoints
```

Field Value

float

changeMaterialAtDamageForDebugging

```
public bool changeMaterialAtDamageForDebugging
```

Field Value

bool

destroyDelay

The delay that should happen when destroying the GameObject

```
[Tooltip("The delay that should happen when destroying the GameObject")]
public float destroyDelay
```

Field Value

float

destroyWhenHealthReachesZero

Destroy the entity when health reaches 0

```
[Tooltip("Destroy the entity when health reaches 0")]
public bool destroyWhenHealthReachesZero
```

Field Value

bool

maxHealth

The maximum health which this component can have

```
[Tooltip("The maximum health which this component can have")]
public float maxHealth
```

Field Value

float

Methods

ApplyDamage(float, GameObject)

Applies damage to this health component

```
public void ApplyDamage(float amount, GameObject damager)
```

Parameters

`amount` float

`damager` GameObject

IncreaseHealth(float, GameObject)

Heals the agent by some amount

```
public void IncreaseHealth(float amount, GameObject healer)
```

Parameters

`amount` float

`healer` GameObject

Exceptions

ArgumentException

OnFullHealth(GameObject)

```
public virtual void OnFullHealth(GameObject healer)
```

Parameters

`healer` GameObject

OnNoHealth(GameObject)

```
public virtual void OnNoHealth(GameObject damager)
```

Parameters

`damager` GameObject

Events

OnHealthChanged

Called when health changes The first parameter is the current health points, the sceond one is the amount of the change and the third is the other GameObject which caused this.

```
public event Action<float, float, GameObject> OnHealthChanged
```

Event Type

Action<float, float, GameObject>

Interface IDamagable

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

Any component which wants to receive damage can implement this interface

```
public interface IDamagable
```

Methods

ApplyDamage(float, GameObject)

Applies damage to the GameObject

```
void ApplyDamage(float amount, GameObject damager)
```

Parameters

`amount` float

`damager` GameObject

IncreaseHealth(float, GameObject)

Heals the GameObject

```
void IncreaseHealth(float amount, GameObject healer)
```

Parameters

`amount` float

`healer` GameObject

Interface INPCZoneMessageReceiver

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

```
public interface INPCZoneMessageReceiver
```

Methods

OnOtherEnteredZone(GameObject)

```
void OnOtherEnteredZone(GameObject other)
```

Parameters

`other` GameObject

OnOtherLeftZone(GameObject)

```
void OnOtherLeftZone(GameObject other)
```

Parameters

`other` GameObject

Class NPCZoneTrigger

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

```
public class NPCZoneTrigger : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← NPCZoneTrigger

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) , Component.SendMessageUpwards(string, object, SendMessageOptions) , Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,

Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,

Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

myNPC

```
public Transform myNPC
```

Field Value

Transform

Methods

GetAgents()

Gets all agents which are in our zone

```
public List<GameObject> GetAgents()
```

Returns

List <GameObject>

Class NavMeshBasedAIMovement

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

Movement component allows you to move an agent on the NavMesh. This component implements the IAIMovement interface so generic actions can use it.

```
[RequireComponent(typeof(NavMeshAgent))]
public class NavMeshBasedAIMovement : MonoBehaviour, IAIMovement
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← NavMeshBasedAIMovement

Implements

IAIMovement

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() ,

Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) ,
Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() ,
Component.GetComponentIndex() , Component.CompareTag(string) ,
Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,

Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

animatorSpeedParameter

```
public string animatorSpeedParameter
```

Field Value

string

Properties

Destination

Should contain the movement destination

```
public Vector3 Destination { get; set; }
```

Property Value

Vector3

Methods

MoveToPosition(GameObject, Action)

```
public bool MoveToPosition(GameObject go, Action callback)
```

Parameters

`go` GameObject

`callback` Action

Returns

bool

MoveToPosition(Vector3, Action)

Moves the agent to a position

```
public bool MoveToPosition(Vector3 position, Action callback)
```

Parameters

`position` Vector3

`callback` Action

Returns

bool

StopMoving()

Stops the movement no matter if we arrived at the set destination or not

```
public void StopMoving()
```

Class ReflectionUtilities

Namespace: NoOpArmy.WiseFeline

Assembly: APIRef.dll

Contains utility methods for working with data types defined as actions and considerations and ... You should not need to use this unless you are making debug/editor tools for Wise Feline

```
public static class ReflectionUtilities
```

Inheritance

object ← ReflectionUtilities

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

GetAllDerivedTypes(Type)

Get all types derived from a specific type defined in the project.

```
public static Type[] GetAllDerivedTypes(Type baseType)
```

Parameters

`baseType` Type

Returns

Type[]

Namespace NoOpArmy.WiseFeline.AITags

Classes

AITags

Attach this component to any object which needs to have tags attached to it

AITagsManager

Add an instance of this to an object in each scene which needs to use AITags components

Class AITags

Namespace: NoOpArmy.WiseFeline.AITags

Assembly: APIRef.dll

Attach this component to any object which needs to have tags attached to it

```
public class AITags : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← AITags

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) ,

Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,

Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

delayBetweenPositionUpdates

How much time should we wait before updating this object's position in the octree

```
public float delayBetweenPositionUpdates
```

Field Value

float

previousUpdate

When did the previous update happen

```
[HideInInspector]
public float previousUpdate
```

Field Value

float

tagsList

The initial list of tags you want to be added to the object (this is used by the inspector)

```
public List<string> tagsList
```

Field Value

List <string>

updatePositionInOctreeAutomatically

Checking this will cause the object to be updated in the octree. If the object moves, you should check this and set the delay based on its speed and the accuracy you require

```
public bool updatePositionInOctreeAutomatically
```

Field Value

bool

Methods

AddItem(string)

Adds a tag at runtime

```
public void AddItem(string tag)
```

Parameters

`tag` string

HasAnyOfTheTags(string[])

```
public bool HasAnyOfTheTags(string[] tags)
```

Parameters

`tags` string[]

Returns

bool

HasTag(string)

Does this object has the tag

```
public bool HasTag(string tag)
```

Parameters

`tag` string

Returns

bool

RemoveItem(string)

Removes a tag at runtime

```
public void RemoveItem(string tag)
```

Parameters

`tag` string

SupportsFilter(string[], string[])

```
public bool SupportsFilter(string[] includeTags, string[] excludeTags)
```

Parameters

`includeTags` string[]

`excludeTags` string[]

Returns

bool

Events

OnTagAdded

Fired when a tag is added to this component

```
public event Action<string> OnTagAdded
```

Event Type

Action<string>

OnTagRemoved

Fired when a tag is removed from this component

```
public event Action<string> OnTagRemoved
```

Event Type

Action<string>

Class AITagsManager

Namespace: NoOpArmy.WiseFeline.AITags

Assembly: APIRef.dll

Add an instance of this to an object in each scene which needs to use AITags components

```
[DefaultExecutionOrder(-10000)]
public class AITagsManager : Singleton<AITagsManager>
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← Singleton<AITagsManager> ← AITagsManager

Inherited Members

Singleton<AITagsManager>.Instance , Singleton<AITagsManager>.Awake() , Singleton<AITagsManager>.OnDestroy() , MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() ,

Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) ,
Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() ,
Component.GetComponentIndex() , Component.CompareTag(string) ,
Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,

Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

worldCenterObject

```
public Transform worldCenterObject
```

Field Value

Transform

worldSize

```
public float worldSize
```

Field Value

float

Properties

tree

```
public PointOctree<AITags> tree { get; }
```

Property Value

PointOctree<AITags>

Methods

GetTagsAroundPoint(Vector3, float)

Returns all of the AITag objects in the sphere

```
public AITags[] GetTagsAroundPoint(Vector3 point, float radius)
```

Parameters

`point` Vector3

`radius` float

Returns

AITags[]

GetTagsAroundPoint(Vector3, float, string, string)

Get objects which match the query

```
public List<AITags> GetTagsAroundPoint(Vector3 point, float radius, string includedTags,
string excludedTags)
```

Parameters

`point` Vector3

`radius` float

`includedTags` string

A tag which the object should include, leave it empty if you don't want to check any tag for inclusion

`excludedTags` string

A tag which the object should not have, leave it empty if you don't want to exclude anything

Returns

List<AITags>

GetTagsAroundPoint(Vector3, float, string[], string[])

Get objects which match the query

```
public List<AITags> GetTagsAroundPoint(Vector3 point, float radius, string[] includedTags,
string[] excludedTags)
```

Parameters

`point` Vector3

`radius` float

`includedTags` string[]

A tag which the object should include, leave it empty if you don't want to check any tag for inclusion

`excludedTags` string[]

A tag which the object should not have, leave it empty if you don't want to exclude anything

Returns

List<AITags>

Remove(AITags)

```
public void Remove(AITags ai)
```

Parameters

`ai` AITags

Namespace NoOpArmy.WiseFeline.BlackBoards

Classes

BlackBoard

This component represents a blackboard which allows you to read and write its key value pairs and use them in your other component as you see fit. Let's say you have an enemy agent which wants to use an influence map to search for health packs, the name of the health pack can be set in a key in the blackboard and then the consideration for searching can read the map name from the blackboard

BlackBoardChange

Describes properties of a change to a blackboard when it comes to key name and type

BlackBoardChangeer

Specifieds the properties of a change to a blackboard when it comes to number of times to do the change and delay between changes and ...

BlackBoardDefinition

This class defines the keys which can exist in a blackboard and their data types. Create instances of this using the unity editor from the create menu and assign them to the blackboard component to use. The blackboard component then will only accept these keys with these data types.

KeyDefinition

This class represents blackboard key definitions.

Enums

BlackBoardChangeer.ChangeType

The type of change to happen

KeyDefinition.KeyType

The possible key types for the blackboard

Class BlackBoard

Namespace: NoOpArmy.WiseFeline.BlackBoards

Assembly: APIRef.dll

This component represents a blackboard which allows you to read and write its key value pairs and use them in your other component as you see fit. Let's say you have an enemy agent which wants to use an influence map to search for health packs, the name of the health pack can be set in a key in the blackboard and then the consideration for searching can read the map name from the blackboard

```
public class BlackBoard : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← BlackBoard

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) ,

Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() ,
Component.GetComponentIndex() , Component.CompareTag(string) ,
Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,

Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

blackBoard

The blackboard definition this component uses

```
[Tooltip("The blackboard definition this component uses")]
public BlackBoardDefinition blackBoard
```

Field Value

BlackBoardDefinition

bools

Dictionary of bool keys for the blackboard

```
public Dictionary<string, bool> bools
```

Field Value

Dictionary<string, bool>

floats

Dictionary of float keys for the blackboard

```
public Dictionary<string, float> floats
```

Field Value

Dictionary<string, float>

gameObjects

Dictionary of GameObject keys for the blackboard

```
public Dictionary<string, GameObject> gameObjects
```

Field Value

Dictionary<string, GameObject>

ints

Dictionary of int keys for the blackboard

```
public Dictionary<string, int> ints
```

Field Value

Dictionary<string, int>

objects

Dictionary of object keys for the blackboard

```
public Dictionary<string, object> objects
```

Field Value

Dictionary<string, object>

strings

Dictionary of string keys for the blackboard

```
public Dictionary<string, string> strings
```

Field Value

Dictionary<string, string>

unityObjects

Dictionary of UnityEngine.Object keys for the blackboard

```
public Dictionary<string, Object> unityObjects
```

Field Value

Dictionary<string, Object>

vector3s

Dictionary of Vector3 keys for the blackboard

```
public Dictionary<string, Vector3> vector3s
```

Field Value

Dictionary<string, Vector3>

Methods

GetBool(string)

Retrieves a boolean value from the blackboard.

```
public bool GetBool(string key)
```

Parameters

`key` string

The key name for the boolean value.

Returns

bool

The boolean value associated with the key.

Exceptions

ArgumentException

Thrown if the key is not defined in the blackboard definition as a boolean.

GetFloat(string)

Retrieves a float value from the blackboard.

```
public float GetFloat(string key)
```

Parameters

`key` string

The key name for the float value.

Returns

float

The float value associated with the key.

Exceptions

ArgumentException

Thrown if the key is not defined in the blackboard definition as a float.

GetGameObject(string)

Retrieves a GameObject value from the blackboard.

```
public GameObject GetGameObject(string key)
```

Parameters

`key` string

The key name for the GameObject value.

Returns

GameObject

The GameObject value associated with the key.

Exceptions

ArgumentException

Thrown if the key is not defined in the blackboard definition as a GameObject.

GetInt(string)

Retrieves an integer value from the blackboard.

```
public int GetInt(string key)
```

Parameters

`key` string

The key name for the integer value.

Returns

int

The integer value associated with the key.

Exceptions

ArgumentException

Thrown if the key is not defined in the blackboard definition as an integer.

GetObject(string)

Retrieves an object value from the blackboard.

```
public object GetObject(string key)
```

Parameters

`key` string

The key name for the object value.

Returns

object

The object value associated with the key.

Exceptions

ArgumentException

Thrown if the key is not defined in the blackboard definition as an object.

GetString(string)

```
public string GetString(string key)
```

Parameters

`key` string

Returns

string

GetUnityObject(string)

Retrieves a Unity Object value from the blackboard.

```
public Object GetUnityObject(string key)
```

Parameters

`key` string

The key name for the Unity Object value.

Returns

Object

The Unity Object value associated with the key.

Exceptions

ArgumentException

Thrown if the key is not defined in the blackboard definition as a Unity Object.

GetVector3(string)

Returns the Vector3 for the key specified

```
public Vector3 GetVector3(string key)
```

Parameters

`key` string

The key to return its value

Returns

Vector3

The Vector3 associated with the key

Exceptions

ArgumentException

Throws if the key is not defined in the definition as a Vector3

HasBool(string)

Does the key name exist in the bool key value pairs?

```
public bool HasBool(string key)
```

Parameters

`key` string

Name of the key to check

Returns

bool

True if the key exists and false otherwise

HasFloat(string)

Does the key name exist in the float key value pairs?

```
public bool HasFloat(string key)
```

Parameters

`key` string

Name of the key to check

Returns

bool

True if the key exists and false otherwise

HasGameObject(string)

Does the key name exist in the GameObject key value pairs?

```
public bool HasGameObject(string key)
```

Parameters

`key` string

Name of the key to check

Returns

bool

True if the key exists and false otherwise

HasInt(string)

Does the key name exist in the int key value pairs?

```
public bool HasInt(string key)
```

Parameters

`key` string

Name of the key to check

Returns

bool

True if the key exists and false otherwise

HasObject(string)

Does the key name exist in the object key value pairs?

```
public bool HasObject(string key)
```

Parameters

`key` string

Name of the key to check

Returns

bool

True if the key exists and false otherwise

HasUnityObject(string)

Does the key name exist in the UnityObject key value pairs?

```
public bool HasUnityObject(string key)
```

Parameters

`key` string

Name of the key to check

Returns

bool

True if the key exists and false otherwise

HasVector3(string)

Does the key name exist in the vector3 key value pairs?

```
public bool HasVector3(string key)
```

Parameters

`key` string

Name of the key to check

Returns

bool

True if the key exists and false otherwise

SetBool(string, bool)

Sets a boolean value in the blackboard.

```
public void SetBool(string key, bool value)
```

Parameters

`key` string

The key name for the boolean value.

`value` bool

The boolean value to set.

Exceptions

ArgumentException

Thrown if the key is not defined in the blackboard definition as a boolean.

SetFloat(string, float)

Sets a float value in the blackboard.

```
public void SetFloat(string key, float value)
```

Parameters

`key` string

The key name for the float value.

`value` float

The float value to set.

Exceptions

ArgumentException

Thrown if the key is not defined in the blackboard definition as a float.

SetGameObject(string, GameObject)

Sets a GameObject value in the blackboard.

```
public void SetGameObject(string key, GameObject value)
```

Parameters

`key` string

The key name for the GameObject value.

`value` GameObject

The GameObject value to set.

Exceptions

ArgumentException

Thrown if the key is not defined in the blackboard definition as a GameObject.

SetInt(string, int)

Sets an integer value in the blackboard.

```
public void SetInt(string key, int value)
```

Parameters

`key` string

The key name for the integer value.

`value` int

The integer value to set.

Exceptions

ArgumentException

Thrown if the key is not defined in the blackboard definition as an integer.

SetObject(string, object)

Sets an object value in the blackboard.

```
public void SetObject(string key, object value)
```

Parameters

`key` string

The key name for the object value.

`value` object

The object value to set.

Exceptions

ArgumentException

Thrown if the key is not defined in the blackboard definition as an object.

SetString(string, string)

Sets a string value in the blackboard.

```
public void SetString(string key, string value)
```

Parameters

`key` string

The key name for the boolean value.

`value` string

The boolean value to set.

Exceptions

ArgumentException

Thrown if the key is not defined in the blackboard definition as a boolean.

SetUnityObject(string, Object)

Sets a Unity Object value in the blackboard.

```
public void SetUnityObject(string key, Object value)
```

Parameters

`key` string

The key name for the Unity Object value.

`value` Object

The Unity Object value to set.

Exceptions

ArgumentException

Thrown if the key is not defined in the blackboard definition as a Unity Object.

SetVector3(string, Vector3)

Sets a Vector3 in the blackboard

```
public void SetVector3(string key, Vector3 value)
```

Parameters

`key` string

The key name for the Vector3

`value` Vector3

The Vector3 to set

Exceptions

ArgumentException

Throws if the key is not defined in the blackboard definition as a Vector3

Class BlackBoardChange

Namespace: NoOpArmy.WiseFeline.BlackBoards

Assembly: APIRef.dll

Describes properties of a change to a blackboard when it comes to key name and type

```
[Serializable]
public class BlackBoardChange
```

Inheritance

object ← BlackBoardChange

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

addInsteadOfSet

If true, adds the value specified to the current value of the key when applicable Otherwise just set the key to the specified value

```
[Tooltip("If true, adds the value specified to the current value of the key when applicable.
otherwise just set the key to the specified value")]
public bool addInsteadOfSet
```

Field Value

bool

boolValue

Bool value to add/set if the type is Bool

```
public bool boolValue
```

Field Value

bool

floatValue

Float value to add/set if the type is Float

```
public float floatValue
```

Field Value

float

gameObjectValue

GameObject value to add/set if the type is GameObject

```
public GameObject gameObjectValue
```

Field Value

GameObject

intValue

Int value to add/set if the type is Int

```
public int intValue
```

Field Value

int

key

The key name and type to change

```
[Tooltip("The key name and type to change")]
public KeyDefinition key
```

Field Value

KeyDefinition

maxFloatValue

max float value to add/set if the type is Float

```
public float maxFloatValue
```

Field Value

float

maxIntValue

Max Int value to set if the type is Int

```
public int maxIntValue
```

Field Value

int

minFloatValue

Min float value to set if the type is Float

```
public float minFloatValue
```

Field Value

float

minIntValue

Min Int value to set if the type is Int

```
public int minIntValue
```

Field Value

int

multiplyByDeltaTimeForAdds

Multiply the value to add by delta time before adding

```
[Tooltip("Multiply the value to add by delta time before adding")]
public bool multiplyByDeltaTimeForAdds
```

Field Value

bool

objectValue

Object value to add/set if the type is Object

```
public object objectValue
```

Field Value

object

stringValue

string value to add/set if the type is Bool

```
public string stringValue
```

Field Value

string

unityObjectValue

UnityEngine.Object value to add/set if the type is UnityObject

```
public Object unityObjectValue
```

Field Value

Object

vectorValue

Vector3 value to add/set if the type is Vector3

```
public Vector3 vectorValue
```

Field Value

Vector3

Methods

ApplyToBlackboard(BlackBoard)

Does the change operation based on the value of the other properties of the class

```
public void ApplyToBlackboard(BlackBoard blackBoard)
```

Parameters

`blackBoard` BlackBoard

The blackboard to do the changes too

Class BlackBoardChangeer

Namespace: NoOpArmy.WiseFeline.BlackBoards

Assembly: APIRef.dll

Specifieds the properties of a change to a blackboard when it comes to number of times to do the change and delay between changes and ...

```
[Serializable]
public class BlackBoardChangeer
```

Inheritance

object ← BlackBoardChangeer

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

change

The key name and type hwich you want the change to happen to

```
[Tooltip("The key name and type which you want the change to happen to")]
public BlackBoardChange change
```

Field Value

BlackBoardChange

changeType

The type of change which you want to happen to the blackboard

```
[Tooltip("The type of change which you want to happen to the blackboard")]
public BlackBoardChangeer.ChangeType changeType
```

Field Value

BlackBoardChangeer.ChangeType

delayInSeconds

The delay between changes

```
[Tooltip("The delay between changes in seconds")]
public float delayInSeconds
```

Field Value

float

maxExecutionCount

Maximum number of times this gets executed in the repeat with delay mode. 0 means unlimited

```
[Tooltip("Maximum number of times this gets executed in the repeat with delay mode. 0
means unlimited")]
public int maxExecutionCount
```

Field Value

int

Methods

Reset()

```
public void Reset()
```

Update(BlackBoard)

```
public void Update(BlackBoard blackBoard)
```

Parameters

`blackBoard` BlackBoard

Enum BlackBoardChangeer.ChangeType

Namespace: NoOpArmy.WiseFeline.BlackBoards

Assembly: APIRef.dll

The type of change to happen

```
public enum BlackBoardChangeer.ChangeType : byte
```

Fields

`EveryFrame = 3`

Do a change every frame and multiply the value change by delta time in the case of additive changes

`None = 0`

No change

`Once = 1`

Only once change the blackBoard

`RepeatWithDelay = 2`

Repeat the change for a specified number of times and with a specific delay

Class BlackBoardDefinition

Namespace: NoOpArmy.WiseFeline.BlackBoards

Assembly: APIRef.dll

This class defines the keys which can exist in a blackboard and their data types. Create instances of this using the unity editor from the create menu and assign them to the blackboard component to use. The blackboard component then will only accept these keys with these data types.

```
[CreateAssetMenu(menuName = "NoOpArmy/Wise Feline/BlackBoard Definition")]
public class BlackBoardDefinition : ScriptableObject
```

Inheritance

object ← Object ← ScriptableObject ← BlackBoardDefinition

Inherited Members

ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) , Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) , Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,

Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

definitions

List of key definitions

```
public List<KeyDefinition> definitions
```

Field Value

List <KeyDefinition>

Class KeyDefinition

Namespace: NoOpArmy.WiseFeline.BlackBoards

Assembly: APIRef.dll

This class represents blackboard key definitions.

```
[Serializable]
public class KeyDefinition
```

Inheritance

object ← KeyDefinition

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

keyName

Unique name of the key in the blackboard

```
public string keyName
```

Field Value

string

keyType

Type of the key

```
public KeyDefinition.KeyType keyType
```

Field Value

KeyDefinition.KeyType

Enum KeyDefinition.KeyType

Namespace: NoOpArmy.WiseFeline.BlackBoards

Assembly: APIRef.dll

The possible key types for the blackboard

```
public enum KeyDefinition.KeyType : byte
```

Fields

`Bool = 4`

A boolean value

`Float = 2`

A floating point number

`GameObject = 6`

A game object

`Int = 3`

A 32 bit integer

`Object = 1`

A System.Object for any type you need and we did not support

`String = 7`

A string

`UnityObject = 5`

A UnityEngine.Object for any unity object which we did not support and you need

`Vector3 = 0`

A UnityEngine.Vector3

Namespace NoOpArmy.WiseFeline.Considerations

Classes

AITagConsideration

This consideration checks if the target or the executing agent has or doesn't have a specific tag.

Enums

AITagConsideration.TagOperation

Class AITagConsideration

Namespace: NoOpArmy.WiseFeline.Considerations

Assembly: APIRef.dll

This consideration checks if the target or the executing agent has or doesn't have a specific tag.

```
public class AITagConsideration : ConsiderationBase
```

Inheritance

object ← Object ← ScriptableObject ← AIObject ← ConsiderationBase ← AITagConsideration

Inherited Members

ConsiderationBase.instantiatePerAgent , ConsiderationBase.minRange , ConsiderationBase.maxRange , ConsiderationBase.Score , ConsiderationBase.Brain , ConsiderationBase.Clone() , ConsiderationBase.OnInitialized() , ConsiderationBase.UpdateRange(float, float) , AIObject.guid , AIObject.Name , AIObject.SetGuid(string, bool) , ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) , Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,

Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

operation

The operation to do on the tag to find the score

```
public AITagConsideration.TagOperation operation
```

Field Value

AITagConsideration.TagOperation

tag

The tag to consider

```
public string tag
```

Field Value

string

Methods

GetValue(Component)

This method should return the current value of the consideration which is used for multiplication with other considerations to calculate the score of the action

```
protected override float GetValue(Component target)
```

Parameters

`target` Component

If the consideration has a target then this value is the target component which the consideration should be calculated for. The actual type of this depends on the type of the component that the action stores in its target list in the UpdateTargets call. If the consideration doesn't have a target, then the Brain component of the executing agent itself is passed to the method.

Returns

float

A value between minRange and maxRange which indicates the current value/score of the consideration

Enum AITagConsideration.TagOperation

Namespace: NoOpArmy.WiseFeline.Considerations

Assembly: APIRef.dll

```
public enum AITagConsideration.TagOperation
```

Fields

`ShouldHaveTag = 0`

The object should have the tag

`ShouldNotHaveTag = 1`

The object should not have the tag

Namespace NoOpArmy.WiseFeline.Data Recorder

Classes

DataRecorder

Record

Each record holds all textual and shape information for a frame of a specific data recorder component

RecordManager

RecordTester

SubRecord

Structs

Shape

Enums

ShapeType

Class DataRecorder

Namespace: NoOpArmy.WiseFeline.DataRecorder

Assembly: APIRef.dll

```
public class DataRecorder : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← DataRecorder

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) , Component.SendMessageUpwards(string, object, SendMessageOptions) , Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,

Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,

Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Properties

records

```
public List<Record> records { get; }
```

Property Value

List <Record>

Methods

AddShape(Shape, int)

```
public void AddShape(Shape shape, int index)
```

Parameters

`shape` Shape

`index` int

AddSubRecord(SubRecord, int)

```
public void AddSubRecord(SubRecord subRecord, int index)
```

Parameters

`subRecord` SubRecord

`index` int

ClearRecords()

```
public void ClearRecords()
```

GetCurrentFrameRecord()

```
public int GetCurrentFrameRecord()
```

Returns

int

RemoveRecord(Record)

```
public void RemoveRecord(Record record)
```

Parameters

`record` Record

Class Record

Namespace: NoOpArmy.WiseFeline.DataRecorder

Assembly: APIRef.dll

Each record holds all textual and shape information for a frame of a specific data recorder component

```
public class Record
```

Inheritance

object ← Record

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

Record()

```
public Record()
```

Record(string, GameObject, long, int, float)

```
public Record(string name, GameObject go, long id, int frameCount, float frameTime)
```

Parameters

`name` string

`go` GameObject

`id` long

`frameCount` int

`frameTime` float

Fields

frameCount

```
public int frameCount
```

Field Value

int

frameTime

```
public float frameTime
```

Field Value

float

go

```
public GameObject go
```

Field Value

GameObject

id

```
public long id
```

Field Value

long

shapes

```
public List<Shape> shapes
```

Field Value

List <Shape>

subRecords

```
public List<SubRecord> subRecords
```

Field Value

List <SubRecord>

Class RecordManager

Namespace: NoOpArmy.WiseFeline.DataRecorder

Assembly: APIRef.dll

```
public class RecordManager : Singleton<RecordManager>
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← Singleton<RecordManager> ← RecordManager

Inherited Members

Singleton<RecordManager>.Instance , Singleton<RecordManager>.Awake() , Singleton<RecordManager>.OnDestroy() , MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) ,

Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,

Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

currentRecord

```
public Record currentRecord
```

Field Value

Record

currentRecorder

```
public DataRecorder currentRecorder
```

Field Value

DataRecorder

Properties

ShouldRecord

```
public bool ShouldRecord { get; set; }
```

Property Value

bool

dataRecorderNames

```
public List<string> dataRecorderNames { get; }
```

Property Value

List<string>

dataRecorders

```
public List<DataRecorder> dataRecorders { get; }
```

Property Value

List<DataRecorder>

Methods

AddDataRecorder(DataRecorder)

```
public void AddDataRecorder(DataRecorder dataRecorder)
```

Parameters

`dataRecorder` DataRecorder

RemoveDataRecorder(DataRecorder)

```
public void RemoveDataRecorder(DataRecorder dataRecorder)
```

Parameters

`dataRecorder` DataRecorder

SetCurrentRecord(int)

```
public void SetCurrentRecord(int selectedIndex)
```

Parameters

`selectedIndex` int

SetCurrentRecorder(DataRecorder)

```
public int SetCurrentRecorder(DataRecorder recorder)
```

Parameters

`recorder` DataRecorder

Returns

int

SetCurrentRecorder(int)

```
public void SetCurrentRecorder(int selectedIndex)
```

Parameters

`selectedIndex` int

Class RecordTester

Namespace: NoOpArmy.WiseFeline.DataRecorder

Assembly: APIRef.dll

```
public class RecordTester : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← RecordTester

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) , Component.SendMessageUpwards(string, object, SendMessageOptions) , Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,

Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,

Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Struct Shape

Namespace: NoOpArmy.WiseFeline.DataRecorder

Assembly: APIRef.dll

```
public struct Shape
```

Inherited Members
ValueType.Equals(object) , ValueType.GetHashCode() , ValueType.ToString() , object.Equals(object, object) , object.GetType() , object.ReferenceEquals(object, object)

Fields

Type

```
public ShapeType Type
```

Field Value

ShapeType

color

```
public Color color
```

Field Value

Color

position

```
public Vector3 position
```

Field Value

Vector3

size

```
public float size
```

Field Value

float

Enum ShapeType

Namespace: NoOpArmy.WiseFeline.DataRecorder

Assembly: APIRef.dll

```
public enum ShapeType : byte
```

Fields

`Cube = 1`

`Sphere = 0`

`WiredCube = 3`

`WiredSphere = 2`

Class SubRecord

Namespace: NoOpArmy.WiseFeline.DataRecorder

Assembly: APIRef.dll

```
public class SubRecord
```

Inheritance

object ← SubRecord

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

indentationLevel

```
public int indentationLevel
```

Field Value

int

record

```
public Record record
```

Field Value

Record

text

```
public string text
```

Field Value

string

Namespace NoOpArmy.WiseFeline.InfluenceMaps

Classes

InfluenceMap

This is the main influence map class of the system which contains the actual map logic and can be created in C# using its constructor and does not depend on any component/MonoBehaviour

InfluenceMapCollection

InfluenceMapComponent

A helper component which makes it easier to use influence maps Attach this component to a GameObject to create an influence map at awake and use it as a reference for InfluencerAgent component. The map field is the actual influence map that you can use to search for values or directly change cell values

InfluenceMapComponentBase

InfluenceMapDecayer

Decays an influence map using the value you gave it and clamps the values to not go above/beyond specified values

InfluenceMapTemplate

This asset type is used as a template for the stamp shape of agents which influence the amp Usually different agents based on their behavior need both different sizes and fall off curve shapes

InfluenceMapView

This class allows you to calculate a map from multiple others. You could previously do this without a view using code but this makes it easier to make maps from other maps without writing a line of code

InfluenceMapView.ViewOperation

InfluencerAgent

This component can be attached to an agent so it influences the map

RectangularInfluencerAgent

This component can be attached to an agent so it influences the map

Enums

InfluenceMapDecayer.OperationMode

The operations done on the map

InfluenceMapView.ViewOperation.Operation

InfluencerAgent.Behavior

The way that this map will add its influence

SearchCondition

The stop condition for search

Class InfluenceMap

Namespace: NoOpArmy.WiseFeline.InfluenceMaps

Assembly: APIRef.dll

This is the main influence map class of the system which contains the actual map logic and can be created in C# using its constructor and does not depend on any component/MonoBehaviour

```
public class InfluenceMap
```

Inheritance

object ← InfluenceMap

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

InfluenceMap(int, int, float)

Creates an influence map

```
public InfluenceMap(int mapWidth, int mapHeight, float cellSize)
```

Parameters

`mapWidth` int

Width of the map in cells

`mapHeight` int

Height of the map in cells

`cellSize` float

Size of each cell in meters

InfluenceMap(float[,], float)

Creates an influence map from a pre-defined array of cells

```
public InfluenceMap(float[,] cells, float cellSize)
```

Parameters

`cells` float[,]

`cellSize` float

Fields

anchorLocation

anchor location of the map in the world. y value is ignored in the main logic it is only useful for drawing gizmos unless you are using the 2d function

```
public Vector3 anchorLocation
```

Field Value

Vector3

Properties

CellSize

The cell size in meters

```
public float CellSize { get; }
```

Property Value

float

Height

Height of the map in cells

```
public int Height { get; }
```

Property Value

int

MaxValue

Maximum value which a cell in the map has at the moment

```
public float MaxValue { get; }
```

Property Value

float

MinValue

Minimum value which a cell in the map has at the moment

```
public float MinValue { get; }
```

Property Value

float

Width

Width of the map in cells

```
public int Width { get; }
```

Property Value

int

Methods

Add(InfluenceMap, InfluenceMap)

Adds two maps with the same size to each other and returns a new map containing the result

```
public InfluenceMap Add(InfluenceMap otherMap, InfluenceMap result = null)
```

Parameters

`otherMap` InfluenceMap

`result` InfluenceMap

If you have a map already allocated for the result, you can supply it here

Returns

InfluenceMap

Add(InfluenceMap, float, InfluenceMap)

Adds two maps with the same size to each other and returns a new map containing the result

```
public InfluenceMap Add(InfluenceMap otherMap, float strength, InfluenceMap result = null)
```

Parameters

`otherMap` InfluenceMap

`strength` float

The other map will be multiplied by this value to increase/decrease its effect

`result` InfluenceMap

If you have a map already allocated for the result, you can supply it here

Returns

InfluenceMap

Add(float, InfluenceMap)

```
public InfluenceMap Add(float value, InfluenceMap result = null)
```

Parameters

`value` float

`result` InfluenceMap

Returns

InfluenceMap

AddAndClamp(float, float, float, InfluenceMap)

```
public InfluenceMap AddAndClamp(float value, float min, float max, InfluenceMap result
= null)
```

Parameters

`value` float

`min` float

`max` float

`result` InfluenceMap

Returns

InfluenceMap

AddInfluence(int, int, InfluenceMapTemplate, float)

Adds an influence stamp to the map

```
public void AddInfluence(int centerX, int centerY, InfluenceMapTemplate template, float
magnitude = 1)
```

Parameters

`centerX` int

`centerY` int

`template` InfluenceMapTemplate

The template which should be propagated

`magnitude` float

Remarks

Templates are scriptable objects and can be created in the project window in the editor or at runtime depending on your needs

AddInfluenceWithLineOfSight(int, int, InfluenceMapTemplate, InfluenceMap, float)

Adds an influence stamp to the map

```
public void AddInfluenceWithLineOfSight(int centerX, int centerY, InfluenceMapTemplate
template, InfluenceMap obstaclesMap, float magnitude = 1)
```

Parameters

`centerX` int

`centerY` int

`template` InfluenceMapTemplate

The template which should be propagated

`obstaclesMap` InfluenceMap

The map which contains obstacles and we check line of sight in

`magnitude` float

Remarks

Templates are scriptable objects and can be created in the project window in the editor or at runtime depending on your needs

AddInverse(InfluenceMap, InfluenceMap)

Adds the inverse of a map to the current map and returns a new map containing the result

```
public InfluenceMap AddInverse(InfluenceMap otherMap, InfluenceMap result = null)
```

Parameters

`otherMap` InfluenceMap

`result` InfluenceMap

If you have a map already allocated for the result, you can supply it here

Returns

InfluenceMap

AddInverse(InfluenceMap, float, InfluenceMap)

Adds the inverse of a map to the current map and returns a new map containing the result

```
public InfluenceMap AddInverse(InfluenceMap otherMap, float strength, InfluenceMap result
= null)
```

Parameters

`otherMap` InfluenceMap

`strength` float

The other map will be multiplied by this value to increase/decrease its effect

`result` InfluenceMap

If you have a map already allocated for the result, you can supply it here

Returns

InfluenceMap

AddRectangularInfluence(int, int, int, int, float)

Adds a rectangular template of a constant value to the map

```
public void AddRectangularInfluence(int startX, int startY, int width, int height, float
value = 1)
```

Parameters

`startX` int

`startY` int

`width` int

`height` int

`value` float

Remarks

This can have negative width and height to move left/down from the starting point

AddValueToCell(int, int, float)

Adds a value to a cell

```
public void AddValueToCell(int x, int y, float value)
```

Parameters

`x` int

`y` int

`value` float

CheckMapForLineOfSight(int, int, int, int)

Checks to see if there is any line of sight between two points. Returns true if there is any and false otherwise.

```
public bool CheckMapForLineOfSight(int x1, int y1, int x2, int y2)
```

Parameters

`x1` int

`y1` int

`x2` int

`y2` int

Returns

bool

GetCellValue(int, int)

Get a cell's value

```
public float GetCellValue(int x, int y)
```

Parameters

`x` int

`y` int

Returns

float

GetCellsArray()

Returns the cells of the map, keep in mind that this is unsafe if you modify the array yourself You need this if you want to do operations on the map's array

```
public float[,] GetCellsArray()
```

Returns

float[,]

A reference to the array that the map uses

Remarks

This method does not return a copy so be careful to not modify the array and just use it to calculate new maps

Invert(InfluenceMap)

Inverts the map values so 0.1 will become 0.9 and 0.6 will become 0.4

```
public InfluenceMap Invert(InfluenceMap result = null)
```

Parameters

`result` InfluenceMap

If you have a map already allocated for the result, you can supply it here

Returns

InfluenceMap

MapToWorldPosition(int, int)

Converts a position from the map coordinates to unity's world space coordinates The y value of the returned vector is always 0 and x and z are set to the converted values

```
public Vector3 MapToWorldPosition(int x, int y)
```

Parameters

`x` int

`y` int

Returns

Vector3

MapToWorldPosition(int, int, Terrain)

Converts a position from the map coordinates to unity's world space coordinates The y value of the returned vector is always 0 and x and z are set to the converted values

```
public Vector3 MapToWorldPosition(int x, int y, Terrain terrain)
```

Parameters

`x` int

x position on the influence map

`y` int

y position on the influence map

`terrain` Terrain

The terrain to get the height from

Returns

Vector3

MapToWorldPosition2d(int, int)

Converts a position from the map coordinates to unity's world space coordinates in 2d plane. The z value of the returned vector is always 0 and x and y are set to the converted values

```
public Vector3 MapToWorldPosition2d(int x, int y)
```

Parameters

`x` int

`y` int

Returns

Vector3

Multiply(InfluenceMap, InfluenceMap)

Multiplies a map to this map and returns a new map containing the result The maps should be the same size

```
public InfluenceMap Multiply(InfluenceMap otherMap, InfluenceMap result = null)
```

Parameters

`otherMap` InfluenceMap

`result` InfluenceMap

If you have a map already allocated for the result, you can supply it here

Returns

InfluenceMap

Exceptions

ArgumentException

Multiply(InfluenceMap, float, InfluenceMap)

Multiplies a map to this map and returns a new map containing the result The maps should be the same size

```
public InfluenceMap Multiply(InfluenceMap otherMap, float strength, InfluenceMap result
= null)
```

Parameters

`otherMap` InfluenceMap

`strength` float

The other map will be multiplied by this value to increase/decrease its effect

`result` InfluenceMap

If you have a map already allocated for the result, you can supply it here

Returns

InfluenceMap

Exceptions

ArgumentException

Multiply(float, InfluenceMap)

Multiplies a map with such a value

```
public InfluenceMap Multiply(float value, InfluenceMap result = null)
```

Parameters

`value` float

The value that all map cells will be multiplied with

`result` InfluenceMap

If you have a map already allocated for the result, you can supply it here

Returns

InfluenceMap

MultiplyAndClamp(float, float, float, InfluenceMap)

```
public InfluenceMap MultiplyAndClamp(float value, float min, float max, InfluenceMap result
= null)
```

Parameters

`value` float

`min` float

`max` float

`result` InfluenceMap

Returns

InfluenceMap

Normalize(InfluenceMap)

Normalizes the values in a map to values between 0 and 1

```
public InfluenceMap Normalize(InfluenceMap result = null)
```

Parameters

`result` InfluenceMap

Returns

InfluenceMap

PropagateInfluence(int, int, int, AnimationCurve, float)

Propagates an influence stamp based on a curve, a radius and a magnitude

```
public void PropagateInfluence(int centerX, int centerY, int radius, AnimationCurve
influenceCurve, float magnitude = 1)
```

Parameters

`centerX` int

`centerY` int

`radius` int

`influenceCurve` AnimationCurve

`magnitude` float

Remarks

This is slower than AddInfluence and is used by the template to bake its values but you can use it if the number of different templates you need is high enough that the amount of memory is used is not acceptable

PropagateInfluenceWithLineOfSight(int, int, int, AnimationCurve, InfluenceMap, float)

Propagates an influence stamp based on a curve, a radius and a magnitude

```
public void PropagateInfluenceWithLineOfSight(int centerX, int centerY, int radius,
AnimationCurve influenceCurve, InfluenceMap obstaclesMap, float magnitude = 1)
```

Parameters

`centerX` int

`centerY` int

`radius` int

`influenceCurve` AnimationCurve

`obstaclesMap` InfluenceMap

`magnitude` float

Remarks

This is slower than AddInfluence and is used by the template to bake its values but you can use it if the number of different templates you need is high enough that the amount of memory is used is not acceptable

SearchForHighestValueClosestToCenter(Vector2Int, int, out Vector2Int)

Searches for the highest value closest to me

```
public float SearchForHighestValueClosestToCenter(Vector2Int centerCell, int radius, out
Vector2Int result)
```

Parameters

`centerCell` Vector2Int

`radius` int

`result` Vector2Int

Returns

float

SearchForHighestValueWithRandomStartingPoint(Vector2Int, int, out Vector2Int)

Searches for the highest value around

```
public float SearchForHighestValueWithRandomStartingPoint(Vector2Int centerCell, int radius,
out Vector2Int result)
```

Parameters

`centerCell` Vector2Int

`radius` int

`result` Vector2Int

Returns

float

SearchForLowestValueClosestToCenter(Vector2Int, int, out Vector2Int)

Searches for the lowest value to me

```
public float SearchForLowestValueClosestToCenter(Vector2Int centerCell, int radius, out
Vector2Int result)
```

Parameters

`centerCell` Vector2Int

`radius` int

`result` Vector2Int

Returns

float

SearchForLowestValueWithRandomStartingPoint(Vector2Int, int, out Vector2Int)

Searches for the lowest value around

```
public float SearchForLowestValueWithRandomStartingPoint(Vector2Int centerCell, int radius,
out Vector2Int result)
```

Parameters

`centerCell` Vector2Int

`radius` int

`result` Vector2Int

Returns

float

SearchForValueWithRandomStartingPoint(float, SearchCondition, Vector2Int, int, out Vector2Int)

Searches for a value in an area specified by a center and a radius and starts at a random point in the area to not return the same value as the result of the flood fill algorithm

```
public bool SearchForValueWithRandomStartingPoint(float searchValue, SearchCondition
condition, Vector2Int centerCell, int radius, out Vector2Int result)
```

Parameters

`searchValue` float

`condition` SearchCondition

`centerCell` Vector2Int

`radius` int

`result` Vector2Int

Returns

bool

SetCellValue(int, int, float)

Sets a cell's value to the specified value

```
public void SetCellValue(int x, int y, float value)
```

Parameters

`x` int

`y` int

`value` float

SetCellsArray(float[,])

Sets the cells of the map to the array you created

```
public void SetCellsArray(float[,] cells)
```

Parameters

`cells` float[,]

Subtract(InfluenceMap, InfluenceMap)

Subtract a map from this map and returns a new map containing the result The maps should be the same size

```
public InfluenceMap Subtract(InfluenceMap otherMap, InfluenceMap result = null)
```

Parameters

`otherMap` InfluenceMap

`result` InfluenceMap

If you have a map already allocated for the result, you can supply it here

Returns

InfluenceMap

Exceptions

ArgumentException

Subtract(InfluenceMap, float, InfluenceMap)

Subtract a map from this map and returns a new map containing the result The maps should be the same size

```
public InfluenceMap Subtract(InfluenceMap otherMap, float strength, InfluenceMap result
= null)
```

Parameters

`otherMap` InfluenceMap

`strength` float

The other map will be multiplied by this value to increase/decrease its effect

`result` InfluenceMap

If you have a map already allocated for the result, you can supply it here

Returns

InfluenceMap

Exceptions

ArgumentException

UpdateMinMaxValue()

```
public void UpdateMinMaxValue()
```

WorldToMapPosition(Vector3)

Converts a position from unity world space to map coordinates. The y component of the vector has no effect and x and z are converted to 2d map coordinates

```
public Vector2Int WorldToMapPosition(Vector3 position)
```

Parameters

`position` Vector3

Returns

Vector2Int

Remarks

Keep in mind that the position can be outside the map depending on where the position is and the map size

WorldToMapPosition2d(Vector3)

Converts a position from unity world space to map coordinates. It uses the 2D x,y plane instead of x,z The z component of the vector has no effect and x and y are converted to 2d map coordinates

```
public Vector2Int WorldToMapPosition2d(Vector3 position)
```

Parameters

`position` Vector3

Returns

Vector2Int

Remarks

Keep in mind that the position can be outside the map depending on where the position is and the map size

Class InfluenceMapCollection

Namespace: NoOpArmy.WiseFeline.InfluenceMaps

Assembly: APIRef.dll

```
public class InfluenceMapCollection : Singleton<InfluenceMapCollection>
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ←
Singleton<InfluenceMapCollection> ← InfluenceMapCollection

Inherited Members

Singleton<InfluenceMapCollection>.Instance , Singleton<InfluenceMapCollection>.Awake() ,
Singleton<InfluenceMapCollection>.OnDestroy() , MonoBehaviour.IsInvoking() ,
MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) ,
MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) ,
MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) ,
MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) ,
MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) ,
MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) ,
MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) ,
MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart ,
MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled ,
Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() ,
Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) ,
Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) ,
Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) ,
Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) ,
Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) ,
Component.GetComponentsInChildren<T>(bool, List<T>) ,
Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) ,
Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) ,
Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() ,
Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) ,
Component.GetComponentsInParent<T>(bool) ,
Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() ,
Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) ,
Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() ,
Component.GetComponentIndex() , Component.CompareTag(string) ,

Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,

Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

mapItems

```
public Dictionary<string, InfluenceMapComponentBase> mapItems
```

Field Value

Dictionary<string, InfluenceMapComponentBase>

Methods

GetMap(string)

Gets a map by its name

```
public InfluenceMapComponentBase GetMap(string name)
```

Parameters

`name` string

Returns

InfluenceMapComponentBase

Register(string, InfluenceMapComponentBase)

Registers a map in the collection. This is called automatically by the influence map component

```
public void Register(string mapName, InfluenceMapComponentBase influenceMap)
```

Parameters

`mapName` string

`influenceMap` InfluenceMapComponentBase

Exceptions

ArgumentException

Unregister(string)

Unregisters a map in the collection. This is called automatically by the influence map component

```
public void Unregister(string mapName)
```

Parameters

`mapName` string

Class InfluenceMapComponent

Namespace: NoOpArmy.WiseFeline.InfluenceMaps

Assembly: APIRef.dll

A helper component which makes it easier to use influence maps Attach this component to a GameObject to create an influence map at awake and use it as a reference for InfluencerAgent component. The map field is the actual influence map that you can use to search for values or directly change cell values

```
public class InfluenceMapComponent : InfluenceMapComponentBase
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← InfluenceMapComponentBase ← InfluenceMapComponent

Inherited Members

InfluenceMapComponentBase.is2dMap , InfluenceMapComponentBase.Map , InfluenceMapComponentBase.IsMapValid() , InfluenceMapComponentBase.AddAndClampValue(float, float, float) , InfluenceMapComponentBase.MultiplyAndClampValue(float, float, float) , InfluenceMapComponentBase.AddInfluence(int, int, InfluenceMapTemplate, float) , InfluenceMapComponentBase.AddInfluenceWithLineOfSight(int, int, InfluenceMapTemplate, InfluenceMapComponentBase, float) , InfluenceMapComponentBase.AddRectangularInfluence(int, int, int, int, float) , InfluenceMapComponentBase.PropagateInfluenceWithLineOfSight(int, int, int, AnimationCurve, InfluenceMapComponentBase, int) , InfluenceMapComponentBase.PropagateInfluence(int, int, int, AnimationCurve, int) , InfluenceMapComponentBase.WorldToMapPosition(Vector3) , InfluenceMapComponentBase.SearchForValueWithRandomStartingPoint(float, SearchCondition, Vector3, int, out Vector3) , InfluenceMapComponentBase.SearchForHighestValueWithRandomStartingPoint(Vector3, int, out Vector3) , InfluenceMapComponentBase.SearchForLowestValueWithRandomStartingPoint(Vector3, int, out Vector3) , InfluenceMapComponentBase.SearchForHighestValueClosestToCenter(Vector3, int, out Vector3) , InfluenceMapComponentBase.SearchForLowestValueClosestToCenter(Vector3, int, out Vector3) , InfluenceMapComponentBase.SearchForValueWithRandomStartingPoint(float, SearchCondition, Vector2Int, int, out Vector2Int) ,

InfluenceMapComponentBase.SearchForHighestValueWithRandomStartingPoint(Vector2Int, int, out Vector2Int) , InfluenceMapComponentBase.SearchForLowestValueWithRandomStartingPoint(Vector2Int, int, out Vector2Int) , InfluenceMapComponentBase.SearchForHighestValueClosestToCenter(Vector2Int, int, out Vector2Int) , InfluenceMapComponentBase.SearchForLowestValueClosestToCenter(Vector2Int, int, out Vector2Int) , InfluenceMapComponentBase.GetCellValue(int, int) , InfluenceMapComponentBase.GetCellValue(Vector3) , InfluenceMapComponentBase.MapToWorldPosition(int, int) , MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) , Component.SendMessageUpwards(string, object, SendMessageOptions) , Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) , Component.SendMessageUpwards(string, SendMessageOptions) , Component.SendMessage(string, object) , Component.SendMessage(string) , Component.SendMessage(string, object, SendMessageOptions) ,

Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,

Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

CreateOnAwake

Create the map on awake

```
public bool CreateOnAwake
```

Field Value

bool

cellSize

Cell size of the map

```
[Tooltip("Cell size of the map")]
public float cellSize
```

Field Value

float

height

Height of the map in cells

```
[Tooltip("Height of the map in cells")]
public int height
```

Field Value

int

mapName

///MapName

```
public string mapName
```

Field Value

string

width

Width of the map in cell count

```
[Tooltip("Width of the map in cell count")]
public int width
```

Field Value

int

Methods

CreateOnStartup()

```
[Obsolete("This method is depricated and will be removed. Use CreateMap() instead")]
public void CreateOnStartup()
```

UpdateAnchorLocationFromTransform()

Updates the anchor location from the transform

```
public void UpdateAnchorLocationFromTransform()
```

Class InfluenceMapComponentBase

Namespace: NoOpArmy.WiseFeline.InfluenceMaps

Assembly: APIRef.dll

```
public class InfluenceMapComponentBase : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← InfluenceMapComponentBase

Derived

InfluenceMapComponent, InfluenceMapView

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) ,

Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,

Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

is2dMap

Is this map used for a 2d game in the x,y plane

```
[Tooltip("Is this map used for a 2d game in the x,y plane")]
public bool is2dMap
```

Field Value

bool

Properties

Map

The actual influence map object which can be used for searching or setting values

```
public InfluenceMap Map { get; protected set; }
```

Property Value

InfluenceMap

Methods

AddAndClampValue(float, float, float)

```
public void AddAndClampValue(float value, float min, float max)
```

Parameters

`value` float

`min` float

`max` float

AddInfluence(int, int, InfluenceMapTemplate, float)

Adds the inverse of a map to the current map and returns a new map containing the result

```
public void AddInfluence(int x, int y, InfluenceMapTemplate myMapTemplate, float magnitude
= 1)
```

Parameters

`x` int

`y` int

`myMapTemplate` InfluenceMapTemplate

`magnitude` float

AddInfluenceWithLineOfSight(int, int, InfluenceMapTemplate, InfluenceMapComponentBase, float)

Adds the inverse of a map to the current map and returns a new map containing the result

```
public void AddInfluenceWithLineOfSight(int x, int y, InfluenceMapTemplate myMapTemplate,
InfluenceMapComponentBase obstaclesMap, float magnitude = 1)
```

Parameters

`x` int

`y` int

`myMapTemplate` InfluenceMapTemplate

`obstaclesMap` InfluenceMapComponentBase

The map to check line of sight in

`magnitude` float

AddRectangularInfluence(int, int, int, int, float)

Adds a rectangular template of a constant value to the map

```
public void AddRectangularInfluence(int x, int y, int width, int height, float value = 1)
```

Parameters

`x` int

`y` int

`width` int

`height` int

`value` float

GetCellValue(int, int)

Get a cell's value

```
public float GetCellValue(int x, int y)
```

Parameters

`x` int

`y` int

Returns

float

GetCellValue(Vector3)

Get a cell's value

```
public float GetCellValue(Vector3 worldPosition)
```

Parameters

`worldPosition` Vector3

Returns

float

IsMapValid()

```
public bool IsMapValid()
```

Returns

bool

MapToWorldPosition(int, int)

Converts a position from the map coordinates to unity's world space coordinates

```
public Vector3 MapToWorldPosition(int x, int y)
```

Parameters

x int

y int

Returns

Vector3

MultiplyAndClampValue(float, float, float)

```
public void MultiplyAndClampValue(float value, float min, float max)
```

Parameters

value float

min float

max float

PropagateInfluence(int, int, int, AnimationCurve, int)

Propagates an influence stamp based on a curve, a radius and a magnitude

```
public void PropagateInfluence(int x, int y, int radius, AnimationCurve curve, int magnitude
= 1)
```

Parameters

x int

y int

`radius` int

`curve` AnimationCurve

`magnitude` int

PropagateInfluenceWithLineOfSight(int, int, int, AnimationCurve, InfluenceMapComponentBase, int)

Propagates an influence stamp based on a curve, a radius and a magnitude

```
public void PropagateInfluenceWithLineOfSight(int x, int y, int radius, AnimationCurve
curve, InfluenceMapComponentBase ObstaclesMap, int magnitude = 1)
```

Parameters

`x` int

`y` int

`radius` int

`curve` AnimationCurve

`ObstaclesMap` InfluenceMapComponentBase

`magnitude` int

SearchForHighestValueClosestToCenter(Vector2Int, int, out Vector2Int)

Searches for the highest value in an area specified by a center and a radius and starts at a random point in the area to not return the same value as the result of the flood fill algorithm

```
public float SearchForHighestValueClosestToCenter(Vector2Int position, int radius, out
Vector2Int result)
```

Parameters

`position` Vector2Int

`radius` int

`result` Vector2Int

Returns

float

SearchForHighestValueClosestToCenter(Vector3, int, out Vector3)

Searches for the highest value in an area specified by a center and a radius and starts at a random point in the area to not return the same value as the result of the flood fill algorithm

```
public float SearchForHighestValueClosestToCenter(Vector3 position, int radius, out Vector3 result)
```

Parameters

`position` Vector3

`radius` int

`result` Vector3

Returns

float

SearchForHighestValueWithRandomStartingPoint(Vector2Int, int, out Vector2Int)

Searches for the highest value in an area specified by a center and a radius and starts at a random point in the area to not return the same value as the result of the flood fill algorithm

```
public float SearchForHighestValueWithRandomStartingPoint(Vector2Int position, int radius, out Vector2Int result)
```

Parameters

`position` Vector2Int

`radius` int

`result` Vector2Int

Returns

float

SearchForHighestValueWithRandomStartingPoint(Vector3, int, out Vector3)

Searches for the highest value in an area specified by a center and a radius and starts at a random point in the area to not return the same value as the result of the flood fill algorithm

```
public float SearchForHighestValueWithRandomStartingPoint(Vector3 position, int radius, out Vector3 result)
```

Parameters

`position` Vector3

`radius` int

`result` Vector3

Returns

float

SearchForLowestValueClosestToCenter(Vector2Int, int, out Vector2Int)

Searches for the lowest value in an area specified by a center and a radius and starts at a random point in the area to not return the same value as the result of the flood fill algorithm

```
public float SearchForLowestValueClosestToCenter(Vector2Int position, int radius, out
Vector2Int result)
```

Parameters

`position` Vector2Int

`radius` int

`result` Vector2Int

Returns

float

SearchForLowestValueClosestToCenter(Vector3, int, out Vector3)

Searches for the lowest value in an area specified by a center and a radius and starts at a random point in the area to not return the same value as the result of the flood fill algorithm

```
public float SearchForLowestValueClosestToCenter(Vector3 position, int radius, out
Vector3 result)
```

Parameters

`position` Vector3

`radius` int

`result` Vector3

Returns

float

SearchForLowestValueWithRandomStartingPoint(Vector2Int, int, out Vector2Int)

Searches for the lowest value in an area specified by a center and a radius and starts at a random point in the area to not return the same value as the result of the flood fill algorithm

```
public float SearchForLowestValueWithRandomStartingPoint(Vector2Int position, int radius,
out Vector2Int result)
```

Parameters

`position` Vector2Int

`radius` int

`result` Vector2Int

Returns

float

SearchForLowestValueWithRandomStartingPoint(Vector3, int, out Vector3)

Searches for the lowest value in an area specified by a center and a radius and starts at a random point in the area to not return the same value as the result of the flood fill algorithm

```
public float SearchForLowestValueWithRandomStartingPoint(Vector3 position, int radius, out
Vector3 result)
```

Parameters

`position` Vector3

`radius` int

`result` Vector3

Returns

float

SearchForValueWithRandomStartingPoint(float, SearchCondition, Vector2Int, int, out Vector2Int)

Searches for a value in an area specified by a center and a radius and starts at a random point in the area to not return the same value as the result of the flood fill algorithm

```
public bool SearchForValueWithRandomStartingPoint(float searchValue, SearchCondition searchCondition, Vector2Int position, int radius, out Vector2Int result)
```

Parameters

`searchValue` float

`searchCondition` SearchCondition

`position` Vector2Int

`radius` int

`result` Vector2Int

Returns

bool

SearchForValueWithRandomStartingPoint(float, SearchCondition, Vector3, int, out Vector3)

Searches for a value in an area specified by a center and a radius and starts at a random point in the area to not return the same value as the result of the flood fill algorithm

```
public bool SearchForValueWithRandomStartingPoint(float searchValue, SearchCondition searchCondition, Vector3 position, int radius, out Vector3 result)
```

Parameters

`searchValue` float

`searchCondition` SearchCondition

`position` Vector3

`radius` int

`result` Vector3

Returns

bool

WorldToMapPosition(Vector3)

Converts a position from unity world space to map coordinates. The y component of the vector has no effect and x and z are converted to 2d map coordinates

```
public Vector2Int WorldToMapPosition(Vector3 position)
```

Parameters

`position` Vector3

Returns

Vector2Int

Remarks

Keep in mind that the position can be outside the map depending on where the position is and the map size

Class InfluenceMapDecayer

Namespace: NoOpArmy.WiseFeline.InfluenceMaps

Assembly: APIRef.dll

Decays an influence map using the value you gave it and clamps the values to not go above/beyond specified values

```
public class InfluenceMapDecayer : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← InfluenceMapDecayer

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) ,

Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,

Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

InitialDelayInSeconds

Initial Delay In Seconds

```
[Tooltip("Initial Delay In Seconds")]
public float InitialDelayInSeconds
```

Field Value

float

decayValue

The value used for decaying

```
public float decayValue
```

Field Value

float

delayBetweenCalculations

Delay between calculations of the map

```
[Tooltip("Delay between calculations of the map")]
public float delayBetweenCalculations
```

Field Value

float

maxValue

The maximum value each map cell should be in

```
[Tooltip("The maximum value each map cell should be in")]
public float maxValue
```

Field Value

float

minValue

The minimum value each map cell should be in

```
[Tooltip("The minimum value each map cell should be in")]
public float minValue
```

Field Value

float

operation

The operation to do

```
public InfluenceMapDecayer.OperationMode operation
```

Field Value

InfluenceMapDecayer.OperationMode

Enum InfluenceMapDecayer.OperationMode

Namespace: NoOpArmy.WiseFeline.InfluenceMaps

Assembly: APIRef.dll

The operations done on the map

```
public enum InfluenceMapDecayer.OperationMode
```

Fields

`Add = 0`

Add the decayValue to the map

`Multiply = 1`

Multiply the decay value by the map

Class InfluenceMapTemplate

Namespace: NoOpArmy.WiseFeline.InfluenceMaps

Assembly: APIRef.dll

This asset type is used as a template for the stamp shape of agents which influence the amp Usually different agents based on their behavior need both different sizes and fall off curve shapes

```
[CreateAssetMenu(menuName = "NoOpArmy/Wise Feline/InfluenceMapTemplate")]
public class InfluenceMapTemplate : ScriptableObject
```

Inheritance

object ← Object ← ScriptableObject ← InfluenceMapTemplate

Inherited Members

ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) , Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) , Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) , Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,

Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) , Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) , Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) , Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

Curve

The fall off curve of the influence

```
[Tooltip("The fall off curve of the influence")]
public AnimationCurve Curve
```

Field Value

AnimationCurve

Radius

Radius of the agent in cell count

```
[Tooltip("Radius of the agent in count")]
public int Radius
```

Field Value

int

baked

The baked version of the animation curve

```
public NativeArray<float> baked
```

Field Value

NativeArray<float>

Properties

Map

```
public InfluenceMap Map { get; }
```

Property Value

InfluenceMap

MapSize

```
public int MapSize { get; }
```

Property Value

int

Methods

Init()

```
public void Init()
```

OnEnable()

```
public void OnEnable()
```

Class InfluenceMapView

Namespace: NoOpArmy.WiseFeline.InfluenceMaps

Assembly: APIRef.dll

This class allows you to calculate a map from multiple others. You could previously do this without a view using code but this makes it easier to make maps from other maps without writing a line of code

```
public class InfluenceMapView : InfluenceMapComponentBase
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← InfluenceMapComponentBase ← InfluenceMapView

Inherited Members

InfluenceMapComponentBase.is2dMap , InfluenceMapComponentBase.Map , InfluenceMapComponentBase.IsMapValid() , InfluenceMapComponentBase.AddAndClampValue(float, float, float) , InfluenceMapComponentBase.MultiplyAndClampValue(float, float, float) , InfluenceMapComponentBase.AddInfluence(int, int, InfluenceMapTemplate, float) , InfluenceMapComponentBase.AddInfluenceWithLineOfSight(int, int, InfluenceMapTemplate, InfluenceMapComponentBase, float) , InfluenceMapComponentBase.AddRectangularInfluence(int, int, int, int, float) , InfluenceMapComponentBase.PropagateInfluenceWithLineOfSight(int, int, int, AnimationCurve, InfluenceMapComponentBase, int) , InfluenceMapComponentBase.PropagateInfluence(int, int, int, AnimationCurve, int) , InfluenceMapComponentBase.WorldToMapPosition(Vector3) , InfluenceMapComponentBase.SearchForValueWithRandomStartingPoint(float, SearchCondition, Vector3, int, out Vector3) , InfluenceMapComponentBase.SearchForHighestValueWithRandomStartingPoint(Vector3, int, out Vector3) , InfluenceMapComponentBase.SearchForLowestValueWithRandomStartingPoint(Vector3, int, out Vector3) , InfluenceMapComponentBase.SearchForHighestValueClosestToCenter(Vector3, int, out Vector3) , InfluenceMapComponentBase.SearchForLowestValueClosestToCenter(Vector3, int, out Vector3) , InfluenceMapComponentBase.SearchForValueWithRandomStartingPoint(float, SearchCondition, Vector2Int, int, out Vector2Int) , InfluenceMapComponentBase.SearchForHighestValueWithRandomStartingPoint(Vector2Int, int, out Vector2Int) ,

InfluenceMapComponentBase.SearchForLowestValueWithRandomStartingPoint(Vector2Int, int, out Vector2Int) ,
InfluenceMapComponentBase.SearchForHighestValueClosestToCenter(Vector2Int, int, out Vector2Int) ,
InfluenceMapComponentBase.SearchForLowestValueClosestToCenter(Vector2Int, int, out Vector2Int) ,
InfluenceMapComponentBase.GetCellValue(int, int) ,
InfluenceMapComponentBase.GetCellValue(Vector3) ,
InfluenceMapComponentBase.MapToWorldPosition(int, int) , MonoBehaviour.IsInvoking() ,
MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) ,
MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) ,
MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) ,
MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) ,
MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) ,
MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) ,
MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) ,
MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart ,
MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled ,
Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() ,
Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) ,
Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) ,
Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) ,
Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) ,
Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) ,
Component.GetComponentsInChildren<T>(bool, List<T>) ,
Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) ,
Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) ,
Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() ,
Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) ,
Component.GetComponentsInParent<T>(bool) ,
Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() ,
Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) ,
Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() ,
Component.GetComponentIndex() , Component.CompareTag(string) ,
Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,

Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,

Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

InitialDelayInSeconds

Initial Delay In Seconds

```
[Tooltip("Initial Delay In Seconds")]
public float InitialDelayInSeconds
```

Field Value

float

cellSize

Cell size of the map

```
[Tooltip("Cell size of the map")]
public float cellSize
```

Field Value

float

delayBetweenCalculations

Delay between calculations of the map

```
public float delayBetweenCalculations
```

Field Value

float

firstMapName

The first map to start operations from

```
public string firstMapName
```

Field Value

string

height

Height of the map in cells

```
[Tooltip("Height of the map in cells")]
public int height
```

Field Value

int

mapName

///MapName

```
public string mapName
```

Field Value

string

operations

The operations which will be done on the map consecutively

```
public List<InfluenceMapView.ViewOperation> operations
```

Field Value

List <InfluenceMapView.ViewOperation>

updatePositionAutomatically

If the object moves, you should check this and set the delay between calculations based on speed and the accuracy you require

```
[Tooltip("If the object moves, you should check this and set the delay between calculations
based on speed and the accuracy you require")]
public bool updatePositionAutomatically
```

Field Value

bool

width

Width of the map in cell count

```
[Tooltip("Width of the map in cell count")]
public int width
```

Field Value

int

Class InfluenceMapView.ViewOperation

Namespace: NoOpArmy.WiseFeline.InfluenceMaps

Assembly: APIRef.dll

```
[Serializable]
public class InfluenceMapView.ViewOperation
```

Inheritance

object ← InfluenceMapView.ViewOperation

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

additionalArgument

The additional argument needed by the operation like the strength of the second map or the value to multiply with in case of MultiplyByValue

```
[Tooltip("The additional argument needed by the operation like the strength of the second
map or the value to multiply with in case of MultiplyByValue")]
public float additionalArgument
```

Field Value

float

mapName

```
public string mapName
```

Field Value

string

operation

```
public InfluenceMapView.ViewOperation.Operation operation
```

Field Value

InfluenceMapView.ViewOperation.Operation

Enum InfluenceMapView.ViewOperation.Operation

Namespace: NoOpArmy.WiseFeline.InfluenceMaps

Assembly: APIRef.dll

```
public enum InfluenceMapView.ViewOperation.Operation
```

Fields

`Add = 0`

`AddInverse = 1`

`AddWithStrength = 7`

`Inverse = 6`

`Multiply = 3`

`MultiplyByValue = 4`

`MultiplyWithStrength = 9`

`Normalize = 5`

`Subtract = 2`

`SubtractWithStrength = 8`

Class InfluencerAgent

Namespace: NoOpArmy.WiseFeline.InfluenceMaps

Assembly: APIRef.dll

This component can be attached to an agent so it influences the map

```
public class InfluencerAgent : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← InfluencerAgent

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) ,

Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,

Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

behavior

The behavior of the map. You can use a single template, a list of them with the ability to change the index of the current one or directly use a radius and a curve

```
[Tooltip("The behavior of the map. You can use a single template, a list of them with the ability to change the index of the current one or directly use a radius and a curve")]
public InfluencerAgent.Behavior behavior
```

Field Value

InfluencerAgent.Behavior

directCurve

The curve to use for the direct value behavior

```
public AnimationCurve directCurve
```

Field Value

directRadius

The radius to use for the direct value behavior

```
public int directRadius
```

Field Value

int

mapComponent

The map that this agent influences

```
[Header("The map to influence")]
[Tooltip("The map that this agent influences")]
public InfluenceMapComponent mapComponent
```

Field Value

InfluenceMapComponent

mapIndex

The index in the list to use when TemplateList behavior is selected. It should be between 0 and list's count - 1

```
public int mapIndex
```

Field Value

int

mapName

The name of the map which should be read from the InfluenceMapCollection This will only be used if the mapComponent is null

```
[Tooltip("The name of the map which should be read from the InfluenceMapCollection. This
will only be used if the mapComponent is null.")]
public string mapName
```

Field Value

string

mapTemplatesList

The list of templates when the TemplateList behavior is selected

```
public List<InfluenceMapTemplate> mapTemplatesList
```

Field Value

List<InfluenceMapTemplate>

myMapTemplate

The template that the agent uses for its influence stamp shape

```
[Tooltip("The template that the agent uses for its influence stamp shape")]
public InfluenceMapTemplate myMapTemplate
```

Field Value

InfluenceMapTemplate

obstaclesMapComponent

The map that this agent considers obstacles for line of sight

```
[Header("The obstacles map for line of sight")]
[Tooltip("The map that this agent considers obstacles for line of sight")]
public InfluenceMapComponent obstaclesMapComponent
```

Field Value

InfluenceMapComponent

obstaclesMapName

The name of the obstacles map which should be read from the InfluenceMapCollection This will only be used if the obstacleMapComponent is null

```
[Tooltip("The name of the obstacles map which should be read from the
InfluenceMapCollection. This will only be used if the mapComponent is null.")]
public string obstaclesMapName
```

Field Value

string

shouldRemoveInfluenceWhenMoving

Should the agent removes its influence from the old place when moving. You might not want this to put breadcrumbs

```
[Tooltip("Should the agent removes its influence from the old place when moving. You might
not want this to put breadcrumbs")]
public bool shouldRemoveInfluenceWhenMoving
```

Field Value

bool

updateInterval

The update interval of the agent. 0.5 means the agent updates its position on the map two times a second.

```
[Tooltip("The update interval of the agent. 0.5 means the agent updates its position on the map two times a second.")]
public float updateInterval
```

Field Value

float

updatePositionAutomatically

Checking this will cause the object influence to be updated in the influence map automatically. If the object moves, you should check this and set the interval based on its speed and the accuracy you require

```
[Tooltip("If the object moves, you should check this and set the interval based on its speed and the accuracy you require")]
public bool updatePositionAutomatically
```

Field Value

bool

useLineOfSightWhenCalculatingInfluence

Set this to true if your agents need to consider obstacles and their line of sight when adding their influence. Usually only range units in shooter games might need this.

```
[Tooltip("Set this to true if your agents need to consider obstacles and their line of sight when adding their influence. Usually only range units in shooter games might need this.")]
public bool useLineOfSightWhenCalculatingInfluence
```

Field Value

bool

Properties

AgentMap

Getter for the agent's map

```
public InfluenceMapComponentBase AgentMap { get; set; }
```

Property Value

InfluenceMapComponentBase

ObstaclesMap

```
public InfluenceMapComponentBase ObstaclesMap { get; set; }
```

Property Value

InfluenceMapComponentBase

Methods

SetMap(InfluenceMapComponent)

Use this to set the map later on if it cannot be set automatically at Start()

```
public void SetMap(InfluenceMapComponent map)
```

Parameters

`map` InfluenceMapComponent

Exceptions

InvalidOperationException

SetObstaclesMap(InfluenceMapComponent)

Use this to set the obstacles map later on if it cannot be set automatically at Start()

```
public void SetObstaclesMap(InfluenceMapComponent map)
```

Parameters

`map` InfluenceMapComponent

Exceptions

InvalidOperationException

Enum InfluencerAgent.Behavior

Namespace: NoOpArmy.WiseFeline.InfluenceMaps

Assembly: APIRef.dll

The way that this map will add its influence

```
public enum InfluencerAgent.Behavior
```

Fields

`DirectValue = 2`

Uses a curve and radius directly This is much slower than the two other approaches since it needs to evaluate the curves but templates evaluate their curves only once at initialization Use this only if using templates is cost prohibitive in terms of memory.

`SingleTemplate = 0`

Uses a single template defined in myMapTemplate

`TemplateList = 1`

Uses a list of tempates defined in templatesList and mapIndex

Class RectangularInfluencerAgent

Namespace: NoOpArmy.WiseFeline.InfluenceMaps

Assembly: APIRef.dll

This component can be attached to an agent so it influences the map

```
public class RectangularInfluencerAgent : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← RectangularInfluencerAgent

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) ,

Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,

Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

TemplateHeight

Height of the rectangle

```
[Tooltip("Height of the rectangle")]
public int TemplateHeight
```

Field Value

int

TemplateWidth

Width of the rectangle

```
[Tooltip("Width of the rectangle")]
public int TemplateWidth
```

Field Value

int

Value

The value to add for this object.

```
public float Value
```

Field Value

float

agentPositionOffset

The offset applied to agent position to calculate starting x and y of the rectangle which then we move to right and up to fill.

```
[Tooltip("The offset applied to agent position to calculate starting x and y of the
rectangle which then we move to right and up to fill.")]
public Vector2Int agentPositionOffset
```

Field Value

Vector2Int

gizmoColor

```
public Color gizmoColor
```

Field Value

Color

mapComponent

The map that this agent influences

```
[Tooltip("The map that this agent influences")]
public InfluenceMapComponent mapComponent
```

Field Value

InfluenceMapComponent

mapName

The name of the map which should be read from the InfluenceMapCollection This will only be used if the mapComponent is null

```
[Tooltip("This will only be used if the mapComponent is null.")]
public string mapName
```

Field Value

string

shouldDrawGizmos

```
public bool shouldDrawGizmos
```

Field Value

bool

updateInterval

The update interval of the agent. 0.5 means the agent updates its position on the map two times a second.

```
[Tooltip("The update interval of the agent. 0.5 means the agent updates its position on the
map two times a second.")]
public float updateInterval
```

Field Value

float

updatePositionAutomatically

Checking this will cause the object influence to be updated in the influence map automatically. If the object moves, you should check this and set the interval based on its speed and the accuracy you require

```
[Tooltip("If the object moves, you should check this and set the interval based on its speed and the accuracy you require")]
public bool updatePositionAutomatically
```

Field Value

bool

Properties

AgentMap

```
public InfluenceMapComponentBase AgentMap { get; set; }
```

Property Value

InfluenceMapComponentBase

Methods

SetMap(InfluenceMapComponent)

Use this to set the map later on if it cannot be set automatically at Start()

```
public void SetMap(InfluenceMapComponent map)
```

Parameters

`map` InfluenceMapComponent

Exceptions

InvalidOperationException

Enum SearchCondition

Namespace: NoOpArmy.WiseFeline.InfluenceMaps

Assembly: APIRef.dll

The stop condition for search

```
public enum SearchCondition : byte
```

Fields

`Equal = 0`

`Greater = 1`

`GreaterOrEqual = 4`

`Less = 2`

`LessOrEqual = 5`

`NotEqual = 3`

Namespace NoOpArmy.WiseFeline.InfluenceMaps.Demo

Classes

AttackAttractor

DefendAttractor

Class AttackAttractor

Namespace: NoOpArmy.WiseFeline.InfluenceMaps.Demo

Assembly: APIRef.dll

```
public class AttackAttractor : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← AttackAttractor

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) , Component.SendMessageUpwards(string, object, SendMessageOptions) , Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,

Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,

Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

mapName

```
public string mapName
```

Field Value

string

Class DefendAttractor

Namespace: NoOpArmy.WiseFeline.InfluenceMaps.Demo

Assembly: APIRef.dll

```
public class DefendAttractor : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← DefendAttractor

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) , Component.SendMessageUpwards(string, object, SendMessageOptions) , Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,

Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,

Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

attractor

```
public Transform attractor
```

Field Value

Transform

mapName

```
public string mapName
```

Field Value

string

Namespace NoOpArmy.WiseFeline.InfluenceMaps.Sample

Classes

FleeFromEnemies

This component uses the influence map to flee from enemies and always tries to put the player in a position with less enemies

MoveToRandomPosition

This component just moves an object to a random position so it can be used as a test enemy

Class FleeFromEnemies

Namespace: NoOpArmy.WiseFeline.InfluenceMaps.Sample

Assembly: APIRef.dll

This component uses the influence map to flee from enemies and always tries to put the player in a position with less enemies

```
public class FleeFromEnemies : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← FleeFromEnemies

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) ,

Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,

Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

EnemiesMap

```
public InfluenceMapComponent EnemiesMap
```

Field Value

InfluenceMapComponent

flee

```
public bool flee
```

Field Value

bool

speed

```
public float speed
```

Field Value

float

Methods

Start()

```
public IEnumerator Start()
```

Returns

IEnumerator

Class MoveToRandomPosition

Namespace: NoOpArmy.WiseFeline.InfluenceMaps.Sample

Assembly: APIRef.dll

This component just moves an object to a random position so it can be used as a test enemy

```
public class MoveToRandomPosition : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← MoveToRandomPosition

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) ,

Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,

Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

area

```
public Rect area
```

Field Value

Rect

Namespace NoOpArmy.WiseFeline.Remembrance

Classes

BrainMemoryRecorder

CircularMemoryContainer<T>

EmotionCollection

EmotionCollection.EmotionData

EmotionComponent

EmotionComponent.UIEmotionData

EmotionList

Memory
: This component allows an agent or any other object to remember things which happend and query about them

MemoryContainer<T>

MemoryData
: This data structure defines a memory entry

SimpleMemory

Enums

EmotionCollection.CombinationType

Class BrainMemoryRecorder

Namespace: NoOpArmy.WiseFeline.Remembrance

Assembly: APIRef.dll

```
public class BrainMemoryRecorder : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← BrainMemoryRecorder

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) , Component.SendMessageUpwards(string, object, SendMessageOptions) , Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,

Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,

Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Class CircularMemoryContainer<T>

Namespace: NoOpArmy.WiseFeline.Remembrance

Assembly: APIRef.dll

```
[Serializable]
public class CircularMemoryContainer<T> where T : MemoryData
```

Type Parameters

T

Inheritance

object ← CircularMemoryContainer<T>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

CircularMemoryContainer(int)

```
public CircularMemoryContainer(int maxCount)
```

Parameters

maxCount int

Fields

memories

```
public Queue<T> memories
```

Field Value

Queue<T>

Methods

AddMemory(T)

```
public T AddMemory(T memory)
```

Parameters

`memory` T

Returns

T

GetAllMemories()

```
public Queue<T> GetAllMemories()
```

Returns

Queue<T>

GetClosestMemoryToTime(Predicate<T>, float, bool)

```
public T GetClosestMemoryToTime(Predicate<T> pred, float time, bool useEndTime = true)
```

Parameters

`pred` Predicate<T>

`time` float

`useEndTime` bool

Returns

T

GetClosestMemoryToTime(float, bool)

```
public T GetClosestMemoryToTime(float time, bool useEndTime = true)
```

Parameters

`time` float

`useEndTime` bool

Returns

T

GetClosestMemoryToTime(string, float, bool)

```
public T GetClosestMemoryToTime(string actionId, float time, bool useEndTime = true)
```

Parameters

`actionId` string

`time` float

`useEndTime` bool

Returns

T

GetClosestMemoryToTime(string, Object, float, bool)

```
public T GetClosestMemoryToTime(string actionId, Object target, float time, bool useEndTime
= true)
```

Parameters

`actionId` string

`target` Object

`time` float

`useEndTime` bool

Returns

T

GetClosestMemoryToTime(Type, float, bool)

```
public T GetClosestMemoryToTime(Type type, float time, bool useEndTime = true)
```

Parameters

`type` Type

`time` float

`useEndTime` bool

Returns

T

GetClosestMemoryToTime(Type, Object, float, bool)

```
public T GetClosestMemoryToTime(Type type, Object target, float time, bool useEndTime
= true)
```

Parameters

`type` Type

`target` Object

`time` float

`useEndTime` bool

Returns

T

GetClosestMemoryToTime(Type, Object, Object, float, bool)

```
public T GetClosestMemoryToTime(Type type, Object initiator, Object target, float time, bool
useEndTime = true)
```

Parameters

`type` Type

`initiator` Object

`target` Object

`time` float

`useEndTime` bool

Returns

T

GetClosestMemoryToTime(Object, float, bool)

```
public T GetClosestMemoryToTime(Object initiator, float time, bool useEndTime = true)
```

Parameters

`initiator` Object

`time` float

`useEndTime` bool

Returns

T

GetClosestMemoryToTime(Object, Object, float, bool)

```
public T GetClosestMemoryToTime(Object initiator, Object target, float time, bool useEndTime
= true)
```

Parameters

`initiator` Object

`target` Object

`time` float

`useEndTime` bool

Returns

T

GetMemoriesByFilter(Predicate<T>)

```
public List<T> GetMemoriesByFilter(Predicate<T> pred)
```

Parameters

`pred` Predicate<T>

Returns

List <T>

Class EmotionCollection

Namespace: NoOpArmy.WiseFeline.Remembrance

Assembly: APIRef.dll

```
[CreateAssetMenu(menuName = "NoOpArmy/Wise Feline/Emotion collection")]
public class EmotionCollection : ScriptableObject
```

Inheritance

object ← Object ← ScriptableObject ← EmotionCollection

Inherited Members

ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) , Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) , Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) , Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) , Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) , Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) , Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,

Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

emotions

```
public List<EmotionCollection.EmotionData> emotions
```

Field Value

List<EmotionCollection.EmotionData>

Enum EmotionCollection.CombinationType

Namespace: NoOpArmy.WiseFeline.Remembrance

Assembly: APIRef.dll

```
public enum EmotionCollection.CombinationType
```

Fields

`Average = 2`

`Max = 1`

`Min = 0`

Class EmotionCollection.EmotionData

Namespace: NoOpArmy.WiseFeline.Remembrance

Assembly: APIRef.dll

```
[Serializable]
public class EmotionCollection.EmotionData
```

Inheritance

object ← EmotionCollection.EmotionData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

name

```
public string name
```

Field Value

string

type

```
public EmotionCollection.CombinationType type
```

Field Value

EmotionCollection.CombinationType

Class EmotionComponent

Namespace: NoOpArmy.WiseFeline.Remembrance

Assembly: APIRef.dll

```
public class EmotionComponent : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← EmotionComponent

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) , Component.SendMessageUpwards(string, object, SendMessageOptions) , Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,

Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,

Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

emotions

Per object emotions of this agent

```
public Dictionary<GameObject, EmotionList> emotions
```

Field Value

Dictionary <GameObject, EmotionList>

generalEmotion

General emotions of this agent

```
public EmotionList generalEmotion
```

Field Value

EmotionList

generalEmotionDefinition

General emotions collection

```
[Tooltip("General emotions collection")]
public EmotionCollection generalEmotionDefinition
```

Field Value

EmotionCollection

objectEmotionDefinition

Per object emotions collection

```
[Tooltip("Per object emotions collection")]
public EmotionCollection objectEmotionDefinition
```

Field Value

EmotionCollection

Methods

AddEmotion(int, float)

Adds a value to a general emotion

```
public void AddEmotion(int index, float value)
```

Parameters

`index` int

`value` float

AddEmotion(GameObject, int, float)

Adds a value to an emotion

```
public void AddEmotion(GameObject go, int index, float value)
```

Parameters

`go` GameObject

`index` int

`value` float

Effect(GameObject)

Makes emotions of two objects toward each other affect the other one's emotion. So if you are angry at me, I'll be angry at you as well.

```
public void Effect(GameObject other)
```

Parameters

`other` GameObject

Exceptions

ArgumentException

GetAll()

```
public List<EmotionComponent.UIEmotionData> GetAll()
```

Returns

List<EmotionComponent.UIEmotionData>

GetEmotion(int)

Gets a general emotion's value

```
public float GetEmotion(int index)
```

Parameters

`index` int

Returns

float

GetEmotion(GameObject, int)

Returns an emotion for an object

```
public float GetEmotion(GameObject go, int index)
```

Parameters

`go` GameObject

`index` int

Returns

float

GetEmotionsForObject(GameObject)

Returns the emotions this object has for another

```
public EmotionList GetEmotionsForObject(GameObject go)
```

Parameters

`go` GameObject

Returns

EmotionList

GetGeneralEmotionIndex(string)

Gets the index of a general emotion based on the name

```
public int GetGeneralEmotionIndex(string name)
```

Parameters

`name` string

Returns

int

GetObjectEmotionIndex(string)

Gets the index of an emotion based on the name for objects

```
public int GetObjectEmotionIndex(string name)
```

Parameters

`name` string

Returns

int

RemoveEmotionsForObject(GameObject)

Removes all emotions related to an object.

```
public void RemoveEmotionsForObject(GameObject go)
```

Parameters

`go` GameObject

Remarks

This is good for not having memory leaks when an NPC meets lots of objects which get destroyed or he/she never meets again.

SetEmotion(int, float)

Sets a general emotion

```
public void SetEmotion(int index, float value)
```

Parameters

`index` int

`value` float

SetEmotion(GameObject, int, float)

Sets the emotion for an object

```
public void SetEmotion(GameObject go, int index, float value)
```

Parameters

`go` GameObject

`index` int

`value` float

Class EmotionComponent.UIEmotionData

Namespace: NoOpArmy.WiseFeline.Remembrance

Assembly: APIRef.dll

```
public class EmotionComponent.UIEmotionData
```

Inheritance

object ← EmotionComponent.UIEmotionData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

emotion

```
public EmotionList emotion
```

Field Value

EmotionList

owner

```
public GameObject owner
```

Field Value

GameObject

Class EmotionList

Namespace: NoOpArmy.WiseFeline.Remembrance

Assembly: APIRef.dll

```
public class EmotionList
```

Inheritance

object ← EmotionList

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

EmotionList()

```
public EmotionList()
```

Fields

emotionNames

```
public List<string> emotionNames
```

Field Value

List<string>

emotions

```
public List<float> emotions
```

Field Value

List<float>

Methods

AddEmotion(int, float)

```
public void AddEmotion(int index, float value)
```

Parameters

`index` int

`value` float

AddEmotion(string, float)

```
public void AddEmotion(string name, float value)
```

Parameters

`name` string

`value` float

GetEmotion(int)

```
public float GetEmotion(int index)
```

Parameters

`index` int

Returns

float

GetIndexFromName(string)

```
public int GetIndexFromName(string name)
```

Parameters

`name` string

Returns

int

SetEmotion(int, float)

```
public void SetEmotion(int index, float value)
```

Parameters

`index` int

`value` float

Class Memory

Namespace: NoOpArmy.WiseFeline.Remembrance

Assembly: APIRef.dll

This component allows an agent or any other object to remember things which happend and query about them

```
public class Memory : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← Memory

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) ,

Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,

Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Fields

everLastingMemories

```
public MemoryContainer<MemoryData> everLastingMemories
```

Field Value

MemoryContainer<MemoryData>

everLastingMemoriesCount

```
public int everLastingMemoriesCount
```

Field Value

int

memories

```
public MemoryContainer<MemoryData> memories
```

Field Value

MemoryContainer<MemoryData>

normalMemoriesCount

```
public int normalMemoriesCount
```

Field Value

int

Methods

AddMemory(MemoryData, bool)

Adds a new memory

```
public void AddMemory(MemoryData memory, bool isEverLasting = false)
```

Parameters

`memory` MemoryData

`isEverLasting` bool

Awake()

```
protected void Awake()
```

CreateAndGetListOfAllMemories()

Gets all memories, both the ones which last for ever and those which can be forgotten/removed due to capacity being reached. This allocates a new list per call. Prefer using GetAllMemories() which returns a tuple

```
public List<MemoryData> CreateAndGetListOfAllMemories()
```

Returns

List <MemoryData>

FindMostRecentMemoryEndTime(string)

Finds the most recent memory with a specific action id This finds finished memories and not in progress ones

```
public float FindMostRecentMemoryEndTime(string guid)
```

Parameters

`guid` string

Returns

float

FindMostRecentMemoryEndTime(string, Object)

Finds the most recent memory with a specific action id and target This finds finished memories and not in progress ones

```
public float FindMostRecentMemoryEndTime(string guid, Object target)
```

Parameters

`guid` string

`target` Object

Returns

float

FindMostRecentMemoryEndTime(Type)

Finds the most recent memory with a specific action type This finds finished memories and not in progress ones

```
public float FindMostRecentMemoryEndTime(Type type)
```

Parameters

`type` Type

Returns

float

GetAllEverLastingMemories()

Gets all memories which will never be forgotten and removed

```
public List<MemoryData> GetAllEverLastingMemories()
```

Returns

List<MemoryData>

GetAllMemories()

```
public (List<MemoryData> normal, List<MemoryData> everLasting) GetAllMemories()
```

Returns

(List<MemoryData> normal, List<MemoryData> everLasting)

GetAllMemoriesByFilter(Predicate<MemoryData>)

```
public (List<MemoryData> normal, List<MemoryData> everLasting)
GetAllMemoriesByFilter(Predicate<MemoryData> pred)
```

Parameters

`pred` Predicate<MemoryData>

Returns

(List<MemoryData> normal, List<MemoryData> everLasting)

GetAllNormalMemories()

Gets all memories which can be forgotten

```
public List<MemoryData> GetAllNormalMemories()
```

Returns

List<MemoryData>

GetClosestMemoryToTime(Predicate<MemoryData>, float)

Gets the closest memory which matches a condition

```
public MemoryData GetClosestMemoryToTime(Predicate<MemoryData> pred, float time)
```

Parameters

`pred` Predicate<MemoryData>

`time` float

Returns

MemoryData

Remarks

Pass Time.time to getting the closest to now and pass 0 to get the earliest

GetClosestMemoryToTime(float)

Gets the closest memory to this time.

```
public MemoryData GetClosestMemoryToTime(float time)
```

Parameters

`time` float

Returns

MemoryData

Remarks

Pass Time.time to getting the closest to now and pass 0 to get the earliest

GetClosestMemoryToTime(string, float)

Get the closest memory with the specified criteria

```
public MemoryData GetClosestMemoryToTime(string actionId, float time)
```

Parameters

`actionId` string

`time` float

Returns

MemoryData

Remarks

Pass Time.time to getting the closest to now and pass 0 to get the earliest

GetClosestMemoryToTime(string, Object, float)

Gets the closest memory with the specified criteria

```
public MemoryData GetClosestMemoryToTime(string actionId, Object target, float time)
```

Parameters

`actionId` string

`target` Object

`time` float

Returns

MemoryData

Remarks

Pass Time.time to getting the closest to now and pass 0 to get the earliest

GetClosestMemoryToTime(Type, float)

Gets the memory closest to the time passed which is of the specified type

```
public MemoryData GetClosestMemoryToTime(Type type, float time)
```

Parameters

`type` Type

`time` float

Returns

MemoryData

Remarks

Pass Time.time to getting the closest to now and pass 0 to get the earliest

GetClosestMemoryToTime(Type, Object, float)

Gets the memory of a specific type and with a specific target which is closest to the passed time.

```
public MemoryData GetClosestMemoryToTime(Type type, Object target, float time)
```

Parameters

`type` Type

`target` Object

`time` float

Returns

MemoryData

Remarks

Pass Time.time to getting the closest to now and pass 0 to get the earliest

GetClosestMemoryToTime(Type, Object, Object, float)

Gets the memory closest to time with the specified time and target and initiator

```
public MemoryData GetClosestMemoryToTime(Type type, Object initiator, Object target,
float time)
```

Parameters

`type` Type

`initiator` Object

`target` Object

`time` float

Returns

MemoryData

Remarks

Pass Time.time to getting the closest to now and pass 0 to get the earliest

GetClosestMemoryToTime(Object, float)

Get the closest memory with the specified initiator. This is good for checking if somebody done anything in a while/what is the last thing he/she has done.

```
public MemoryData GetClosestMemoryToTime(Object initiator, float time)
```

Parameters

`initiator` Object

`time` float

Returns

MemoryData

GetClosestMemoryToTime(Object, Object, float)

Gets the closest memory with the specified criteria

```
public MemoryData GetClosestMemoryToTime(Object initiator, Object target, float time)
```

Parameters

`initiator` Object

`target` Object

`time` float

Returns

MemoryData

Remarks

Pass Time.time to getting the closest to now and pass 0 to get the earliest

GetCurrentActionRuntime()

Gets the amount of time passed since the last action got started

```
public float GetCurrentActionRuntime()
```

Returns

float

GetEverLastingMemoriesByFilter(Predicate<MemoryData>)

```
public List<MemoryData> GetEverLastingMemoriesByFilter(Predicate<MemoryData> pred)
```

Parameters

`pred` Predicate<MemoryData>

Returns

List<MemoryData>

GetMemoryAtIndex(int, bool)

Gets the memory at index

```
public MemoryData GetMemoryAtIndex(int index, bool isEverLasting)
```

Parameters

`index` int

`isEverLasting` bool

Returns

MemoryData

GetNormalMemoriesByFilter(Predicate<MemoryData>)

```
public List<MemoryData> GetNormalMemoriesByFilter(Predicate<MemoryData> pred)
```

Parameters

`pred` Predicate<MemoryData>

Returns

List<MemoryData>

RemoveMemory(MemoryData)

Removes a memory

```
public void RemoveMemory(MemoryData memory)
```

Parameters

`memory` MemoryData

Update()

```
protected void Update()
```

Events

OnMemoriesForgotten

```
public event Action<List<MemoryData>> OnMemoriesForgotten
```

Event Type

Action<List<MemoryData>>

OnMemoryAdded

```
public event Action<MemoryData> OnMemoryAdded
```

Event Type

Action<MemoryData>

OnMemoryRemoved

```
public event Action<MemoryData> OnMemoryRemoved
```

Event Type

Action<MemoryData>

Class MemoryContainer<T>

Namespace: NoOpArmy.WiseFeline.Remembrance

Assembly: APIRef.dll

```
[Serializable]
public class MemoryContainer<T> where T : MemoryData
```

Type Parameters

T

Inheritance

object ← MemoryContainer<T>

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MemoryContainer(int)

```
public MemoryContainer(int maxCount)
```

Parameters

maxCount int

Fields

memories

```
public List<T> memories
```

Field Value

List<T>

Methods

AddMemory(T)

```
public T AddMemory(T memory)
```

Parameters

`memory` T

Returns

T

ForGetOldThings(float)

```
public List<T> ForGetOldThings(float currentTime)
```

Parameters

`currentTime` float

Returns

List<T>

GetAllMemories()

```
public List<T> GetAllMemories()
```

Returns

List<T>

GetClosestMemoryToTime(Predicate<T>, float, bool)

```
public T GetClosestMemoryToTime(Predicate<T> pred, float time, bool useEndTime = true)
```

Parameters

`pred` Predicate<T>

`time` float

`useEndTime` bool

Returns

T

GetClosestMemoryToTime(float, bool)

```
public T GetClosestMemoryToTime(float time, bool useEndTime = true)
```

Parameters

`time` float

`useEndTime` bool

Returns

T

GetClosestMemoryToTime(string, float, bool)

```
public T GetClosestMemoryToTime(string actionId, float time, bool useEndTime = true)
```

Parameters

`actionId` string

`time` float

`useEndTime` bool

Returns

T

GetClosestMemoryToTime(string, Object, float, bool)

```
public T GetClosestMemoryToTime(string actionId, Object target, float time, bool useEndTime
= true)
```

Parameters

`actionId` string

`target` Object

`time` float

`useEndTime` bool

Returns

T

GetClosestMemoryToTime(Type, float, bool)

```
public T GetClosestMemoryToTime(Type type, float time, bool useEndTime = true)
```

Parameters

`type` Type

time float

useEndTime bool

Returns

T

GetClosestMemoryToTime(Type, Object, float, bool)

```
public T GetClosestMemoryToTime(Type type, Object target, float time, bool useEndTime
= true)
```

Parameters

type Type

target Object

time float

useEndTime bool

Returns

T

GetClosestMemoryToTime(Type, Object, Object, float, bool)

```
public T GetClosestMemoryToTime(Type type, Object initiator, Object target, float time, bool
useEndTime = true)
```

Parameters

type Type

initiator Object

target Object

time float

useEndTime bool

Returns

T

GetClosestMemoryToTime(Object, float, bool)

```
public T GetClosestMemoryToTime(Object initiator, float time, bool useEndTime = true)
```

Parameters

initiator Object

time float

useEndTime bool

Returns

T

GetClosestMemoryToTime(Object, Object, float, bool)

```
public T GetClosestMemoryToTime(Object initiator, Object target, float time, bool useEndTime
= true)
```

Parameters

initiator Object

target Object

time float

useEndTime bool

Returns

T

GetMemoriesByFilter(Predicate<T>)

Gets all memories which pass a filter

```
public List<T> GetMemoriesByFilter(Predicate<T> pred)
```

Parameters

`pred` Predicate<T>

Returns

List<T>

GetMemoryAtIndex(int)

```
public T GetMemoryAtIndex(int index)
```

Parameters

`index` int

Returns

T

RemoveMemory(T)

```
public void RemoveMemory(T memory)
```

Parameters

```
memory T
```

Class MemoryData

Namespace: NoOpArmy.WiseFeline.Remembrance

Assembly: APIRef.dll

This data structure defines a memory entry

```
[Serializable]
public class MemoryData
```

Inheritance

object ← MemoryData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

MemoryData(int, float, float, Vector3, Type, Object, Object, string, object, float)

Constructs a memory data. If endTime is 0, you should call End later to say when the action is done

```
public MemoryData(int memoryType, float startTime, float forgetAfterSeconds, Vector3
position, Type actionType, Object initiator, Object target, string actionId, object
additionalData, float endTime = 0)
```

Parameters

`memoryType` int

`startTime` float

`forgetAfterSeconds` float

`position` Vector3

`actionType` Type

`initiator` Object

`target` Object

`actionId` string

`additionalData` object

`endTime` float

Properties

ActionId

The unique id of the ction happenning to identify instances of a System.Type

```
public string ActionId { get; set; }
```

Property Value

string

ActionType

The Type of the class initiaing this action. The initiator is the GameObject/component which caused this and this is the type of utility AI action or state machine state or ... which caused this

```
public Type ActionType { get; set; }
```

Property Value

Type

AdditionalData

Any additional data you need for this memory to be casted based on memoryType

```
public object AdditionalData { get; set; }
```

Property Value

object

EndTime

The time this finished

```
public float EndTime { get; set; }
```

Property Value

float

ForgetAfterSeconds

The number of seconds which this will be forgotten after. This will never be forgotten if the value is less than or equal to 0

```
public float ForgetAfterSeconds { get; set; }
```

Property Value

float

Initiator

The object which caused this memory to happen. Can be null.

```
public Object Initiator { get; set; }
```

Property Value

Object

IsEnded

Is this memory ended or still happenning

```
public bool IsEnded { get; }
```

Property Value

bool

MemoryType

Type of the memory, usually an enum in your project which help you cast any additional data or in general find out what memory type is this

```
public int MemoryType { get; set; }
```

Property Value

int

OnEnded

Called when this memory is ended and the agent is moving to another action

```
public Action OnEnded { get; set; }
```

Property Value

Action

OnStarted

Called when this memory is created and started

```
public Action OnStarted { get; set; }
```

Property Value

Action

Position

The position this memory happend

```
public Vector3 Position { get; set; }
```

Property Value

Vector3

StartTime

The time this started

```
public float StartTime { get; set; }
```

Property Value

float

Target

The target of the memory. Can be null.

```
public Object Target { get; set; }
```

Property Value

Methods

End(float)

Ends the action. Like if memory is combat with giant then this should be called when the battle ends.

```
public void End(float time)
```

Parameters

`time` float

GetTime(bool)

```
public float GetTime(bool useEndTime = true)
```

Parameters

`useEndTime` bool

Returns

float

Class SimpleMemory

Namespace: NoOpArmy.WiseFeline.Remembrance

Assembly: APIRef.dll

```
public class SimpleMemory : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← SimpleMemory

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) , Component.SendMessageUpwards(string, object, SendMessageOptions) , Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,

Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,

Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

everLastingMemories

```
public CircularMemoryContainer<MemoryData> everLastingMemories
```

Field Value

CircularMemoryContainer<MemoryData>

everLastingMemoriesCount

```
public int everLastingMemoriesCount
```

Field Value

int

memories

```
public CircularMemoryContainer<MemoryData> memories
```

Field Value

CircularMemoryContainer<MemoryData>

normalMemoriesCount

```
public int normalMemoriesCount
```

Field Value

int

Methods

AddMemory(MemoryData, bool)

```
public void AddMemory(MemoryData memory, bool isEverLasting = false)
```

Parameters

`memory` MemoryData

`isEverLasting` bool

CreateAndGetListOfAllMemories()

This call allocates a new queue each time you call it to merge the two queues. Prefer using GetAllMemories() which returns a tuple.

```
public Queue<MemoryData> CreateAndGetListOfAllMemories()
```

Returns

Queue<MemoryData>

FindMostRecentMemoryEndTime(string)

```
public float FindMostRecentMemoryEndTime(string guid)
```

Parameters

`guid` string

Returns

float

FindMostRecentMemoryEndTime(string, Object)

```
public float FindMostRecentMemoryEndTime(string guid, Object target)
```

Parameters

`guid` string

`target` Object

Returns

float

FindMostRecentMemoryEndTime(Type)

```
public float FindMostRecentMemoryEndTime(Type type)
```

Parameters

`type` Type

Returns

float

GetAllEverLastingMemories()

```
public Queue<MemoryData> GetAllEverLastingMemories()
```

Returns

Queue<MemoryData>

GetAllMemories()

```
public (Queue<MemoryData> normal, Queue<MemoryData> everLasting) GetAllMemories()
```

Returns

(Queue<MemoryData> normal, Queue<MemoryData> everLasting)

GetAllMemoriesByFilter(Predicate<MemoryData>)

```
public (List<MemoryData> normal, List<MemoryData> everLasting)
GetAllMemoriesByFilter(Predicate<MemoryData> pred)
```

Parameters

`pred` Predicate<MemoryData>

Returns

(List<MemoryData> normal, List<MemoryData> everLasting)

GetAllNormalMemories()

```
public Queue<MemoryData> GetAllNormalMemories()
```

Returns

Queue<MemoryData>

GetClosestMemoryToTime(Predicate<MemoryData>, float)

```
public MemoryData GetClosestMemoryToTime(Predicate<MemoryData> pred, float time)
```

Parameters

`pred` Predicate<MemoryData>

`time` float

Returns

MemoryData

GetClosestMemoryToTime(float)

```
public MemoryData GetClosestMemoryToTime(float time)
```

Parameters

`time` float

Returns

MemoryData

GetClosestMemoryToTime(string, float)

```
public MemoryData GetClosestMemoryToTime(string actionId, float time)
```

Parameters

`actionId` string

`time` float

Returns

MemoryData

GetClosestMemoryToTime(string, Object, float)

```
public MemoryData GetClosestMemoryToTime(string actionId, Object target, float time)
```

Parameters

`actionId` string

`target` Object

`time` float

Returns

MemoryData

GetClosestMemoryToTime(Type, float)

```
public MemoryData GetClosestMemoryToTime(Type type, float time)
```

Parameters

`type` Type

`time` float

Returns

MemoryData

GetClosestMemoryToTime(Type, Object, float)

```
public MemoryData GetClosestMemoryToTime(Type type, Object target, float time)
```

Parameters

`type` Type

`target` Object

`time` float

Returns

MemoryData

GetClosestMemoryToTime(Type, Object, Object, float)

```
public MemoryData GetClosestMemoryToTime(Type type, Object initiator, Object target,
float time)
```

Parameters

`type` Type

`initiator` Object

`target` Object

`time` float

Returns

MemoryData

GetClosestMemoryToTime(Object, float)

```
public MemoryData GetClosestMemoryToTime(Object initiator, float time)
```

Parameters

`initiator` Object

`time` float

Returns

MemoryData

GetClosestMemoryToTime(Object, Object, float)

```
public MemoryData GetClosestMemoryToTime(Object initiator, Object target, float time)
```

Parameters

`initiator` Object

`target` Object

`time` float

Returns

MemoryData

GetCurrentActionRuntime()

```
public float GetCurrentActionRuntime()
```

Returns

float

GetEverLastingMemoriesByFilter(Predicate<MemoryData>)

```
public List<MemoryData> GetEverLastingMemoriesByFilter(Predicate<MemoryData> pred)
```

Parameters

`pred` Predicate<MemoryData>

Returns

List<MemoryData>

GetNormalMemoriesByFilter(Predicate<MemoryData>)

```
public List<MemoryData> GetNormalMemoriesByFilter(Predicate<MemoryData> pred)
```

Parameters

`pred` Predicate<MemoryData>

Returns

List<MemoryData>

Namespace NoOpArmy.WiseFeline.Remembrance.Demo

Classes

EmotionBoxDemo

NPCBehavior

Enums

MemoryType

This enum is used as memory type so we can find out what each memory is

Class EmotionBoxDemo

Namespace: NoOpArmy.WiseFeline.Remembrance.Demo

Assembly: APIRef.dll

```
public class EmotionBoxDemo : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← EmotionBoxDemo

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) , Component.SendMessageUpwards(string, object, SendMessageOptions) , Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,

Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,

Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Enum MemoryType

Namespace: NoOpArmy.WiseFeline.Remembrance.Demo

Assembly: APIRef.dll

This enum is used as memory type so we can find out what each memory is

```
public enum MemoryType
```

Fields

BoughtStuff = 1

GaveMeFood = 0

GiveProductForFree = 3

HappyToSeeyou = 4

TriedToDamageMe = 2

Class NPCBehavior

Namespace: NoOpArmy.WiseFeline.Remembrance.Demo

Assembly: APIRef.dll

```
public class NPCBehavior : MonoBehaviour, INPCZoneMessageReceiver
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← NPCBehavior

Implements

INPCZoneMessageReceiver

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) ,

Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,

Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

shouldExecutePlayerInput

If true then you can execute actions with 1,2 and 3 on the NPC

```
public bool shouldExecutePlayerInput
```

Field Value

bool

textPrefab

We use this to show what action the NPC has done

```
public GameObject textPrefab
```

Field Value

GameObject

Methods

ExecuteAction(MemoryType, string, GameObject, Color)

```
public void ExecuteAction(MemoryType type, string text, GameObject other, Color textColor)
```

Parameters

`type` MemoryType

`text` string

`other` GameObject

`textColor` Color

OnOtherEnteredZone(GameObject)

```
public void OnOtherEnteredZone(GameObject other)
```

Parameters

`other` GameObject

OnOtherLeftZone(GameObject)

```
public void OnOtherLeftZone(GameObject other)
```

Parameters

`other` GameObject

Namespace NoOpArmy.WiseFeline.SmartObjects

Classes

SlotDefinition

SlotSpecificSmartObjectBehavior
: This class represents a behavior assignment to a smart object for a specific slot

SmartObject
: Attach this component to any object which you want to declare as smart object

SmartObject.FilterResult
: Represents the data which shows if a behavior is supported by an object

SmartObject.SmartObjectSlotData
: Represents runtime data about the status of a slot

SmartObjectBehaviourBase
: Behaviors of smart objects inherit from this class. A smart object behavior operates on the agent which reserves, uses or frees a slot in a smart objects. These events are defined in this base class as virtual methods to override.

SmartObjectDefinition
: Defines the characteristics of a smart object

SmartObjectSlotOwnerSetter
: Sets the owners of the smart object slots to the elements of the array

SmartObjectsManager
: Manages smart objects and allows you to query them based on their behaviors, tags and position An instance of this is created automatically if one cannot be found in the scene

Enums

SlotState
: Different states a slot can be in

SmartObjectsManager.SearchQueryOptions
: Use this enum to customize a search query to return objects which you own or are free or all objects

Class SlotDefinition

Namespace: NoOpArmy.WiseFeline.SmartObjects

Assembly: APIRef.dll

```
[Serializable]
public class SlotDefinition
```

Inheritance

object ← SlotDefinition

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

agentOffsetFromPosition

The agent's ideal position to use the slot from the slot's position

```
[Tooltip("The agent's ideal position to use the slot from the slot's position")]
public Vector3 agentOffsetFromPosition
```

Field Value

Vector3

agentRotation

The agent's ideal rotation to use the slot

```
[Tooltip("The agent's ideal rotation to use the slot")]
public Vector3 agentRotation
```

Field Value

positionOffset

Position offset of the slot relative to the object itself

```
[Tooltip("Position offset of the slot relative to the object itself")]
public Vector3 positionOffset
```

Field Value

Vector3

Class SlotSpecificSmartObjectBehavior

Namespace: NoOpArmy.WiseFeline.SmartObjects

Assembly: APIRef.dll

This class represents a behavior assignment to a smart object for a specific slot

```
[Serializable]
public class SlotSpecificSmartObjectBehavior
```

Inheritance

object ← SlotSpecificSmartObjectBehavior

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

behaviors

The behavior to use

```
public List<SmartObjectBehaviourBase> behaviors
```

Field Value

List<SmartObjectBehaviourBase>

slotIndex

The slot index which uses this behavior

```
public int slotIndex
```

Field Value

int

Enum SlotState

Namespace: NoOpArmy.WiseFeline.SmartObjects

Assembly: APIRef.dll

Different states a slot can be in

```
public enum SlotState
```

Fields

`Claimed = 1`

Reserved by an agent

`Free = 0`

Free to be used by any agent

`InUse = 2`

In use by an agent

Class SmartObject

Namespace: NoOpArmy.WiseFeline.SmartObjects

Assembly: APIRef.dll

Attach this component to any object which you want to declare as smart object

```
public class SmartObject : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← SmartObject

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) ,

Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,

Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

behaviors

The behaviors which all slots offer

```
[Tooltip("The behaviors which all slots offer")]
public List<SmartObjectBehaviourBase> behaviors
```

Field Value

List <SmartObjectBehaviourBase>

definition

The smart object definition this component will use

```
[Tooltip("The smart object definition this component will use")]
public SmartObjectDefinition definition
```

Field Value

SmartObjectDefinition

delayBetweenPositionUpdates

If we should update the position of the octree, how many seconds of delay should be between the updates

```
[Tooltip("If we should update the position of the octree, how many seconds of delay should be between the updates")]
public float delayBetweenPositionUpdates
```

Field Value

float

previousUpdate

The last time this object's position has updated on the octree

```
[HideInInspector]
public float previousUpdate
```

Field Value

float

priorityMultiplier

Multiplies the definition's priority by this number

```
public float priorityMultiplier
```

Field Value

float

slotBehaviors

Slot specific behaviors

```
[Tooltip("Slot specific behaviors")]
public List<SlotSpecificSmartObjectBehavior> slotBehaviors
```

Field Value

List<SlotSpecificSmartObjectBehavior>

tags

Smart object tags

```
[Tooltip("Smart object tags")]
public List<string> tags
```

Field Value

List<string>

updatePositionInOctreeAutomatically

Should the position of this object be updated on the octree. If the object doesn't move this should be false

```
[Tooltip("Should the position of this object be updated on the octree. If the object doesn't
move this should be false")]
public bool updatePositionInOctreeAutomatically
```

Field Value

bool

Methods

AddSlot(SlotDefinition)

Adds a slot to the smart object

```
public void AddSlot(SlotDefinition def)
```

Parameters

`def` SlotDefinition

AddSlot(Vector3, Vector3, Vector3)

Adds a slot to the smart object

```
public void AddSlot(Vector3 positionOffset, Vector3 agentOffsetPosition,
Vector3 agentRotation)
```

Parameters

`positionOffset` Vector3

Offset of the slot from the object's position

`agentOffsetPosition` Vector3

Offset of the ideal agent position from the slot

`agentRotation` Vector3

Idle agent rotation for the slot

Claim(int, GameObject, bool)

Tries to claim the slot for the agent

```
public bool Claim(int slot, GameObject user, bool forceOwnership = false)
```

Parameters

`slot` int

`user` GameObject

`forceOwnership` bool

Returns

bool

Free(int, GameObject, bool)

Frees the object from its agent

```
public bool Free(int slot, GameObject user, bool forceOwnership = false)
```

Parameters

`slot` int

`user` GameObject

`forceOwnership` bool

Returns

bool

Free(GameObject)

Release the first slot owned by the provided user

```
public bool Free(GameObject user)
```

Parameters

`user` GameObject

Returns

bool

Return true if freeing up a slot.

FreeAllSlots()

Frees all slots by their users

```
public void FreeAllSlots()
```

GetAgentSlotPosition(int)

Gets the position and rotation of the agent from the slot

```
public (Vector3 position, Quaternion rotation) GetAgentSlotPosition(int index)
```

Parameters

`index` int

Returns

(Vector3 position, Quaternion rotation)

GetAgentSlotPosition(GameObject)

Gets the position and rotation of the agent from the slot

```
public (Vector3 position, Quaternion rotation) GetAgentSlotPosition(GameObject owner)
```

Parameters

`owner` GameObject

Returns

(Vector3 position, Quaternion rotation)

GetClosestSlotPosition(GameObject, SearchQueryOptions)

Find closest slot position to target based on the search query option

```
public Vector3 GetClosestSlotPosition(GameObject target,
SmartObjectsManager.SearchQueryOptions queryOption)
```

Parameters

`target` GameObject

`queryOption` SmartObjectsManager.SearchQueryOptions

Returns

Vector3

The function returns Vector3.positiveInfinity when no slot meets the search query conditions.

GetFirstOwnedSlotIndex(GameObject)

Returns the first slot we own

```
public int GetFirstOwnedSlotIndex(GameObject potentialOwner)
```

Parameters

`potentialOwner` GameObject

Returns

int

GetFreeSlotIndex()

Finds and returns a free slot

```
public int GetFreeSlotIndex()
```

Returns

int

Index of the first free slot it finds, -1 if no free slot can be found

GetFreeWithoutOwnerSlotIndex()

Finds and returns a free slot

```
public int GetFreeWithoutOwnerSlotIndex()
```

Returns

int

Index of the first free slot it finds, -1 if no free slot can be found

GetOwner(int)

Gets the owner of the object

```
public GameObject GetOwner(int slotIndex)
```

Parameters

`slotIndex` int

Returns

GameObject

GetPriority()

Returns the priority of this object which says how good it can do the job. The value is arbitrary and can be used by your code to determine which of the objects are better to use. This alongside tags and other features can help you choose a better object. for example a ready bomb to demolate somewhere instead of the material to make one and a bomb making kit.

```
public float GetPriority()
```

Returns

float

GetSlotCount()

Returns the number of slots the smart object has

```
public int GetSlotCount()
```

Returns

int

GetSlotData(int)

Gets a copy of the slot's runtime data

```
public SmartObject.SmartObjectSlotData GetSlotData(int index)
```

Parameters

`index` int

Returns

SmartObject.SmartObjectSlotData

GetSlotDefinition(int)

Gets the slot definition

```
public SlotDefinition GetSlotDefinition(int index)
```

Parameters

`index` int

Returns

SlotDefinition

GetSlotDefinition(GameObject)

Gets the slot definition

```
public SlotDefinition GetSlotDefinition(GameObject owner)
```

Parameters

`owner` GameObject

Returns

SlotDefinition

GetSlotPosition(int)

Gets the position of the slot

```
public Vector3 GetSlotPosition(int index)
```

Parameters

`index` int

Returns

Vector3

GetSlots()

returns the slot data of the smart object which you can use to get slot states, claiming/using GameObject and definition

```
public List<SmartObject.SmartObjectSlotData> GetSlots()
```

Returns

List <SmartObject.SmartObjectSlotData>

RemoveSlotAtIndex(int)

Removes a slot from the smart object

```
public void RemoveSlotAtIndex(int index)
```

Parameters

`index` int

SetOwner(GameObject, int)

Sets the owner of the object

```
public void SetOwner(GameObject go, int slotIndex)
```

Parameters

`go` GameObject

`slotIndex` int

SetOwnerOfAllSlots(GameObject)

Sets the owner of all slots to the gameobject

```
public void SetOwnerOfAllSlots(GameObject go)
```

Parameters

`go` GameObject

SupportsBehavior(Type, SearchQueryOptions, GameObject, string[], string[])

Does this object support our requirements of behaviors and tags

```
public SmartObject.FilterResult SupportsBehavior(Type behavior,
SmartObjectsManager.SearchQueryOptions options = SearchQueryOptions.None, GameObject
potentialOwner = null, string[] tagsToInclude = null, string[] tagsToExclude = null)
```

Parameters

`behavior` Type

The behavior type the object should support

`options` SmartObjectsManager.SearchQueryOptions

`potentialOwner` GameObject

`tagsToInclude` string[]

Should we check if the object has any of these tags, pass null

`tagsToExclude` string[]

Tags to exclude so an object with any of these tags doesn't count. Pass null if you don't want to check any tags

Returns

SmartObject.FilterResult

Returns if this object supports the query and if yes is it slot specific and with what slots

SupportsTagFilter(string[], string[])

```
public bool SupportsTagFilter(string[] includeTags, string[] excludeTags)
```

Parameters

`includeTags` string[]

`excludeTags` string[]

Returns

bool

Use(int, GameObject, bool)

Tries to reserve the object for the agent

```
public bool Use(int slot, GameObject user, bool forceOwnership = false)
```

Parameters

`slot` int

`user` GameObject

`forceOwnership` bool

Returns

bool

Class SmartObject.FilterResult

Namespace: NoOpArmy.WiseFeline.SmartObjects

Assembly: APIRef.dll

Represents the data which shows if a behavior is supported by an object

```
public class SmartObject.FilterResult
```

Inheritance

object ← SmartObject.FilterResult

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

isSlotSpecific

Is the support slot specific or all slots support the behavior

```
public bool isSlotSpecific
```

Field Value

bool

isSupported

Is the type supported

```
public bool isSupported
```

Field Value

bool

slots

List of the slots in case of the slot specific support

```
public List<int> slots
```

Field Value

List <int>

Class SmartObject.SmartObjectSlotData

Namespace: NoOpArmy.WiseFeline.SmartObjects

Assembly: APIRef.dll

Represents runtime data about the status of a slot

```
public class SmartObject.SmartObjectSlotData
```

Inheritance

object ← SmartObject.SmartObjectSlotData

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

definition

```
public SlotDefinition definition
```

Field Value

SlotDefinition

owner

Owner of the slot

```
[SerializeField]
public GameObject owner
```

Field Value

GameObject

state

```
public SlotState state
```

Field Value

SlotState

user

```
public GameObject user
```

Field Value

GameObject

Class SmartObjectBehaviourBase

Namespace: NoOpArmy.WiseFeline.SmartObjects

Assembly: APIRef.dll

Behaviors of smart objects inherit from this class. A smart object behavior operates on the agent which reserves, uses or frees a slot in a smart objects. These events are defined in this base class as virtual methods to override.

```
public class SmartObjectBehaviourBase : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← SmartObjectBehaviourBase

Derived

BlackboardSmartObjectBehavior, DrivingBehaviour, EatFoodBehaviour

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) ,

Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() ,
Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) ,
Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() ,
Component.GetComponentIndex() , Component.CompareTag(string) ,
Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,

Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Methods

OnClaimedByAgent(GameObject, SmartObject, int)

Called when a slot of the smart object is successfully claimed by an object

```
public virtual void OnClaimedByAgent(GameObject agent, SmartObject smartObject, int slot)
```

Parameters

`agent` GameObject

`smartObject` SmartObject

`slot` int

OnFreedByAgent(GameObject, SmartObject, int)

Called when an agent frees the smart object

```
public virtual void OnFreedByAgent(GameObject agent, SmartObject smartObject, int slot)
```

Parameters

`agent` GameObject

`smartObject` SmartObject

`slot` int

OnUsedByAgent(GameObject, SmartObject, int)

Called when a slot of the object is successfully used by an agent

```
public virtual void OnUsedByAgent(GameObject agent, SmartObject smartObject, int slot)
```

Parameters

`agent` GameObject

`smartObject` SmartObject

`slot` int

Class SmartObjectDefinition

Namespace: NoOpArmy.WiseFeline.SmartObjects

Assembly: APIRef.dll

Defines the characteristics of a smart object

```
[CreateAssetMenu(menuName = "NoOpArmy/Wise Feline/SmartObject Definition")]
public class SmartObjectDefinition : ScriptableObject
```

Inheritance

object ← Object ← ScriptableObject ← SmartObjectDefinition

Inherited Members

ScriptableObject.SetDirty() , ScriptableObject.CreateInstance(string) , ScriptableObject.CreateInstance(Type) , ScriptableObject.CreateInstance<T>() , Object.GetInstanceID() , Object.GetHashCode() , Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) , Object.InstantiateAsync<T>(T, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) , Object.InstantiateAsync<T>(T, int, Transform) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) , Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, CancellationToken) , Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) , Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>, InstantiateParameters, CancellationToken) , Object.Instantiate(Object, Vector3, Quaternion) , Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) , Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) , Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) , Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) , Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,

Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) , Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) , Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) , Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) , Object.FindObjectsByType(Type, FindObjectsSortMode) , Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) , Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) , Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) , Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() , Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) , Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) , Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) , Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() , Object.FindFirstObjectByType<T>(FindObjectsInactive) , Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) , Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) , Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) , Object.FindFirstObjectByType(Type, FindObjectsInactive) , Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name , Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Fields

priority

How good this smart object is at doing its job

```
[Tooltip("How good this smart object is at doing its job. Agents use this to decide if they should use this object or another")]
public float priority
```

Field Value

float

slots

The slots of the smart object

```
[Tooltip("The slots of the smart object")]
public SlotDefinition[] slots
```

Field Value

SlotDefinition[]

Class SmartObjectSlotOwnerSetter

Namespace: NoOpArmy.WiseFeline.SmartObjects

Assembly: APIRef.dll

Sets the owners of the smart object slots to the elements of the array

```
public class SmartObjectSlotOwnerSetter : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← SmartObjectSlotOwnerSetter

Inherited Members

MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() , Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) , Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() , Component.GetComponentIndex() , Component.CompareTag(string) , Component.CompareTag(TagHandle) ,

Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,
Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,

Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Class SmartObjectsManager

Namespace: NoOpArmy.WiseFeline.SmartObjects

Assembly: APIRef.dll

Manages smart objects and allows you to query them based on their behaviors, tags and position An instance of this is created automatically if one cannot be found in the scene

```
public class SmartObjectsManager : Singleton<SmartObjectsManager>
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← Singleton<SmartObjectsManager> ← SmartObjectsManager

Inherited Members

Singleton<SmartObjectsManager>.Instance , Singleton<SmartObjectsManager>.Awake() , Singleton<SmartObjectsManager>.OnDestroy() , MonoBehaviour.IsInvoking() , MonoBehaviour.CancelInvoke() , MonoBehaviour.Invoke(string, float) , MonoBehaviour.InvokeRepeating(string, float, float) , MonoBehaviour.CancelInvoke(string) , MonoBehaviour.IsInvoking(string) , MonoBehaviour.StartCoroutine(string) , MonoBehaviour.StartCoroutine(string, object) , MonoBehaviour.StartCoroutine(IEnumerator) , MonoBehaviour.StartCoroutine_Auto(IEnumerator) , MonoBehaviour.StopCoroutine(IEnumerator) , MonoBehaviour.StopCoroutine(Coroutine) , MonoBehaviour.StopCoroutine(string) , MonoBehaviour.StopAllCoroutines() , MonoBehaviour.print(object) , MonoBehaviour.destroyCancellationToken , MonoBehaviour.useGUILayout , MonoBehaviour.didStart , MonoBehaviour.didAwake , MonoBehaviour.runInEditMode , Behaviour.enabled , Behaviour.isActiveAndEnabled , Component.GetComponent(Type) , Component.GetComponent<T>() , Component.TryGetComponent(Type, out Component) , Component.TryGetComponent<T>(out T) , Component.GetComponent(string) , Component.GetComponentInChildren(Type, bool) , Component.GetComponentInChildren(Type) , Component.GetComponentInChildren<T>(bool) , Component.GetComponentInChildren<T>() , Component.GetComponentsInChildren(Type, bool) , Component.GetComponentsInChildren(Type) , Component.GetComponentsInChildren<T>(bool) , Component.GetComponentsInChildren<T>(bool, List<T>) , Component.GetComponentsInChildren<T>() , Component.GetComponentsInChildren<T>(List<T>) , Component.GetComponentInParent(Type, bool) , Component.GetComponentInParent(Type) , Component.GetComponentInParent<T>(bool) , Component.GetComponentInParent<T>() , Component.GetComponentsInParent(Type, bool) , Component.GetComponentsInParent(Type) , Component.GetComponentsInParent<T>(bool) , Component.GetComponentsInParent<T>(bool, List<T>) , Component.GetComponentsInParent<T>() ,

Component.GetComponents(Type) , Component.GetComponents(Type, List<Component>) ,
Component.GetComponents<T>(List<T>) , Component.GetComponents<T>() ,
Component.GetComponentIndex() , Component.CompareTag(string) ,
Component.CompareTag(TagHandle) ,
Component.SendMessageUpwards(string, object, SendMessageOptions) ,
Component.SendMessageUpwards(string, object) , Component.SendMessageUpwards(string) ,
Component.SendMessageUpwards(string, SendMessageOptions) ,
Component.SendMessage(string, object) , Component.SendMessage(string) ,
Component.SendMessage(string, object, SendMessageOptions) ,
Component.SendMessage(string, SendMessageOptions) ,
Component.BroadcastMessage(string, object, SendMessageOptions) ,
Component.BroadcastMessage(string, object) , Component.BroadcastMessage(string) ,
Component.BroadcastMessage(string, SendMessageOptions) , Component.transform ,
Component.gameObject , Component.tag , Object.GetInstanceID() , Object.GetHashCode() ,
Object.Equals(object) , Object.InstantiateAsync<T>(T) , Object.InstantiateAsync<T>(T, Transform) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, Transform, Vector3, Quaternion) , Object.InstantiateAsync<T>(T, int) ,
Object.InstantiateAsync<T>(T, int, Transform) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion) ,
Object.InstantiateAsync<T>(T, int, Transform, Vector3, Quaternion, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>) ,
Object.InstantiateAsync<T>(T, int, Transform, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
CancellationToken) ,
Object.InstantiateAsync<T>(T, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, Vector3, Quaternion, InstantiateParameters, CancellationToken) ,
Object.InstantiateAsync<T>(T, int, ReadOnlySpan<Vector3>, ReadOnlySpan<Quaternion>,
InstantiateParameters, CancellationToken) ,
Object.Instantiate(Object, Vector3, Quaternion) ,
Object.Instantiate(Object, Vector3, Quaternion, Transform) , Object.Instantiate(Object) ,
Object.Instantiate(Object, Scene) , Object.Instantiate<T>(T, InstantiateParameters) ,
Object.Instantiate<T>(T, Vector3, Quaternion, InstantiateParameters) ,
Object.Instantiate(Object, Transform) , Object.Instantiate(Object, Transform, bool) ,
Object.Instantiate<T>(T) , Object.Instantiate<T>(T, Vector3, Quaternion) ,
Object.Instantiate<T>(T, Vector3, Quaternion, Transform) , Object.Instantiate<T>(T, Transform) ,
Object.Instantiate<T>(T, Transform, bool) , Object.Destroy(Object, float) , Object.Destroy(Object) ,
Object.DestroyImmediate(Object, bool) , Object.DestroyImmediate(Object) ,

Object.FindObjectsOfType(Type) , Object.FindObjectsOfType(Type, bool) ,
Object.FindObjectsByType(Type, FindObjectsSortMode) ,
Object.FindObjectsByType(Type, FindObjectsInactive, FindObjectsSortMode) ,
Object.DontDestroyOnLoad(Object) , Object.DestroyObject(Object, float) ,
Object.DestroyObject(Object) , Object.FindSceneObjectsOfType(Type) ,
Object.FindObjectsOfTypeIncludingAssets(Type) , Object.FindObjectsOfType<T>() ,
Object.FindObjectsByType<T>(FindObjectsSortMode) , Object.FindObjectsOfType<T>(bool) ,
Object.FindObjectsByType<T>(FindObjectsInactive, FindObjectsSortMode) ,
Object.FindObjectOfType<T>() , Object.FindObjectOfType<T>(bool) ,
Object.FindFirstObjectByType<T>() , Object.FindAnyObjectByType<T>() ,
Object.FindFirstObjectByType<T>(FindObjectsInactive) ,
Object.FindAnyObjectByType<T>(FindObjectsInactive) , Object.FindObjectsOfTypeAll(Type) ,
Object.FindObjectOfType(Type) , Object.FindFirstObjectByType(Type) ,
Object.FindAnyObjectByType(Type) , Object.FindObjectOfType(Type, bool) ,
Object.FindFirstObjectByType(Type, FindObjectsInactive) ,
Object.FindAnyObjectByType(Type, FindObjectsInactive) , Object.ToString() , Object.name ,
Object.hideFlags , object.Equals(object, object) , object.GetType() , object.MemberwiseClone() ,
object.ReferenceEquals(object, object)

Methods

GetSmartObjectsAroundPoint(Vector3, float)

Returns all of the objects in the sphere

```
public SmartObject[] GetSmartObjectsAroundPoint(Vector3 point, float radius)
```

Parameters

`point` Vector3

`radius` float

Returns

SmartObject[]

GetSmartObjectsAroundPoint(Vector3, float, Type, SearchQueryOptions, GameObject, string[], string[])

Get objects which match the query

```
public List<(SmartObject smartObject, SmartObject.FilterResult data)>
GetSmartObjectsAroundPoint(Vector3 point, float radius, Type behavior = null,
SmartObjectsManager.SearchQueryOptions queryOptions = SearchQueryOptions.None, GameObject
potentialOwner = null, string[] tagsToInclude = null, string[] tagsToExclude = null)
```

Parameters

`point` Vector3

`radius` float

`behavior` Type

`queryOptions` SmartObjectsManager.SearchQueryOptions

`potentialOwner` GameObject

`tagsToInclude` string[]

`tagsToExclude` string[]

Returns

List<(SmartObject smartObject, SmartObject.FilterResult data)>

GetSmartObjectsAroundPoint<T>(Vector3, float, SearchQueryOptions, GameObject, string[], string[])

Get objects which match the query

```
public List<(SmartObject smartObject, SmartObject.FilterResult data)>
GetSmartObjectsAroundPoint<T>(Vector3 point, float radius,
SmartObjectsManager.SearchQueryOptions queryOptions = SearchQueryOptions.None, GameObject
potentialOwner = null, string[] tagsToInclude = null, string[] tagsToExclude = null) where T
: SmartObjectBehaviourBase
```

Parameters

`point` Vector3

`radius` float

`queryOptions` SmartObjectsManager.SearchQueryOptions

`potentialOwner` GameObject

`tagsToInclude` string[]

`tagsToExclude` string[]

Returns

List<(SmartObject smartObject, SmartObject.FilterResult data)>

Type Parameters

`T`

Enum SmartObjectsManager.SearchQueryOptions

Namespace: NoOpArmy.WiseFeline.SmartObjects

Assembly: APIRef.dll

Use this enum to customize a search query to return objects which you own or are free or all objects

```
public enum SmartObjectsManager.SearchQueryOptions : byte
```

Fields

`Free = 2`

Returns all objects which meet other criteria of the search and have free slots

`FreeOrOwned = 3`

Returns all objects which meet other criteria of the search and either have free slots or slots you own

`None = 0`

Returns all objects meeting the other criteria of the search

`Owned = 1`

Returns all objects meeting other criteria of the search which at least have a slot which you own